

CHAPTER 3

LANDING GEAR

CHAPTER OVERVIEW

SECTION	TITLE
3-1	Equipment Description And Data
3-2	Troubleshooting
3-3	On Aircraft Inspections
3-4	Maintenance Procedures
3-5	Maintenance Procedures (BENCH)

3-1/(3-2 Blank)

SECTION 3-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
3-1-1	Landing Gear System Description And Data
3-1-2	Parking Brake System Description And Data
3-1-3	Tailwheel Lock System Description And Data

3-1-1 LANDING GEAR SYSTEM DESCRIPTION AND DATA.

3-1-1.1 Landing Gear System.

The landing gear system consists of two fixed main landing gears, one mounted on each side of the helicopter fuselage midsection, a tail landing gear mounted on lower rear section of tail cone, a wheel and self-adjusting brake assembly on each main landing gear, and the landing gear brake system. The landing gear system enables the helicopter to maneuver during ground operations, absorbs landing loads, and insulates the airframe and occupants from shock (Figure 3-1-1-1).

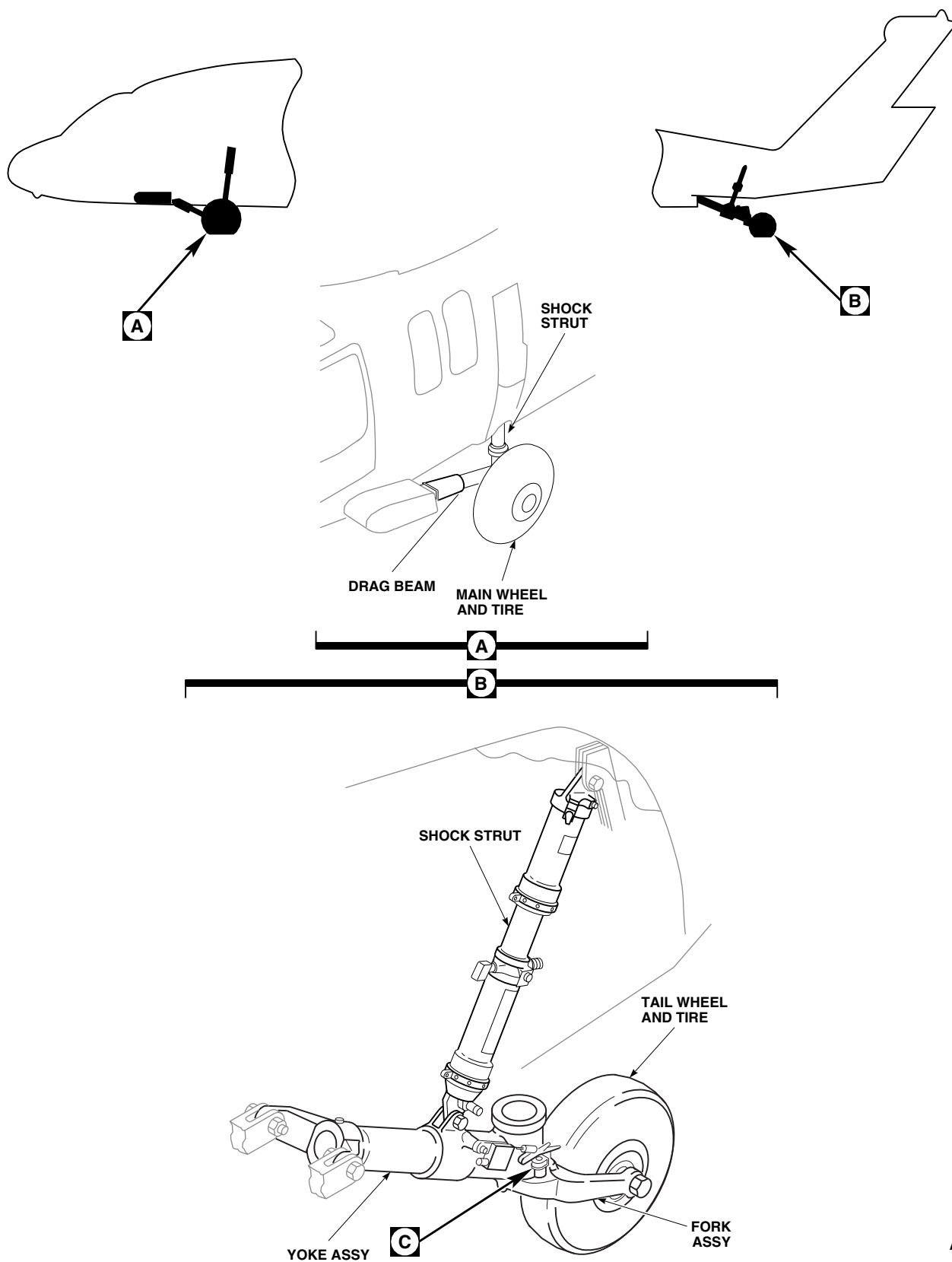
3-1-1.2 Main Landing Gear.

A fixed main landing gear, mounted rear of the crew chief window on each side of the helicopter fuselage midsection, supports the helicopter while it is on the ground. Each main landing gear consists of a shock strut assembly, drag beam and axle assembly, wheel and tire, and hydraulic brake. The shock strut supports the helicopter during ground operations and absorbs impact loads when landing. It consists of two floating pistons, one in the upper cylinder and one in the lower cylinder, which separate nitrogen from hydraulic fluid, and a servicing valve for each floating piston cylinder. The lower cylinder absorbs the normal landing loads. During a hard landing (rate of descent above 600 feet per minute), the upper and lower cylinders work together to absorb the shock, preventing damage to the helicopter. The drag beam and axle is a cylinder attached to the airframe at one end and to the shock strut at the other, which transmits landing loads to the airframe and shock strut, and to which the wheel is attached. Each wheel consists of a 26 x

10.0-11 tubeless tire, a two section wheel rim, bearings, and self-adjusting disc-type brake assembly.

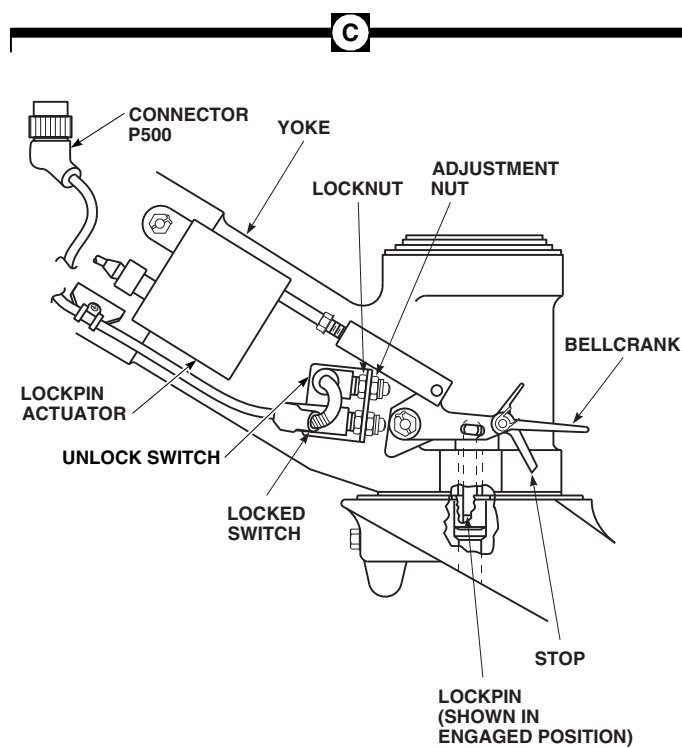
3-1-1.3 Tail Landing Gear.

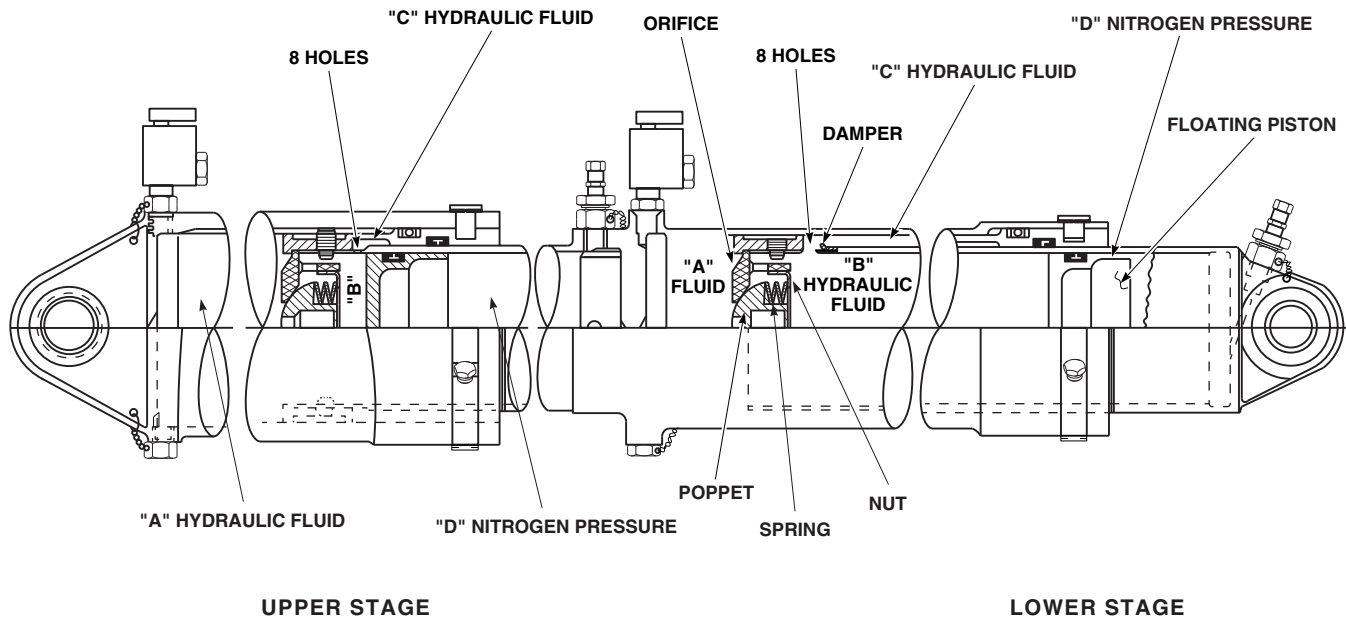
A fixed tail landing gear, secured to structural attachment fittings in the rear tail cone structure, provides rear support of the helicopter (Figure 3-1-1-1). The tail landing gear consists of shock strut, fork assembly, tailwheel lock and actuator, yoke assembly, and wheel and tire. The shock strut supports the helicopter rear structure during ground operations and cushions impact loads when landing. The shock strut is a two-stage air-oil type with two floating pistons separating air from hydraulic fluid, with a servicing valve for each stage. The fork assembly, secured to the yoke assembly, is the attachment point for the tailwheel and allows the wheel to swivel through 360° for ground control. The wheel lock and electrically operated actuator, secured to the yoke assembly, secure the tailwheel in the trailing position when the helicopter is parked or in-flight. Two tailwheel lock switches are installed near the tailwheel lockpin. When the pin is locked or unlocked, one of the switches is actuated, giving a cockpit indication as to position of pin. The unlock switch is located above the lock switch. The yoke assembly, attached to the tail cone structure and the shock strut, transmits landing loads to the helicopter airframe and shock strut. The wheel, installed in the fork assembly, consists of a 15 x 6.00-6 tube type tire, a 6.00-6 tube, and a two piece wheel rim and bearings.



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FIGURE 3-1-1-1. LANDING GEAR SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

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SA**FIGURE 3-1-1-1. LANDING GEAR SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)**



NOTES

1. SHOWN PROPERLY SERVICED WITH WEIGHT OF HELICOPTER ON GEAR.
2. UPPER STRUT SHOWN IN ITS NORMAL POSITION CHAMBERS "C" AND "B" DECREASED TO A MINIMUM.
3. NITROGEN IN CHAMBER "D" PROVIDES SPRING ACTION AS STRUT COMPRESSES.
4. RATE OF COMPRESSION IS REGULATED BY HYDRAULIC FLUID IN CHAMBER "A" BEING FORCED THRU POPPET VALVE INTO CHAMBER "B".
5. RATE OF EXTENSION REGULATED BY HYDRAULIC FLUID CHAMBER "C" BEING FORCED INTO CHAMBERS "B" AND "A".

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FIGURE 3-1-1-2. LANDING GEAR SHOCK STRUT - CUTAWAY

3-1-2 PARKING BRAKE SYSTEM DESCRIPTION AND DATA.

3-1-2.1 Parking Brake System.

- a. The parking brake system is actuated by a parking brake switch located on the pilot's side of the center console (Figure 3-1-2-1). The parking brake system prevents the helicopter from moving when parked.
 - b. This system consists of two brake controls on each pilot's and copilot's directional control pedals, four master brake cylinders, two slave mixer valves, a parking brake handle, and two wheel brakes. When the pilot's or copilot's brake pedals are pressed, hydraulic pressure builds up in the master brake cylinder attached to the pedal. The pressurized hydraulic fluid flows from the master brake cylinder through the slave mixer valves and the parking brake valve below the cockpit floor, to each brake at each main landing gear wheel.
 - c. During the very short duration of helicopter braking, kinetic energy is transformed into heat by friction. This heat is stored in the disk stack, which has enough mass to absorb heat without reaching very high temperatures. High temperatures can cause torque decay, high wear rate, warping, dishing, and cracking. It can also cause deterioration of piston seals and hydraulic brake fluid. Heat is dissipated by the rotating and stationary disks and the wear pads. There is friction material on all rubbing surfaces of the brake.
- b. The master cylinders have two displacement pistons. During the first portion of piston travel, a large volume of fluid is displaced. This is at low pressure to initiate brake puck motion. During the remaining piston travel, the output volume decreases and brake pressure increases. A poppet valve provides a smooth transition from low pressure to high pressure. This is basically a relief valve.
 - c. When the pilot presses on the right brake pedal, pressure from the master brake cylinder is applied to the slave mixer valve. The piston in the slave mixer valve is pushed to the left, blocking the copilot pressure port. Fluid, under pressure, is forced out the slave mixer valve pressure port to the parking brake valve and wheel brake cylinder (Figure 3-1-2-2).
 - d. There are two wear indicator devices built into the brake cylinder housing. They are the press-to-test pin type. The wear indicator devices are adjusted so that when they are pressed flush with the bushing the discs need replacing.

3-1-2.1.1 Electrical Power.

Electrical power of 28 vdc for the parking brake system is supplied from the No. 1 dc primary bus through the LIGHTS ADVSY circuit breaker on the copilot's circuit breaker panel. When the parking brake handle is pulled up, a microswitch closes thus lighting the PARKING BRAKE ON advisory capsule on the caution/advisory panel.

3-1-2.1.2 Master Brake Cylinder.

- a. There are four identical master brake cylinders. There are also two pistons operating manually in series and in two separate stages. Reservoir

and pistons function from the same fluid connection in either flow direction. The first stage operating pressure is 75 psi. The second stage operating pressure is 800 psi. Maximum operating pressure is 1020 psi.

3-1-2.1.3 Slave Mixer Valve.

The slave mixer valve contains a piston and spring-loaded pin, and is located between the master brake cylinders and the parking brake valve. It controls hydraulic flow to the wheel brakes from either the pilot or copilot pedals. The slave mixer valve isolates the fluid in the copilot brake system from the pilot brake system. A bleed plug on the slave mixer valve allows air to be bled from the brake lines.

3-1-2.1.4 Brake Assembly.

The brake assembly consists of two steel rotating discs, brake pads, parking brake valve, and a housing that contains the pistons. The brake has six pistons symmetrically located about the vertical

centerline, operating within a forged aluminum cylinder housing. The brake assembly also contains a heat sink.

3-1-2.1.4.1 Brake Discs.

Three rotating discs are keyed to and driven by the wheel. There are also two stationary discs, which are keyed to the brake torque tube. The discs are 0.140 inch thick with about 0.010 inch of friction material sintered to each rubbing surface. The rotating discs are sandwiched between the two stationary discs and 12 one-inch diameter steel pads. The steel pads have sintered friction material wearing surfaces. The stationary discs are between the three rotating discs. Like the rotor discs, they are also a one-piece construction. The discs have a high static friction coefficient for parking in bad conditions. The discs will shrink with usage and are subject to adverse thermal stress. The discs are key-slotted to engage the drive keys of the main wheel and have relief slots and terminal holes to minimize excessive dishing, warping, and cracking. The disc turns the throat between the brake housing. The brake's rotating disc surface can be seen at the cutout area of the cylinder housing. This provides continuous indication of the brake's condition.

3-1-2.1.4.2 Brake Pads.

Brake pads are fitted into recesses in the housing and pistons. One wear brake pad is at each of the six piston positions. Also, one wear pad is recessed into the backing plate at six locations, which are opposite the piston positions. The pistons produce braking action by clamping the discs between the brake pads. The torque tubes and backplate are separate pieces, in order to provide improved accessibility to the friction discs.

3-1-2.1.4.3 Parking Brake Valve.

The parking brake valve is actuated by the parking brake handle on the pilot's side of the center console. The parking brake valve then traps fluid pressure to the brakes and closes a microswitch, lighting the PARKING BRAKE ON advisory light on the caution/advisory panel. Pressing the brake pedals produces pressure in the master cylinder to actuate the parking brake valve release mechanism

and release the parking brake. Thermal compensation is provided in the parking brake valve when the brakes are at PARK, and in the master brake cylinders when the brakes are OFF. The parking brake handle is connected to linkage, which rotates a camshaft in the parking brake valve. The pin from chamber G blocks movement of this linkage unless pressure is applied to chamber G. This pressure will retract the pin, allowing the linkage free movement. The camshaft has two positions. With the parking brake on, valves in chambers C and D are open and valves in chambers A and B are closed. With the parking brake off, valves in chambers C and D are closed and valves in chambers A and B are open (Figure 3-1-2-2).

3-1-2.1.4.4 Brake Pistons.

The pistons are hollow and contain cylindrically-shaped, high-strength phenolic insulators. These provide a socket for containing wear pads. They are needed to shield the piston and fluid from the high pad temperature. When the copilot presses the left brake pedal, the piston in the master brake cylinder applies pressure in the circuit. This pressure will cause the left brake piston to apply pressure to the slave mixer valve. The slave mixer valve contains a spring-loaded piston and pin. The piston is forced to the right by pressure from the copilot master brake cylinder. This creates pressure in the right side of the slave mixer valve. The pressure is prevented from being applied up to the pilot left pedal master brake cylinder by the pin (Figure 3-1-2-3). The pin closes off the pilot slave mixer valve pressure port. Pressure from the slave mixer valve is then applied out to the parking brake valve, and left wheel brake. The copilot's right brake and slave mixer valves operate the same way.

3-1-2.1.5 Setting the Parking Brake Valve.

With the parking brake off, and the right and left pedals applied, hydraulic lines are pressurized and both brakes are applied. Chamber G is also pressurized to retract the pin, allowing free movement of the parking brake linkage. When the parking brake handle is raised, the camshaft is turned. This closes valves in chambers A and B and opens valves in chambers C and D. The parking brake switch lights the PARKING BRAKE ON capsule on the caution/advisory panel when the parking

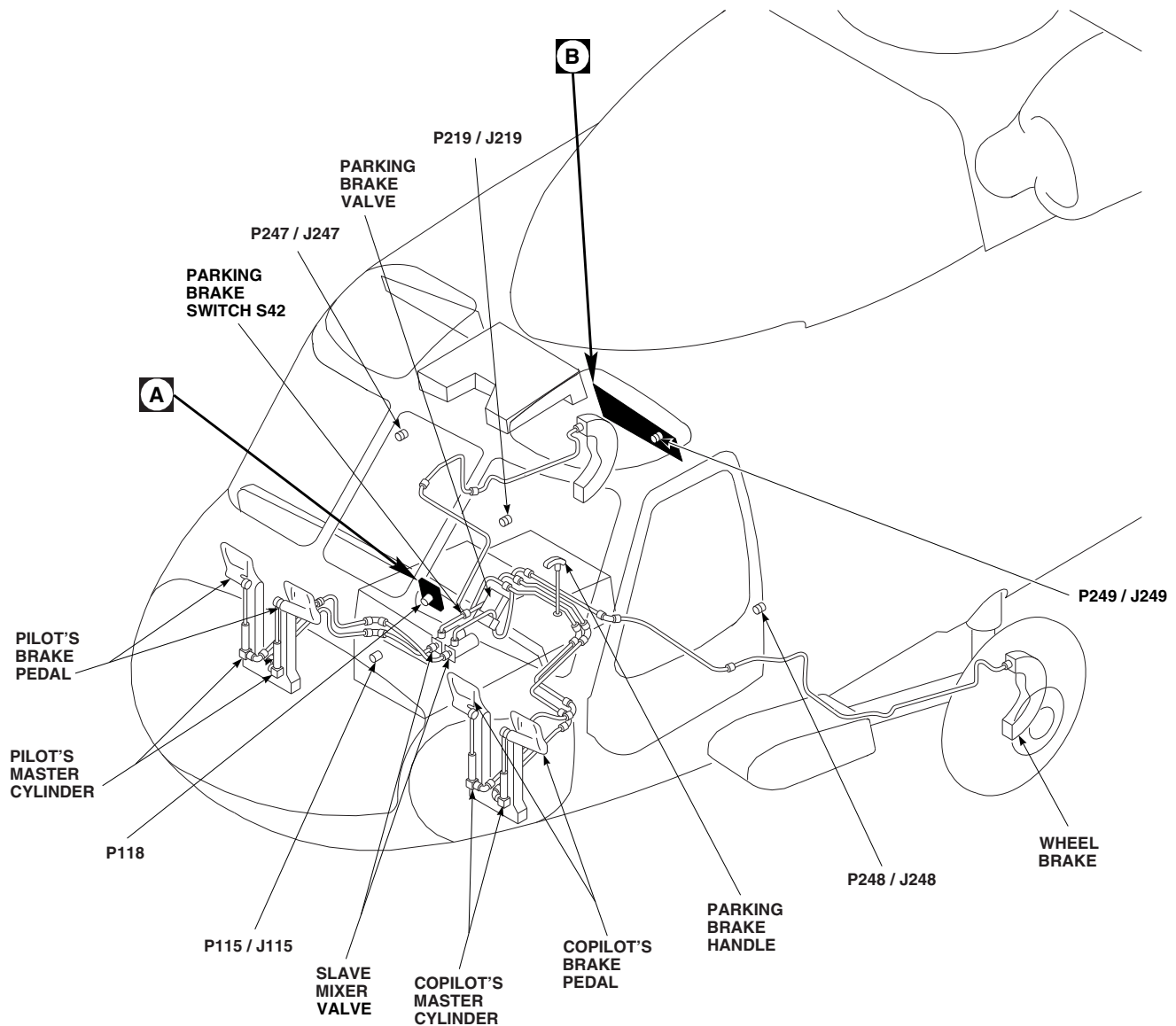
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brake handle is raised. Releasing the pedals allows the pin to extend and locks the linkage in place.

3-1-2.1.6 Releasing the Parking Brake Valve.

When the left brake pedal is applied, chamber G is pressurized, retracting the pin and allowing the

parking brake to release. The camshaft then turns, allowing chamber A and B to be open for normal braking. Chambers E and F act as small accumulators to compensate for expansion and contraction in the parking brake hydraulic circuit.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P115 / J115	COCKPIT CONSOLE BL 7.5 RH, STA 210
P118	CAUTION / ADVISORY PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH

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FIGURE 3-1-2-1. PARKING BRAKE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

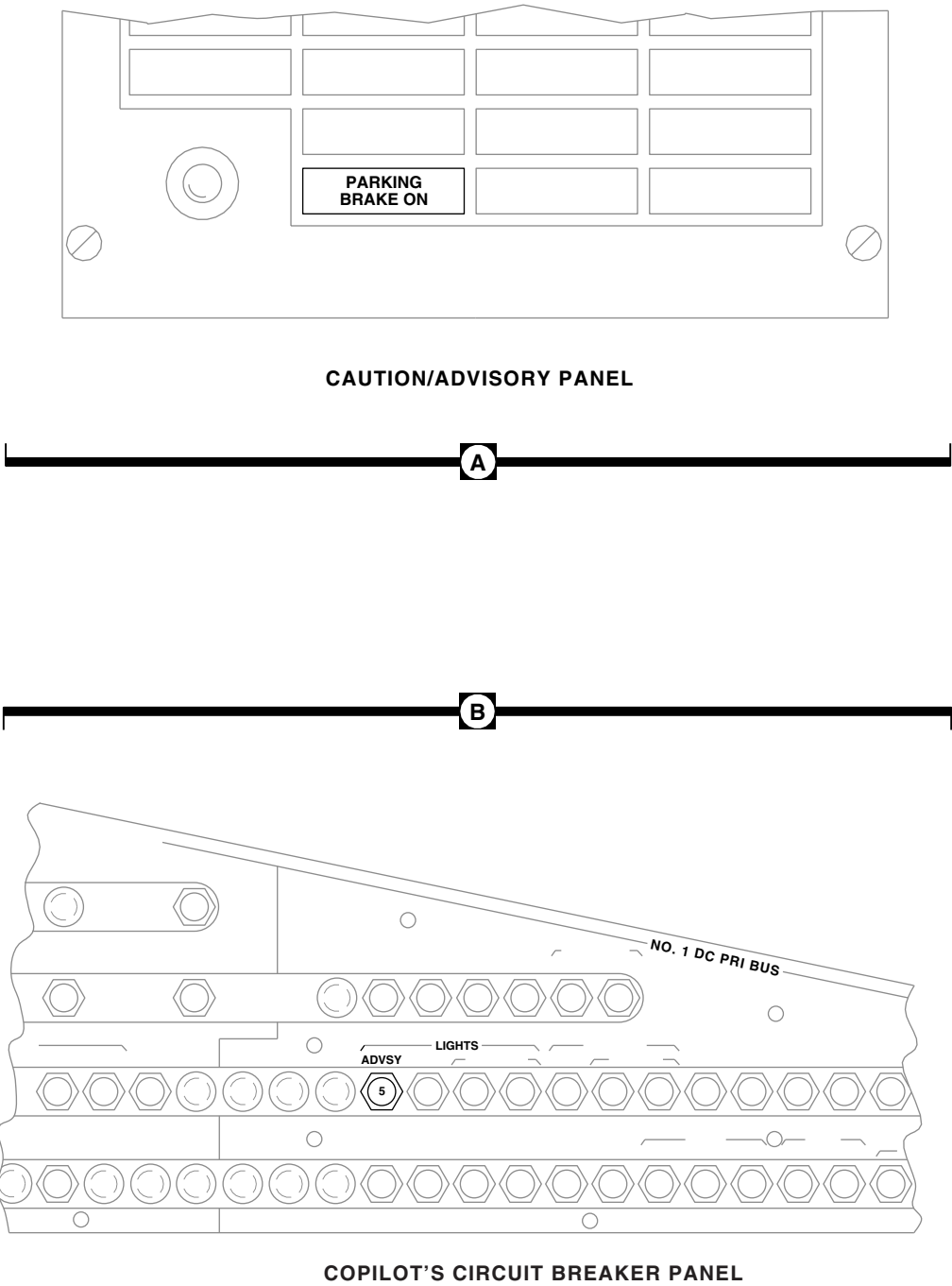


FIGURE 3-1-2-1. PARKING BRAKE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

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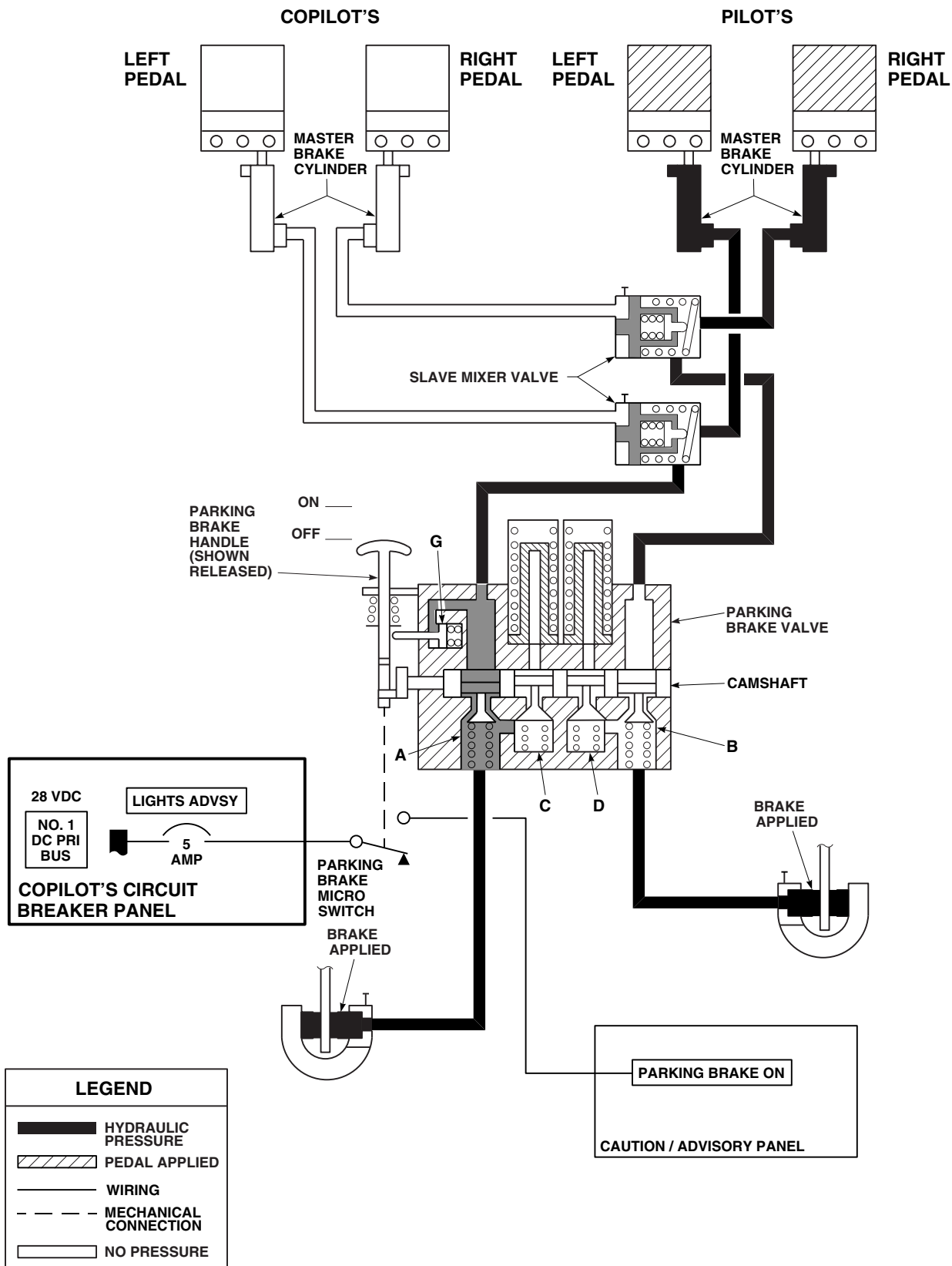


FIGURE 3-1-2-2. PARKING BRAKE SYSTEM BLOCK DIAGRAM

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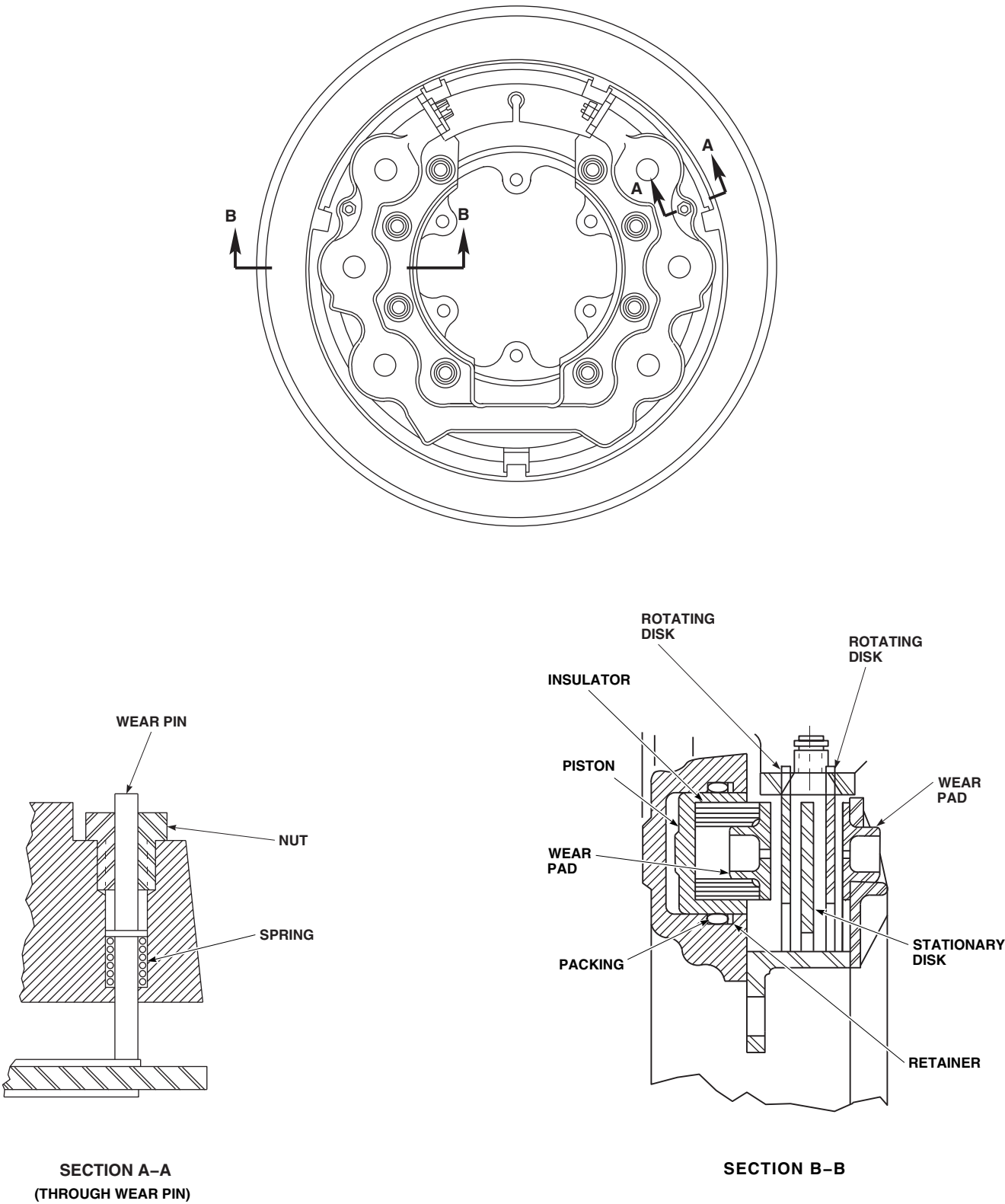


FIGURE 3-1-2-3. WHEEL BRAKES - CUTAWAY

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3-1-3 TAILWHEEL LOCK SYSTEM DESCRIPTION AND DATA.

3-1-3.1 Tailwheel Lock System Overview.

The tailwheel lock system consists of a tailwheel lock actuator, bellcrank, and lock and unlock switches. The tailwheel lock system enables the helicopter to maneuver during ground operations, (Figure 3-1-3-1).

3-1-3.2 Tailwheel Lock System.

3-1-3.2.1 Power Distribution.

The dc essential bus supplies 28 vdc through the TAIL WHEEL LOCK circuit breaker to the tailwheel switch in the miscellaneous relay panel which then routes 28 vdc to the tailwheel lock actuator.

3-1-3.2.2 Manual Tailwheel Unlock Operation.

The manual tailwheel unlock on the tail landing gear provides a way of unlocking the tailwheel should the tailwheel lock actuator fail. This is done by positioning a manual override lever down. This allows the use of a tow bar or steering bar. If the

locking actuator fails, pushing the lever up locks the tailwheel. This allows the helicopter to be parked in a secured position.

3-1-3.2.3 Tailwheel Lock Actuator.

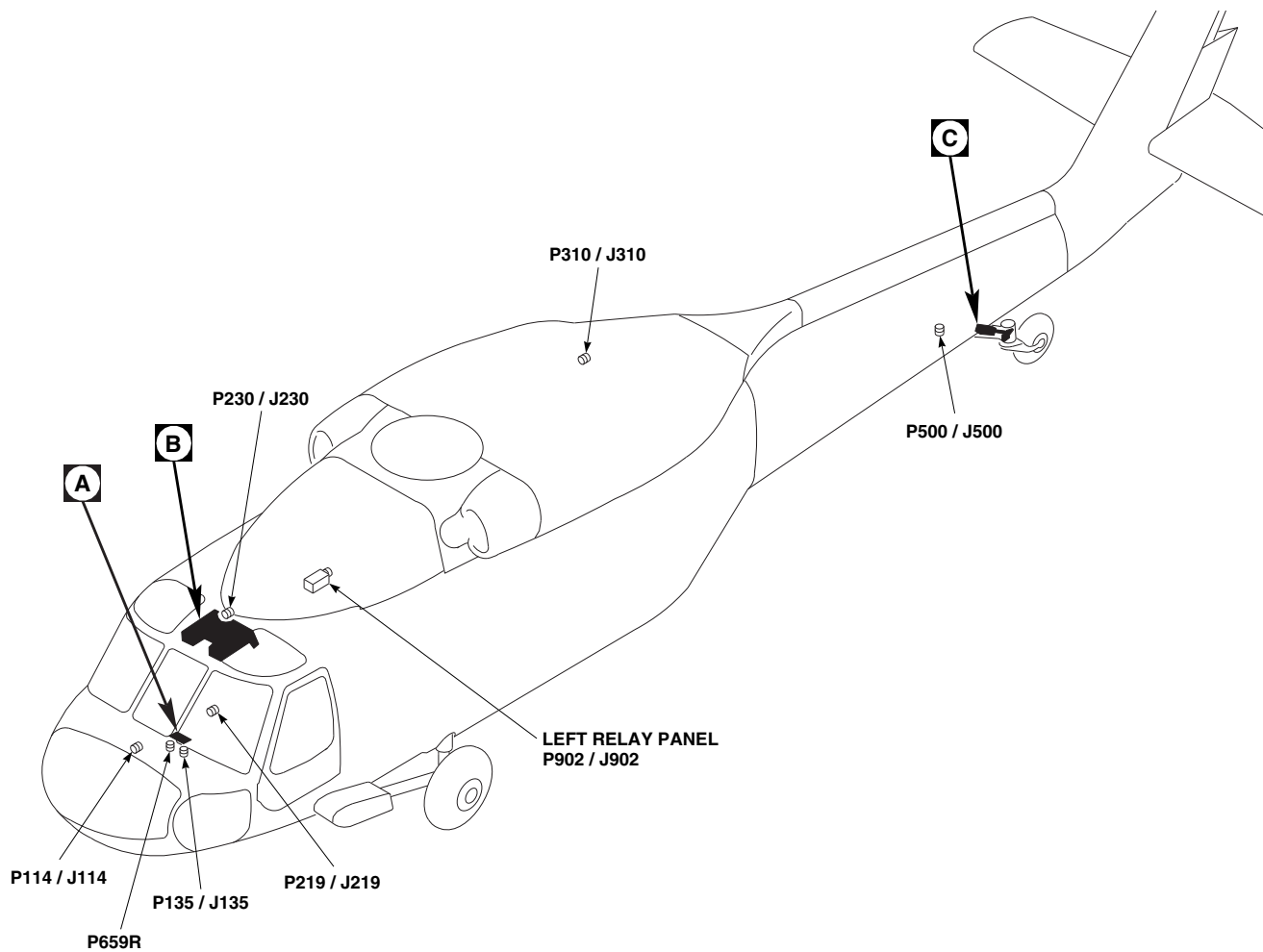
The tailwheel lock actuator, on the tailwheel strut, is controlled through the use of the tailwheel lock switch. The tailwheel lock actuator's extension or retraction causes the tailwheel lockpin to move.

3-1-3.2.4 Tailwheel Lockpin.

The tailwheel lockpin, on the tailwheel strut, is used to lock or unlock the tailwheel. The lockpin is controlled by the tailwheel lock actuator. With the tailwheel lock actuator retracted, the lockpin is unlocked. With the tailwheel lock actuator extended, the lockpin is locked.

3-1-3.2.5 Tailwheel Switch.

The TAIL WHEEL switch on the miscellaneous switch panel has two positions, UNLK - LOCK. The switch is used to control the tailwheel lock switch S32, the unlock switch S48, and the tailwheel lock actuator.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P114 / J114	COCKPIT BL 8 RH, STA 200
P135 / J135	MISCELLANEOUS SWITCH PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P310 / J310	CABIN CEILING BL 16.7 RH, STA 390
P500 / J500	TAIL CONE BL 0, STA 609
P507	TAIL WHEEL LOCK ACTUATOR
P659R / J659R	MISCELLANEOUS SWITCH PANEL
P902 / J902	LEFT RELAY PANEL

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FIGURE 3-1-3-1. TAILWHEEL LOCK SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

3-1-3-2

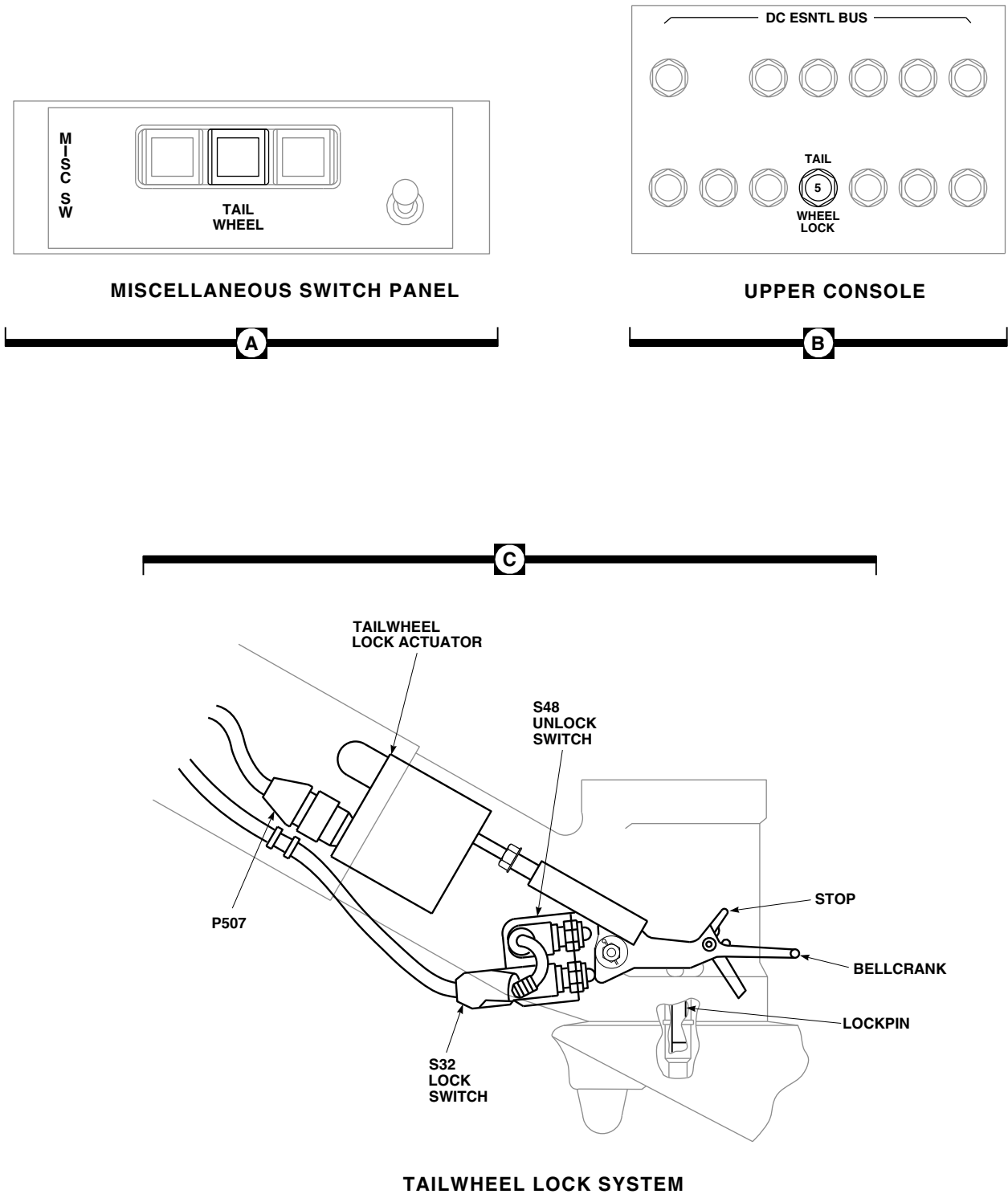


FIGURE 3-1-3-1. TAILWHEEL LOCK SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

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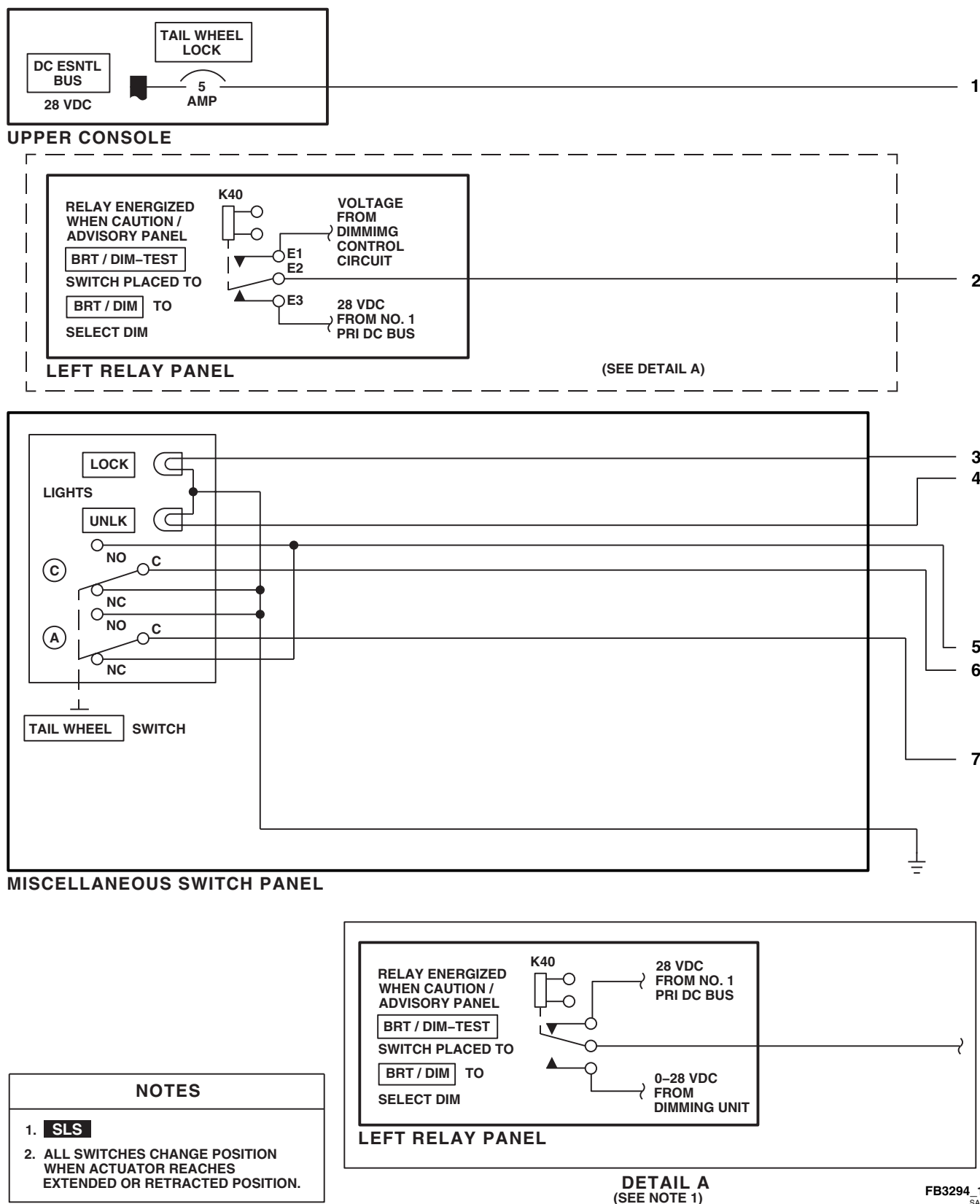
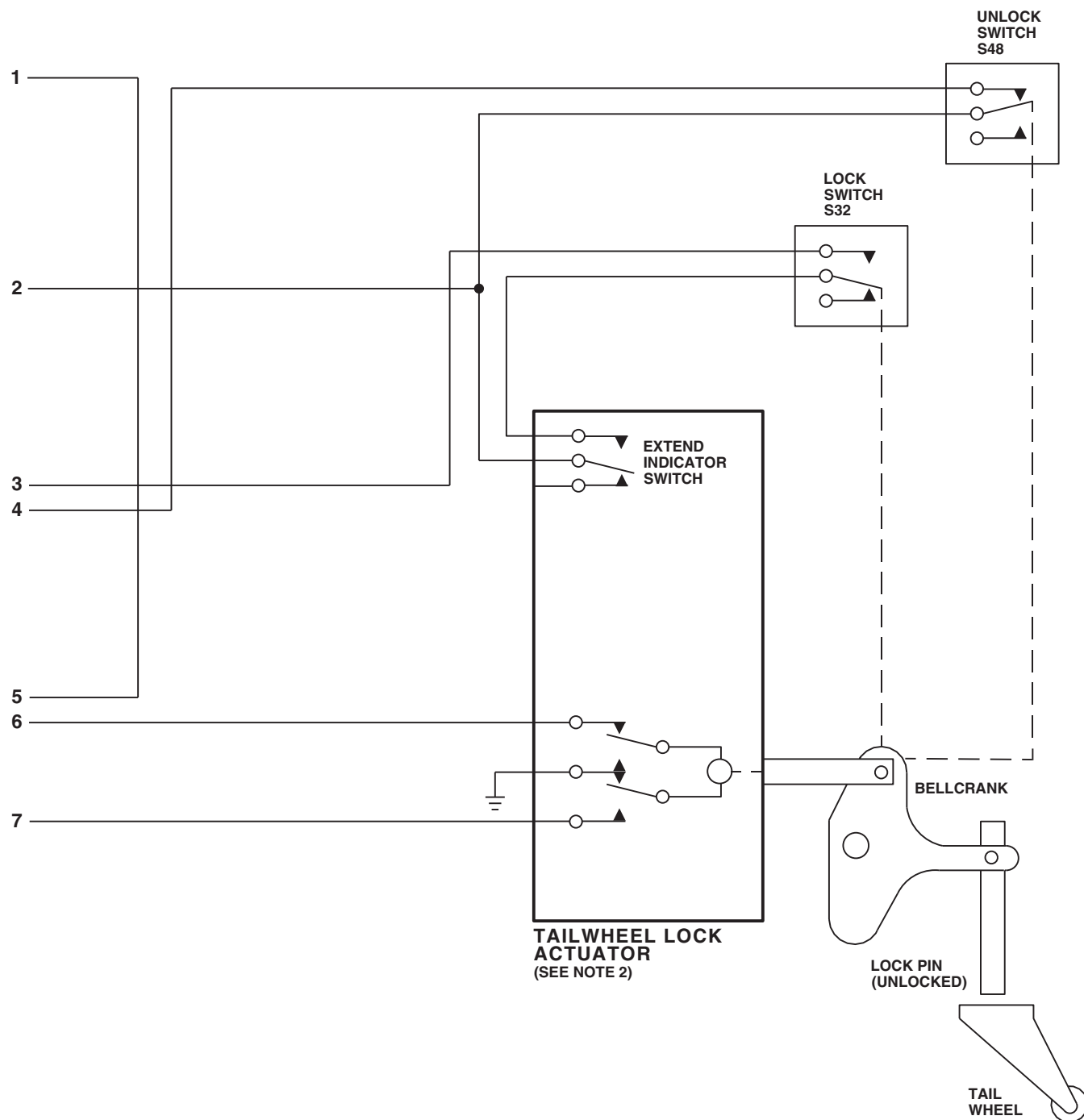


FIGURE 3-1-3-2. TAILWHEEL LOCK SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)



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FIGURE 3-1-3-2. TAILWHEEL LOCK SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

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SECTION 3-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
3-2-1	Parking Brake System
3-2-2	Tailwheel Lock System

3-2-1 PARKING BRAKE SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Power Available And/Or APU Operational

NOTE

If a circuit breaker pops out during the Parking Brake System Operational/Troubleshooting Procedure, a short circuit is indicated (PARA 9-2-3).

3-2-1.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Make sure wheels of helicopter are chocked.
- Turn on electrical power (PARA 1-6-16).
- Momentarily press top of pilot's or copilot's yaw pedals to release parking brake.

RESULT: Pedals shall have little movement and feel firm.

- If result is not as specified, check for air in Parking Brake System or leaks.
 - If air is present, bleed Parking Brake System (PARA 3-4-20).
 - If leaks are present, repair/replace leaking component (PARA 3-4-14 through 3-4-19).
- If yaw pedals bottom out (no resistance felt), go to TABLE NO. 3-2-1-1.

- Manually disengage tailwheel lock.

- Press and hold top of pilot's yaw pedals.
- Pull PARKING BRAKE handle up.
- Release yaw pedals.

RESULT:

- PARKING BRAKE ON capsule (caution/advisory panel) shall go on.
 - PARKING BRAKE handle shall stay up.
- If result (1) is not as specified, go to TABLE NO. 3-2-1-2.
 - If result (2) is not as specified, check linkage between handle and valve.

CAUTION

To prevent damage to helicopter, have personnel sit in pilot's seat to be a safety brakeman.

- Remove right wheel chock.
- Push side of tail cone near tailwheel.

RESULT: Helicopter shall not move.

- If result is not as specified, go to TABLE NO. 3-2-1-3.

k. Chock right wheel.

l. Press and hold top of pilot's yaw pedals.

m. Pull PARKING BRAKE handle up.

n. Release yaw pedals.

o. Momentarily press top of pilot's left yaw pedal.

RESULT:

(1) PARKING BRAKE ON capsule on caution/advisory panel shall go off.

(2) PARKING BRAKE handle shall move down.

(3) Brakes shall release.

- If result (1) is not as specified, replace parking brake switch (PARA 3-4-18).
- If result (2) is not as specified, check that linkage is intact and does not bind at cockpit floor and console, repair/replace as necessary (PARA 3-4-21).
- If trouble remains, replace parking brake valve (PARA 3-4-17).
- If result (3) is not as specified, check for air in Parking Brake System or leaks.
 - (a) If air is present, bleed Parking Brake System (PARA 3-4-20).
 - (b) If leaks are present, repair/replace leaking component (PARA 3-4-14 through 3-4-19).

p. Press and hold top of copilot's yaw pedals.

q. Pull PARKING BRAKE handle up.

r. Release yaw pedals.

s. Momentarily press top of copilot's left yaw pedal.

RESULT:

(1) PARKING BRAKE ON capsule on caution/advisory panel shall go off.

(2) PARKING BRAKE handle shall move down.

(3) Brakes shall release.

- If result (1) is not as specified, replace parking brake switch (PARA 3-4-18).
- If result (2) is not as specified, check that linkage is intact and does not bind at cockpit floor and console, repair/replace as necessary (PARA 3-4-21).
- If trouble remains, replace parking brake valve (PARA 3-4-17).
- If result (3) is not as specified, check for air in Parking Brake System or leaks.
 - (a) If air is present, bleed Parking Brake System (PARA 3-4-20).
 - (b) If leaks are present, repair/replace leaking component (PARA 3-4-14 through 3-4-19).

t. Manually engage tailwheel lock.

u. Turn off electrical power (PARA 1-6-16).

TABLE NO. 3-2-1-1. Brake Pedals Bottom Out (No Resistance Felt).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check lines and Parking Brake System components for leaks.	Step 1. If leaks are found, repair/replace components as necessary (PARA 3-4-14 through 3-4-19). Go to 3. Step 2. If leaks are not found, go to 2.
2. Check master cylinders for proper fluid level.	Step 1. If level is proper, replace master cylinders as required (PARA 3-4-15). Go to 3. Step 2. If level is not proper, service master cylinders as required (PARA 3-4-20). Go to 3.
3. Procedure completed.	

TABLE NO. 3-2-1-2. PARKING BRAKE ON Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Make sure linkage between brake handle and parking brake valve is installed.	Step 1. If linkage installed, go to 2. Step 2. If linkage not installed, connect linkage as required (PARA 3-4-21). Go to 9.
2. Place and hold BRT/DIM-TEST switch on caution/advisory panel to TEST.	Go to 3.
3. Check that PARKING BRAKE ON capsule goes on.	Step 1. If capsule goes on, go to 4. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9.
4. Remove and check Pin Filter Adapter (PFA) P118 (PARA 9-4-98).	Step 1. If PFA is good, go to 5. Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.

TABLE NO. 3-2-1-2. PARKING BRAKE ON Capsule Does Not Go On. (Cont)

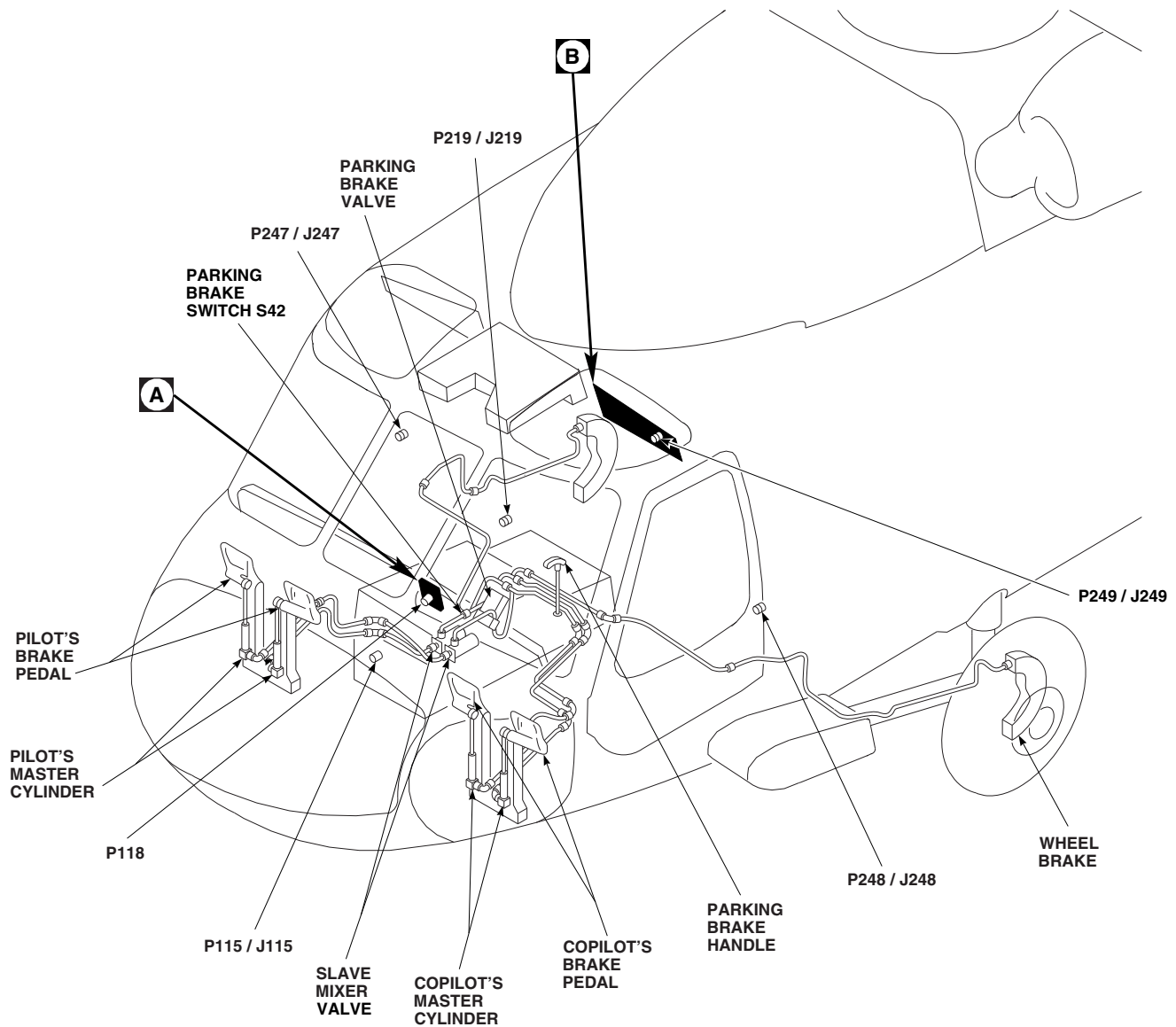
TEST OR INSPECTION	CORRECTIVE ACTION
	5. Install PFA (PARA 9-4-98). Go to 6.
	6. Check for 28 VDC between P118-H and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9.
	Step 2. If voltage is not as specified, go to 7.
	7. Check for 28 VDC between terminal 1 (LIGHTS ADVSY circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.
	Step 1. If voltage is as specified, go to 8.
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
	8. Check continuity between: S42-COM and terminal 1 (LIGHTS ADVSY circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) P118-H and S42-NC
	Step 1. If continuity is present, replace parking brake valve switch (PARA 3-4-18). Go to 9.
	Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 9.
	9. Procedure completed.

TABLE NO. 3-2-1-3. Brakes Do Not Hold.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check lines and system components for leaks.
	Step 1. If leaks are found, repair/replace components as necessary (PARA 3-4-14 through 3-4-19). Go to 4.
	Step 2. If leaks are not found, go to 2.
	2. Push wear indicator pin in on wheel brake housing. Go to 3.

TABLE NO. 3-2-1-3. Brakes Do Not Hold. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Is wear pin flush with housing?	Step 1. If wear pin is flush with housing, replace brake pads (PARA 3-4-14). Go to 4. Step 2. If wear pin is not flush with housing, replace parking brake valve (PARA 3-4-17). Go to 4.
4. Procedure completed.	



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P115 / J115	COCKPIT CONSOLE BL 7.5 RH, STA 210
P118	CAUTION / ADVISORY PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH

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FIGURE 3-2-1-1. PARKING BRAKE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

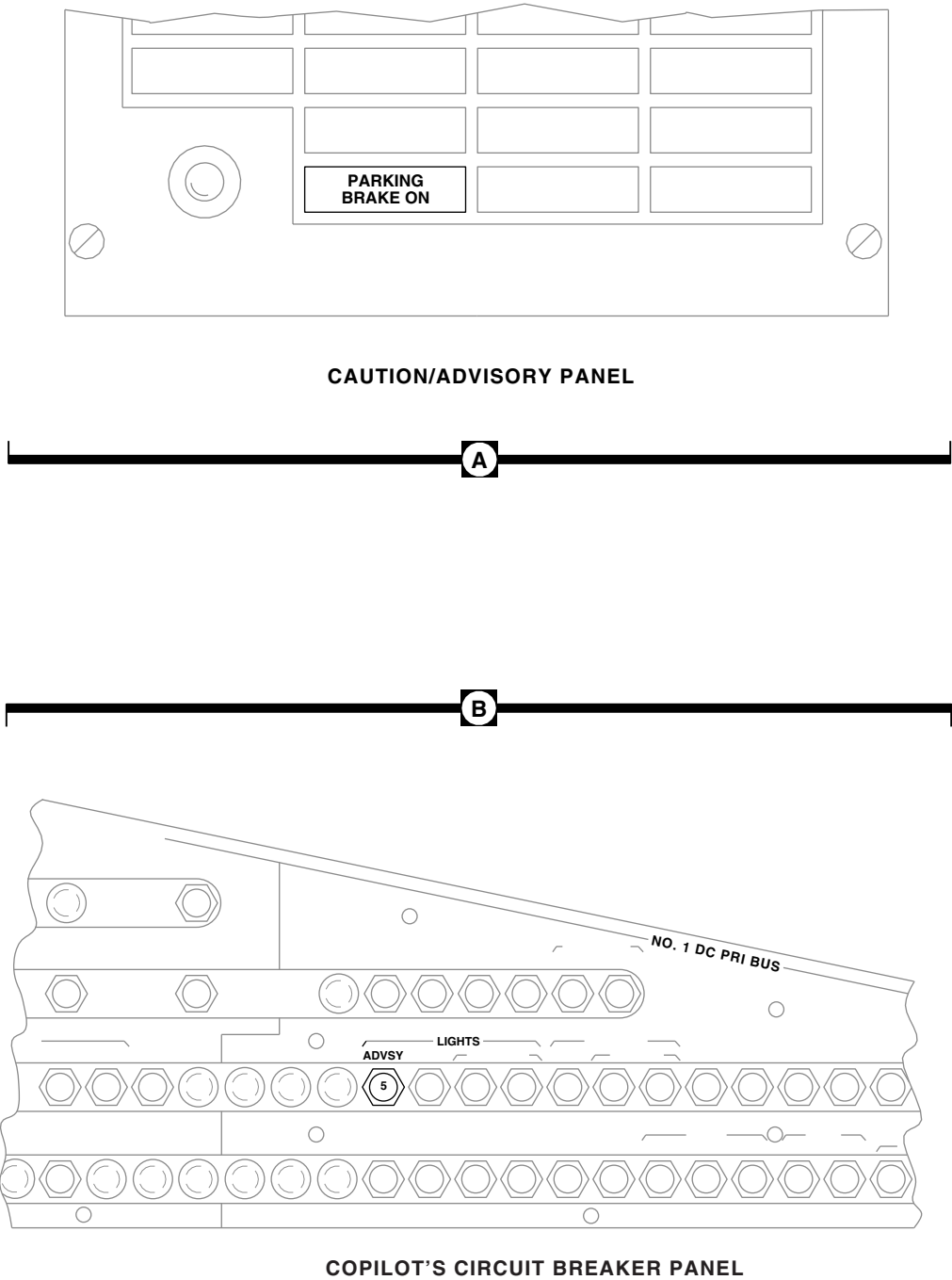


FIGURE 3-2-1-1. PARKING BRAKE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

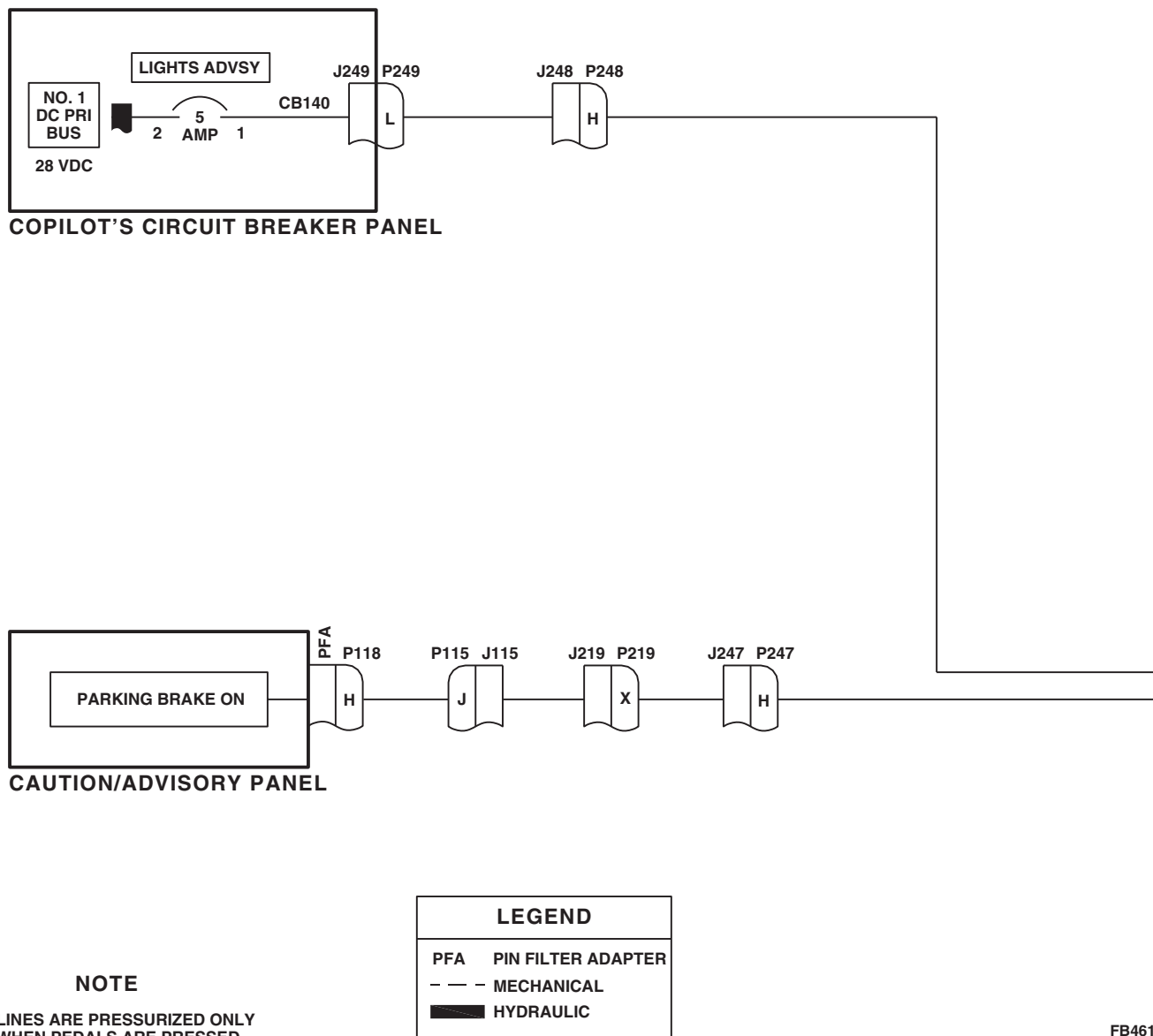


FIGURE 3-2-1-2. PARKING BRAKE SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

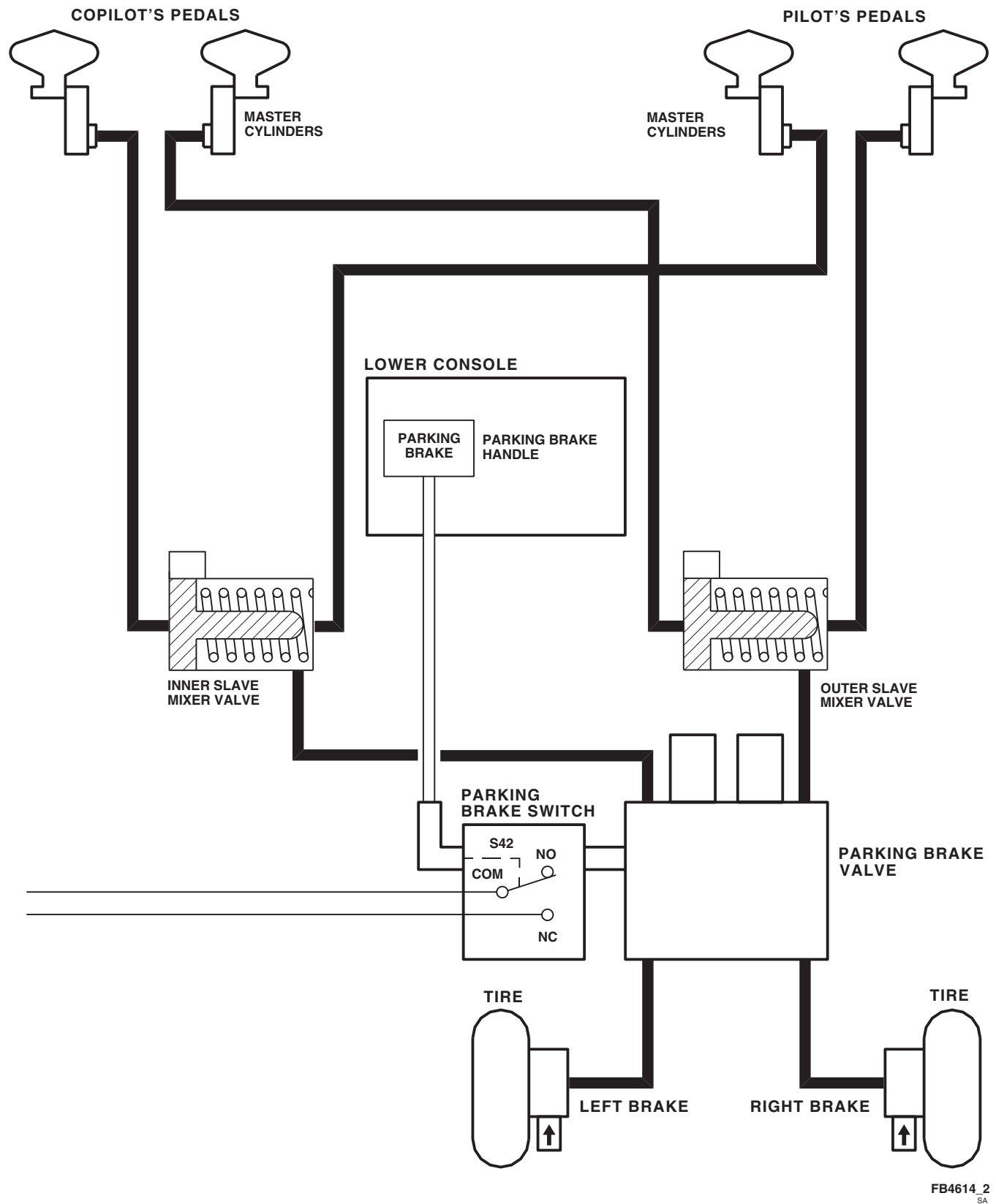


FIGURE 3-2-1-2. PARKING BRAKE SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

3-2-2 TAILWHEEL LOCK SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available Or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

3-2-2.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

NOTE

- Battery power (BATT switch ON) will only operate tailwheel lock actuator. TAIL WHEEL switch light will not go on.
- Make sure stop is left in position as shown in Figure 3-2-2-1, or pin will not go up or down.

- Visually check to see if lockpin is lined up with hole.

RESULT: Lockpin shall be lined up with hole.

- If result is not as specified, push bellcrank down and push side of tail cone as required to line up hole and pin.
- Turn on electrical power (PARA 1-6-16).

- Check miscellaneous switch panel and position of tailwheel lockpin for agreement.
- Push TAIL WHEEL switch on miscellaneous switch panel.

RESULT:

- Tailwheel lockpin shall engage or disengage.
 - TAIL WHEEL switch shall likewise indicate LOCK or UNLK.
- If tailwheel lockpin does not disengage and TAIL WHEEL switch indicates LOCK, go to TABLE NO. 3-2-2-1.
 - If tailwheel lockpin disengages, but TAIL WHEEL switch indicates LOCK, go to TABLE NO. 3-2-2-2.
 - If tailwheel lockpin does not engage and TAIL WHEEL switch indicates UNLK, go to TABLE NO. 3-2-2-3.
 - If tailwheel lockpin engages, but TAIL WHEEL switch indicates UNLK, go to TABLE NO. 3-2-2-4.

- Turn off electrical power (PARA 1-6-16).

TABLE NO. 3-2-2-1. Tailwheel Lockpin Does Not Disengage And TAIL WHEEL Switch Indicates LOCK.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check actuator for proper position (retracted).	
Step 1. If actuator in proper position, go to 2 . Step 2. If actuator not in proper position, go to 6 .	
2. Check linkage from actuator to lockpin for proper installation or freedom from damage.	
Step 1. If linkage is good, go to 3 . Step 2. If linkage is not good, replace/install linkage or spring as required (PARA 3-4-27). Go to 11 .	
3. Check that lockpin becomes disengaged using manual release.	
Step 1. If lockpin becomes disengaged, check lockpin for damage or hole for signs of corrosion. Repair/replace as necessary (PARA 3-4-27). Go to 11 . Step 2. If lockpin does not become disengaged, go to 4 .	
4. Push side of tail cone. Go to 5.	
5. Check that lockpin disengages.	
Step 1. If lockpin disengages, system is operational. Go to 11 . Step 2. If lockpin does not disengage, replace lockpin (PARA 3-4-27). Go to 11 .	
6. Check for 28 vdc between P507-B and P507-C.	
Step 1. If voltage is as specified, replace tailwheel lock actuator (PARA 3-4-27). Go to 11 . Step 2. If voltage is not as specified, go to 7 .	
7. Check for 28 vdc between P507-B and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P507-C and ground (Appendix F). Go to 11 . Step 2. If voltage is not as specified, go to 8 .	
8. Check continuity between J135-A and J135-D.	
Step 1. If continuity is present, go to 9 .	

TABLE NO. 3-2-2-1. Tailwheel Lockpin Does Not Disengage And TAIL WHEEL Switch Indicates LOCK. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 2. If continuity is not present, replace miscellaneous switch panel (PARA 9-4-7). Go to 11 .	
9. Check for 28 vdc between P135-A and ground.	
Step 1. If voltage is as specified, repair/replace wiring between (Appendix F), then go to 11 :	
P135-D and P507-B	
P135-F and P507-A	
Step 2. If voltage is not as specified, go to 10 .	
10. Check for 28 vdc between terminal 1 (TAIL WHEEL LOCK circuit breaker) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P135-A (Appendix F). Go to 11 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 11 .	
11. Procedure completed.	

TABLE NO. 3-2-2-2. Tailwheel Lockpin Disengages, But TAIL WHEEL Switch Indicates LOCK.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P135-G and P659R-H.	
Step 1. If voltage is as specified, replace miscellaneous switch panel (PARA 9-4-7). Go to 6 .	
Step 2. If voltage is not as specified, go to 2 .	
2. Check for 28 vdc between P135-G and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P659R-H and ground (Appendix F). Go to 6 .	
Step 2. If voltage is not as specified, go to 3 .	
3. Check continuity between P135-G and J500-6.	

TABLE NO. 3-2-2-2. Tailwheel Lockpin Disengages, But TAIL WHEEL Switch Indicates LOCK. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
4. Check for 28 vdc between J500-4 and ground.	Step 1. If voltage is as specified, replace electrical cable (PARA 3-4-25). Go to 6. Step 2. If voltage is not as specified, go to 5.
5. Check for 28 vdc between J902-X and ground.	Step 1. If voltage is as specified, repair/replace wiring, as required between (Appendix F), then go to 6: P902-X and J500-4 P902-X and J500-12 Step 2. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming system (PARA 9-2-10). Go to 6.
6. Procedure completed.	

TABLE NO. 3-2-2-3. Tailwheel Lockpin Does Not Engage And TAIL WHEEL Switch Indicates UNLK.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check actuator for proper position (extended).	Step 1. If actuator is extended, go to 2. Step 2. If actuator is not extended, go to 5.
2. Check linkage from actuator to lockpin for proper installation or freedom from damage.	Step 1. If linkage is good, go to 3. Step 2. If linkage is not good, repair/replace parts as required (PARA 3-4-27). Go to 8.
3. Push side of tail cone to line up lockpin with hole. Go to 4.	
4. Check that lockpin engages.	

**TABLE NO. 3-2-2-3. Tailwheel Lockpin Does Not Engage And TAIL WHEEL Switch Indicates UNLK.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If lockpin engages, system is operational. Go to 8 .
	Step 2. If lockpin does not engage, check lockpin hole for damage. Replace lockpin if necessary (PARA 3-4-27). Go to 8 .
5. Check for 28 vdc between P507-A and P507-C.	Step 1. If voltage is as specified, replace tailwheel lock actuator (PARA 3-4-27). Go to 8 . Step 2. If voltage is not as specified, go to 6 .
6. Check continuity between P507-C and ground.	Step 1. If continuity is present, go to 7 . Step 2. If continuity is not present, repair/replace wiring between J500-9 and ground (Appendix F). Go to 8 . Step 3. If trouble remains, replace electrical cable (PARA 3-4-25). Go to 8 .
7. Check continuity between P135-F and P507-A.	Step 1. If continuity is present, replace miscellaneous switch panel (PARA 9-4-7). Go to 8 . Step 2. If continuity is not present, repair/replace wiring between P135-F and J500-7 (Appendix F). Go to 8 . Step 3. If trouble remains, replace electrical cable (PARA 3-4-25). Go to 8 .
8. Procedure completed.	

TABLE NO. 3-2-2-4. Tailwheel Lockpin Engages, But TAIL WHEEL Switch Indicates UNLK.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J507-J and J507-H.	Step 1. If continuity is present, go to 2 . Step 2. If continuity is not present, replace tailwheel lock actuator (PARA 3-4-27). Go to 4 .

TABLE NO. 3-2-2-4. Tailwheel Lockpin Engages, But TAIL WHEEL Switch Indicates UNLK. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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2. Check for 28 vdc between P507-J and ground.

Step 1. If voltage is as specified, go to **3**.

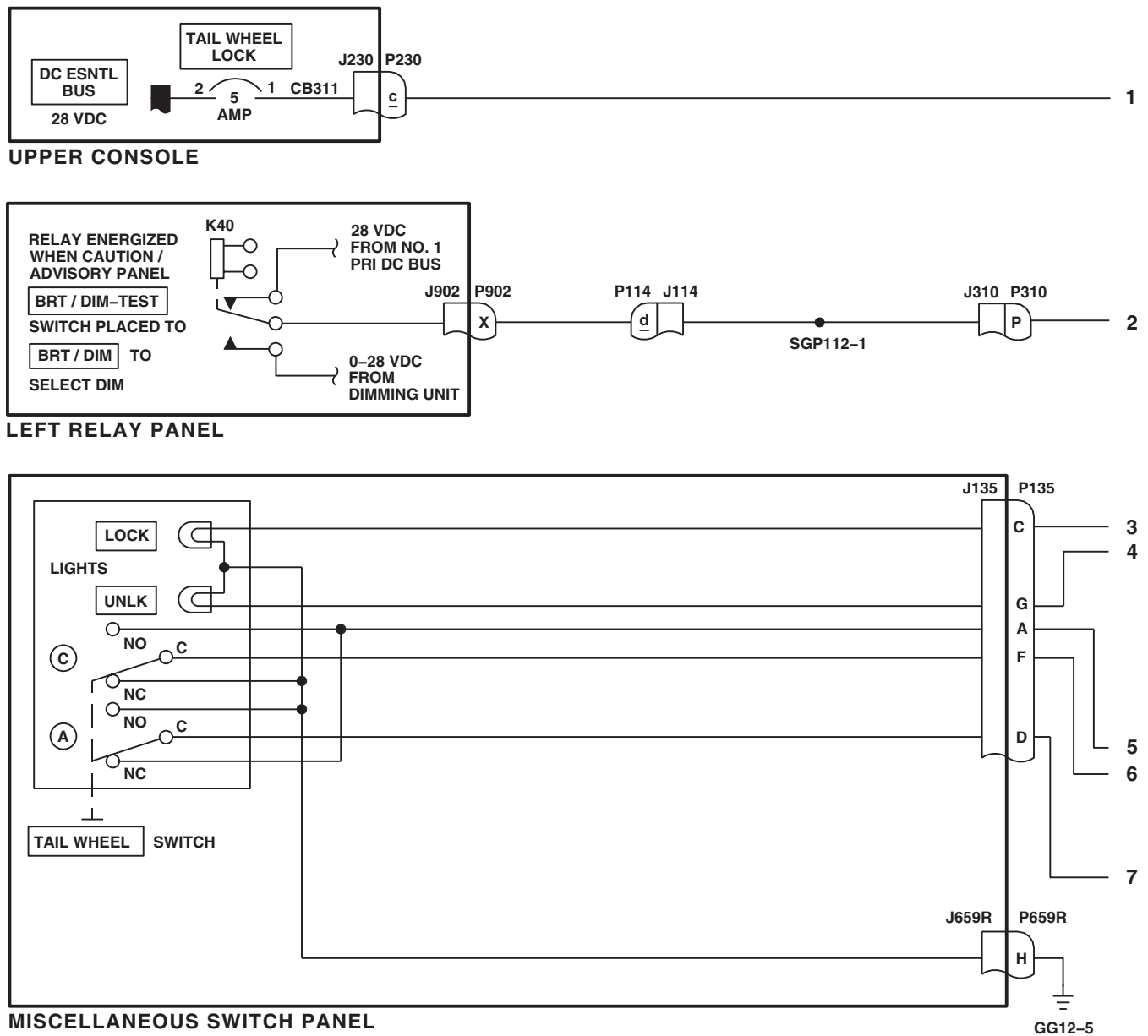
Step 2. If voltage is not as specified, go to **TABLE NO. 3-2-2-2**.

3. Check continuity between P500-1 and P500-3.

Step 1. If continuity is present, repair/replace wiring between J500-1 and J500-11 (Appendix F). Go to **4**.

Step 2. If continuity is not present, replace electrical cable (PARA 3-4-25). Go to **4**.

4. Procedure completed.



MISCELLANEOUS SWITCH PANEL

NOTE

ALL SWITCHES CHANGE POSITION WHEN ACTUATOR REACHES EXTENDED OR RETRACTED POSITION.

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FIGURE 3-2-2-1. TAILWHEEL LOCK SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

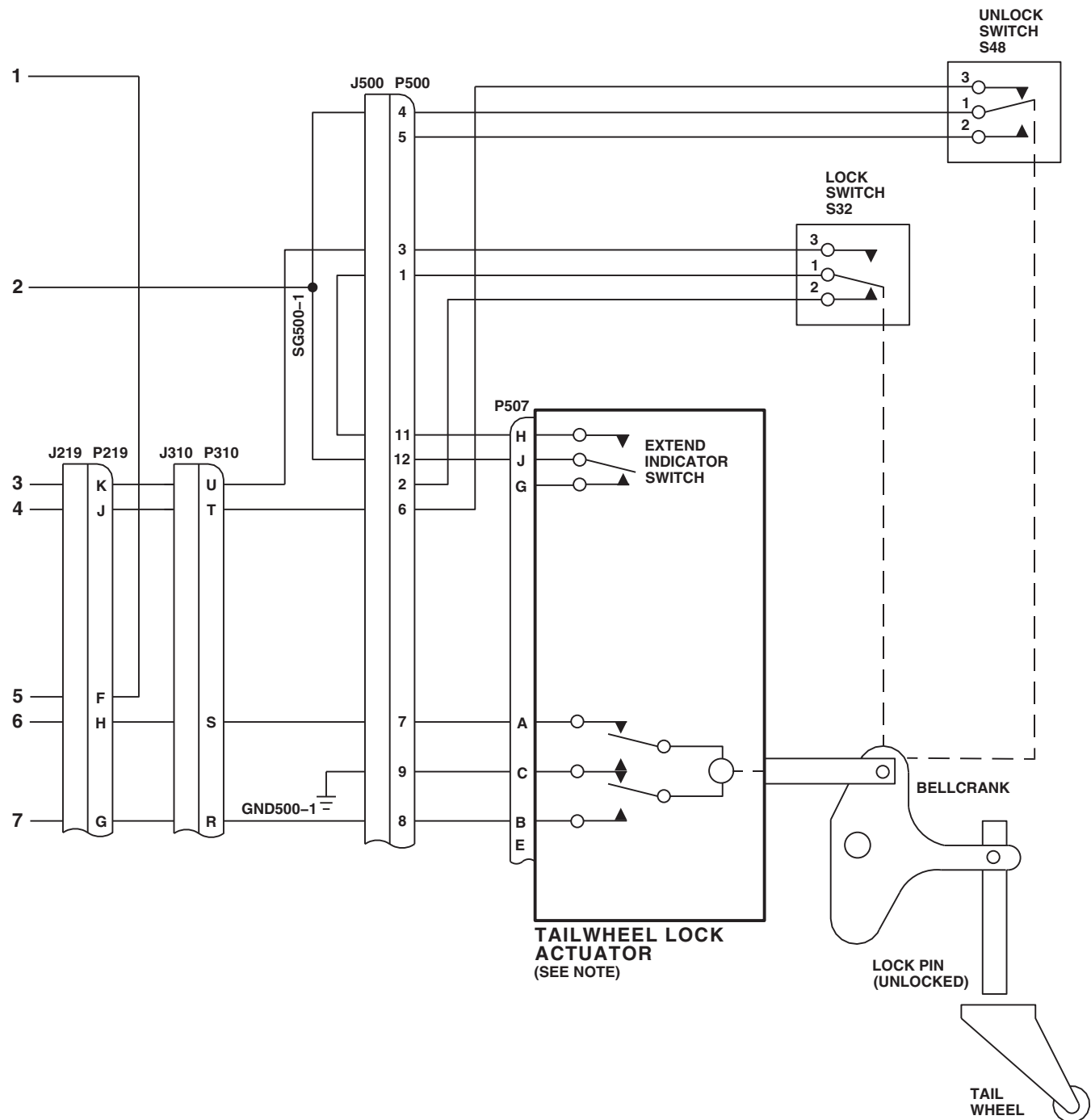
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FIGURE 3-2-2-1. TAILWHEEL LOCK SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

3-2-2-8

SECTION 3-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
3-3-1	Main Landing Gear Drag Beam and Axle Inspection
3-3-2	Main Landing Gear Inspection Shock Strut Quick Service Check
3-3-3	Main Landing Gear Tire Inspection
3-3-4	Main Landing Gear Wheel Brake Inspection
3-3-5	Parking Brake Rod Inspection
3-3-6	Tail Landing Gear Yoke Inspection
3-3-7	Tail Landing Gear Tire Inspection

3-3-1/(3-3-2 Blank)

3-3-1 MAIN LANDING GEAR DRAG BEAM AND AXLE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-3-1.1	Main Landing Gear Drag Beam Inspection	3-3-1-1	3-3-1.2	Main Landing Gear Axle Inspection	3-3-1-4

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D

- Cleaning Compound, Solvent, Item 99, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Sealing Compound, Item 295A, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 3-4-1

3-3-1.1 Main Landing Gear Drag Beam Inspection.



FSCAP

Shotpeening of exterior surfaces of main landing gear drag beam is a critical characteristic. Cadmium-plating and baked resin coating of axle bore are critical characteristics. Use extreme care to

avoid damage to these surfaces during repair of main landing gear axle.

- Inspect drag beam pivot pin holes, attachment bolthole, and jackpad hole for nicks, pits, or scratches. **NONE ALLOWED.** If damage is found, replace main landing gear drag beam (PARA 3-4-1).
- On helicopters with main landing gear, 70250-12050-047, -049, -052, and -053, inspect area around jackpad and tiedown ring mounting holes as follows (Figure 3-3-1-1, Detail A):

NOTE

Make sure area is clean prior to inspection. Sealing compound is not to extend more than 0.380-inch beyond edge of washers found under jackpad and retaining nut.

- (1) If excess sealing compound is found, remove excess compound using nonmetallic scraper.

NOTE

A crack will appear as a thin line in painted surface radiating outwards from under the bead of sealing compound.

- (2) Using 10X magnifying glass, visually inspect 3-inch radius around upper and lower jackpad holes for evidence of cracking. If crack is suspected, fluorescent penetrant inspect using fluorescent penetrant inspection kit (ASTM E1417). **NONE ALLOWED.** If crack is found, replace main landing gear drag beam (PARA 3-4-1).
- c. Inspect main landing gear drag beam for dents. **ROUND SMOOTH DENTS MUST NOT EXCEED 1-INCH IN DIAMETER AND 0.010-INCH DEEP.** If dents exceed limits, replace main landing gear drag beam (PARA 3-4-1).
- d. Inspect main landing gear drag beam for nicks, pits, and scratches. **NONE ALLOWED.** If necessary, using abrasive cloth, Item 8, Appendix D, repair as follows:

CAUTION

Damage to drag beam will result if machine grinder is used for repairs. Machine grinder will go beyond repair limits. Hand polish all repairs to drag beam.

- (1) Repair drag beam area, which includes fuselage attachment lugs, as follows:

- (a) **BLEND NICKS, PITS, OR SCRATCHES UP TO 0.010-INCH DEEP. NO MORE THAN 1 REPAIR ON EACH LUG ALLOWED. BLEND NO MORE THAN 20 % OF ORIGINAL MATERIAL THICKNESS.**

- (b) If repairs exceed limits, replace main landing gear drag beam (PARA 3-4-1).

- (2) Repair drag beam main body as follows:

- (a) **BLEND LOCAL NICKS, PITS, OR SCRATCHES UP TO 0.010-INCH DEEP. REWORK AREA IS LIMITED TO 20 % OF TOTAL AREA OF SURFACE WHERE IMPERFECTIONS ARE LOCATED.**

- (b) **BLEND SCRATCHES WHICH ENCIRCLE DRAG BEAM MAIN BODY NO MORE THAN 20% OF DRAG BEAM CIRCUMFERENCE UP TO 0.003-INCH DEEP.**

- (c) **BLEND SCRATCHES PARALLEL TO CENTERLINE NO MORE THAN 3-INCHES LONG UP TO 0.003-INCH DEEP.**

- (d) If repairs exceed limits, replace main landing gear drag beam (PARA 3-4-1).

NOTE

Repair of diameter that includes jackpad on drag beam main body is only applicable for helicopters with main landing gear, 70250-12050-047, -049, -052, and -053.

- (3) Repair drag beam in diameter that includes jackpad as follows:

- (a) **BLEND NICKS, PITS, OR SCRATCHES UP TO 0.015-INCH DEEP.**

- (b) If repairs exceed limit, replace main landing gear drag beam (PARA 3-4-1).
- (4) On helicopters with main landing gear, 70250-12050-047, -049, -052, and -053, repair drag beam flat surfaces adjacent to jackpad holes as follows:
 - (a) **BLEND NICKS, PITS, OR SCRATCHES UP TO 0.005-INCH DEEP.**
 - (b) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect reworked area for cracks. **NONE ALLOWED.**
 - (c) Inspect drag beam flat surfaces for worn plating. If worn plating is found, apply two coats of epoxy primer, using epoxy primer coating kit, Item 135, Appendix D.
 - (d) If repairs exceed limits, replace main landing gear drag beam (PARA 3-4-1).
- (5) Repair drag beam area which includes shock strut attachment lugs as follows:
 - (a) **BLEND NICKS, PITS, OR SCRATCHES UP TO 0.010-INCH DEEP. NO MORE THAN 1 REPAIR ON EACH LUG ALLOWED.**
 - (b) If repairs exceed limits, replace main landing gear drag beam (PARA 3-4-1).
- (6) Using abrasive cloth, Item 8, Appendix D, remove reddish-brown corrosion without evidence of pitting, from drag beam. Blend edges of repair. **EVIDENCE OF PITTING IS CAUSE FOR REMOVAL OF DRAG BEAM.**
- e. **TOTAL SUM OF REPAIRED AREAS ALLOWED ON DRAG BEAM MUST NOT EXCEED 3 SQUARE-INCHES.** If repairs exceed limit, replace main landing gear drag beam (PARA 3-4-1).
- f. Using 1-inch masking tape, Item 212, Appendix D, mask pivot pin holes, attachment bolthole, and jackpad hole.
- g. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean repaired areas.
- h. Using corrosion compound, Item 108, Appendix D, and acid brush, Item 84, Appendix D, apply compound to repaired areas. Allow to dry for 30 seconds. Wipe dry with machinery towel, Item 211, Appendix D.
- i. Touch up repaired areas (PARA 2-6-5).
- j. On helicopters with main landing gear, 70250-12050-047, -049, -052, and -053, inspect jackpad nut torque stripe for slippage. If slippage has occurred, torque nut as follows:
 - (1) Using nonmetallic scraper, remove sealing compound from nut and surrounding area (upper surface of main landing gear drag beam).

NOTE

Make sure to restrain jackpad prior to loosening nut. Do not allow jackpad to rotate. If movement occurs, sealing compound must be removed from lower surface of jackpad and drag beam, and replaced.

- (2) Restrain jackpad and loosen nut.
- (3) **TORQUE NUT TO 540 - 600 INCH-POUNDS.**
- (4) Apply sealing compound, Item 295A, Appendix D, over threads, nut, and washers. **SEALING COMPOUND MUST NOT EXTEND MORE THAN 0.380-INCH BEYOND EDGE OF SEALING WASHERS.**
- (5) If jackpad slippage occurred in step j. (2), apply sealing compound, Item 295A, Appendix D, over washer on bottom side of jackpad. **SEALING COMPOUND MUST**

NOT EXTEND MORE THAN 0.380-INCH BEYOND EDGE OF SEALING WASHERS.

- (6) Apply torque stripe over nut (PARA 2-6-5).

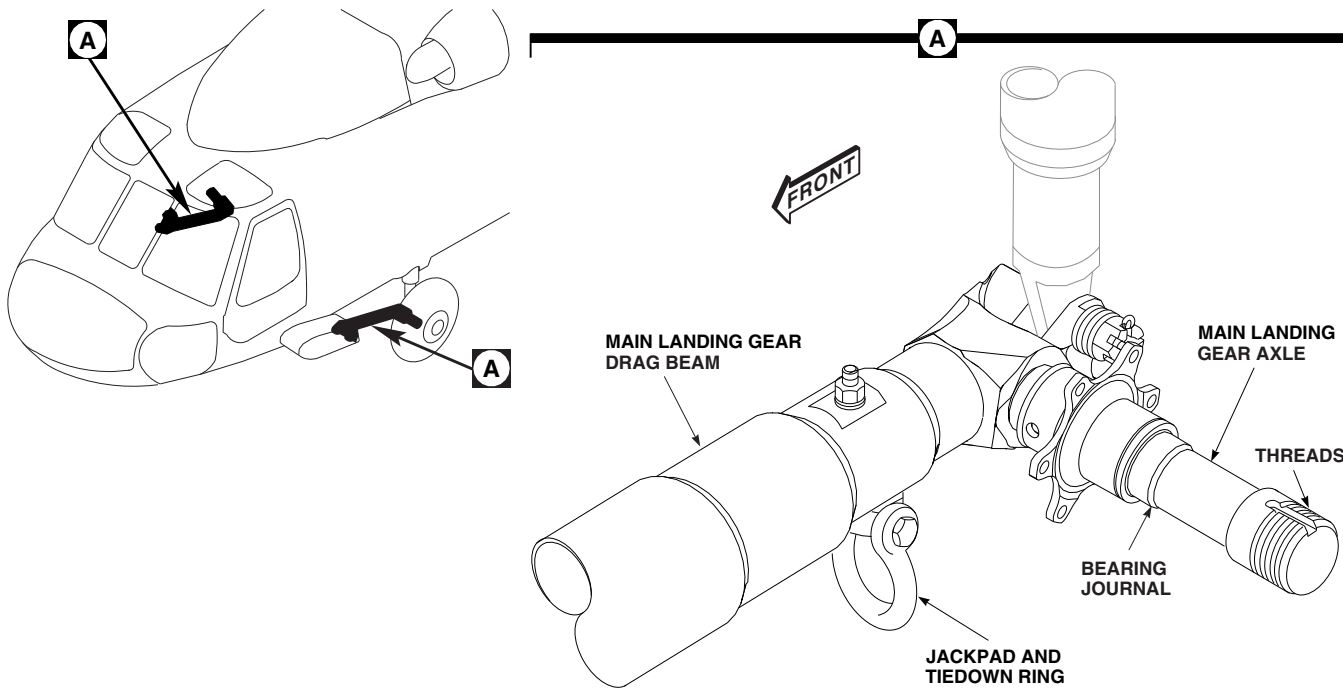
3-3-1.2 Main Landing Gear Axle Inspection.

NOTE

Repair of damaged areas is covered in step f.

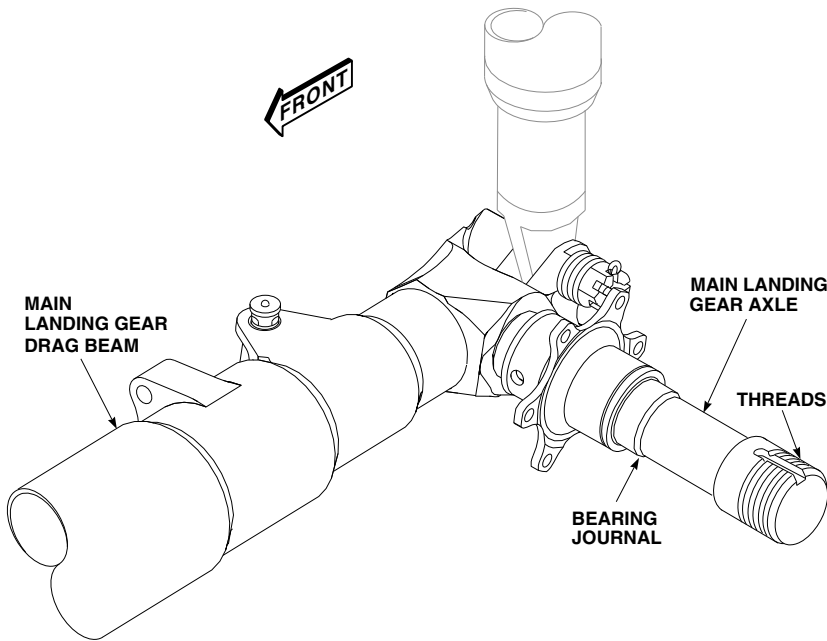
- a. Visually inspect main landing gear axle threads for stripped, crossed, or flattened threads (Figure 3-3-1-1, Detail A). **NONE ALLOWED.**
- b. Visually inspect bearing journals for missing, scored, or damaged nickel-plating. **NONE ALLOWED.**
- c. Using 10X magnifying glass, inspect main landing gear axle for cracks. **NONE ALLOWED.**

- d. Visually inspect main landing gear axle surface for nicks and scratches perpendicular to centerline. **SCRATCHES PERPENDICULAR TO CENTERLINE MUST NOT EXCEED 20% OF CIRCUMFERENCE AND BE NO DEEPER THAN 0.005-INCH AFTER BLENDING.**
- e. Visually inspect main landing gear axle for isolated nicks, scratches, gouges, or corrosion. **DEPTH AFTER BLENDING MUST NOT EXCEED 0.010-INCH.**
- f. If damaged, repair as follows:
 - (1) Using abrasive cloth, Item 1A or Item 7, Appendix D, blend out damaged area. **DO NOT GRIND.**
 - (2) Using solvent cleaning compound, Item 99, Appendix D, clean area.
 - (3) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer, allowing two hours between applications.



EFFECTIVITY

ON HELICOPTERS WITH MAIN LANDING GEAR, 70250-12050-047, -049, -052, AND -053.



EFFECTIVITY

ON HELICOPTERS WITH MAIN LANDING GEAR, 70250-22009-041 AND -042.

FIGURE 3-3-1-1. INSPECT MAIN LANDING GEAR AXLE

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3-3-2 MAIN LANDING GEAR INSPECTION SHOCK STRUT QUICK SERVICE CHECK.

PARAGRAPH BREAKDOWN

	PAGE
3-3-2.1 Main Landing Gear Shock Strut Quick Service Check	3-3-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- Appropriate Flight Manual
- PARA 1-3-2

3-3-2.1 Main Landing Gear Shock Strut Quick Service Check.

NOTE

This procedure is a quick way to check if shock strut needs servicing.

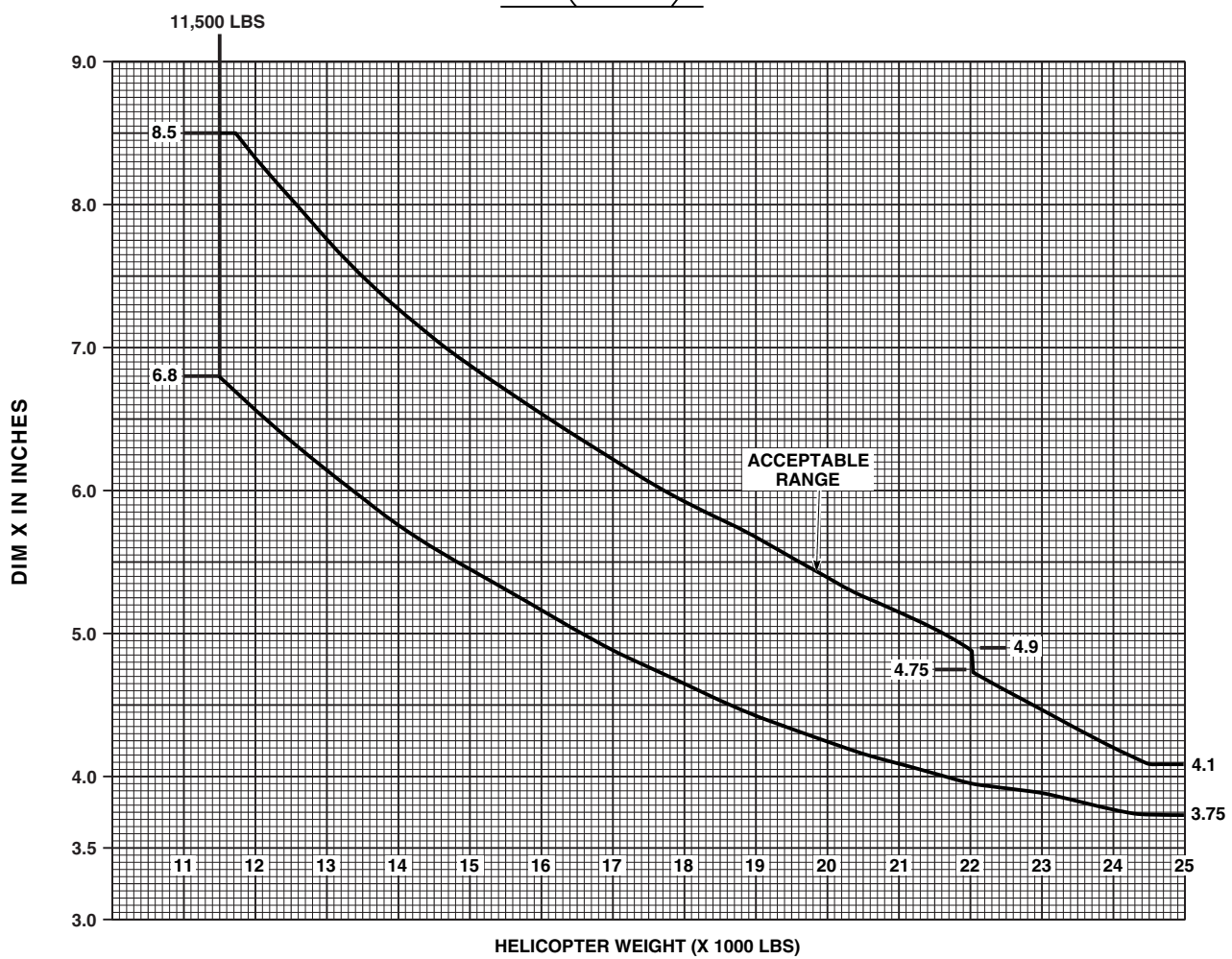
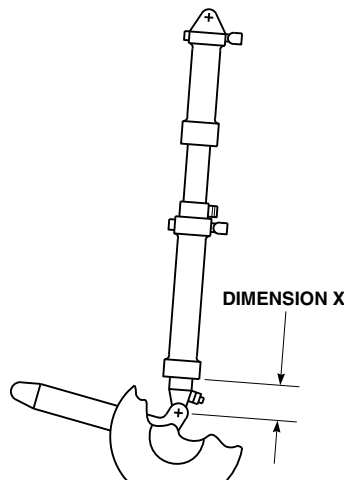
- Measure shock strut for dimension X (Figure 3-3-2-1).
- Determine weight of helicopter (Appropriate Flight Manual).

- Using Figure 3-3-2-1, check if dimension X is within acceptable range.

Example: Helicopter weight is 16500 lbs. Dimension X is 5-3/4 inches. Acceptable range is 5 to 6-3/8 inches.

Result: Dimension X is within acceptable range. No need to service shock strut.

- If dimension X is above or below acceptable range, service shock strut (PARA 1-3-2).



SHOCK STRUT DIMENSION X VERSUS HELICOPTER WEIGHT

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FIGURE 3-3-2-1. MAIN LANDING GEAR SHOCK STRUT QUICK SERVICE CHECK

3-3-3 MAIN LANDING GEAR TIRE INSPECTION.

PARAGRAPH BREAKDOWN

	PAGE
3-3-3.1 Main Landing Gear Tire Inspection	3-3-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-3-3
- PARA 3-4-13

3-3-3.1 Main Landing Gear Tire Inspection.

WARNING

Injury to personnel will result if foreign objects embedded in tire are removed while tire is inflated. Deflate tire before removing embedded foreign objects.

- a. Inspect tire sidewall for blisters. **NONE ALLOWED.** If blisters are found, replace tire (PARA 3-4-13).
- b. Inspect tire for wear. **WEAR SHALL NOT REACH FIRST FABRIC PLY OF TREAD GROOVE OR BE WORN FOR A CONTINUOUS LENGTH OF 12-INCHES AT BOTTOM OF TREAD GROOVE.** If wear exceeds limits, replace tire (PARA 3-4-13).
- c. Inspect tire tread and shoulder for cuts or damage. **CUTS OR DAMAGE MAY BE ANY LENGTH PROVIDED DAMAGE DOES NOT CUT FIRST LAYER OF CORD BODY. IF CUT PENETRATES FIRST LAYER OF**

CORD BODY AND IS LESS THAN 1-INCH LONG, DEPTH OF CUT SHALL NOT EXCEED CUT LIMIT SHOWN ON SIDEWALL OF TIRE. If damage exceeds limits, replace tire (PARA 3-4-13). **IF TIRE GROOVE NEAREST CUT IS WORN SMOOTH AND CUT PENETRATES CORD BODY, REPLACE TIRE** (PARA 3-4-13).

- d. Inspect tire sidewalls for cuts, scratches, abrasions, or scuffs. **DAMAGE SHALL NOT PENETRATE FIRST LAYER OF CORD BODY.** If damage exceeds limits, replace tire (PARA 3-4-13).
- e. Inspect tire surface for weather checking. **NO SEPARATION OF RUBBER BETWEEN CORDS ALLOWED. MAXIMUM DEPTH OF CRACK BENEATH TREAD RIB IS 1/8-INCH DEEP AND 1/4-INCH LONG.** If weather checking exceeds limits, replace tire (PARA 3-4-13).
- f. Inspect tire for proper pressure. **PRESSURE SHALL BE 130 - 140 PSI.** If pressure is not within limits, service tire (PARA 1-3-3).

3-3-4 MAIN LANDING GEAR WHEEL BRAKE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
3-3-4.1	Main Landing Gear Wheel Brake Inspection	3-3-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 3-4-14

3-3-4.1 Main Landing Gear Wheel Brake Inspection.

- a. Check brake for wear as follows:
- Apply parking brake.
 - Press on wear pin (Figure 3-3-4-1).
WEAR PIN MUST PROTRUDE

ABOVE HOUSING NUT. If wear pin is flush with housing nut, replace brake (PARA 3-4-14).

- Release parking brake.

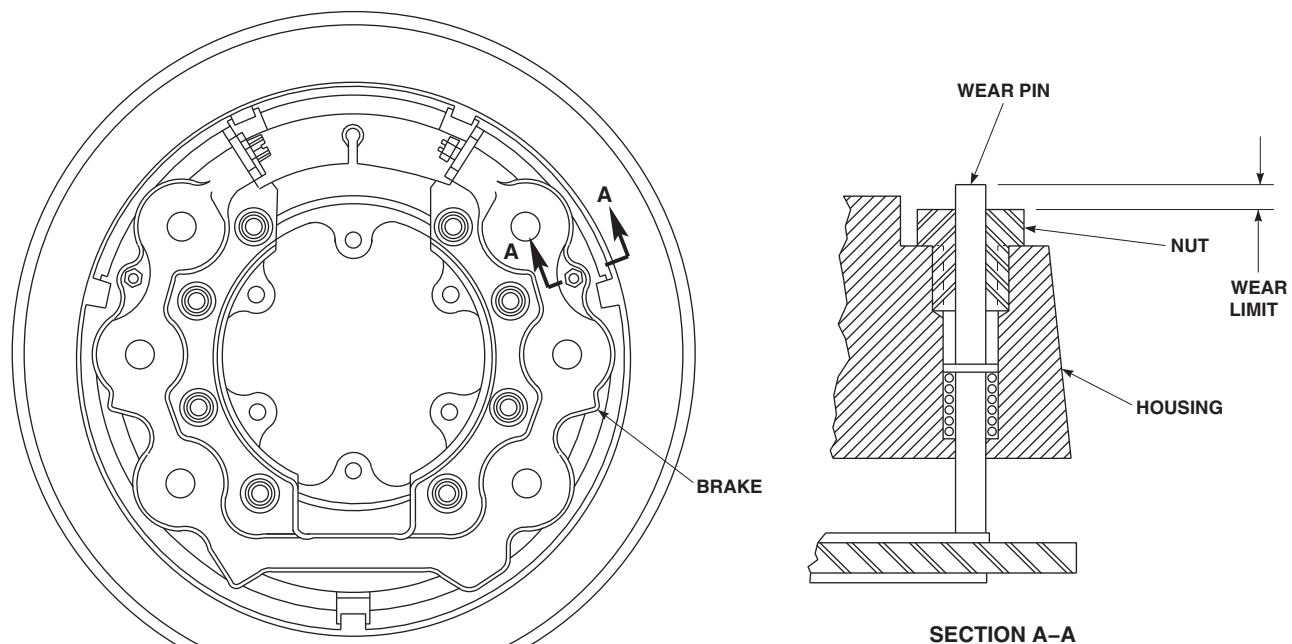


FIGURE 3-3-4-1. INSPECT MAIN LANDING GEAR WHEEL BRAKE

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3-3-5 PARKING BRAKE ROD INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
3-3-5.1	Parking Brake Rod Inspection	3-3-5-1

INITIAL SETUP Tools - Aircraft Mechanic’s Toolkit Materials/Parts - Corrosion Compound, Item 108, Appendix D	- Corrosion Preventive Compound, Item 113, Appendix D - Scouring Pad, Item 270, Appendix D References - Appendix D - PARA 3-4-21
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3-3-5.1 Parking Brake Rod Inspection.

NOTE

Corrosion is indicated by red film or pitting.

- a. Visually inspect rod for corrosion. **PITS SHALL NOT EXCEED 0.050-INCH DEEP.** If pits exceed limit, replace parking brake rod (PARA 3-4-21).
- b. Remove corrosion manually from rod, using scouring pad, Item 270, Appendix D.
- c. Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.
- d. Apply corrosion compound solution to corroded areas of rod.
- e. Rinse thoroughly with clean water.
- f. Coat rod with corrosion preventive compound, Item 113, Appendix D.

3-3-6 TAIL LANDING GEAR YOKE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
3-3-6.1	Tail Landing Gear Yoke Inspection	3-3-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Black Light
- Dark Cloth Sheet
- Dark Plastic Sheet
- Fluorescent Penetrant Inspection Kit
- Inspection Mirror
- Magnifying Glass, 12X
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Brush, Paint, Item 86, Appendix D
- Cloth, Cleaning, Item 102A, Appendix D

- Corrosion Resistant Coating, Item 118, Appendix D
- Cotton Swab, Item 119, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Pressure Sensitive Adhesive Tape, Item 249, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 3-4-25
- PARA 3-5-9

3-3-6.1 Tail Landing Gear Yoke Inspection.

- Using 12X magnifying glass, visually inspect entire yoke for evidence of cracks. Pay particular attention to AREA A (Figure 3-3-6-1). If cracks are suspected, perform the following:
 - Remove tail landing gear yoke (PARA 3-4-25).

- Remove paint from yoke.
- Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect yoke for cracks. **NONE ALLOWED.** If crack is found, replace tail landing gear yoke (PARA 3-4-25).
- Check lateral movement of yoke assembly. **LATERAL MOVEMENT MUST NOT EX-**

CEED 0.047-INCH. If lateral movement exceeds limit, perform the following:

- (1) Remove tail landing gear yoke (PARA 3-4-25).
- (2) Replace yoke attachment bushings (PARA 3-5-9).

- c. If yoke attachment bushings were replaced, **LATERAL MOVEMENT MUST NOT EXCEED 0.039-INCH.** If lateral movement exceeds limit, replace fuselage/yoke fitting bushing (PARA 3-5-9).

NOTE

- The following inspection is applicable for yoke assembly, 70250-13158-042, only.
 - Inspection should be performed every 500 flight hours.
- d. Inspect yoke assembly as follows (Figure 3-3-6-2):
- (1) Remove tail landing gear yoke (PARA 3-4-25).
 - (2) Using pressure sensitive adhesive tape, Item 249, Appendix D, mask off diameter approximately 2-inches from transition blend radius between diameter and forked end of yoke.

NOTE

If paint is present, allow paint remover to remain on surface until paint bubbles.

- (3) Using paint brush, Item 86, Appendix D, apply paint remover, Item 230, Appendix D, to inspection area. Using machinery towel, Item 211, Appendix D, or nonmetallic scraper, remove paint remover.
- (4) Using machinery towel, Item 211, Appendix D, saturated with water, thoroughly remove all remains of paint remover. Repeat if necessary.

- (5) Using machinery towel, Item 211, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D, clean inspection area.

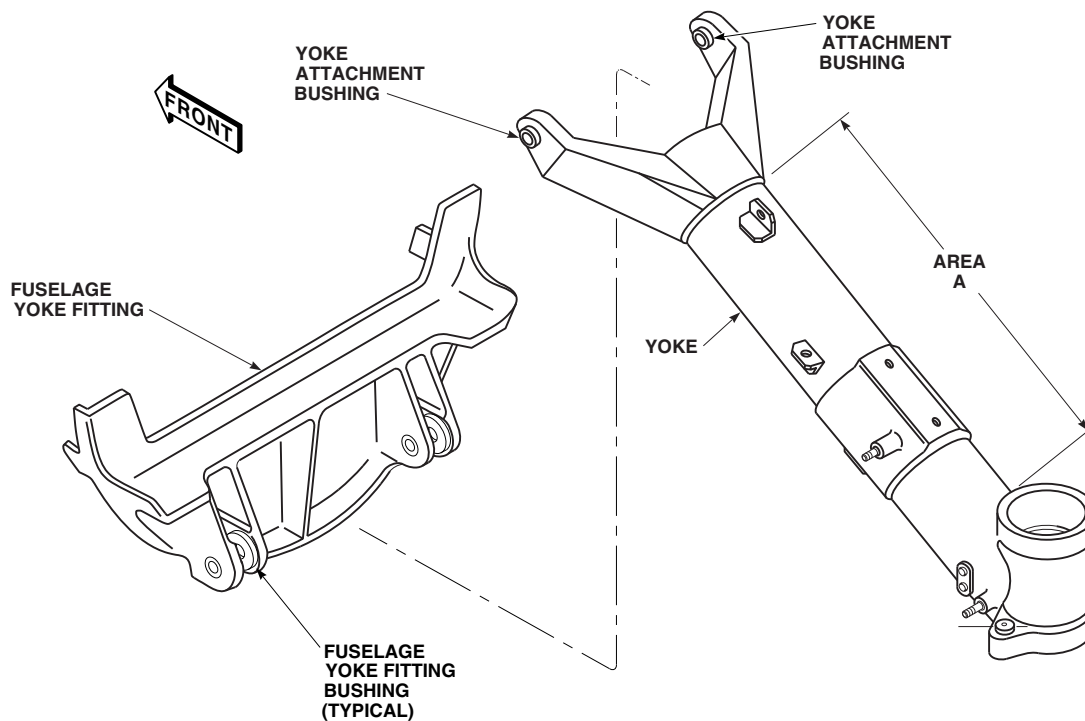
NOTE

- Temperature of yoke and surrounding area must be between 40° - 120°F (104° - 248°C).
 - Personnel performing fluorescent penetrant inspection must be certified Level II penetrant inspectors.
 - Penetrant inspection materials (penetrant, developer, solvent cleaner) must be from the same manufacturer. Mixing inspection materials from different manufacturers is not permitted.
- (6) Using fluorescent penetrant inspection kit, Type I, Method C, Level 3 Sensitivity, fluorescent penetrant inspect cleaned area as follows:
- (a) Clean inspection area with solvent remover provided with fluorescent penetrant inspection kit.
 - (b) Using cotton swab, Item 119, Appendix D, apply penetrant to inspection area.
 - (c) Using inspection mirror and black light, make sure entire inspection area is coated with penetrant.

NOTE

If inspection is performed at or near minimum temperature of 40°F (104°C), allow penetrant to dwell for 60 minutes due to change in penetrant viscosity.

- (d) Allow penetrant to dwell for 30 minutes.
- (e) Wipe excess penetrant from inspection area using cleaning cloth, Item 102A, Appendix D.



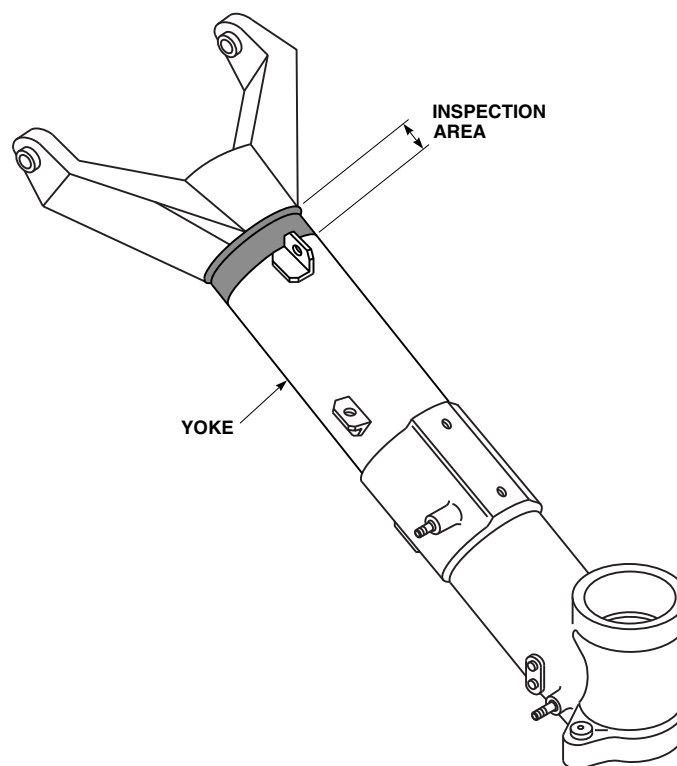
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FIGURE 3-3-6-1. INSPECT TAIL LANDING GEAR YOKE

CAUTION

Do not spray solvent remover directly on the inspection area.

- (f) Remove remaining penetrant from inspection area using cleaning cloth, Item 102A, Appendix D, lightly moistened with solvent remover. Using inspection mirror and black light, make sure all background fluorescence has been removed.
- (g) Apply developer to inspection area and allow to dwell for 10 minutes.
- (h) Block ambient white light from inspection area using dark cloth or dark plastic sheet.
- (i) Using inspection mirror and black light, inspect area for any indications of cracks. **NONE ALLOWED.** If crack is found, replace yoke and contact Sikorsky Aircraft ILS manager for disposition.
- (j) Remove residual penetrant materials from inspection area by thoroughly cleaning with solvent remover.
- (7) Remove pressure sensitive tape and touch up inspection area with corrosion resistant coating, Item 118, Appendix D. Apply primer and paint (PARA 2-6-5).
- (8) Install tail landing gear yoke (PARA 3-4-25).
- (9) Record time of inspection on log card (SA Form 7343-21).

**EFFECTIVITY**

TAIL LANDING GEAR YOKE,
70250-13158-042.

FB3655
SA

FIGURE 3-3-6-2. INSPECT TAIL LANDING GEAR YOKE

3-3-7 TAIL LANDING GEAR TIRE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
3-3-7.1	Tail Landing Gear Tire Inspection	3-3-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-3-5
- PARA 3-4-28

3-3-7.1 Tail Landing Gear Tire Inspection.

WARNING

Injury to personnel will result if foreign objects embedded in tire are removed while tire is inflated. Deflate tire before removing embedded foreign objects.

- Inspect tire sidewall for blisters. **NONE ALLOWED.** If blisters are found, replace tire (PARA 3-4-28).
- Inspect tire for wear. **WEAR SHALL NOT EXTEND TO BOTTOM OF GROOVE AT ANY ONE SPOT.** If wear exceeds limits, replace tire (PARA 3-4-28).
- Inspect tire tread and shoulder for cuts or damage. **CUTS OR DAMAGE MAY BE ANY**

LENGTH PROVIDED DAMAGE DOES NOT CUT FIRST LAYER OF CORD BODY. IF CUT PENETRATES FIRST LAYER OF CORD BODY AND IS LESS THAN 1-INCH LONG, DEPTH OF CUT SHALL NOT EXCEED CUT LIMIT SHOWN ON SIDEWALL OF TIRE. If damage exceeds limits, replace tire (PARA 3-4-28). **IF TIRE GROOVE NEAREST CUT IS WORN SMOOTH AND CUT PENETRATES CORD BODY, REPLACE TIRE** (PARA 3-4-28).

- Inspect tire sidewalls for cuts, scratches, abrasions, or scuffs. **DAMAGE SHALL NOT PENETRATE FIRST LAYER OF CORD BODY.** If damage exceeds limit, replace tire (PARA 3-4-28).
- Check tire for proper pressure. **PRESSURE SHOULD BE 115 - 125 PSI.** If pressure is not within limits, service tire (PARA 1-3-5).

SECTION 3-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
3-4-1	Main Landing Gear Drag Beam/Axle
3-4-2	Main Landing Gear Axle
3-4-3	Brake Flange Spacers
3-4-4	Drag Beam Support Fitting Bearings
3-4-5	Main Landing Gear Drag Beam Jackpad and Tiedown Ring
3-4-6	Main Landing Gear Drag Beam Switch
3-4-7	Main Landing Gear Shock Strut
3-4-8	Main Landing Gear Shock Strut Kneeling Valve
3-4-9	Main Landing Gear Shock Strut Air Valve
3-4-10	Main Landing Gear Wheel and Tire
3-4-11	Main Landing Gear Wheel Valve
3-4-12	Main Landing Gear Wheel Disc Drive Keys
3-4-13	Main Landing Gear Tire
3-4-14	Main Landing Gear Wheel Brake
3-4-15	Master Brake Cylinder
3-4-16	Main Landing Gear Wheel Brake Lines and Hoses
3-4-17	Parking Brake Valve
3-4-18	Parking Brake Valve Switch
3-4-19	Slave Mixer Valve
3-4-20	Bleed Main Landing Gear Wheel Brake System

PARAGRAPH	TITLE
3-4-21	Parking Brake Handle and Mechanism
3-4-22	Tail Landing Gear Shock Strut
3-4-23	Tail Landing Gear Shock Strut Kneeling Valve
3-4-24	Tail Landing Gear Shock Strut Air Valve
3-4-25	Tail Landing Gear Yoke
3-4-26	Tail Landing Gear Fork
3-4-27	Tail Landing Gear Lock Mechanism
3-4-28	Tail Landing Gear Wheel and Tire

3-4-1 MAIN LANDING GEAR DRAG BEAM/AXLE.

PARAGRAPH BREAKDOWN

		PAGE
3-4-1.1	Replace Main Landing Gear Drag Beam/Axle	3-4-1-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 194, Appendix D
- Lockwire, Item 198, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Plugs and Caps
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-84
- PARA 2-6-5
- PARA 2-6-12
- PARA 3-3-1
- PARA 3-4-2
- PARA 3-4-3
- PARA 3-4-4
- PARA 3-4-5
- PARA 3-4-6
- PARA 3-4-7
- PARA 3-4-14
- PARA 3-4-20

3-4-1.1 Replace Main Landing Gear Drag Beam/Axle.

NOTE

Replacement of both main landing gear drag beams is the same, except as noted.

3-4-1.1.1 Remove.

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of main landing gear drag beam/axle.

- a. Remove wire strike main landing gear cutter clamps (PARA 2-4-84).
- b. Remove main landing gear wheel brake (PARA 3-4-14).
- c. Disconnect brake hose at quick-disconnect coupling on drag beam (Figure 3-4-1-1, Sheet 1, Detail A, Sheet 2, Detail A, or Sheet 3, Detail A).
- d. Remove bonding jumper as follows:
 - (1) On left drag beam, remove screw and washer securing bonding jumper to brake line support.
 - (2) On right drag beam, remove screw and washer securing bonding jumper to striker and bracket (Figure 3-4-1-1, Sheet 4, Detail E).
- e. Disconnect main landing gear shock strut from drag beam (PARA 3-4-7).
- f. Remove retaining rings and laminated washers from pivot pins.
- g. Remove pivot pins and remove drag beam and axle from fuselage support fitting.

3-4-1.1.2 Disassemble.

- a. On helicopters with main landing gear drag beam/axle, 70250-12050-047 and -049, remove brake line as follows (Figure 3-4-1-1, Sheet 1, Detail A):

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of main landing gear drag beam/axle.

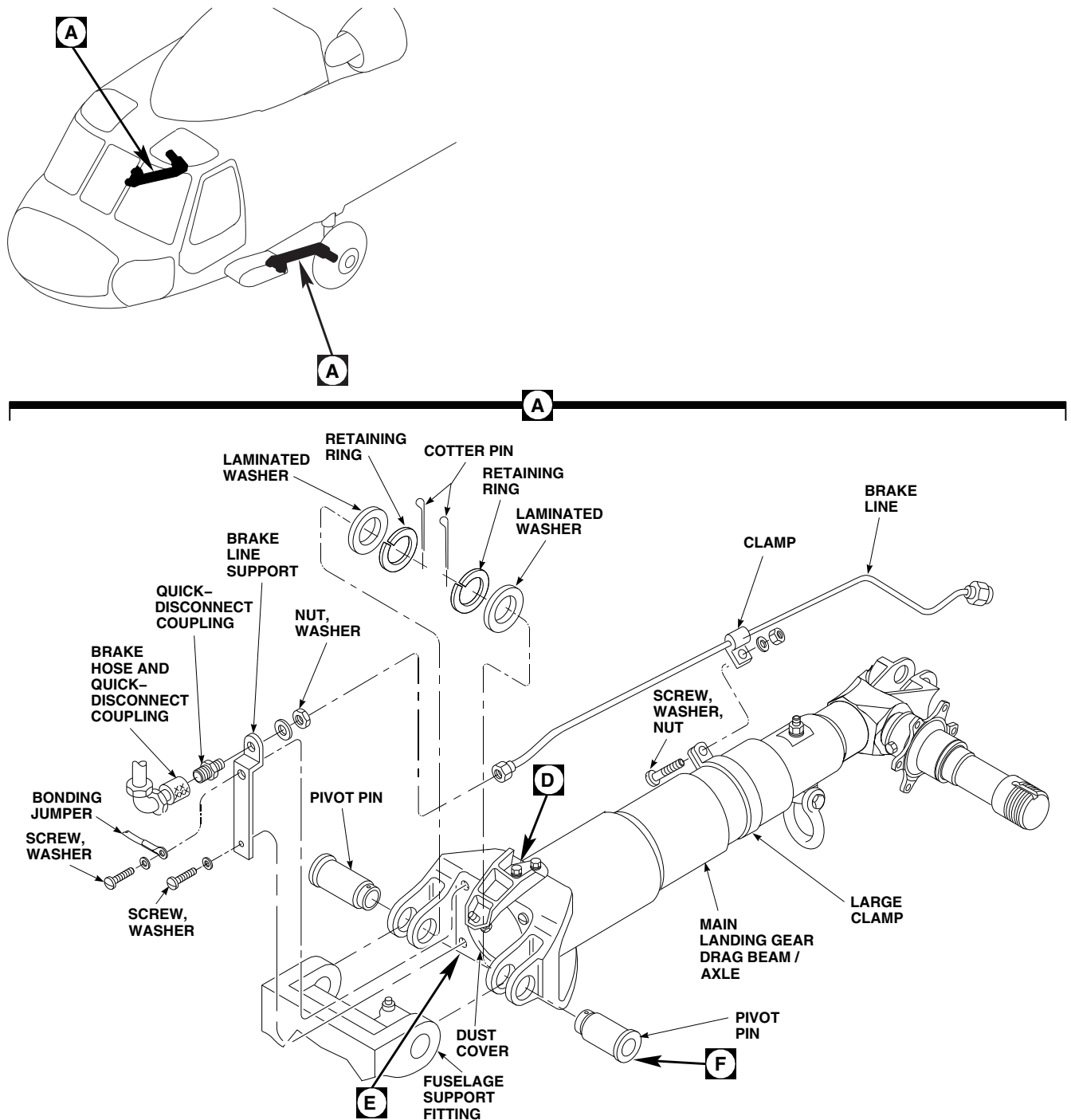
- (1) Disconnect brake line from quick-disconnect coupling.
- (2) Remove screws, washers, and nuts from clamp. Remove clamp from brake line. Cap brake line.
- (3) Remove large clamp from drag beam.
- (4) Remove nut and washer securing quick-disconnect coupling to brake line support. Remove quick-disconnect coupling. Cap coupling.
- b. On helicopters with main landing gear drag beam/axle, 70250-12050-052 and -053, or 70250-22009-041 and -042, remove brake line as follows:

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of main landing gear drag beam/axle.

- (1) Disconnect brake line from quick-disconnect coupling (Figure 3-4-1-1, Sheet 2, Detail A, or Sheet 3, Detail A).



LEFT MAIN LANDING GEAR DRAG BEAM (SHOWN)

EFFECTIVITY

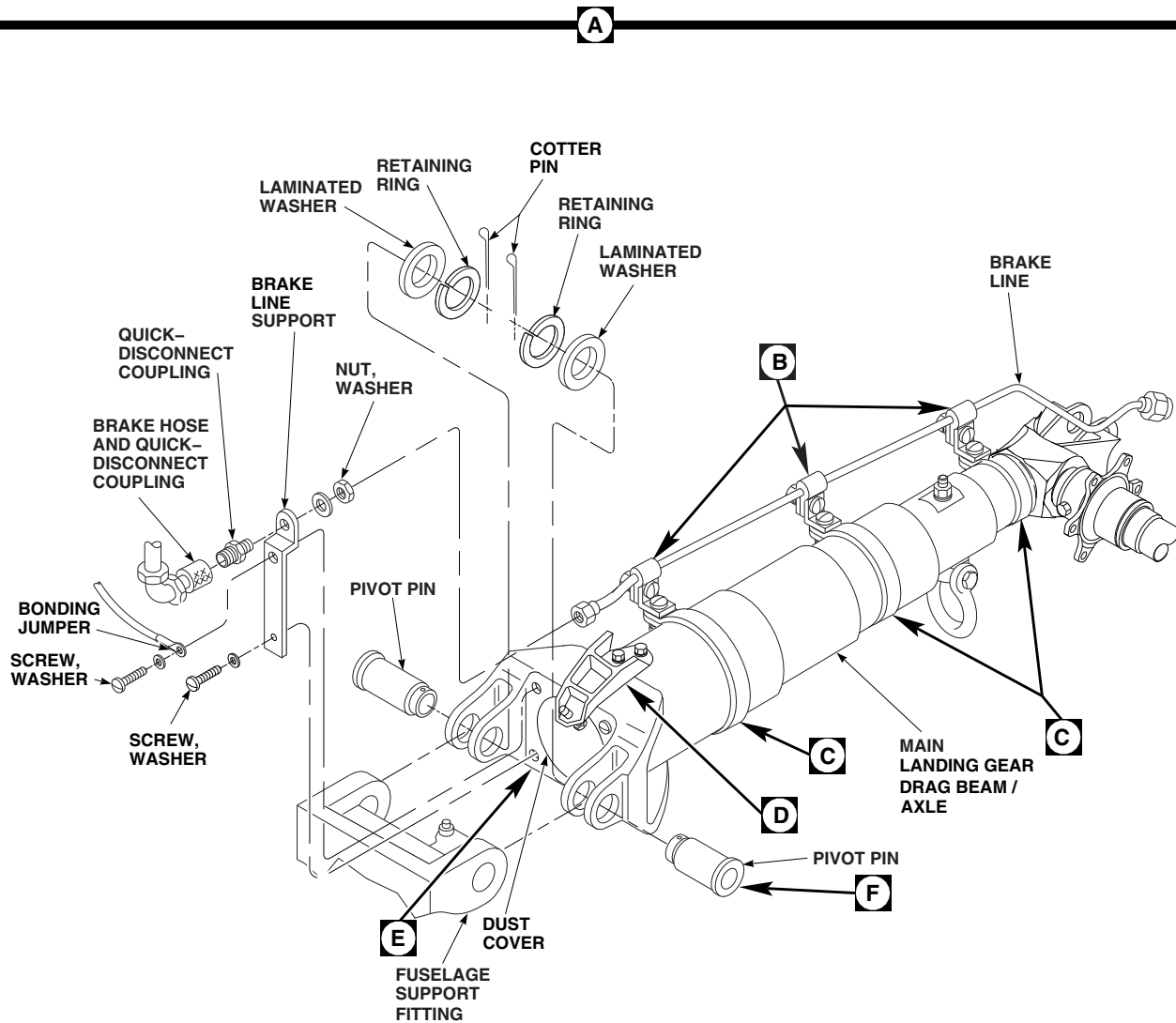
ON HELICOPTERS WITH MAIN LANDING GEAR, 70250-12050-047 AND -049.

NOTE

ON 70250-12050-052 AND -053 ONLY.

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FIGURE 3-4-1-1. REPLACE MAIN LANDING GEAR DRAG BEAM/AXLE (SHEET 1 OF 4)



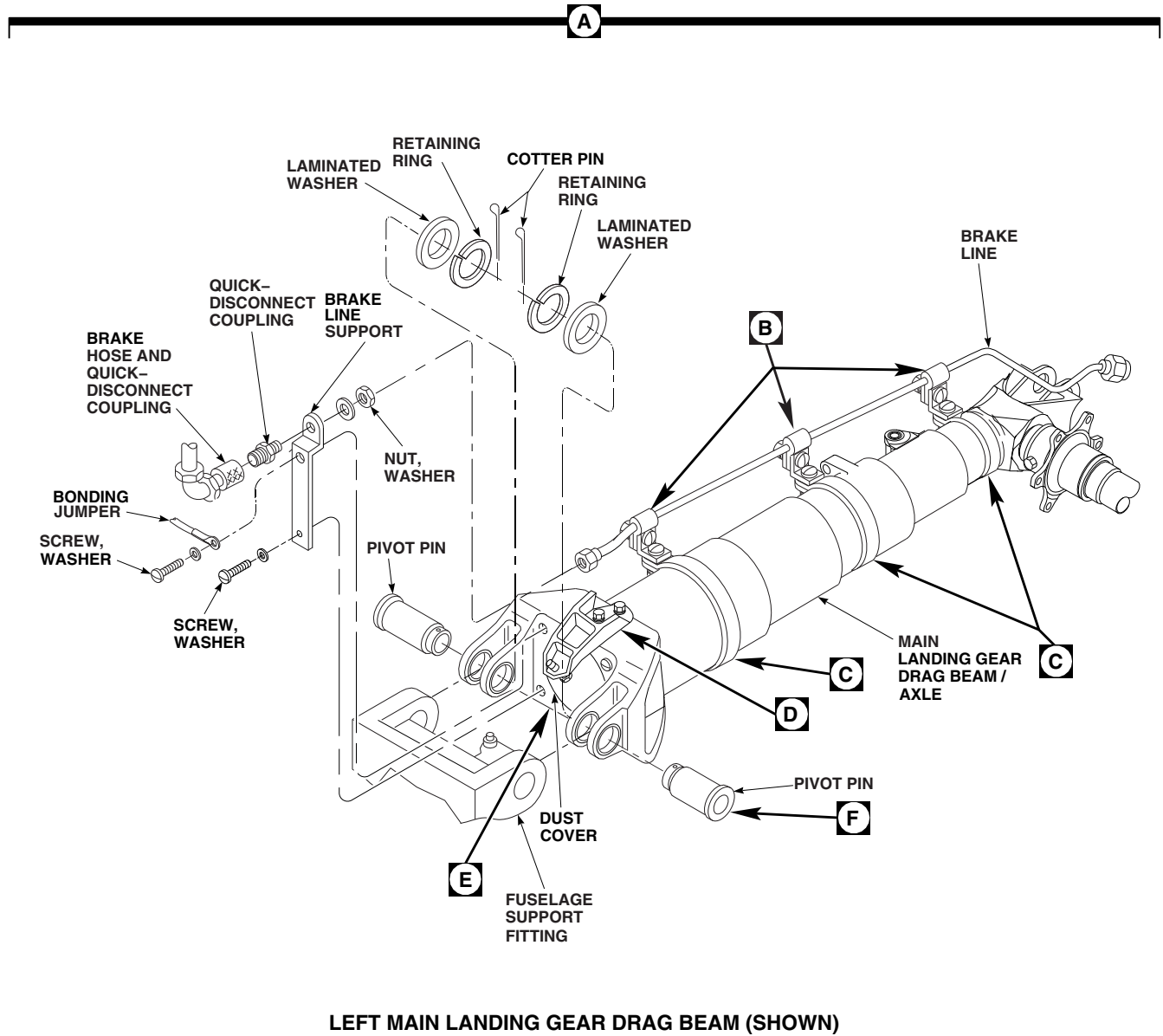
LEFT MAIN LANDING GEAR DRAG BEAM (SHOWN)

EFFECTIVITY

ON HELICOPTERS WITH MAIN LANDING GEAR, 70250-12050-052 AND -053.

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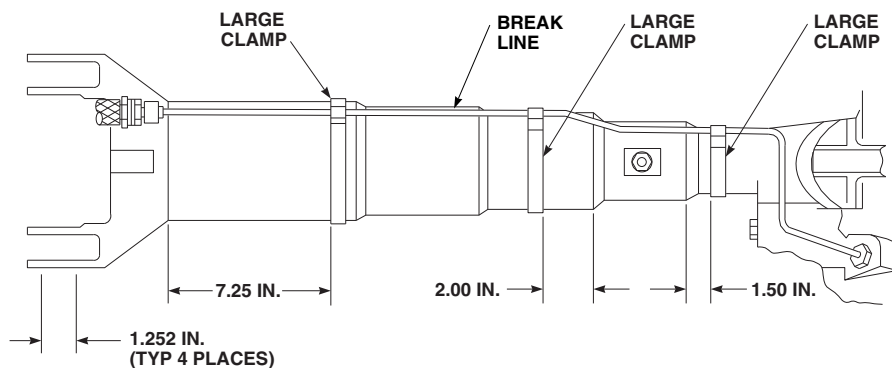
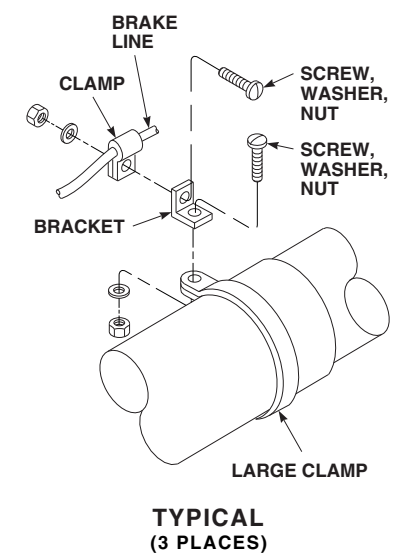
FIGURE 3-4-1-1. REPLACE MAIN LANDING GEAR DRAG BEAM/AXLE (SHEET 2 OF 4)



EFFECTIVITY
ON HELICOPTERS WITH MAIN LANDING
GEAR, 70250-22009-041 AND -042.

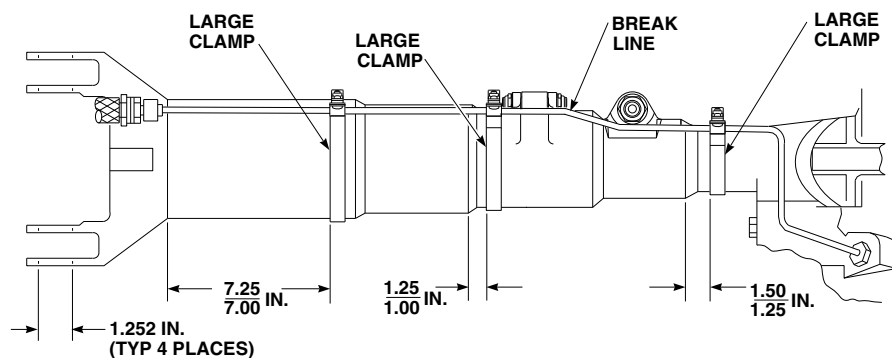
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FIGURE 3-4-1-1. REPLACE MAIN LANDING GEAR DRAG BEAM/AXLE (SHEET 3 OF 4)



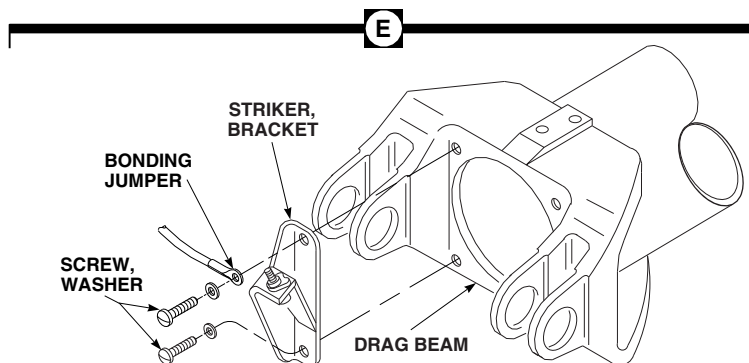
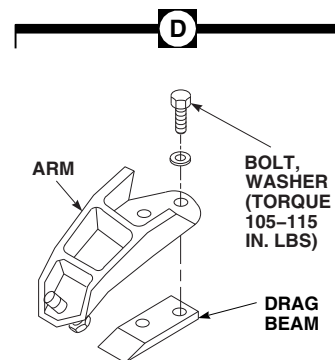
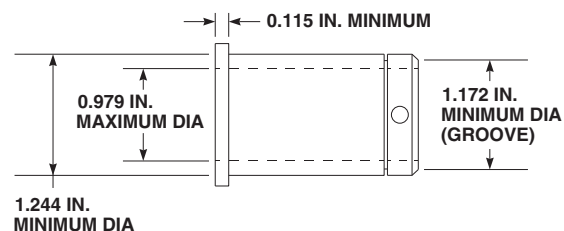
EFFECTIVITY

ON HELICOPTERS WITH MAIN LANDING
GEAR DRAG BEAM / AXLE, 70250-12050-047
AND -049, AND 70250-12050-052 AND -053.



EFFECTIVITY

ON HELICOPTERS WITH MAIN LANDING
GEAR DRAG BEAM / AXLE, 70250-22009-041
AND -042.



RIGHT MAIN LANDING GEAR DRAG BEAM/AXLE

LEFT MAIN LANDING GEAR DRAG BEAM/AXLE

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FIGURE 3-4-1-1. REPLACE MAIN LANDING GEAR DRAG BEAM/AXLE (SHEET 4 OF 4)

- (2) Remove screws, washers, and nuts from clamps. Remove clamps from brackets and brake line (Figure 3-4-1-1, Sheet 4, Detail B). Cap brake line.
- (3) Remove screws, washers, and nuts and remove three large clamps and brackets from drag beam (Figure 3-4-1-1, Sheet 4, Detail B and Detail C).
- (4) Remove nut and washer securing quick-disconnect coupling to brake line support. Remove quick-disconnect coupling. Cap coupling (Figure 3-4-1-1, Sheet 2, Detail A, or Sheet 3, Detail A).
- c. On left drag beam, remove bolts and washers and remove arm (Figure 3-4-1-1, Sheet 4, Detail D).
- d. On right drag beam, remove remaining screw and washer and remove bracket with striker (Figure 3-4-1-1, Sheet 4, Detail E).
- e. On left drag beam, remove remaining screw and washer and remove brake line support (Figure 3-4-1-3, Sheet 1, Detail A).
- f. Remove main landing gear axle (PARA 3-4-2).
- g. Remove main landing gear drag beam/axle jackpad and tiedown ring (PARA 3-4-5).
- h. Remove brake flange spacers, as required (PARA 3-4-3).

3-4-1.1.3 Inspect.

- a. Inspect main landing gear drag beam/axle (PARA 3-3-1).
- b. Inspect fuselage support fittings, 70219-02526-041 and -042, for cracks, nicks, scratches, and corrosion (PARA 2-6-12).
- c. Check bearing for wear in fuselage support fitting. **WEAR MUST NOT EXCEED 0.010-INCH OVERSIZE AND/OR OUT-OF-ROUND. BEARING BORE DIAMETER MUST NOT EXCEED 1.2615-INCHES AT ANY POINT IN BORE.** If limits are exceeded, replace bearing (PARA 3-4-4).

- d. Inspect Teflon coating on bearings. **TEFLON MUST NOT BE MISSING OR SHOW SIGNS OF METAL CONTACT.** If Teflon is missing or shows signs of metal contact, replace bearings (PARA 3-4-4).
- e. Visually inspect pivot pins for corrosion. **NONE ALLOWED.** Remove corrosion from pivot pins as follows:

NOTE

Corrosion is indicated by red film or pitting.

- (1) Using abrasive cloth, Item 8, Appendix D, blend out minor corrosion or pits.
- (2) After corrosion removal, inspect pivot pins for dimensional limits. **DIMENSIONAL LIMITS SHOWN IN FIGURE 3-4-1-1, SHEET 4, DETAIL F, MUST NOT BE EXCEEDED.** If dimensions are exceeded, replace pin.
- f. Measure pivot pin bore in drag beam (Figure 3-4-1-1, Sheet 4, Detail C). **DIMENSION LIMIT OF PIVOT PIN BORE IN DRAG BEAM MUST NOT EXCEED 1.252-INCHES.** If limit is exceeded, replace drag beam.
- g. Using epoxy primer coating kit, Item 135, Appendix D, touch up nonmating surfaces of pivot pin (PARA 2-6-5).
- h. After epoxy primer coating kit touch up, apply corrosion preventive compound, Item 114, Appendix D.

3-4-1.1.4 Assemble.

- a. On left drag beam, install arm as follows (Figure 3-4-1-1, Sheet 4, Detail D):
 - (1) Position arm on drag beam and secure with bolts and washers.
 - (2) **TORQUE TO 105 - 115 INCH-POUNDS.**
 - (3) Using lockwire, Item 198, Appendix D, lockwire bolts.

- b. On right drag beam, install bracket with striker with one screw and washer in lower hole at this time (Figure 3-4-1-1, Sheet 4, Detail E).
- c. Install main landing gear axle (PARA 3-4-2).
- d. Install main landing gear drag beam/axle jack-pad and tiedown ring (PARA 3-4-5).
- e. Install brake flange spacers, as required (PARA 3-4-3).

NOTE

Upper brake line support screw and washer are installed with bonding jumper during main landing gear drag beam/axle install procedure (PARA 3-4-1.1.5).

- f. Install brake line support on left drag beam with screw and washer in lower, inboard hole. Do not disturb dust cover. Use thin washers on inboard screws and thick washers on outboard screws (Figure 3-4-1-1, Sheet 1, Detail A, Sheet 2, Detail A, or Sheet 3, Detail A).
- g. On helicopters with main landing gear drag beam/axle, 70250-12050-047 and -049, install brake line as follows (Figure 3-4-1-1, Sheet 1, Detail A):

WARNING**FSCAP**

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of brake line.

- (1) Remove cap and install quick-disconnect coupling in brake line support, and secure with washer and nut.
- (2) Install large clamp on drag beam.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (3) Remove caps and blow brake line clear with dry compressed air. Replace cap at wheel brake end.
- (4) Using hydraulic fluid, Item 154, or Item 155, Appendix D, coat threads of quick-disconnect coupling and connect brake line to quick-disconnect coupling.
- (5) Clamp brake line to large clamp with screw, washer, and nut.
- h. On helicopters with main landing gear drag beam/axle, 70250-12050-052 and -053, or 70250-22009-041 and -042, install brake line as follows:

WARNING**FSCAP**

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of brake line.

- (1) Remove cap and install quick-disconnect coupling in brake line support, and secure with washer and nut (Figure 3-4-1-1, Sheet 2, Detail A, or Sheet 3, Detail A).
- (2) Install three large clamps and brackets on drag beam with screws, washers, and

nuts. Connect each clamp to correct diameter on drag beam and correct distance from shoulder (Figure 3-4-1-1, Sheet 4, Detail B and Detail C).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (3) Remove caps and blow brake line clear with dry compressed air. Replace cap at wheel brake end.
- (4) Using hydraulic fluid, Item 154, or Item 155, Appendix D, coat threads of quick-disconnect coupling and connect brake line to quick-disconnect coupling.
- (5) Install clamps on brake line and secure to brackets with screws, washers, and nuts (Figure 3-4-1-1, Sheet 4, Detail B).

3-4-1.1.5 Install.

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of main landing gear drag beam/axle.

- a. Position drag beam and axle on fuselage support fitting. Install pivot pins and retaining rings (Figure 3-4-1-1, Sheet 1, Detail A, Sheet 2, Detail A, or Sheet 3, Detail A).

- b. Using feeler gage, measure endplay between retaining rings and drag beam lugs. **PEEL LAMINATED WASHERS AS REQUIRED TO OBTAIN 0.001 - 0.005-INCH TOTAL ENDPLAY.**
- c. Remove retaining rings and install laminated washers between lugs and retaining rings, with laminations facing lugs.
- d. Install cotter pins in pivot pins.
- e. Connect main landing gear shock strut to drag beam (PARA 3-4-7).
- f. Dampen machinery towel, Item 211, Appendix D, with dry-cleaning solvent, Item 124, Appendix D, and clean area where bonding jumper will be connected.
- g. Install bonding jumper as follows:
 - (1) On left drag beam, install screw and washer to secure bonding jumper to brake line support.
 - (2) On right drag beam, install screw and washer to secure bonding jumper to striker and bracket (Figure 3-4-1-1, Sheet 4, Detail E).
- h. Using lockwire, Item 194, Appendix D, lockwire top screws together.
- i. Using sealing compound, Item 292, Appendix D, coat head of screw, washer, and terminal of bonding jumper.
- j. Connect brake hose at quick-disconnect coupling on drag beam (Figure 3-4-1-1, Sheet 1, Detail A, Sheet 2, Detail A, or Sheet 3, Detail A).
- k. Check adjustment of drag beam switch (PARA 3-4-6).
- l. Install main landing gear wheel brake (PARA 3-4-14).
- m. Bleed main landing gear wheel brake system (PARA 3-4-20).

WARNING**FSCAP**

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Use

extreme care to avoid damage to these surfaces during installation of main landing gear cutter and clamps.

- n. Install wire strike main landing gear cutter clamps (PARA 2-4-84).

3-4-2 MAIN LANDING GEAR AXLE.

PARAGRAPH BREAKDOWN

		PAGE
3-4-2.1	Replace Main Landing Gear Axle	3-4-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Dial Indicator
- Magnifying Glass, 10X
- Mechanical Puller Kit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 3-4-1
- PARA 3-4-14

3-4-2.1 Replace Main Landing Gear Axle.

NOTE

Replacement of both main landing gear axles is the same, except as noted.

3-4-2.1.1 Remove.

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Cadmium-plating and baked resin coating of axle bore are critical characteris-

tics. Use extreme care to avoid damage to these surfaces during removal of main landing gear axle.

- Remove main landing gear wheel brake (PARA 3-4-14).
- Break sealing compound around axle bolt and nut. Using nonmetallic scraper, remove sealing compound.
- Remove axle bolt, nut, and washer(s). On left drag beam, also remove clip and ground wire. If necessary, remove ground wire from clip as follows (Figure 3-4-2-1, Detail B):
 - Loosen nut at bottom of clip.
 - Pull ground wire out through bottom of clip.

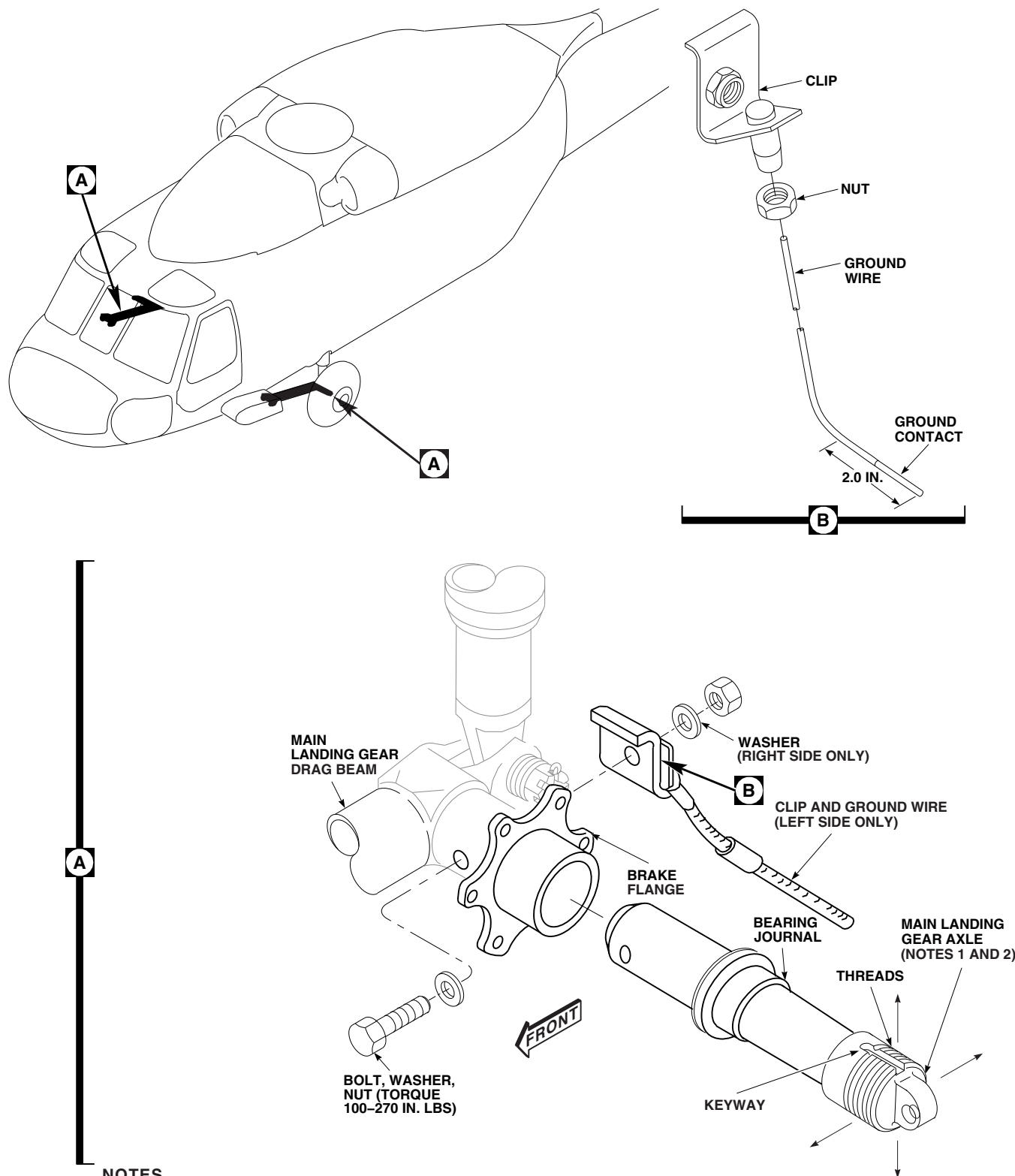
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FIGURE 3-4-2-1. REPLACE MAIN LANDING GEAR AXLE

- d. Using nonmetallic scraper, remove sealing compound around brake flange.
- e. Attach mechanical puller to axle and remove axle from brake flange.

3-4-2.1.2 Inspect/Repair.

- a. With axle removed, inspect interior surface of drag beam as follows:
 - (1) For drag beams with baked resin, inspect interior surface of drag beam for breaks in baked resin coating. **NONE ALLOWED.** If any break in baked resin coating is visible, replace main landing gear drag beam (PARA 3-4-1).
 - (2) For drag beams with epoxy primer, inspect interior surface of drag beam axle bore for missing epoxy primer coating. **NONE ALLOWED.** If epoxy primer coating is missing, perform the following:
 - (a) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer to areas missing epoxy primer.
 - (b) Allow two hours between applications.
 - (c) **MINIMUM BORE DIAMETER MUST BE 2.0-INCH AFTER PRIMER APPLICATION.** If diameter is less than 2.0-inch after application, replace drag beam (PARA 3-4-1).
- b. Inspect axle for stripped, crossed, or flattened threads (Figure 3-4-2-1, Detail A). **NONE ALLOWED.** If threads are damaged, replace main landing gear axle.
- c. Inspect axle bearing journals for missing, scored, or damaged nickel-plating. **NONE ALLOWED.** If bearing journals are damaged, replace main landing gear axle.
- d. Inspect inside diameter of axle for damage or deteriorated foam. **NONE ALLOWED.** If damaged or deteriorated foam is present, replace main landing gear axle.

- e. Using 10X magnifying glass, inspect axle for fractures and cracks. **NONE ALLOWED.** If axle is fractured or cracked, replace main landing gear axle.
- f. Inspect axle for nicks and scratches perpendicular to centerline. **NICKS AND SCRATCHES MUST NOT EXCEED 20 % OF CIRCUMFERENCE.** If limit is exceeded, replace main landing gear axle. If damage is within limit, repair as follows:
 - (1) Using abrasive cloth, Item 8, Appendix D, blend nicks and scratches. **BLENDING MUST NOT EXCEED 0.005-INCH AFTER REPAIR.** If blending exceeds limit, replace main landing gear axle.
 - (2) Using dry-cleaning solvent, Item 124, Appendix D, clean blended area.
 - (3) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer. **BLENDING MUST NOT EXCEED 0.005-INCH AFTER REPAIR.** If blending exceeds limit, replace main landing gear axle.

3-4-2.1.3 Install.

WARNING

FSCAP

Shotpeening of exterior surfaces of drag beam is a critical characteristic. Cadmium-plating and baked resin coating of axle bore are critical characteristics. Use extreme care to avoid damage to these surfaces during installation of main landing gear axle.

- a. Using nonmetallic scraper, remove old sealing compound from drag beam flange and bolthole. Using dry-cleaning solvent, Item 124, Appendix D, loosen sealing compound and scrape off.
- b. Insert main landing gear axle into drag beam axle bore with keyway in axle threads at top

and boltholes lined up (Figure 3-4-2-1, Detail A).

- c. Using dial indicator, inspect main landing gear axle for radial movement at tip of axle. **MAXIMUM RADIAL MOVEMENT IS 0.024-INCH.**

- d. Install bolt, clip and ground wire (left drag beam only), washer(s), and nut. **TORQUE NUT TO 100 - 270 INCH-POUNDS.** Make sure axle tip is angled slightly downward.

- e. On left drag beam, if previously removed, install ground wire as follows (Figure 3-4-2-1, Detail B):

- (1) Loosen ground wire nut on clip.
- (2) Pull ground wire in through bottom of clip.
- (3) Adjust length of ground wire so that 2-inches of ground wire contacts ground.
- (4) Tighten ground wire nut.

- f. On left drag beam, adjust ground wire length as follows:

- (1) Loosen ground wire nut on clip.
- (2) Adjust length of ground wire so that 2-inches of ground wire contacts ground.

- (3) Tighten ground wire nut.

- g. Apply sealing compound, Item 292, Appendix D, around bolthead, nut, and brake flange.

WARNING

The 7 lug wheel, 5003889-2 or 5003889-3, may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14-lug wheel, 040-25300, should be utilized when operating at or above gross weights of 20,500 pounds.

- h. Install main landing gear wheel brake (PARA 3-4-14).

3-4-3 BRAKE FLANGE SPACERS.

PARAGRAPH BREAKDOWN

		PAGE
3-4-3.1	Replace Brake Flange Spacers	3-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Adhesive, Item 32, Appendix D

- Adhesive, Item 60, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 254, Appendix D

References

- Appendix D
- PARA 3-4-1

3-4-3.1 Replace Brake Flange Spacers.

NOTE

Replacement of both right and left side brake flange spacers is the same, except as noted.

3-4-3.1.1 Remove.

- a. If necessary, remove main landing gear drag beam (PARA 3-4-1).

NOTE

Note orientation of original brake flange spacers prior to removal.

- b. Using sharp nonmetallic scraper, remove brake flange spacers from brake flange. Carefully tap around bonded edges of spacer to prevent dam-

age to spacer or flange (Figure 3-4-3-1, Detail A).

- c. Remove old adhesive from brake flange spacers and brake flange with methyl ethyl ketone, Item 217, Appendix D, and abrasive mat, Item 14, Appendix D. Wipe clean and dry with machinery towel, Item 211, Appendix D.

3-4-3.1.2 Inspect/Repair.

- a. Brake flange spacers may be repaired if inspection shows damage to be within the following limits:

- (1) **CORROSION UP TO 0.005-INCH DEEP AND 20% OF TOTAL AREA.**
- (2) **SCRATCHES UP TO 0.010-INCH DEEP, 0.040-INCH WIDE, AND 0.050-INCH LONG.**

(3) **NICKS AND GOUGES UP TO 0.020-INCH DEEP AND 1/4-INCH IN DIAMETER.**

(4) If limits are exceeded, replace brake flange spacer.

- b. Use abrasive cloth, Item 7, Appendix D, to blend damage to a smooth contour, removing as little material as possible.
- c. Using corrosion resistant coating, Item 118, Appendix D, touch up repaired surfaces.

3-4-3.1.3 Install.

NOTE

- When rebonding brake flange spacers while main landing gear drag beam is installed on helicopter, use same procedure as listed, except as follows: directions "UP" and "DOWN" used in positioning spacers are determined in relationship with drag beam centerline. When the helicopter is supported on jacks, the axle end of drag beam rotates downward 35°. The keyway in axle may be used as a reference to position spacers, regardless of the drag beam angle.
- When changing drag beam from left/right to right/left, the spacers, 70250-12098-101 through 70250-12098-103, on the brake flange must be repositioned (Figure 3-4-3-1, Detail A).

a. Position main landing gear drag beam as follows:

- (1) On helicopters with main landing gear, 70250-12050-045 and 70250-12050-047, place drag beam horizontally on bench with axle bore to the right and facing self.

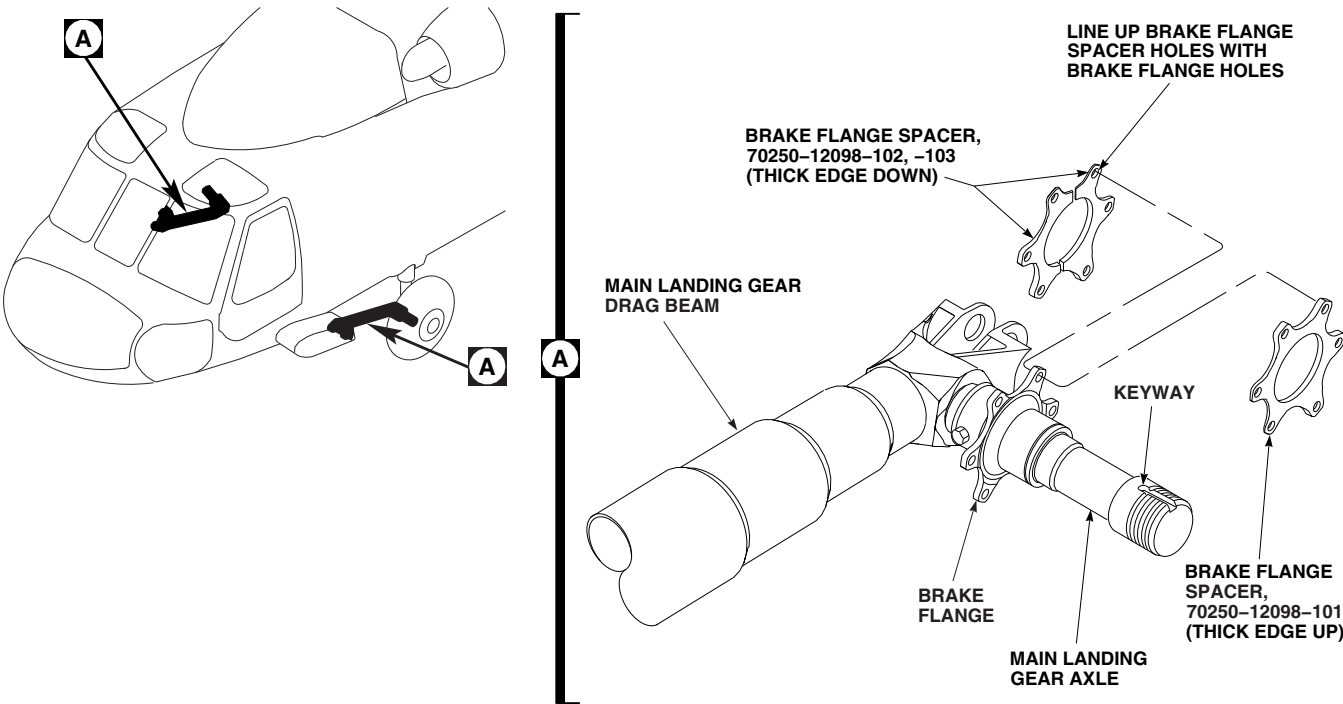
- (2) On helicopters with main landing gear, 70250-12050-046, 70250-12050-048, and 70250-12050-049, place drag beam horizontally on bench with axle bore to the left and facing self.

b. Bond brake flange spacers to brake flange as follows:

- (1) Apply even coat of primer, Item 254, Appendix D, to each surface to be bonded. Allow to dry thoroughly for at least 3 minutes at room temperature.

NOTE

- Spacers, 70250-12098-102 and -103, are to be positioned on inboard surface of brake flange. Locate spacers with thick edge down and part number exposed. Gap between spacers, 70250-12098-102 and 70250-12098-103, should line up with keyway in axle.
- Spacer, 70250-12098-101, is to be positioned on outboard side of brake flange with thick edge up and part number exposed.
- If spacers are correctly positioned, the exposed edges should be parallel to each other.
- (2) Apply bead of adhesive, Item 32 or Item 60, Appendix D, on one bonding surface only. Position spacer and bond in place immediately. Use slight contact pressure to hold spacer in place and allow bond to cure.
- c. If previously removed, install main landing gear drag beam (PARA 3-4-1).



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FIGURE 3-4-3-1. REPLACE BRAKE FLANGE SPACERS

3-4-4 DRAG BEAM SUPPORT FITTING BEARINGS.

PARAGRAPH BREAKDOWN

	PAGE
3-4-4.1 Replace Drag Beam Support Fitting Bearings	3-4-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drag Beam Fuselage Support Fitting Bearing Drift, Locally-Made, Figure H-178, Appendix H
- Fluorescent Penetrant Inspection Kit
- Inside Micrometer
- Nonmetallic Hammer
- Puller Kit, Mechanical
- Slide Hammer

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D
- Primer, Item 257, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 3-4-1

3-4-4.1 Replace Drag Beam Support Fitting Bearings.

NOTE

Replacement of both left and right drag beam support fitting bearings is the same.

3-4-4.1.1 Remove.

- Remove main landing gear drag beam (PARA 3-4-1).
- Locally make drag beam fuselage support fitting bearing drift (Figure H-178, Appendix H).

NOTE

- Puller jaws may require some alteration to allow jaws to fit gap (about 1/16-inch) between inner and outer bearings.
- Remove outer bearing from outboard lug of fuselage support fitting with slide hammer and internal pulling jaws from puller kit (Figure 3-4-4-1, Detail A).
 - Using drag beam fuselage support fitting bearing drift and nonmetallic hammer, drive inner bearing from outboard lug.

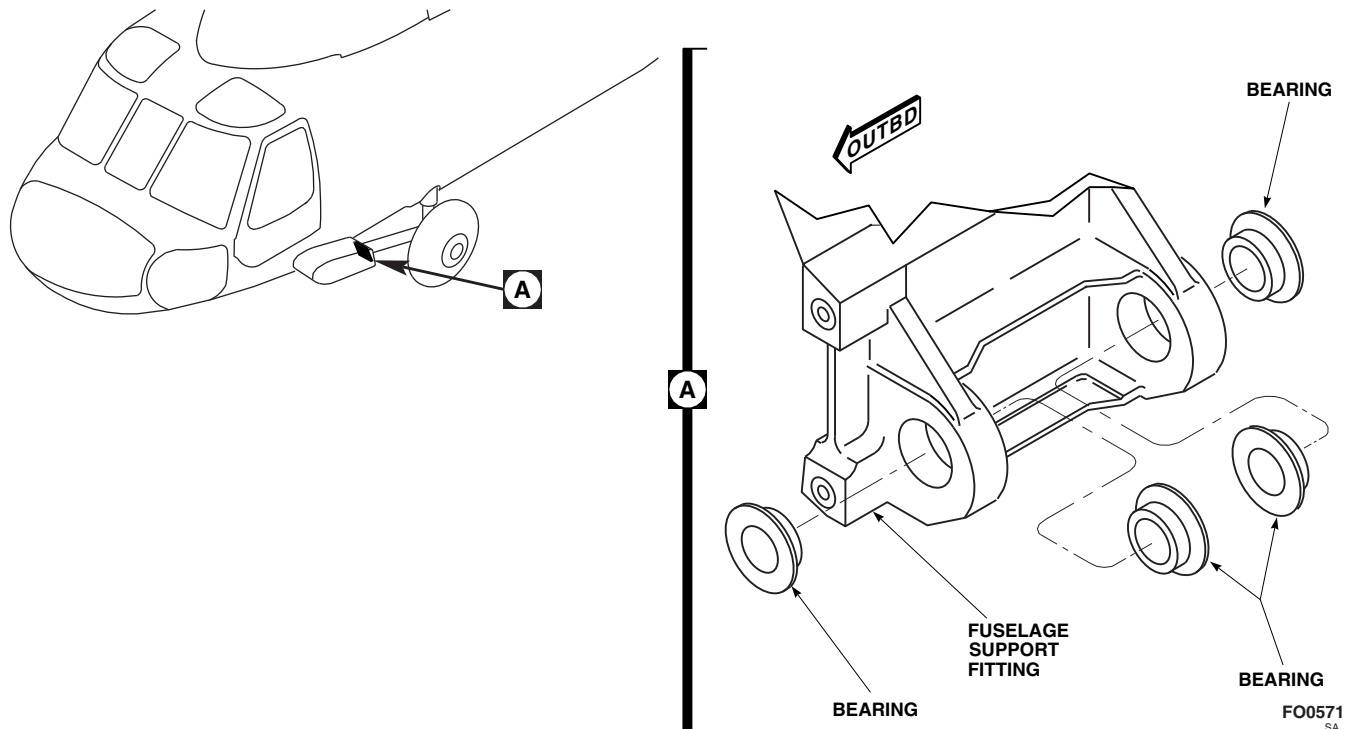


FIGURE 3-4-4-1. REPLACE DRAG BEAM SUPPORT FITTING BEARINGS

- e. For inboard lug of fitting, repeat step c. and step d.

3-4-4.1.2 Inspect/Repair.

- a. Using paint remover, Item 230, Appendix D, strip paint from lug area of fitting.
- b. Check bearing bores in fuselage support fitting for scoring and corrosion. Using abrasive cloth, Item 8, Appendix D, blend out any damage. **DAMAGE MUST NOT BE DEEPER THAN 0.010-INCH OR BLENDED AREA MUST NOT EXCEED 25% OF BORE AREA.** If damage or blending exceeds limits, replace fitting.
- c. Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect lug area of fuselage support fitting for cracks. **NONE ALLOWED.** If cracks are found, replace fitting.
- d. Using inside micrometer, measure inside diameter of bearing bores in fuselage support fitting. **BORE DIAMETER MUST NOT EXCEED**

1.4385-INCHES. If bore diameter exceeds limit, replace fitting.

- e. If fuselage support fitting damage is within acceptable limits, clean repaired area with machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe with clean machinery towel until dry.
- f. Using pressure sensitive adhesive tape, Item 247, Appendix D, mask fitting bearing bores. Touch up repaired area with three coats of primer, Item 257, Appendix D. Allow primer to dry and remove pressure sensitive adhesive tape.

3-4-4.1.3 Install.

- a. Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean bearing bores in fitting and outside surface of new bearings. Wipe with clean machinery towel until dry.
- b. Apply thin even coat of sealing compound, Item 292, Appendix D, to outside surface of

new bearings. Keep sealing compound off Teflon coating of bearing.

- c. While sealing compound, Item 292, Appendix D, is wet, line up bearings with bearing bore in lug and drive in with nonmetallic hammer (Figure 3-4-4-1, Detail A). Make sure bearing is fully seated.



Damage to bearing will result if sealing compound is allowed to remain on

Teflon coating. Do not allow sealing compound to remain on Teflon coating.

- d. Using sealing compound, Item 292, Appendix D, seal edge between bearing and fitting lug, and gap between inner ends of bearings.
- e. Install main landing gear drag beam (PARA 3-4-1).

3-4-5 MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING.

PARAGRAPH BREAKDOWN

		PAGE
3-4-5.1	Replace Main Landing Gear Drag Beam Jackpad and Tiedown Ring	3-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Drift, Locally-Made
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Mat, Item 14, Appendix D

- Corrosion Preventive Compound, Item 110, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 295A, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 3-3-1

3-4-5.1 Replace Main Landing Gear Drag Beam Jackpad and Tiedown Ring.

NOTE

Replacement of both main landing gear drag beam jackpad and tiedown rings is the same.

3-4-5.1.1 Remove.

- Using nonmetallic scraper, remove sealing compound from around nut and jackpad.

NOTE

Jackpad removal may require driving the jackpad shank out of main landing

gear drag beam with nonmetallic drift, due to foam adhering to jackpad.

- On helicopters with main landing gear, 70250-12050-047 and -049,-052 and -053, perform the following (Figure 3-4-5-1, Detail A):
 - Remove dust plug, nut, and sealing washer from jackpad.
 - Remove jackpad from main landing gear drag beam.
 - Remove remaining sealing washer from jackpad. Discard nut.

- (4) Remove nut, washers, spring washer, and bolt, and remove tiedown ring from jackpad.
- c. On helicopters with main landing gear, 70250-22009-041 and -042, perform the following (Figure 3-4-5-1, Detail A):
 - (1) Remove dust plug, nut, and sealing washer from jackpad.
 - (2) Remove jackpad from main landing gear drag beam.
 - (3) Remove remaining sealing washer from jackpad. Discard nut.
 - (4) Remove nut, washers, spring washer, and bolt, and remove tiedown ring from main landing gear drag beam.
- d. Using nonmetallic scraper, methyl ethyl ketone, Item 217, Appendix D, and abrasive mat, Item 14, Appendix D, clean up remaining sealing compound on main landing gear drag beam.

3-4-5.1.2 Inspect.

- a. Inspect jackpad hole and adjacent flat surfaces (PARA 3-3-1).
- b. Inspect cadmium-plated circular shank portion of jackpad for scratches, nicks, or pits. **SCRATCHES MUST NOT PENETRATE THROUGH CADMIUM PLATING.** If scratches penetrate through cadmium plating, replace jackpad.
- c. Inspect remaining portion of jackpad for scratches, nicks, or pits and using abrasive cloth, Item 8, Appendix D, repair as follows:
 - (1) Repair flat surfaces of jackpad as follows:
 - (a) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.020-INCH DEEP.**
 - (b) If repair exceeds limit, replace jackpad.
 - (c) Touch up finish on all repaired surfaces (PARA 2-6-5).

- (2) Repair spherical ball portion of jackpad as follows:
 - (a) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.025-INCH DEEP. BLENDED AREAS MUST NOT EXCEED 25 % OF TOTAL SPHERICAL SURFACE.**
 - (b) If repairs exceed limits, replace jackpad.
 - (c) Touch up finish on all repaired surfaces (PARA 2-6-5).

3-4-5.1.3 Install.

- a. On helicopters with main landing gear, 70250-12050-047, -049, -052, and -053, perform the following:
 - (1) Apply corrosion preventive compound, Item 110, Appendix D, to bolt.
 - (2) Position washers between tiedown ring and jackpad (Figure 3-4-5-1, Detail A). While bolt is wet, secure tiedown ring with bolt, spring washer, and nut. **TORQUE NUT TO 150 - 170 INCH-POUNDS.**
 - (3) Using sealing compound, Item 295A, Appendix D, fill hole in main landing gear drag beam. While sealing compound is wet, install jackpad as follows:
 - (a) Apply bead of sealing compound, Item 295A, Appendix D, to both sides of sealing washer. Position sealing washer on bolt end of jackpad.
 - (b) Insert bolt end of jackpad through drag beam with rotating motion, pushing out extra sealing compound. Position jackpad angled back toward axle end of drag beam.
 - (c) Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean

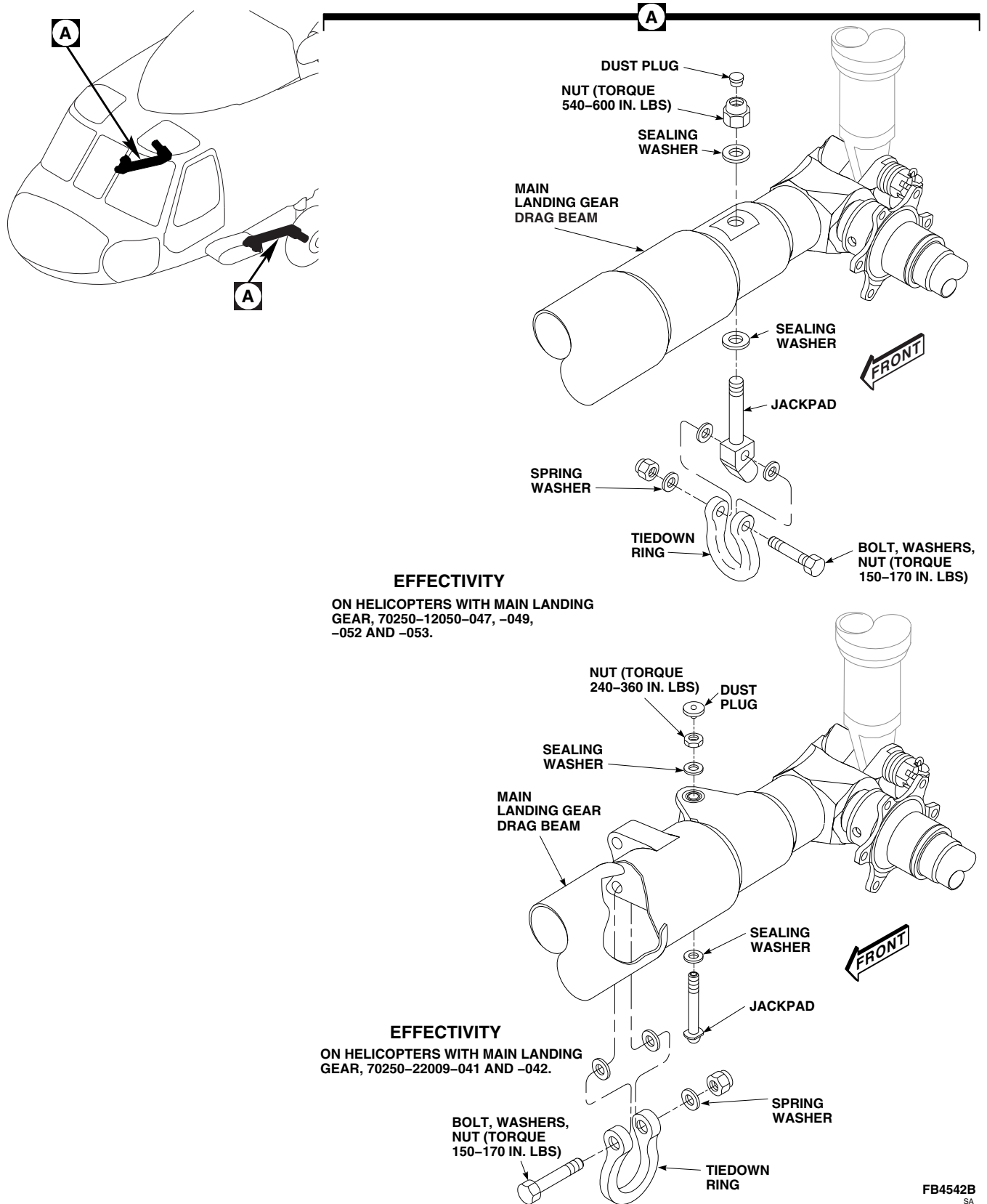


FIGURE 3-4-5-1. REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING

off sealing compound forced out of drag beam. Wipe area clean, including threaded end of jackpad.

- (d) Apply bead of sealing compound, Item 295A, Appendix D, to one side of sealing washer. With sealing compound facing drag beam, install sealing washer over jackpad.

- (e) Install new nut on jackpad.
TORQUE NUT TO 540 - 600 INCH-POUNDS.

- (4) Using sealing compound, Item 295A, Appendix D, apply over washer on bottom side of jackpad and over threads, nut, and washers on top side. **SEALING COMPOUND MUST NOT EXTEND MORE THAN 0.380-INCH BEYOND EDGE OF SEALING WASHERS.**

- (5) Install dust plug on threaded end of jackpad.

- (6) Using sealing compound, Item 295A, Appendix D, cover dust plug.

- (7) Apply torque stripe (PARA 2-6-5).

- b. On helicopters with main landing gear, 70250-22009-041 and -042, perform the following:

- (1) Apply corrosion preventive compound, Item 110, Appendix D, to bolt.

- (2) Position washers between tiedown ring (Figure 3-4-5-1, Detail A). While bolt is wet, secure tiedown ring to main landing gear drag beam with bolt, spring washer, and nut. **TORQUE NUT TO 150 - 170 INCH-POUNDS.**

- (3) Using sealing compound, Item 295A, Appendix D, fill hole in main landing gear

drag beam. While sealing compound is wet, install jackpad as follows:

- (a) Apply bead of sealing compound, Item 295A, Appendix D, to both sides of sealing washer. Position sealing washer on bolt end of jackpad.

- (b) Insert bolt end of jackpad through drag beam with rotating motion, pushing out extra sealing compound. Position jackpad angled back toward axle end of drag beam.

- (c) Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean off sealing compound forced out of drag beam. Wipe area clean, including threaded end of jackpad.

- (d) Apply bead of sealing compound, Item 295A, Appendix D, to one side of sealing washer. With sealing compound facing drag beam, install sealing washer over jackpad.

- (e) Install new nut on jackpad.
TORQUE NUT TO 240 - 360 INCH-POUNDS.

- (4) Using sealing compound, Item 295A, Appendix D, apply over washer on bottom side of jackpad and over threads, nut, and washers on top side.

- (5) Install dust plug on threaded end of jackpad.

- (6) Using sealing compound, Item 295A, Appendix D, cover dust plug.

- (7) Apply torque stripe (PARA 2-6-5).

3-4-6 MAIN LANDING GEAR DRAG BEAM SWITCH.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-6.1	Replace Main Landing Gear Drag Beam Switch	3-4-6-1	3-4-6.2	Adjust Main Landing Gear Drag Beam Switch	3-4-6-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Multimeter
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 194, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-3
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-94

3-4-6.1 Replace Main Landing Gear Drag Beam Switch.

NOTE

- Replacement of left and right switches is the same, except as noted.
- Drag beam switch is part of a molded cable assembly. If switch malfunctions, entire assembly must be replaced.

3-4-6.1.1 Remove.

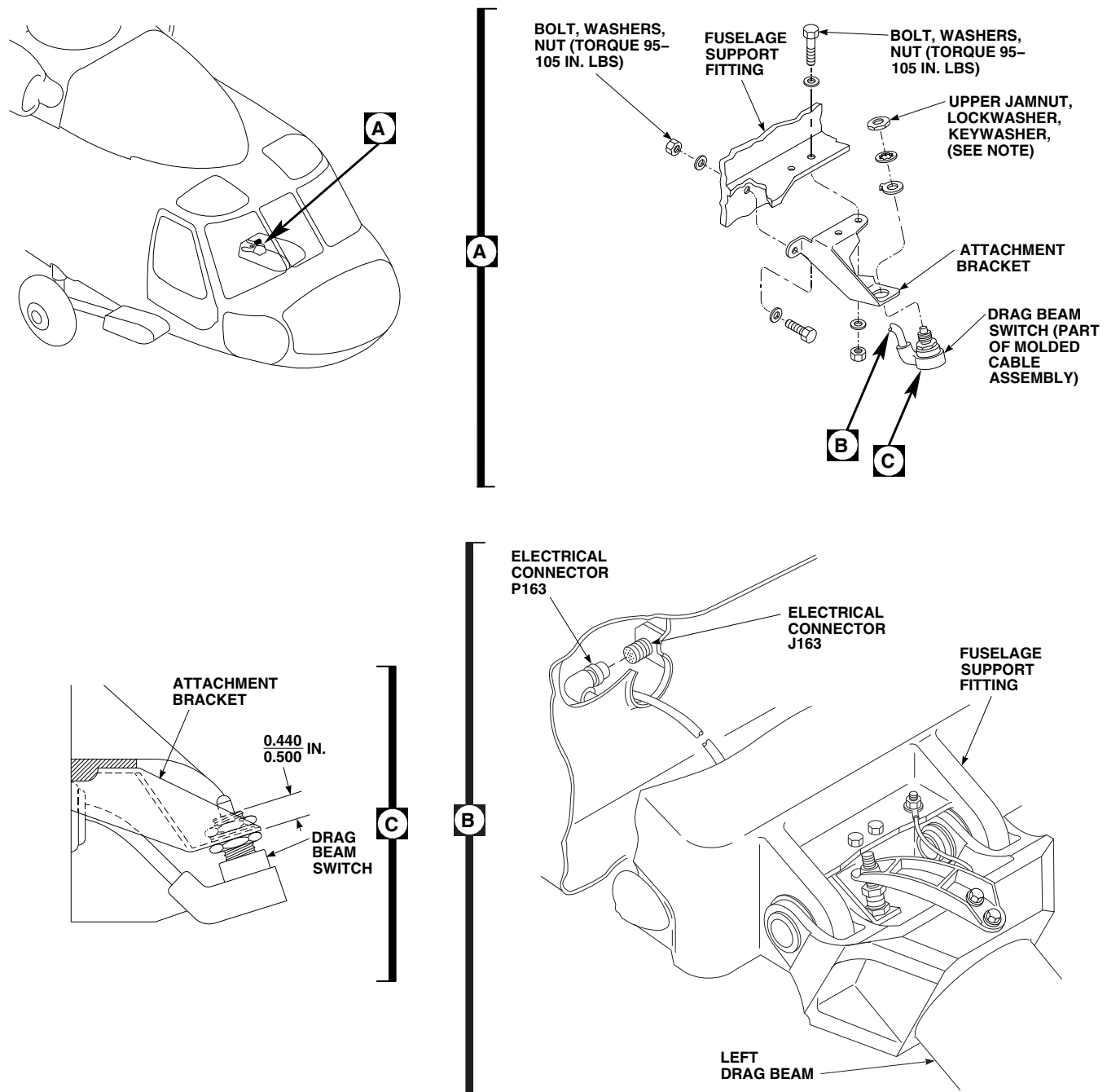
- Remove drag beam support fairings (PARA 2-4-94).
- Remove bolts, washers, nuts, and radius block (if installed on right switch) securing attachment bracket to fuselage support fitting. Remove attachment bracket to access drag

beam switch (Figure 3-4-6-1, Detail A or Figure 3-4-6-2, Detail A).

- Remove upper jamnut, lockwasher, and key-washer from drag beam switch. Remove switch from attachment bracket.
- Disconnect electrical connector, P163 (left) or P164 (right), from electrical connector, J163 (left) or J164 (right), on bracket at front of fuselage support fitting (Figure 3-4-6-1, Detail B or Figure 3-4-6-2, Detail B).
- Install protective caps and plugs on electrical connectors.
- Remove molded cable assembly from fuselage support.

3-4-6.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

**NOTE**

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 3-4-6-1. REPLACE LEFT MAIN LANDING GEAR DRAG BEAM SWITCH

- b. Remove protective caps and plugs from electrical connectors.
- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. Feed replacement molded cable assembly through fuselage support fitting and connect electrical connector, P163 (left) or P164 (right), to electrical connector, J163 (left) or J164 (right) (Figure 3-4-6-1, Detail B or Figure 3-4-6-2, Detail B).
- e. On replacement drag beam switch, remove upper jamnut, lockwasher, and keywasher from switch.
- f. Position switch in attachment bracket and install keywasher, lockwasher, and upper jamnut (Figure 3-4-6-1, Detail A or Figure 3-4-6-2, Detail A).
- g. Adjust drag beam switch in attachment bracket as follows:
 - (1) If installing left switch, **ADJUST SWITCH POSITION WITHIN ATTACHMENT BRACKET TO OBTAIN 0.440 - 0.500-INCH MEASUREMENT BETWEEN BASE OF BRACKET AND END OF THREADED PART OF SWITCH** (Figure 3-4-6-1, Detail C).
 - (2) If installing right switch, **ADJUST SWITCH POSITION, WITHIN ATTACHMENT BRACKET TO OBTAIN 0.515 - 0.575-INCH MEASUREMENT BETWEEN BASE OF BRACKET AND END OF THREADED PART OF SWITCH** (Figure 3-4-6-2, Detail C).
 - (3) **TIGHTEN UPPER JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 3-4-6-1, Detail A or Figure 3-4-6-2, Detail A).
- h. If installing left switch, position left attachment bracket (with switch installed) on fuselage support and secure with with bolts, washers, and nuts (Figure 3-4-6-1, Detail A). **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

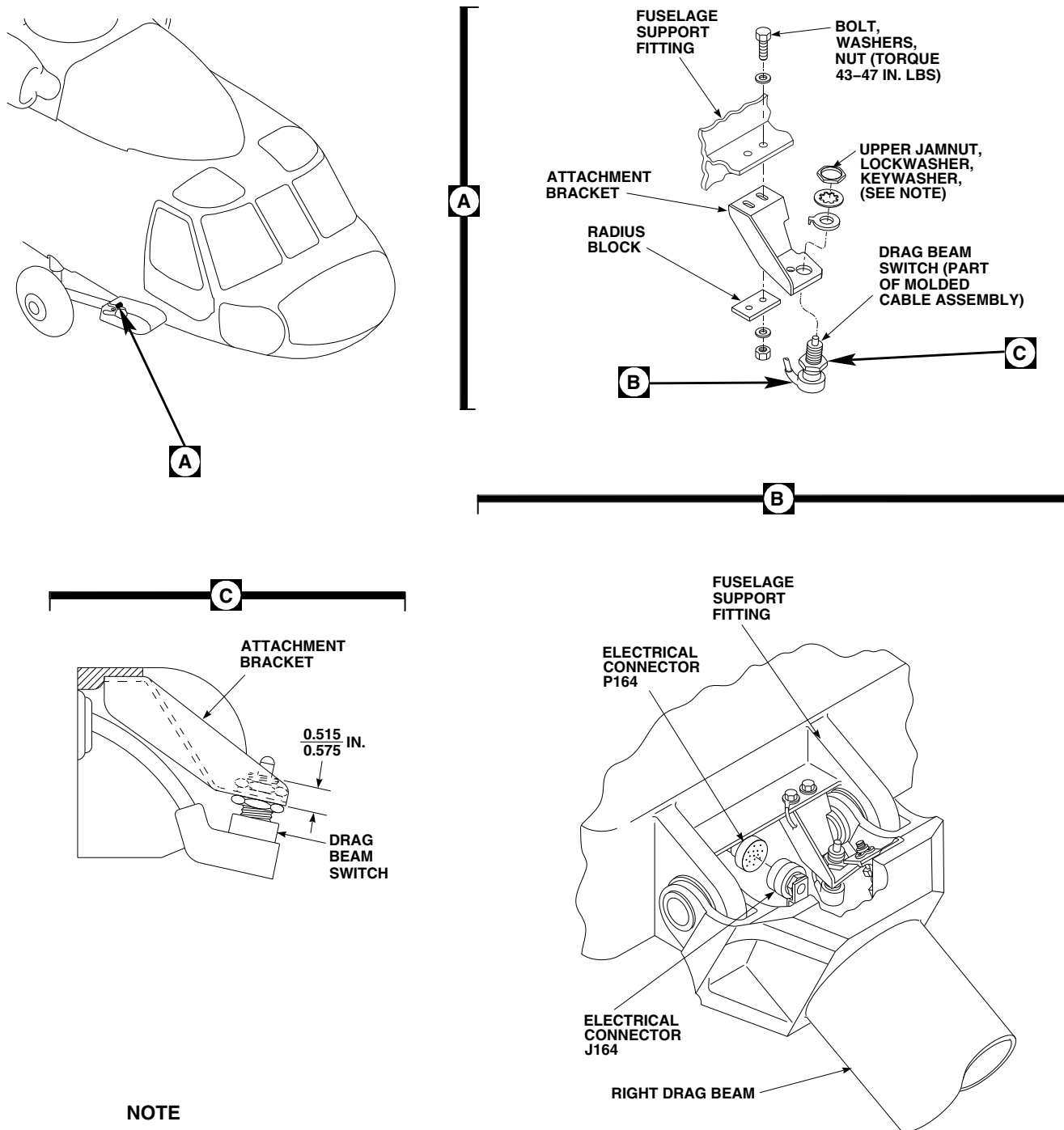
- i. If installing right switch, position right attachment bracket (with switch installed) on fuselage support and secure with bolts, radius block (if required), washers, and nuts (Figure 3-4-6-2, Detail A). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- j. Lockwire jamnut to attachment bracket with lockwire, Item 194, Appendix D.

3-4-6.1.3 Test.

NOTE

Testing of left or right drag beam switch is the same, except as noted.

- a. Disconnect electrical connector, P163 (left), or P164 (right), from electrical connector, J163 (left), or J164 (right) (Figure 3-4-6-1, Detail B or Figure 3-4-6-2, Detail B).
- b. Test drag beam switch (ground position) as follows:
 - (1) Simulate switch ground position by extending switch plunger.
 - (2) Using multimeter, check for continuity between pins A and B and E and D of electrical connector, P163 (left) or P164 (right).
 - (3) Using multimeter, check for integrity of the ground of both left and right switches by checking continuity from pin A of J164 and pin D of J163 to aircraft ground.
 - (4) If continuity is not shown, replace drag beam switch (PARA 3-4-6.1).
- c. Test drag beam switch (flight position) as follows:
 - (1) Simulate switch flight position by compressing switch plunger.
 - (2) Using multimeter, check for continuity between pins A and C, and F and D of electrical connector, P163 (left) or P164 (right).



NOTE

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 3-4-6-2. REPLACE RIGHT MAIN LANDING GEAR DRAG BEAM SWITCH

- (3) Using multimeter, check for integrity of the ground of both left and right switches by checking continuity from pin A of J164 and pin D of J163 to aircraft ground.
- (4) If continuity is not shown, replace drag beam switch (PARA 3-4-6.1).
- d. Connect electrical connector, P163 (left), or P164 (right), to electrical connector, J163 (left), or J164 (right).
- e. Adjust main landing gear drag beam switch (PARA 3-4-6.2).

3-4-6.2 Adjust Main Landing Gear Drag Beam Switch.

NOTE

Adjustment of left and right switches is the same, except as noted.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Place jack under stub wing jackpad on side where drag beam switch is being adjusted.
- c. Jack helicopter until shock strut is fully extended (PARA 1-6-3).

WARNING

Injury to personnel and damage to equipment will result if shock strut is stuck or not fully extended. Kick or shake main landing gear wheel to make sure shock strut is fully extended.

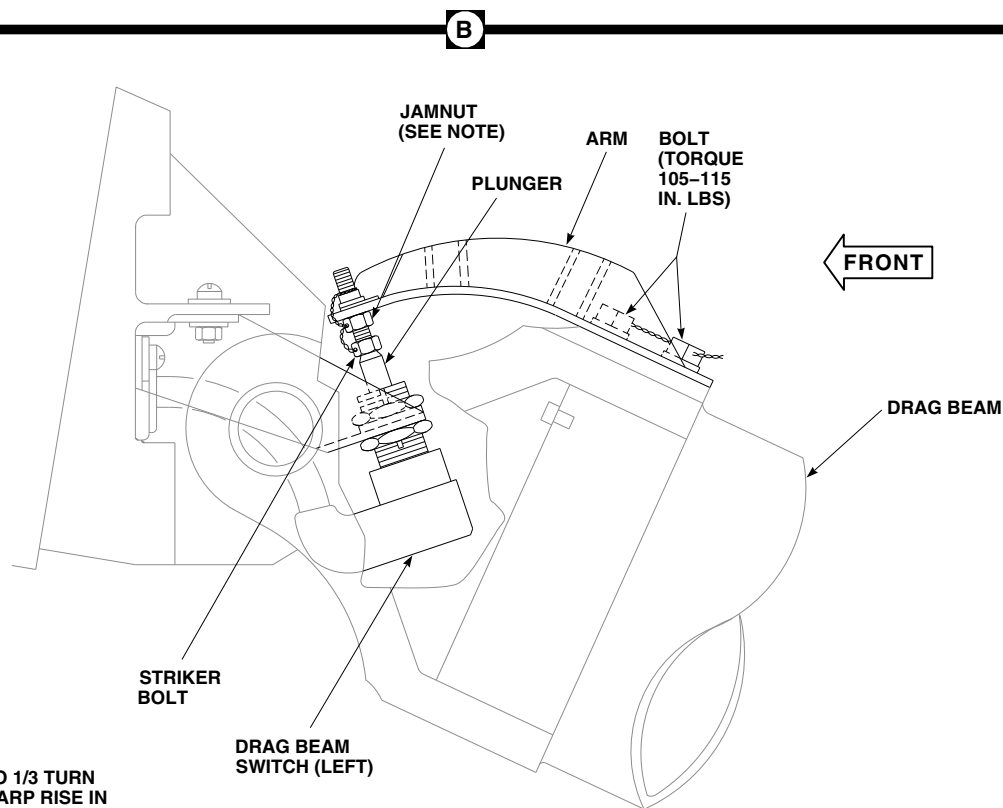
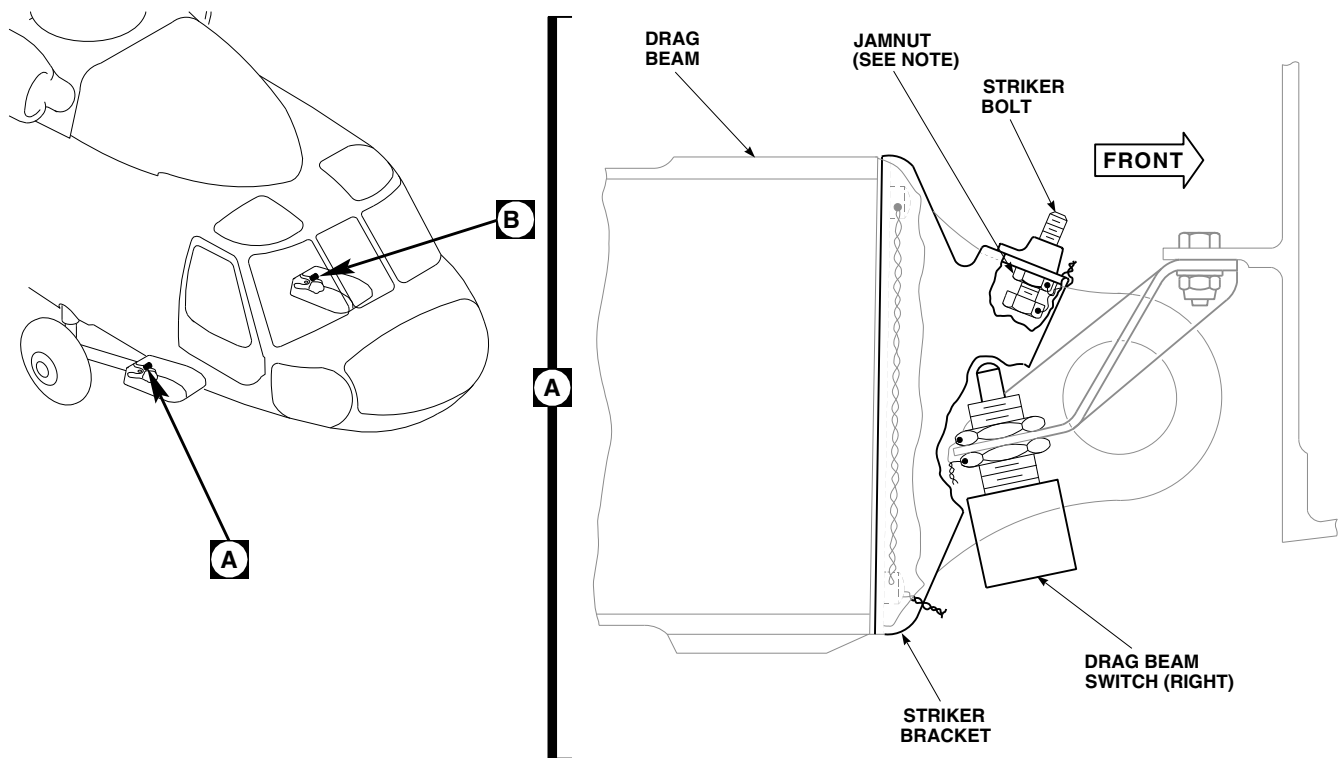
- d. On left main landing gear, perform the following (Figure 3-4-6-3, Detail B):

- (1) Loosen two bolts attaching arm to drag beam.
- (2) Position arm so that striker bolt (on arm) lines up with plunger of drag beam switch.
- (3) **TIGHTEN TWO BOLTS ATTACHING ARM TO DRAG BEAM. TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

NOTE

On right main landing gear, a fixed-position striker bracket is attached to front face of drag beam. Switch and striker bolt are adjustable up and down (Figure 3-4-6-3, Detail A).

- e. Loosen jamnut of striker bolt and adjust bolt so it does not touch drag beam switch (Figure 3-4-6-3, Detail A or Detail B).
- f. **ADJUST STRIKER BOLT UNTIL HEAD OF BOLT TOUCHES DRAG BEAM SWITCH. TURN BOLT TOWARD DRAG BEAM SWITCH ANOTHER FOUR FULL TURNS.**
- g. **TIGHTEN JAMNUT ON STRIKER BOLT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- h. Lockwire jamnut, starting at head of striker bolt, to jamnut, to threaded shank of bolt with lockwire, Item 194, Appendix D.
- i. On left main landing gear, lockwire two bolts attaching arm to drag beam with lockwire, Item 194, Appendix D (Figure 3-4-6-3, Detail B).
- j. Lower helicopter and remove jack (PARA 1-6-3).
- k. Install drag beam support fairings (PARA 2-4-94).

**NOTE**

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

FIGURE 3-4-6-3. ADJUST MAIN LANDING GEAR DRAG BEAM SWITCH

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3-4-7 MAIN LANDING GEAR SHOCK STRUT.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-7.1	Replace Main Landing Gear Shock Strut	3-4-7-1	3-4-7.2	Repair Main Landing Gear Shock Strut	3-4-7-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beakers, 800 ml (2)
- Container, 2-Gallon
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 3, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Petrolatum, Item 234, Appendix D

- Tubing, Nonmetallic, Item 344, Appendix D
- Union, AN815-4

References

- Appendix D
- MIL-PRF-46170
- PARA 1-3-2
- PARA 1-6-3
- PARA 2-4-95
- PARA 2-4-96
- PARA 2-5-9
- PARA 2-6-5
- PARA 3-4-8

3-4-7.1 Replace Main Landing Gear Shock Strut.

NOTE

Replacement of both main landing gear shock struts is the same.

3-4-7.1.1 Prepare to Remove.

- Remove upper and lower main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).

- Jack helicopter until shock strut is fully extended (PARA 1-6-3).

3-4-7.1.2 Inspect.

- Use beakers to collect fluid from upper and lower stages, identify as upper or lower.
- Remove upper/lower stage valve caps (Figure 3-4-7-1, Detail B).
- Slip nonmetallic tubing, Item 344, Appendix D, over valve stem. Place free end in correct

container to capture hydraulic fluid as nitrogen is released from shock strut chambers.

- d. Turn outer hexnut counterclockwise to release nitrogen from upper/lower shock strut chamber.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- e. After all nitrogen pressure is released, turn inner hexnut counterclockwise to remove.
- f. Allow any remaining hydraulic fluid in upper/lower shock strut chambers to drain into containers.
- g. **IF DISCHARGE FROM UPPER OR LOWER STAGE IS GREATER THAN 200-MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT.**
- h. **IF DISCHARGE IS BETWEEN 100 AND 200-MILLILITERS (CUBIC CENTIMETERS), SERVICE SHOCK STRUT (PARA 1-3-2). REINSPECT SHOCK STRUT AT 200 FLIGHT-HOURS.**
- i. **IF DISCHARGE IS LESS THAN 100-MILLILITERS (CUBIC CENTIMETERS), SERVICE SHOCK STRUT (PARA 1-3-2). REINSPECT SHOCK STRUT AT 500 FLIGHT-HOURS.**

3-4-7.1.3 Remove.

WARNING

Injury to personnel and damage to equipment will result if shock strut is not supported during removal. Shock

strut weighs 60 pounds. Make sure shock strut is supported during removal.

- a. Remove bolts, countersunk washers, washers, and nuts from upper and lower end of shock strut (Figure 3-4-7-1, Detail A).
- b. Remove shock strut.
- c. Check upper fuselage fitting assembly spherical bearing for maximum radial and axial play. **FUSELAGE FITTING ASSEMBLY SPHERICAL BEARING MAY HAVE UP TO 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY.** Replace fitting assembly spherical bearing if out of tolerance (PARA 2-5-9).

3-4-7.1.4 Install.

- a. If replacement shock strut assembly is filled with hydraulic preservative fluid, MIL-PRF-46170, drain as follows:
 - (1) Remove cap from upper valve (Figure 3-4-7-1, Detail B).
 - (2) Release nitrogen pressure in shock strut by turning outer hexnut counterclockwise. A hissing sound can be heard when releasing nitrogen.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- (3) When valve stops hissing, turn inner hexnut counterclockwise and remove valve.
- (4) Remove bleeder plug and packing from upper shock strut (Figure 3-4-7-1, Detail A). Retain packing for use on union.
- (5) Install union with packing off bleeder plug, in bleeder port. Connect nonmetallic

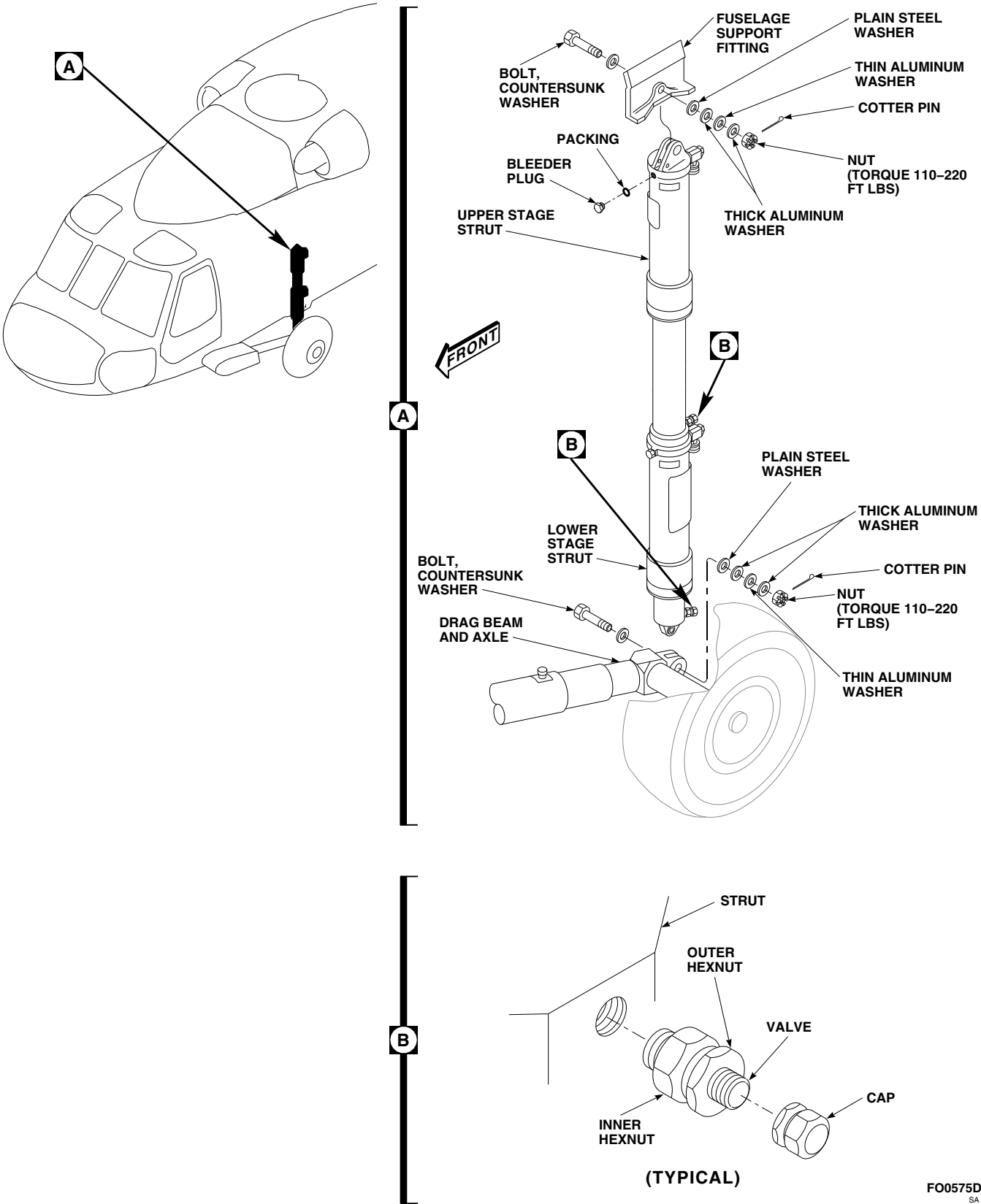


FIGURE 3-4-7-1. REPLACE MAIN LANDING GEAR SHOCK STRUT

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tubing, Item 344, Appendix D, to union. Place overflow line from bleeder port into 2-gallon container.

- (6) Install kneeling valve in shock strut upper stage (PARA 3-4-8).
- (7) Attach servicing chuck from bottled nitrogen, Item 221, Appendix D, to upper valve.
- (8) Lay shock strut on side with bleeder plug facing down.

WARNING

Injury to personnel and damage to equipment will result if an unregulated nitrogen source is used. Apply pressure gradually from a regulated source of nitrogen.

NOTE

- Do not pressurize strut with more than 100 psi pressure.
 - There are 72.94-inches between center of upper and lower attachment boltholes when both stages of shock strut are fully extended.
- (9) Increase nitrogen pressure until upper stage strut is fully extended.
 - (10) Disconnect nitrogen bottle from valve and continue draining shock strut until hydraulic fluid cannot be seen coming from bleeder port.
 - (11) Remove union from bleeder port.
 - (12) Install packing, lubricated with petrolatum, Item 234, Appendix D, on bleeder plug.
 - (13) Screw bleeder plug into cylinder until contact is made with cylinder surface.
 - (14) Drain preservative fluid from lower stage of strut using the same method as with upper stage.

WARNING

To prevent injury to personnel and damage to equipment, when using shock struts, 70250-12051-042, make sure both left and right sides have same part number. Shock struts, 70250-12051-043 through 70250-12051-045, are fully interchangeable, and may be mixed with each other on left and right sides.

- b. Position upper and lower shock strut lugs on fuselage and drag beam support fittings (Figure 3-4-7-1, Detail A).
- c. Bolt upper end of shock strut to fuselage support fitting and lower end of shock strut to drag beam fitting as follows:
 - (1) Install countersunk washer on bolt with countersunk side toward head.
 - (2) Install bolt with head facing inboard.
 - (3) Install plain steel washer on bolt.
 - (4) Install thick aluminum washer on bolt.
 - (5) Install thin aluminum washer on bolt.
 - (6) Install thick aluminum washer on bolt.
 - (7) Install nut.
- d. **TORQUE NUT TO 110 - 220 FOOT-POUNDS.** Install cotter pin.

NOTE

Minor seepage in service is not considered cause for replacement.

- e. Service main landing gear shock strut upper and lower stages (PARA 1-3-2).
- f. After replacing shock strut with new strut, perform static leak check as follows:
 - (1) Wipe polished piston of shock strut dry using machinery towel, Item 211, Appendix D.

- (2) Wait 1 hour and check polished piston for hydraulic fluid.
- (3) If there is any evidence of hydraulic fluid, replace shock strut.
- (4) If no fluid is present, soak a clean machinery towel, Item 211, Appendix D, with hydraulic fluid, Item 154, Appendix D, and wipe all exposed polished portions of piston.
- g. Make sure area is clean and free of foreign material.
- h. Lower helicopter and remove jack (PARA 1-6-3).
- i. Install upper and lower main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).

3-4-7.2 Repair Main Landing Gear Shock Strut.

- a. Repair scratches found on shock strut as follows:
 - (1) Blend out any minor scratches on housing with abrasive cloth, Item 3, Appendix D. **SCRATCHES ORIENTATED AT RIGHT ANGLES TO CENTERLINE MUST NOT BE OVER 20% OF HOUSING CIRCUMFERENCE. SCRATCHES PARALLEL TO CENTERLINE MUST NOT BE MORE THAN 1.5-INCHES LONG. DEEPEST PART OF SCRATCH MUST NOT BE MORE THAN 0.005-INCH AFTER BLENDING.** If scratches exceed limits, replace shock strut (PARA 3-4-7.1).

- (2) Blend out all other scratches or nicks on upper stage housing and lug with abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.020-INCH AFTER BLENDING.** If scratches exceed limits, replace shock strut (PARA 3-4-7.1).
- (3) Blend out all other scratches or nicks on lower stage housing with abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.** If scratches exceed limits, replace shock strut (PARA 3-4-7.1).
- b. No repair is allowed to upper or lower piston surface. **ALLOWABLE DEPTH OF DAMAGE IS 0.010-INCH, LENGTH NOT MORE THAN 1-INCH, NOT OVER 20 % OF PISTON CIRCUMFERENCE.** If damage exceed limits, replace shock strut (PARA 3-4-7.1).
- c. **ALLOWABLE WEAR IS LIMITED TO 0.0015-INCH OR SMALLEST OUTSIDE DIAMETER OF 3.4935-INCHES FOR UPPER PISTON, AND 2.4935-INCHES FOR LOWER PISTON.** If wear exceeds limits, replace shock strut (PARA 3-4-7.1).
- d. Lower piston lug scratches or nicks may be blended out using abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.** If scratches exceed limits, replace shock strut (PARA 3-4-7.1).
- e. Touch up paint as required (PARA 2-6-5).

3-4-8 MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-8.1	Replace Main Landing Gear Shock Strut Kneeling Valve	3-4-8-1	3-4-8.2	Repair Main Landing Gear Shock Strut Kneeling Valve	3-4-8-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hydraulic Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-2
- PARA 1-6-3
- PARA 2-4-95
- PARA 2-4-96

3-4-8.1 Replace Main Landing Gear Shock Strut Kneeling Valve.

NOTE

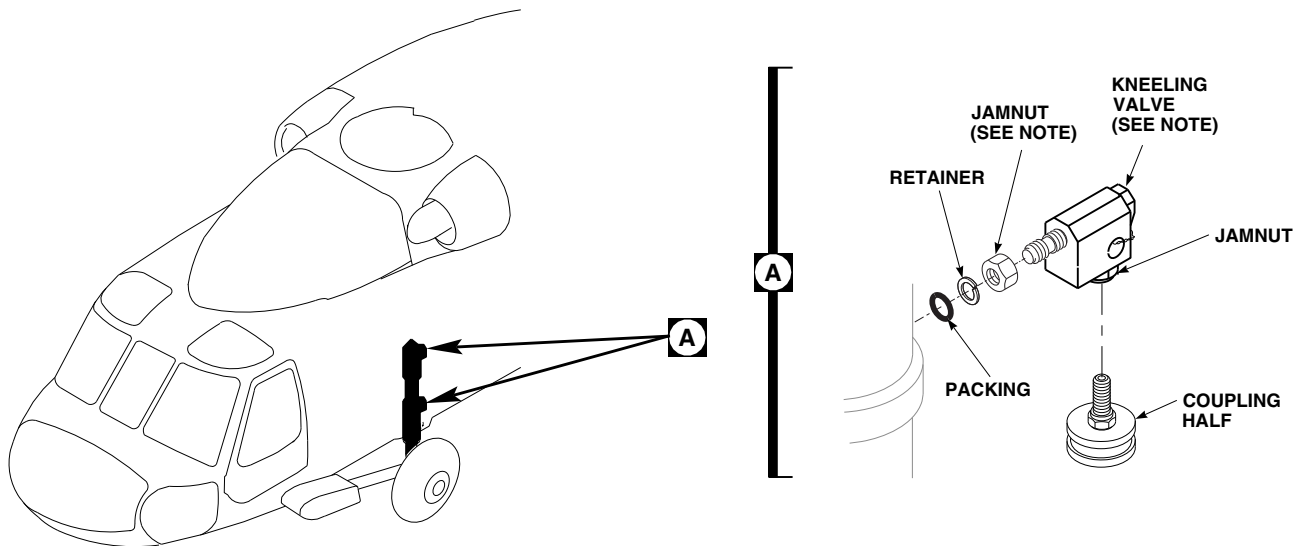
Replacement of upper and lower kneeling valves on both shock struts is the same.

3-4-8.1.1 Remove.



Damage to equipment will result if both lower and upper stages of main landing gear shock strut are collapsed. The rear,

- lower ESSS hard point will contact the main landing gear tire.
- Remove upper and lower main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).
 - Jack helicopter with jack beneath pad on drag beam support (PARA 1-6-3).
 - Deservice upper stage shock strut (PARA 1-3-2).
 - Loosen jamnut and remove coupling half from kneeling valve (Figure 3-4-8-1, Detail A).
 - Loosen jamnut and remove kneeling valve from strut. Remove packing, retainer, and jamnut from valve. Keep jamnut and retainer for installation of replacement valve. Discard packing.

**NOTE**

TIGHTEN KNEELING VALVE AND JAMNUT 1-1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN.

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FIGURE 3-4-8-1. REPLACE MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE

3-4-8.1.2 Install.

- Lubricate packing, retainer, and male threads of replacement kneeling valve with hydraulic fluid, Item 154, Appendix D.
- Install jamnut, retainer, and packing onto kneeling valve. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of kneeling valve (Figure 3-4-8-1, Detail A).
- Install kneeling valve in strut until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN KNEELING VALVE**

AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN.

- Install coupling half in kneeling valve and tighten jamnut.
- Service upper stage shock strut (PARA 1-3-2).
- Make sure area is free of foreign material before closing up.

3-4-8-2 Change 8

- g. Lower helicopter and remove jack (PARA 1-6-3).
- h. Install upper and lower main landing gear fairings (PARA 2-4-95 and PARA 2-4-96).

3-4-8.2 Repair Main Landing Gear Shock Strut Kneeling Valve.

NOTE

Repair of both main landing gear shock strut kneeling valves is the same.

3-4-8.2.1 Disassemble.

- a. Remove knob stop pin from valve body (Figure 3-4-8-2).
- b. Remove shaft by turning knob counterclockwise.
- c. Remove packing and retainers from shaft. Discard packing and retainers.
- d. Loosen jamnut and remove coupling half from valve body.
- e. Remove packing, retainer, and jamnut from coupling half. Discard packing and retainer.

3-4-8.2.2 Inspect.

- a. Inspect shaft for corrosion or deformation. **NONE ALLOWED.** If corroded or deformed, replace shaft.
- b. Inspect shaft bore in valve body for corrosion and deformation. **NONE ALLOWED.** If corroded or deformed, replace body.

3-4-8.2.3 Assemble.

- a. Lubricate packing, retainer, and male threads of coupling half with hydraulic fluid, Item 154, Appendix D.
- b. Install jamnut, retainer, and packing onto coupling half. Turn nut and retainer down until packing is pushed firmly against the lower threaded section of coupling half (Figure 3-4-8-2).
- c. Install coupling half into valve body until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN COUPLING HALF AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- d. **PREVENT COUPLING HALF FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
- e. Lubricate packing and retainers with hydraulic fluid, Item 154, Appendix D.
- f. Install retainer, packing, and second retainer on shaft.
- g. Insert shaft carefully into valve body.
- h. Turn knob clockwise to set specified clearance between knob and valve body. **CLEARANCE MUST RANGE FROM 0.060 TO 0.130-INCH.**
- i. Install knob stop pin in valve body.
- j. Using lockwire, Item 197, Appendix D, lockwire knob to hole in valve body.

3-4-8.2.4 Test.

Test for operation on helicopter, no bench test required.

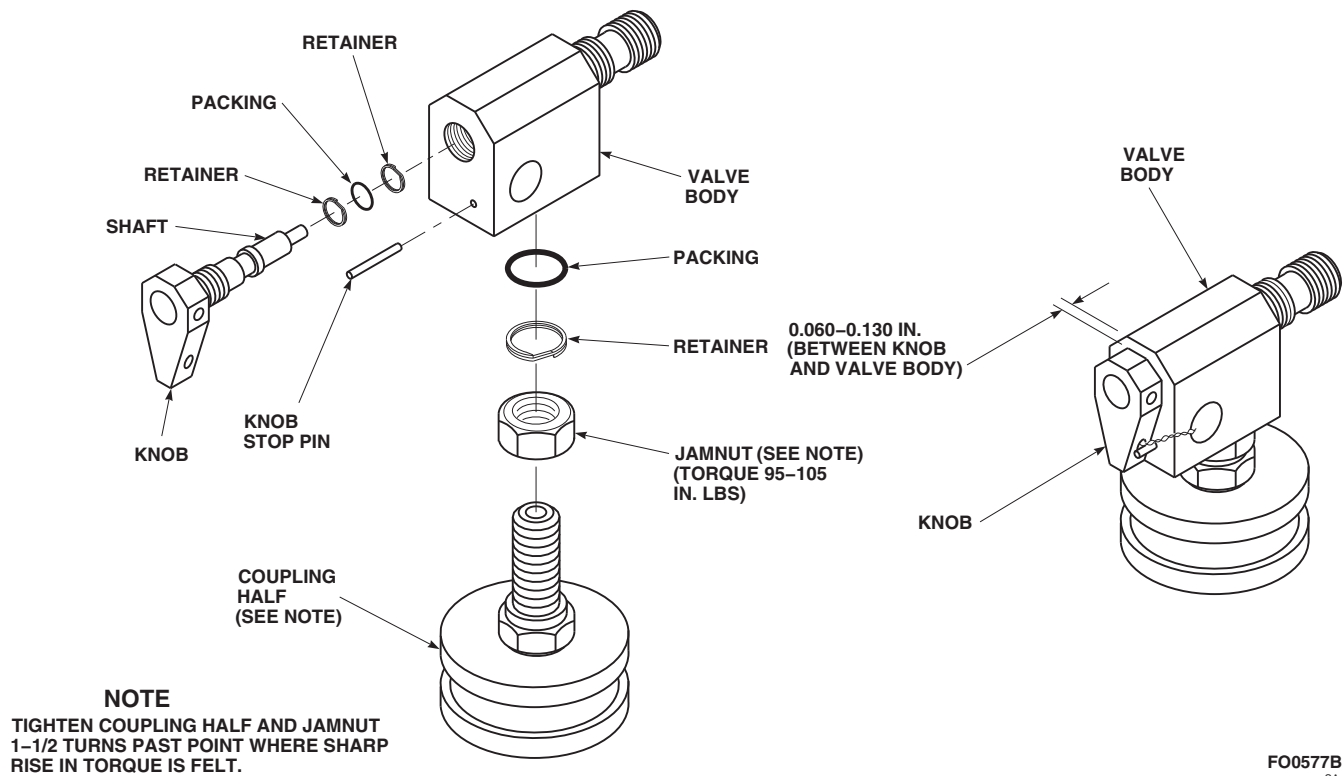


FIGURE 3-4-8-2. REPAIR MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE

3-4-9 MAIN LANDING GEAR SHOCK STRUT AIR VALVE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-9.1 Replace Main Landing Gear Shock Strut Air Valve	3-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-2
- PARA 2-4-95

3-4-9.1 Replace Main Landing Gear Shock Strut Air Valve.

NOTE

Replacement of upper and lower main landing gear shock strut air valves is the same.

3-4-9.1.1 Remove.

CAUTION

Damage to equipment will result if both lower and upper stages of main landing gear shock strut are collapsed. The rear, lower ESSS hard point will contact the main landing gear tire.

- Remove lower main landing gear fairing (PARA 2-4-95).
- Remove cap from air valve being replaced (Figure 3-4-9-1, Detail A).

- Turn outer hexnut counterclockwise until nitrogen hissing from valve is heard.

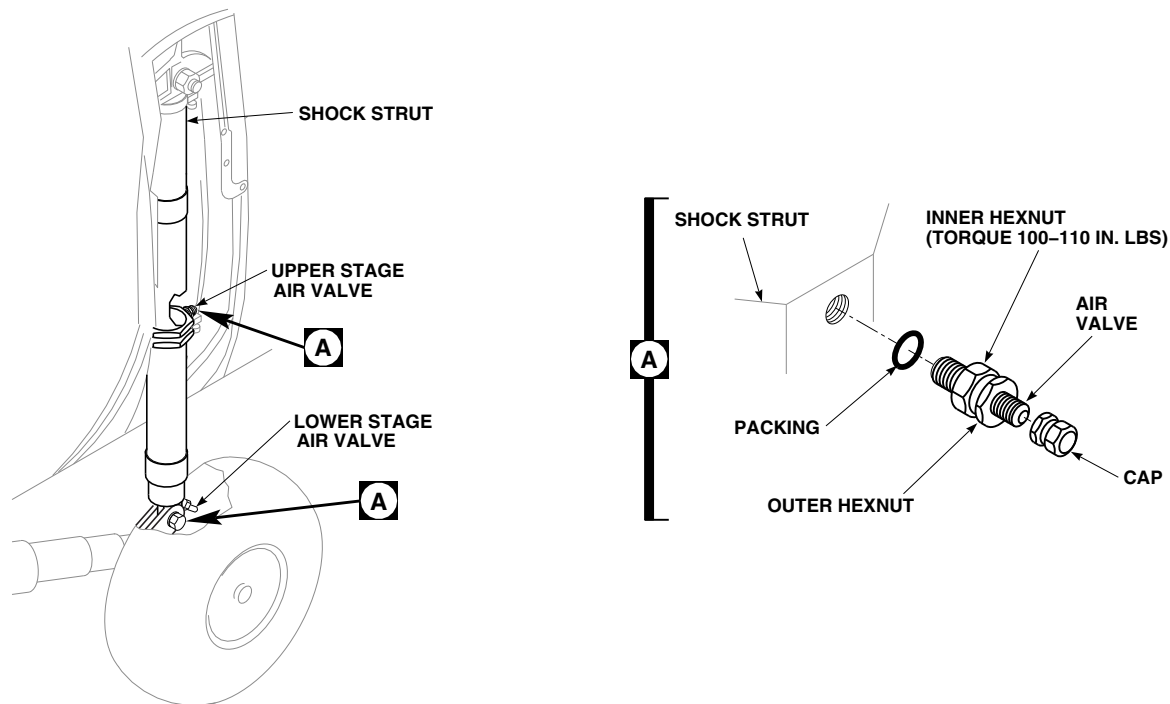
WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valves. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valves.

- When air valve stops hissing, turn inner hexnut counterclockwise and remove air valve and packing.

3-4-9.1.2 Inspect.

- Inspect packing for damage. If packing is determined to be source of air leak, replace packing.

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SA**FIGURE 3-4-9-1. REPLACE MAIN LANDING GEAR SHOCK STRUT AIR VALVE****NOTE**

Replacement air valve includes new packing.

- b. Inspect valve for damage. If valve is determined to be malfunctioning, replace valve and discard packing.

3-4-9.1.3 Install.

- a. Install air valve and packing in shock strut. (Figure 3-4-9-1, Detail A).
- b. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.**
- c. Using lockwire, Item 197, Appendix D, lockwire inner hexnut to shock strut.
- d. Service stage of shock strut from which valve was replaced (PARA 1-3-2).
- e. Make sure area is clean and free of foreign material before closing up.
- f. Install lower main landing gear fairing (PARA 2-4-95).

3-4-10 MAIN LANDING GEAR WHEEL AND TIRE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-10.1 Replace Main Landing Gear Wheel and Tire	3-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Extension, 5-in.
- Handle, Drive, ½-in.
- Protection, Eye
- Tire Inflator Kit
- Torque Wrench, 30 - 150 ft lbs
- Wrench, Crowfoot, 2 ¼-in.

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 149, Appendix D

- Leak Detection Fluid, Item 189, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-3
- PARA 3-4-11
- PARA 3-4-13

3-4-10.1 Replace Main Landing Gear Wheel and Tire.

NOTE

Replacement of both main landing gear wheels and tires is the same.

3-4-10.1.1 Remove.

- Jack helicopter until shock strut is fully extended (PARA 1-6-3).
- Set parking brake. Leave brake on until wheel is replaced.



- To prevent injury to personnel, completely release pressure from tire before removing valve core from valve stem.
- Do not remove valve core from valve stem to shorten time required to deflate tire.
- Remove valve cap from valve stem. Completely deflate tire, then remove valve core

from valve stem. Attach deflation tag (Figure 3-4-10-1, Detail A).

CAUTION

Damage to bearings will result if care is not taken while removing wheel. Bearings will drop from wheel. Make sure bearings do not drop while removing wheel.

- d. Remove cotter pin, nut, and keywasher using 2 1/4-inch crowfoot wrench, 5-inch extension, and 1/2-inch drive handle. Remove outer seal and outer bearing from wheel.

NOTE

In the following step, inner bearing and seal may remain on axle when removing wheel. Make sure bearing does not fall on ground.

- e. Carefully slide wheel and tire assembly from axle.
- f. Remove inner bearing and inner seal from axle or wheel.

3-4-10.1.2 Inspect.

- a. Check bearing seals for damaged reinforcing ring, visible damage, and deformation. **NO DAMAGE OR DEFORMATION ALLOWED.** If any seal is found to be unserviceable, replace seal.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure

all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Do not spin bearings while drying.

- b. Using dry-cleaning solvent, Item 124, Appendix D, clean bearings. Dry bearings with compressed air, if available, or let air-dry.
- c. Inspect bearing for evidence of wear, corrosion, galling, scratches, pits, water stains, heat discoloration, or cracked and distorted bearing cage. **NONE ALLOWED.** If damaged, replace bearings.
- d. If bearing is determined serviceable, repack bearing using grease, Item 149, Appendix D.

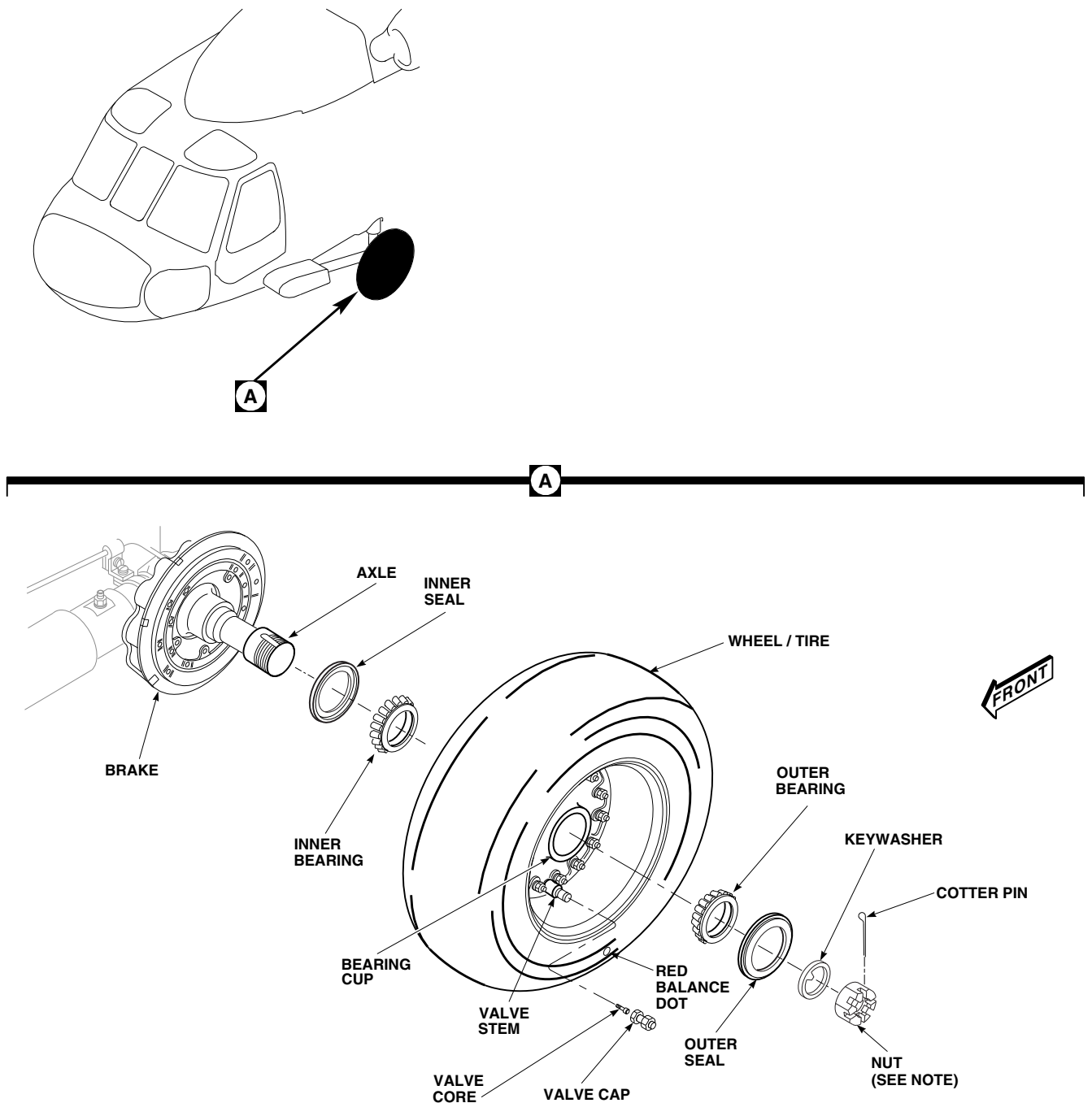
3-4-10.1.3 Install.

WARNING

Injury to personnel and damage to equipment will result if tire pressure exceeds 20 psi prior to installation. Make sure tire pressure is 20 psi or less prior to installation.

NOTE

- 10 ply and 12 ply tires are fully interchangeable and may be mixed with each other on left and right sides.
 - If there is no red balance dot present on tire sidewall, align serial number with valve stem.
- a. Make sure red balance dot on tire sidewall lines up with valve stem on wheel (Figure 3-4-10-1, Detail A).
 - b. Inspect wheel for obvious damage before installing.



NOTE

USING 2-1/4 IN. CROWFOOT WRENCH, 5 IN. EXTENSION, AND 1/2 IN. DRIVE HANDLE, TURN WHEEL TIGHTENING NUT WITH TORQUE WRENCH UNTIL AN INCREASE IN ROTATIONAL WHEEL DRAG IS FELT, THEN TORQUE NUT 70-90 FT LBS, THEN BACK OFF NUT AND SLOWLY TORQUE NUT 35-45 FT LBS.

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FIGURE 3-4-10-1. REPLACE MAIN LANDING GEAR WHEEL AND TIRE

- c. Using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D, clean axle and bearing cups.
- d. Using grease, Item 149, Appendix D, lubricate axle. Do not lubricate threads.

CAUTION

The part number for the seal and inner bearing is different, depending on whether a 7 lug wheel, 5003889-2 or 5003889-3, or 14 lug wheel, 040-25300, configuration is being installed. Make sure correct part number for the seal and inner bearing is installed.

- e. Install inner seal and inner bearing on axle. Lip of inner seal is toward inner bearing.
- f. Using grease, Item 149, Appendix D, lubricate bearing cups in wheel and lips of seals.
- g. Support wheel and tire assembly and line up brake discs.

WARNING

The 7 lug wheel, 5003889-2 or 5003889-3, may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel, 040-25300, should be utilized when operating at or above gross weights of 20,500 pounds.

- h. Carefully slide wheel and tire assembly on axle. Make sure wheel disc lugs are through brake discs and inner bearing is properly seated.

CAUTION

The part number for the seal and outer bearing is different depending on whether a 7 lug wheel, 5003889-2 or 5003889-3, or 14 lug wheel, 040-25300, configuration is being installed. Make sure correct part number for the seal and outer bearing is installed.

- i. Place outer bearing and outer seal on axle. Install bearing in bearing cup in wheel. Install seal in bearing cup. Lip of seal is toward bearing.
- j. Install keywasher and nut on axle.
- k. Release parking brake.
- l. **USING 2 1/4-INCH CROWFOOT WRENCH, 5-INCH EXTENSION, AND 1/2-INCH DRIVE HANDLE, TURN WHEEL WHILE TIGHTENING NUT WITH TORQUE WRENCH UNTIL AN INCREASE IN ROTATIONAL WHEEL DRAG IS FELT.**
- m. While continuing to turn wheel, **TORQUE NUT TO 70 - 90 FOOT-POUNDS**, to make sure full seating of wheel bearings is achieved.
- n. Back out nut until it no longer contacts wheel.
- o. While continuously turning wheel, slowly **TORQUE NUT TO 35 - 45 FOOT-POUNDS.**
- p. Back off to next castellation.
- q. Push wheel inboard and outboard to make sure there is no free play in bearings. If play exists, repeat step l. through step q. until no free play exists in bearings.
- r. If play still exists, advance nut to next castellation. Wheel must rotate freely with no binding. Install cotter pin.

- s. Remove valve cap from valve stem.

WARNING

- To prevent injury to personnel or damage to equipment, use a tire inflator kit containing a pressure relief valve or similar device. Before inflating tire, check tire inflator to make sure that nitrogen source is vented at 160 psi maximum.
- If required equipment is inoperative or unavailable, tire inflation must not be attempted.
- To prevent injury to personnel or damage to equipment, do not inflate tire above its rated inflation pressure.
- To prevent injury to personnel, do not use an unregulated nitrogen pressure source to service tire.

- To prevent injury to personnel, never stand facing sidewall of tire being serviced. Stand facing tire tread.
- t. Using regulated nitrogen pressure source connected to tire inflator, **INFLATE TIRE TO 130 - 140 PSI WITH NITROGEN, ITEM 221, APPENDIX D.**
- u. After a 5 - 10 minute waiting period, check wheel for leakage as follows:
 - (1) Apply leak detection fluid, Item 189, Appendix D, to valve stem area and around tire beads.
 - (2) If bubbles appear, correct at point of leakage (PARA 3-4-11 or PARA 3-4-13).
- v. Using sealing compound, Item 292, Appendix D, seal around nut and axle.
- w. Install valve cap on valve stem.
- x. Release parking brake.
- y. Lower helicopter and remove jack (PARA 1-6-3).

3-4-11 MAIN LANDING GEAR WHEEL VALVE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-11.1 Replace Main Landing Gear Wheel Valve	3-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Tire Inflator Assembly
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Grease, Item 149, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 1-6-3
- PARA 3-4-13

3-4-11.1 Replace Main Landing Gear Wheel Valve.

NOTE

Replacement of both main landing gear wheel valves is the same, except as noted.

3-4-11.1.1. Remove.

- Jack helicopter until shock strut is fully extended (PARA 1-6-3).
- Apply parking brake.

WARNING

- To prevent injury to personnel, completely release pressure from tire before removing valve core from valve stem.

- Do not remove valve core from valve stem to shorten time required to deflate tire.
- Remove valve cap from valve stem. Completely deflate tire, then remove valve core from valve stem (Figure 3-4-11-1, Detail A).
 - Unscrew and remove valve stem from wheel.
 - Remove and discard packing from valve stem.

3-4-11.1.2. Install.

- Lightly lubricate new packing with grease, Item 149, Appendix D.
- Install packing on valve stem (Figure 3-4-11-1, Detail A).
- Install valve stem into outer wheel half.
- On wheel, 5003889-2 or 5003889-3, **TORQUE VALVE STEM TO 60 - 75 INCH-POUNDS.**

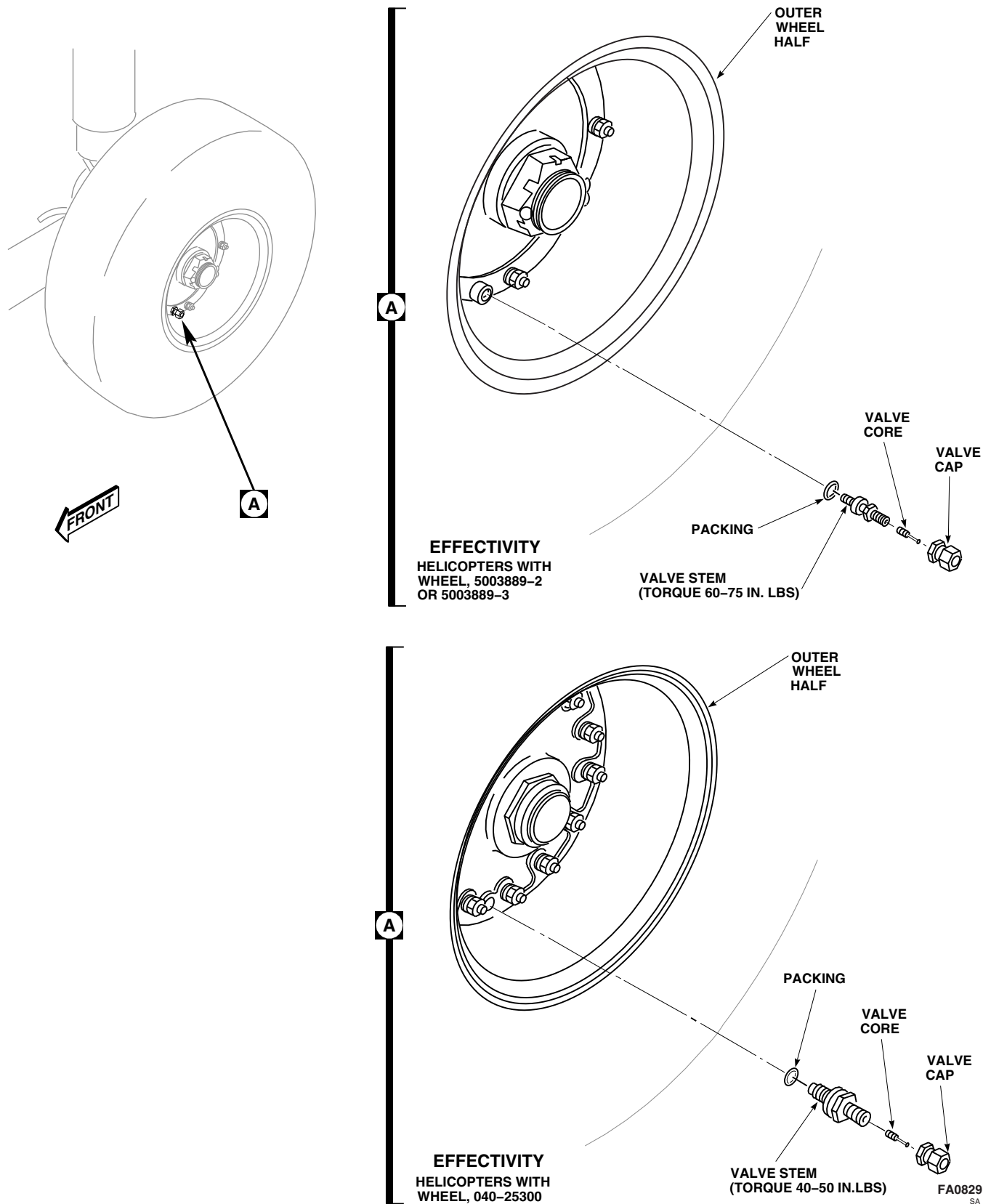


FIGURE 3-4-11-1. REPLACE MAIN LANDING GEAR WHEEL VALVE

3-4-11-2

- e. On wheel, 040-25300, **TORQUE VALVE STEM TO 40 - 50 INCH-POUNDS.**
- f. Install new valve core in valve stem.

WARNING

- To prevent injury to personnel or damage to equipment, check tire inflator assembly to make sure that nitrogen source is vented at 160 psi maximum.
- To prevent injury to personnel or damage to equipment, do not inflate tire above its rated inflation pressure.
- To prevent injury to personnel, do not use an unregulated nitrogen pressure source.
- To prevent injury to personnel, never stand facing sidewall of tire being serviced. Stand facing tire tread.

- g. **INFLATE TIRE TO 130 - 140 PSI WITH NITROGEN, ITEM 221, APPENDIX D.** Use regulated nitrogen pressure source connected to tire inflator assembly.
- h. After a 5 to 10 minute waiting period, check valve stem for leakage as follows:
 - (1) Moisten valve stem area and around tire beads.
 - (2) If bubbles appear, deflate, disassemble, and correct at point of leakage (PARA 3-4-13).
- i. Install valve cap on valve stem.
- j. Release parking brake.
- k. Lower helicopter and remove jack (PARA 1-6-3).

3-4-12 MAIN LANDING GEAR WHEEL DISC DRIVE KEYS.

PARAGRAPH BREAKDOWN

	PAGE
3-4-12.1 Replace Main Landing Gear Wheel Disc Drive Keys	3-4-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill
- Drill Bit, 1⁄8-inch diameter

- Torque Wrench, 30 - 150 in. lbs

References

- PARA 3-4-10
- PARA 3-5-3

3-4-12.1 Replace Main Landing Gear Wheel Disc Drive Keys.

NOTE

Replacement of both main landing gear wheel disc drive keys is the same.

- b. Check disc drive keys for wear. **ONE OR MORE KEYS MUST NOT MEASURE LESS THAN 0.696-INCH BETWEEN DISC CONTACTING SURFACES.** If one or more keys is below limit, replace all keys on inner wheel half.

3-4-12.1.1 Remove.

- a. Remove main landing gear wheel and tire (PARA 3-4-10).
- b. On inner wheel half of 7 lug wheel, 5003889-2 or 5003889-3, remove screw staking using 1⁄8-inch diameter drill bit and drill.
- c. On 7 lug wheel, 5003889-2 or 5003889-3, and 14 lug wheel, 040-25300, remove screws and disc drive keys (Figure 3-4-12-1).

3-4-12.1.2 Inspect.

- a. Inspect and repair inner wheel half as required (PARA 3-5-3).

3-4-12.1.3 Install.

- a. On inner wheel half, place disc drive keys on wheel key bosses (Figure 3-4-12-1).
- b. On inner wheel half of 7 lug wheel, 5003889-2 or 5003889-3, install and stake screws.
- c. On inner wheel half of 14 lug wheel, 040-25300, install screws. **TORQUE SCREWS TO 30 - 35 INCH-POUNDS.**
- d. Install main landing gear wheel and tire (PARA 3-4-10).

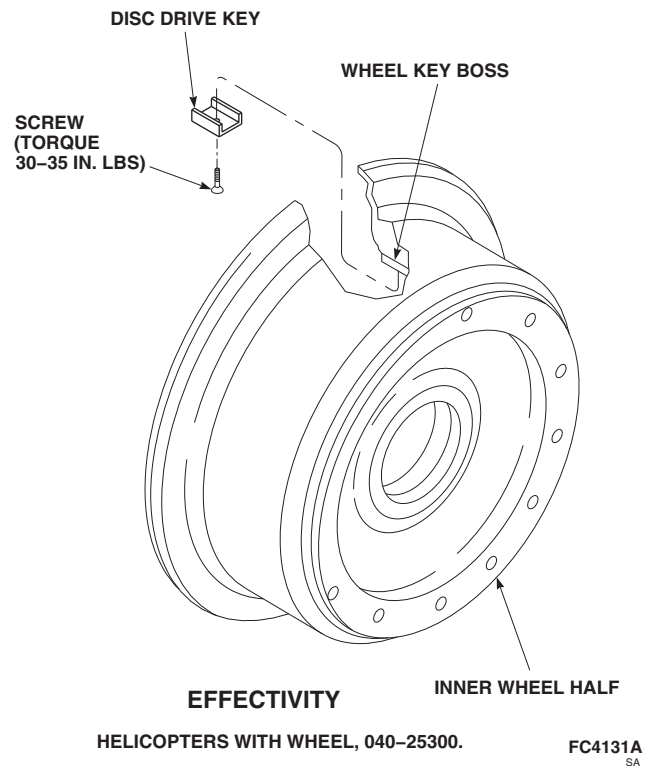
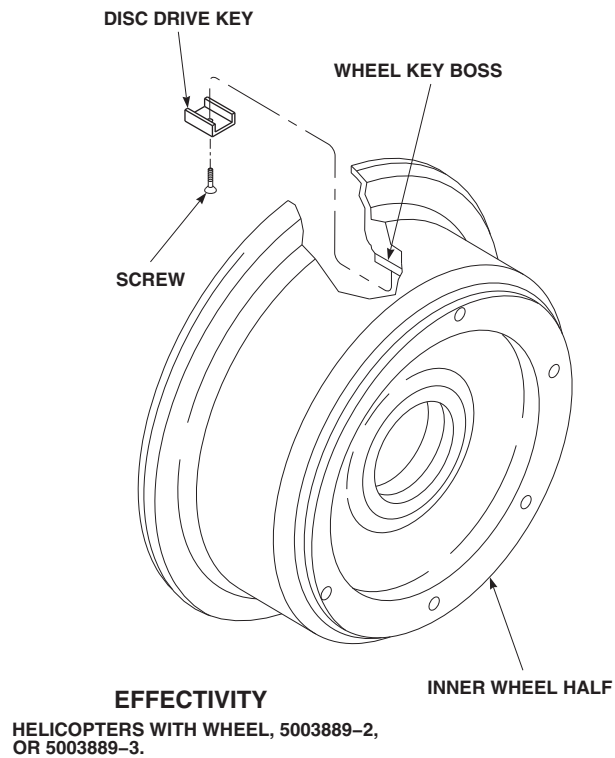


FIGURE 3-4-12-1. REPLACE MAIN LANDING GEAR WHEEL DISC DRIVE KEYS

3-4-13 MAIN LANDING GEAR TIRE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-13.1 Replace Main Landing Gear Tire	3-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Aircraft Tire Bead Breaker
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Antiseize Compound, Item 75, Appendix D
- Bolt, NAS625-96 (2)

- Bolt, NAS626-96 (2)
- Grease, Item 149, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nut, 0.375-24 (2)

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 3-4-10
- PARA 3-4-11
- PARA 3-5-3

3-4-13.1 Replace Main Landing Gear Tire.

NOTE

Replacement of both main landing gear tires is the same, except as noted.

3-4-13.1.1 Remove.

- Remove main landing gear wheel and tire (PARA 3-4-10).



- To prevent injury to personnel, completely release pressure from tire before removing valve core from valve stem.
- Do not remove valve core from valve stem to shorten time required to deflate tire.

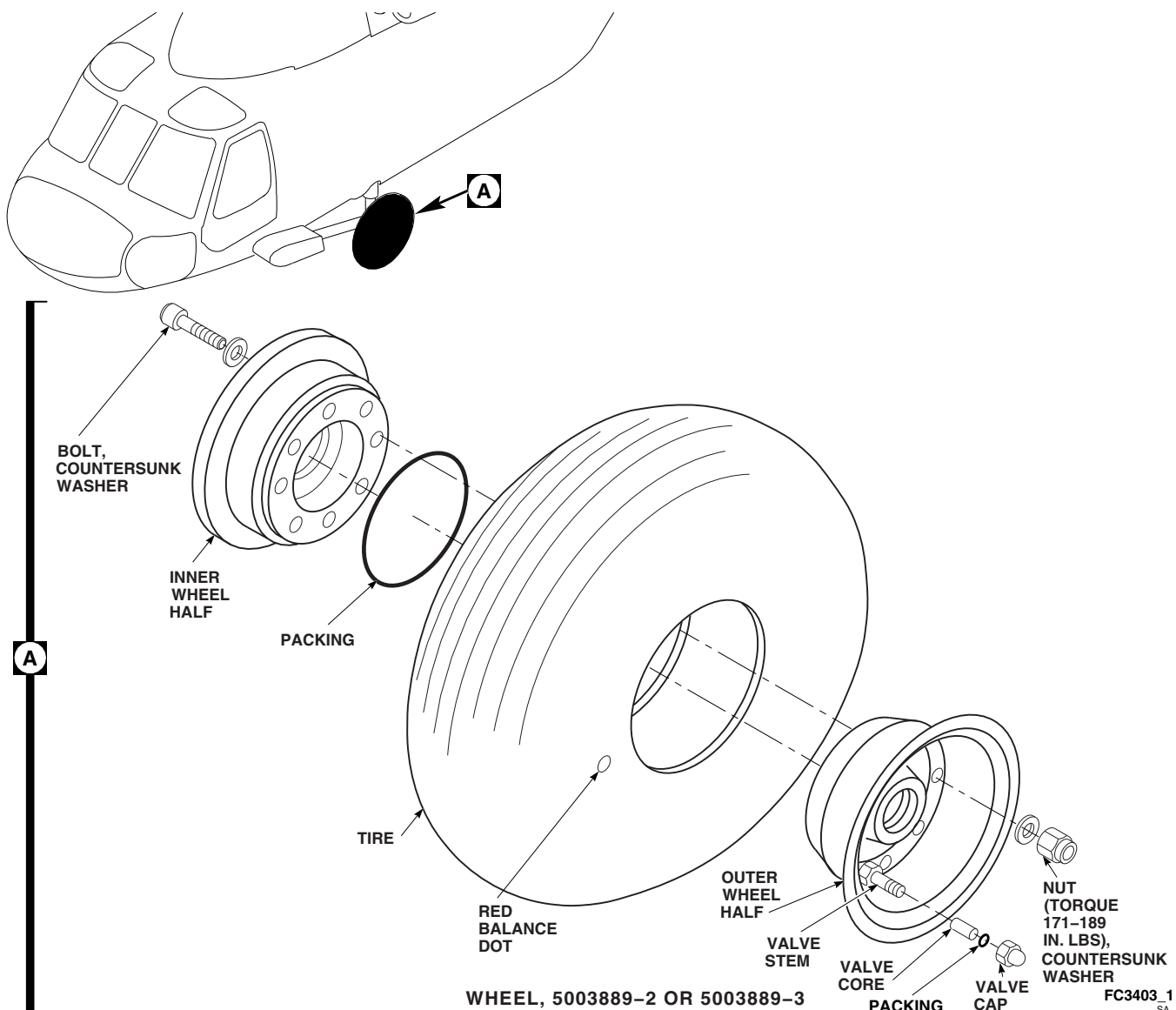


FIGURE 3-4-13-1. REPLACE MAIN LANDING GEAR TIRE (SHEET 1 OF 2)

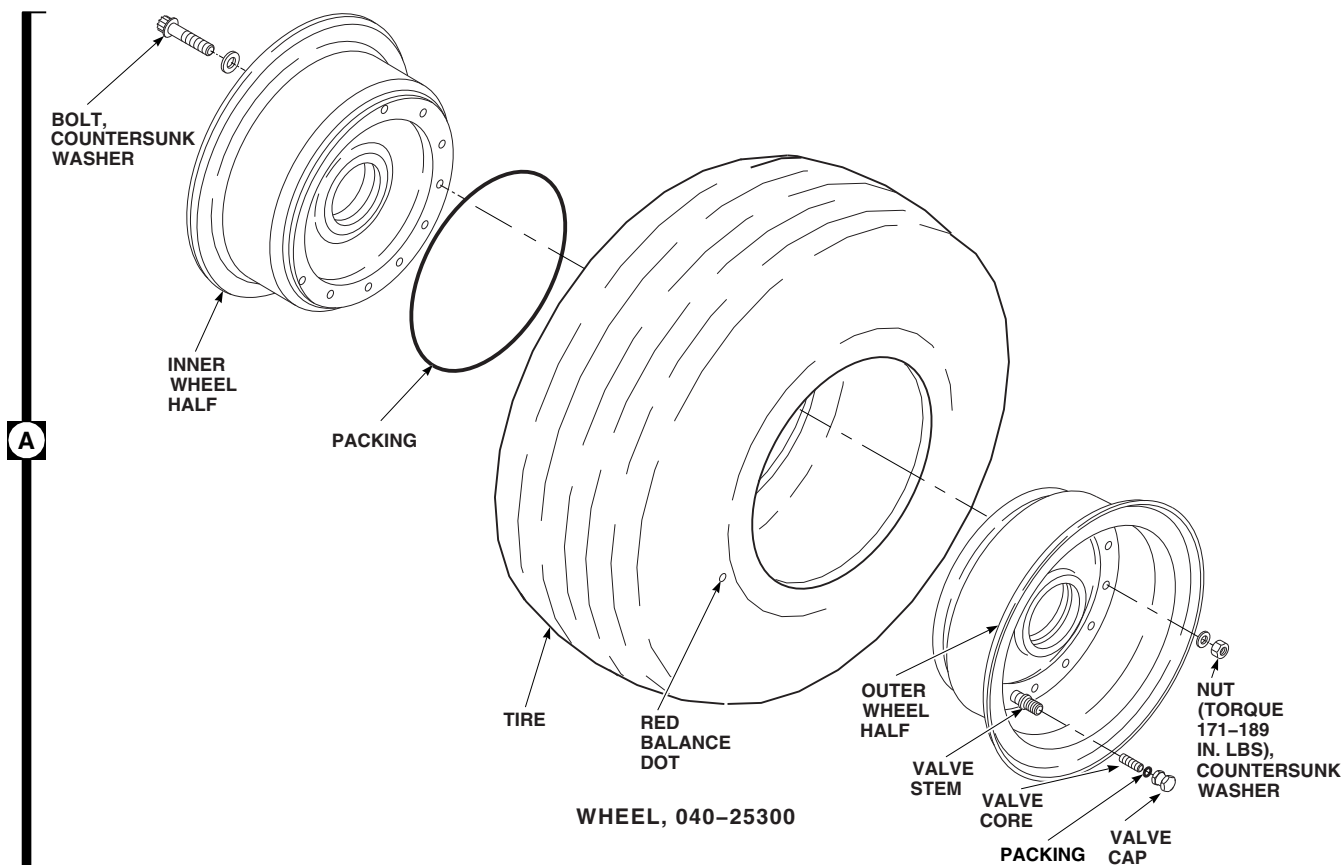
- b. Remove valve cap, packing, and valve core (Figure 3-4-13-1, Sheet 1, Detail A, or Sheet 2, Detail A).
- c. Using aircraft tire bead breaker, break tire beads away from wheel flange by applying even pressure to sides.

CAUTION

- Use aircraft tire bead breaker on clean flat surface to disassemble and assemble wheel. Use care not to damage protective finish.
- Do not pry between tire bead and wheel with hand tools.

WARNING

Injury to personnel or damage to equipment may occur if hardware is reused on wheel, 5003889-2 or 5003889-3. Replace all bolts and nuts on wheel, 5003889-2 or 5003889-3.



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FIGURE 3-4-13-1. REPLACE MAIN LANDING GEAR TIRE (SHEET 2 OF 2)

CAUTION

Damage to equipment will result if impact or power wrenches are used to remove bolts and nuts. Wheel assembly will be damaged. Do not use impact or power wrenches to remove bolts and nuts.

NOTE

Hardware on wheel, 040-25300, may be reused if no cracks are found.

- d. Remove bolts, countersunk washers, and nuts. Separate wheel halves and remove tire. Discard hardware on wheel, 5003889-2 or 5003889-3.
- e. Remove and discard packing from inner wheel half.

3-4-13.1.2 Inspect.

- a. Inspect tire sidewall for blisters. **NONE ALLOWED.** If blisters are found, replace tire.
- b. Inspect tire for wear. **WEAR MUST NOT REACH FIRST FABRIC PLY ON TREAD GROOVE OR BE WORN FOR A**

CONTINUOUS LENGTH OF 12-INCHES AT BOTTOM OF TREAD GROOVE. If damage exceeds limits, replace tire.

- c. Inspect tire tread and shoulder for cuts or damage. **CUTS OR DAMAGE MAY BE ANY LENGTH PROVIDED DAMAGE DOES NOT CUT FIRST LAYER OF CORD BODY. IF CUT PENETRATES FIRST LAYER OF CORD BODY AND IS LESS THAN 1-INCH LONG, DEPTH OF CUT MUST NOT EXCEED CUT LIMIT SHOWN ON SIDEWALL OF TIRE. IF TIRE GROOVE NEAREST CUT IS WORN SMOOTH AND CUT PENETRATES CORD BODY REPLACE TIRE.** If damage exceeds limits, replace tire.
- d. Inspect tire sidewalls for cuts, scratches, abrasions, or scuffs. **DAMAGE MUST NOT PENETRATE FIRST LAYER OF CORD BODY.** If damage exceeds limit, replace tire.
- e. Inspect tire surface for weather checking. **NO SEPARATION OF RUBBER BETWEEN CORDS ALLOWED. MAXIMUM DEPTH OF CRACK BENEATH TREAD RIB IS 1/8-INCH DEEP AND 1/4-INCH LONG.** If damage exceeds limits, replace tire.
- f. Check valve for damage. **NONE ALLOWED.** If damaged, replace main landing gear wheel valve (PARA 3-4-11).
- g. Inspect new tire beads for damage. Clean tire beads with machinery towel, Item 211, Appendix D, dampened with denatured alcohol, Item 70, Appendix D.

CAUTION

Injury to personnel or damage to equipment will result if hardware is reused on wheel, 5003889-2 or 5003889-3. Replace all bolts and nuts on wheel, 5003889-2 or 5003889-3.

NOTE

Hardware on wheel, 040-25300, may be reused if no cracks are found.

- h. On wheel, 040-25300, using fluorescent penetrant inspection kit, fluorescent penetrant inspect bolts for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace bolts.

WARNING

Injury to personnel and damage to equipment will result if wheel halves are not inspected whenever wheel is disassembled. Inspect wheel halves whenever wheel is disassembled.

NOTE

Fluorescent penetrant inspection is not required on wheel halves, 040-25300.

- i. Inspect and repair main landing gear wheel (PARA 3-5-3).

3-4-13.1.3 Install.

WARNING

The 7 lug wheel, 5003889-2 or 5003889-3, may experience lug failure when operating at or above gross weights of 20500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel, 040-25300, should be utilized when operating at or above gross weights of 20500 pounds.

- a. Clean wheel bead seat with machinery towel, Item 211, Appendix D, dampened with denatured alcohol, Item 70, Appendix D.
- b. Lubricate packing with light coat of grease, Item 149, Appendix D.
- c. Install packing in groove on inner wheel half (Figure 3-4-13-1, Sheet 1, Detail A, or Sheet 2, Detail A).
- d. Place tire on inner wheel half.

- e. Place outer wheel half in tire and line up bolt-holes in both wheel halves. Line up red balance dot on sidewall with valve stem.
- f. Apply antiseize compound, Item 75, Appendix D, on new bolt and nut threads, face of nut, under boltheads, and both sides of washers.

WARNING

Injury to personnel or damage to equipment may occur if hardware is reused on wheel, 5003889-2 or 5003889-3. Replace all bolts and nuts on wheel, 5003889-2 or 5003889-3.

CAUTION

Damage to equipment will result if impact or power wrenches are used to tighten or torque bolts or nuts. Wheel assembly and hardware will be damaged. Do not use impact or power wrenches when tightening or torquing bolts or nuts.

NOTE

- Hardware on wheel, 040-25300, may be reused if no cracks are found.
- Short bolts and stiff tires sometimes make it hard to draw wheel halves together enough to start nuts.

- g. Secure wheel halves together with bolts, countersunk washers, and nut (with boltheads on inner side of wheel). Install countersunk washers so countersunk sides are toward boltheads and nut faces respectively. If nuts cannot be started on bolts of wheel, perform the following:

- (1) Temporarily install two bolts, NAS625-96, NAS626-96 (for wheel, 5003889-2 or 5003889-3), or NAS625-96 (for wheel, 040-25300) 180° apart, through both wheel halves. Install washers and nuts, 0.375-24.

- (2) **TO DRAW WHEEL HALVES TOGETHER, TIGHTEN NUTS EVENLY, A LITTLE AT A TIME.**

- (3) Once wheel is together, install regularly used bolts, washers, and nuts, with bolt-heads on inner side of wheel. Install countersunk washers so countersunk sides are toward boltheads and nut faces respectively.

- (4) Remove temporary hardware and repeat step g. (3) for last two holes.

- h. **TIGHTEN NUTS IN A CRISSCROSS PATTERN, IN EVEN INCREMENTS OF 25 INCH-POUNDS, UNTIL A FINAL TORQUE OF 171 - 189 INCH-POUNDS IS OBTAINED.**

- i. Apply torque stripes to wheel nuts (PARA 2-6-5).
- j. Install valve core, packing, and valve cap.
- k. Install main landing gear wheel and tire (PARA 3-4-10).

3-4-14 MAIN LANDING GEAR WHEEL BRAKE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-14.1 Replace Main Landing Gear Wheel Brake	3-4-14-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Fluorescent Penetrant Inspection Kit
- Protection, Eye and Skin
- Pry Tool, Phenolic, Plastic, or Wood, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Antiseize Compound, Item 76, Appendix D
- Brush, Paint, Item 86, Appendix D

- Corrosion Removing Compound, Item 116, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1 ½-inches, Item 213, Appendix D
- Paint, Heat Resistant Aluminum, Item 221A, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- ASTM E1417
- PARA 3-2-1
- PARA 3-4-10
- PARA 3-4-20

3-4-14.1 Replace Main Landing Gear Wheel Brake.

NOTE

Replacement of both main landing gear wheel brakes is the same, except as noted.

3-4-14.1.1 Remove.

- Remove main landing gear wheel and tire (PARA 3-4-10).
- Release parking brake.
- Disconnect brake hose at quick-disconnect coupling on drag beam (Figure 3-4-14-1, Sheet 1, Detail A).
- Disconnect brake line from elbow of brake and drag beam. Cap and plug ports. Position 1-quart container to catch fluid under elbow.
- Remove bolts, countersunk washers, washers, and nuts securing brake to drag beam. Carefully slide brake off axle.
- If brake is being replaced, remove elbow, jam nut, retainer, and packing from lower inlet port (Figure 3-4-14-1, Sheet 2, Detail B). Discard packing.
- Remove shield from axle and retain shield for installation (Figure 3-4-14-1, Sheet 1, Detail A).

3-4-14.1.2 Disassemble.

- Place brake on a clean, flat surface with housing down.

CAUTION

Damage to brake will result if impact or power wrenches are used to remove brake bolts or nuts. Do not use impact or power wrenches to remove brake bolts or nuts.

- Remove eight nuts securing brake assembly together. Do not remove bolts now (Figure 3-4-14-2).
- Remove backing plate, with six wear pads attached.
- Remove six wear pads from backing plate.
- Remove rotating disc (outboard).
- Remove stationary disc.
- Remove rotating disc (inboard).
- Remove torque tube from housing.
- Remove eight bolts and countersunk washers.

3-4-14.1.3 Clean.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to discs will result if cleaned with solvent. Do not clean discs with solvent.

- Using compressed air (if available) and paint brush, Item 86, Appendix D, clean rotating and stationary discs.
- Using dry-cleaning solvent, Item 124, Appendix D, and paint brush, Item 86, Appendix D, clean backing plate, torque tube, and housing.

ing. Dry parts with compressed air or by wiping off with machinery towel, Item 211, Appendix D.

3-4-14.1.4 Inspect.

- a. Check backing plate for obvious damage and warpage. **NO DAMAGE ALLOWED. WARPAGE MUST NOT EXCEED 0.015-INCH.** If damaged or warpage exceeds limit, replace backing plate and install new wear pads.
- b. Check stationary disc for cracks, especially at key slot corners. **NONE ALLOWED.** If crack is found, replace disc.
- c. Check disc for thickness. **WEAR ON OUTER EDGE MUST NOT EXCEED 0.140-INCH.** If damage exceeds limit, replace disc.
- d. Check stationary disc key slots for wear and battering. **WIDTH OF ANY KEY SLOT MUST BE A MINIMUM OF 0.515-INCH AND A MAXIMUM OF 0.560-INCH.** If wear or battering exceeds limits, replace disc.
- e. Check stationary disc for shrinkage and distortion. Replace disc if:
 - (1) **DISHED 0.015-INCHES OR MORE.**
 - (2) **1/8-INCH OR MORE OUT-OF-ROUND.**
 - (3) **SHRUNK TO AN INSIDE DIAMETER OF 6.32-INCHES OR LESS, MEASURED BETWEEN ENDS OF KEY SLOT TANGS.**
- f. Check rotating discs for wear and chipping. Replace disc if:
 - (1) **WORN TO 1/8-INCH OR LESS.**
 - (2) **UNCHIPPED MIX IS WORN TO 0.010-INCH THICK AT ANY POINT ON EITHER FACE OF DISC. AS MUCH AS 20 % OF MIX MAY BE LOST THROUGH CHIPPING.**
 - (3) **MIX LOSS IS SCATTERED. NO MORE THAN 25% LOSS OF MIX IS ALLOWED IN ANY SEGMENT.**
- g. Check rotating discs for shrinkage and distortion. Replace disc if:
 - (1) **DISHED 0.015-INCHES OR MORE.**
 - (2) **1/8-INCH OR MORE OUT-OF-ROUND.**
 - (3) **SHRUNK TO AN INSIDE DIAMETER OF 6.940-INCHES OR LESS AT ANY POINT.**
- h. Check rotating discs for corrosion at relief slot terminal holes. **NONE ALLOWED.** If corrosion is found, repair as follows:
 - (1) Using 1 1/2-inch masking tape, Item 213, Appendix D, cover face of discs.
 - (2) Using abrasive cloth, Item 7, Appendix D, or abrasive wheel, Item 22, Appendix D, remove corrosion. **CORROSION REMOVAL MUST NOT EXCEED 0.015-INCH FOR RELIEF SLOTS AND 0.281-INCH FOR TERMINAL HOLES.** If corrosion removal for relief slots or terminal holes exceeds limits, replace rotating disc.
 - (3) Clean area with corrosion removing compound, Item 116, Appendix D.
 - (4) Using water, rinse discs and dry.
 - (5) Paint repair area with heat resistant aluminum paint, Item 221A, Appendix D.
- i. Check rotating disc drive slots for wear and battering. Replace disc if:
 - (1) **IF ONE OR MORE TANGS ARE BENT.**
 - (2) **ANY SLOT IS 0.780-INCH OR MORE IN WIDTH.**
- j. Check rotating discs for cracks, especially at relief slot terminal holes. **NONE ALLOWED.** If crack is found, replace disc.
- k. Locally make pry tool out of phenolic, plastic, wood, or other similar material.

- l. Mark location of each wear pad as it is removed, so it can be installed in same location.
- m. Remove six wear pads and piston insulators on pistons.
- n. Using pry tool, remove wear pads on piston insulators.
- o. Inspect piston insulators for deterioration and wear. **REPLACE PISTON INSULATORS WORN TO 0.635-INCH OR LESS IN LENGTH, OR WITH BLISTERS AND RAISED AREAS, OR WITH END FACES NOT PARALLEL. BLISTERS LESS THAN 0.010-INCH CAN BE FILED DOWN PROVIDING 0.635-INCH LENGTH IS MAINTAINED.** Replace any cracked or gouged piston insulator.

CAUTION

Damage to brake will result if new wear pads are mixed with used pads. If any wear pad on piston or on backing plate is worn beyond limits, replace that set of six wear pads.

- p. Check thickness of backing plate wear pads. **WEAR PAD THICKNESS MUST NOT BE 0.380-INCH OR LESS.** If any one pad is worn to 0.380-inch thick or less, replace all six.
- q. Check thickness of piston wear pads. **WEAR PAD THICKNESS MUST NOT BE 0.380-INCH OR LESS.** If any one pad is worn to 0.380-inch thick or less, replace all six.
- r. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect brake bolts specifically for cracks at radius under bolthead and within threads next to bolt shank (ASTM E1417). **NONE ALLOWED.** If crack is found, replace brake bolts.

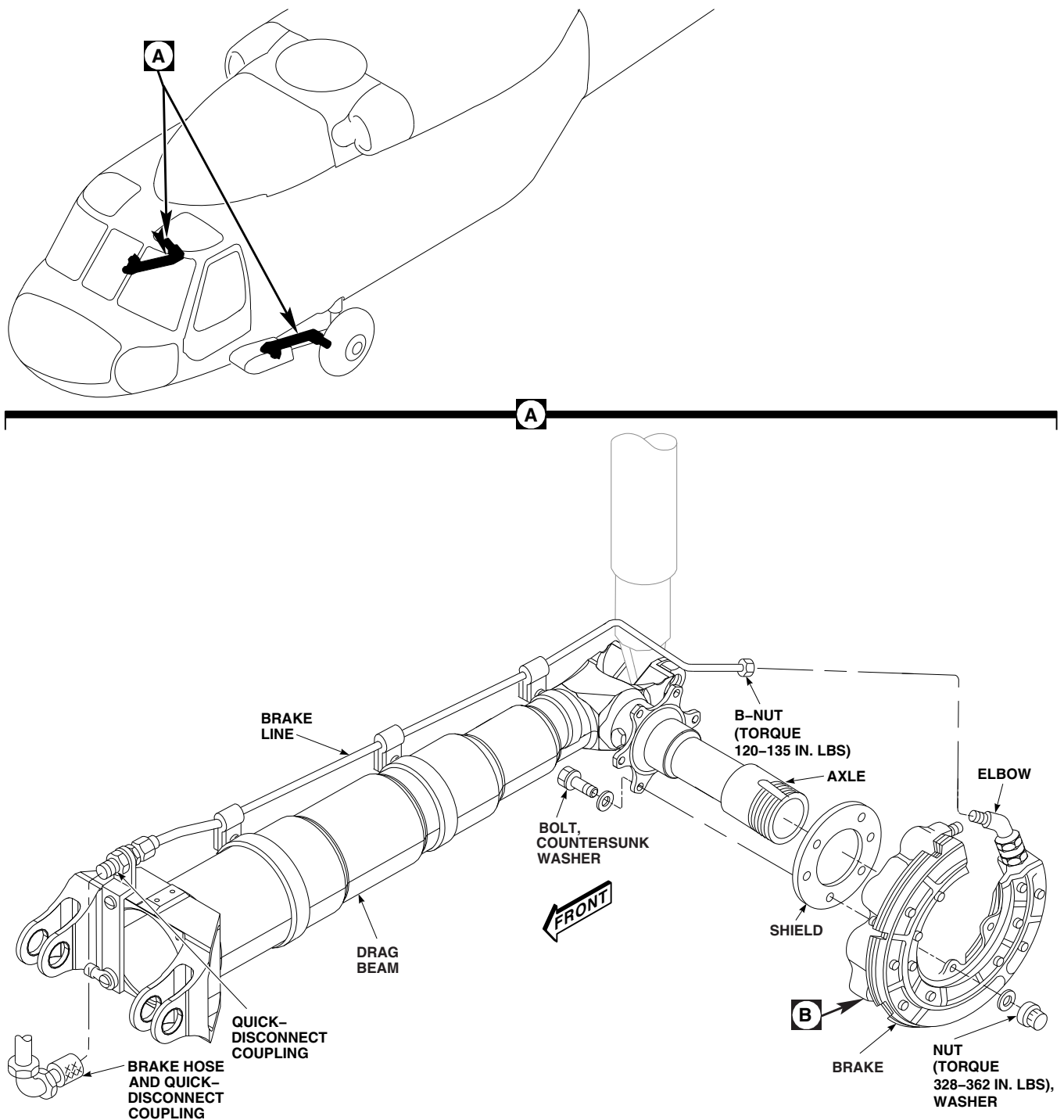
- s. Check torque tube for cracks. If cracked, replace tube.
- t. Check width of torque tube key. **WIDTH OF KEY MUST NOT BE 0.470-INCH OR LESS AT ANY POINT ALONG LENGTH OF ANY KEY.** If width exceeds limit, replace torque tube key.

3-4-14.1.5 Assemble.

- a. Using antiseize compound, Item 76, Appendix D, coat threads of eight bolts and self-locking nuts, and bearing surfaces of boltheads, washers, and self-locking nuts.
- b. Place a countersunk washer under each bolt-head (countersunk side faces bolthead) and insert bolt through housing (Figure 3-4-14-2).
- c. Place brake on a clean, flat surface with housing down.
- d. Place torque tube over bolts. Make sure torque tube scallops face housing.
- e. Insert wear pad into chamfered end of piston insulator.
- f. Install six piston insulators and six wear pads in housing.
- g. Install inboard rotating disc over torque tube. Add stationary disc and outboard rotating disc on top of tube.
- h. Install backing plate and six wear pads, flat face down, against inboard rotating disc, with bolts sticking through backing plate.
- i. Install self-locking nuts on eight bolts. **TIGHTEN EACH NUT A LITTLE AT A TIME UNTIL SNUG. TORQUE NUTS TO 85 - 95 INCH-POUNDS.**

3-4-14.1.6 Install.

- a. If replacing right brake with new brake, perform the following:

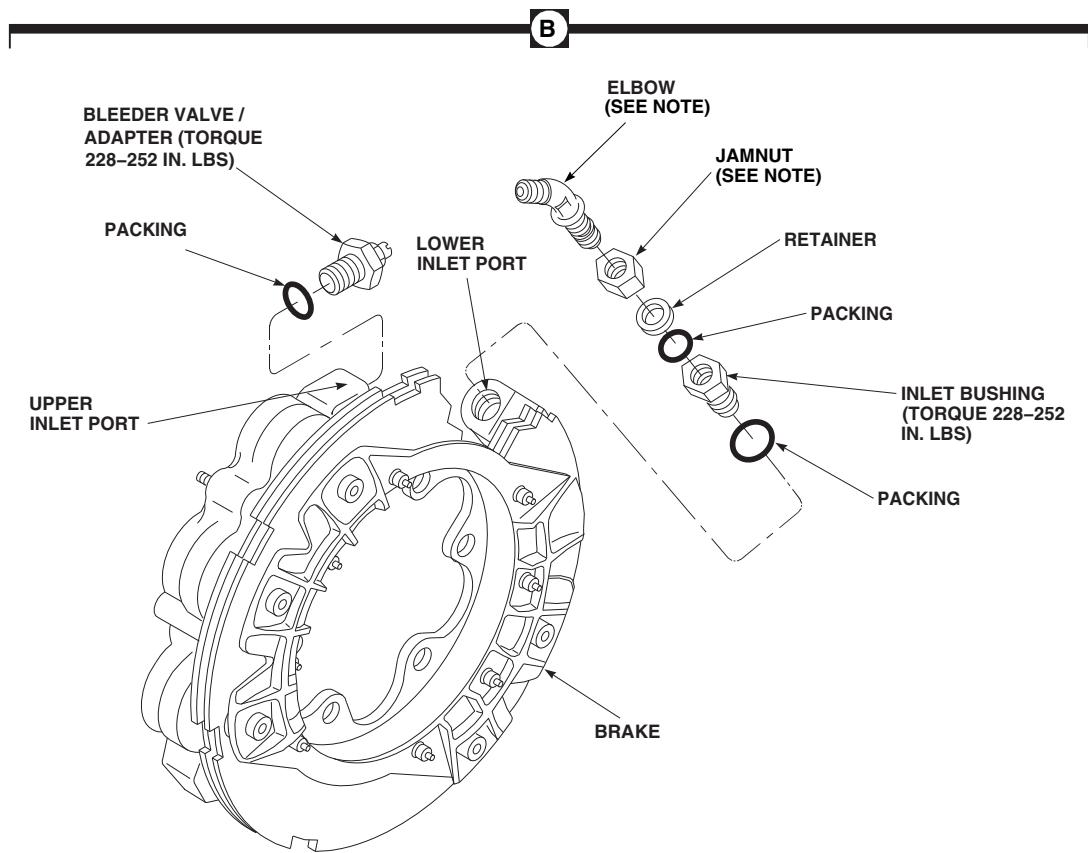


NOTE

TIGHTEN ELBOW AND JAMNUT 1-1/2 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FACING BRAKE LINE, THEN TORQUE JAMNUT TO 95-105 IN. LBS.

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FIGURE 3-4-14-1. REPLACE MAIN LANDING GEAR WHEEL BRAKE (SHEET 1 OF 2)

FO0582_2A
SA**FIGURE 3-4-14-1. REPLACE MAIN LANDING GEAR WHEEL BRAKE (SHEET 2 OF 2)****NOTE**

Brakes are received in left configuration with bleeder valve adapter, bleeder valve, and bleeder screw in lower inlet port. If replacing right brake assembly, bleeder valve adapter, bleeder valve, and bleeder screw must be moved to upper inlet port.

- (1) Remove bleeder valve/adapter as an assembly from lower inlet port of brake (Figure 3-4-14-1, Sheet 2, Detail B).

Remove packing from bleeder valve adapter. Discard packings.

- (2) Remove inlet bushing from upper inlet port of brake. Remove packing from bushing. Discard packings.
- (3) Using hydraulic fluid, Item 154, Appendix D, lubricate new packing. Install packing on bleeder valve adapter.
- (4) Install bleeder valve/adapter as an assembly in upper inlet port of brake.

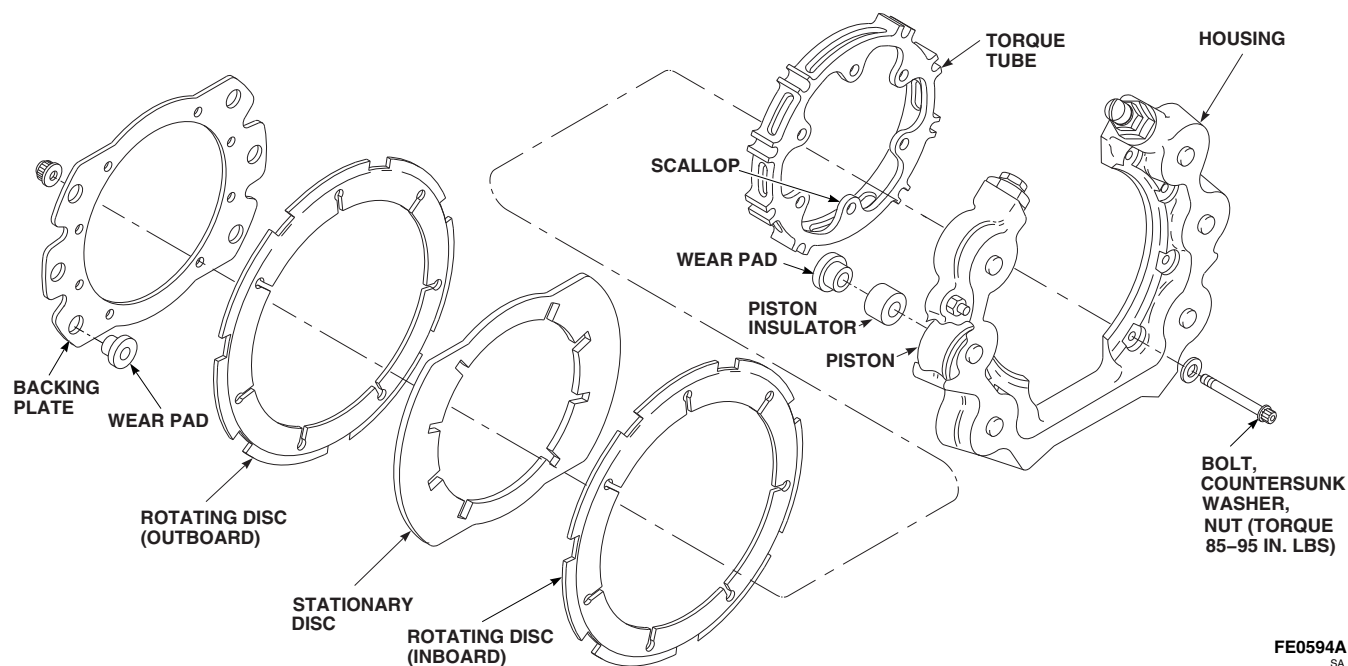


FIGURE 3-4-14-2. REPAIR MAIN LANDING GEAR WHEEL BRAKE

TORQUE BLEEDER VALVE/ ADAPTER TO 228 - 252 INCH-POUNDS.

- (5) Using hydraulic fluid, Item 154, Appendix D, lubricate new packing. Install packing on inlet bushing.
- (6) Install inlet bushing in lower inlet port of brake. **TORQUE INLET BUSHING TO 228 - 252 INCH-POUNDS.**
- (7) Using hydraulic fluid, Item 154, Appendix D, lubricate packing, retainer, and male threads of elbow.
- (8) Install jamnut, retainer, and packing onto elbow. Turn nut and retainer down until packing is pushed firmly against lower threaded section of elbow.
- (9) Install elbow in inlet bushing until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN ELBOW AND JAMNUT 1 ½ TURNS PAST POINT**

WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FACING BRAKE LINE.

- (10) **PREVENT ELBOW FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
 - b. Position shield over axle and against drag beam flange (Figure 3-4-14-1, Sheet 1, Detail A).
 - c. Place brake on drag beam flange. Line up brake so brake line can be connected and shield is properly positioned.
 - d. Make sure countersunk side of washers face bolthead. Install bolts, washers, and nuts, making sure boltheads are on inner side of brake.
 - e. **TORQUE NUTS IN A CRISSCROSS MANNER, IN 100 INCH-POUND INCREMENTS, TO 328 - 362 INCH-POUNDS.**
 - f. Connect brake line to elbow. **TORQUE B-NUT TO 120 - 135 INCH-POUNDS.**

- g. Connect hose at quick-disconnect coupling in drag beam.

WARNING

The 7 lug wheel, 5003889-2 or 5003889-3, may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire

failure. The 14 lug wheel, 040-25300, should be utilized when operating at or above gross weights of 20,500 pounds.

- h. Install main landing gear wheel and tire (PARA 3-4-10).
- i. Bleed appropriate pilot's brake (PARA 3-4-20).
- j. Perform operational check of parking brake system (PARA 3-2-1).

3-4-15 MASTER BRAKE CYLINDER.

PARAGRAPH BREAKDOWN

	PAGE
3-4-15.1 Replace Master Brake Cylinder	3-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Crowfoot Attachment
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 194, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 2-4-43
- PARA 2-6-5
- PARA 3-2-1
- PARA 3-4-20
- PARA 11-4-7

3-4-15.1 Replace Master Brake Cylinder.

NOTE

Replacement of all master brake cylinders is the same.

3-4-15.1.1 Remove.

- Chock helicopter wheels.
- Release parking brake.
- Remove pilot’s/copilot’s yaw control pedal boots (PARA 11-4-7).

- Remove pilot’s/copilot’s access panels (PARA 2-4-43).
- Disconnect hose from swivel fitting on master cylinder. Use machinery towel, Item 211, Appendix D, to catch any fluid that might leak or drip. Plug hose with protective plug (Figure 3-4-15-1, Detail A).
- Remove self-retaining bolts, washers, and nuts securing master cylinder to pedal assembly and support assembly. Remove master cylinder.
- If master cylinder is being replaced, remove swivel fitting and packing from master cylinder. Discard packing.

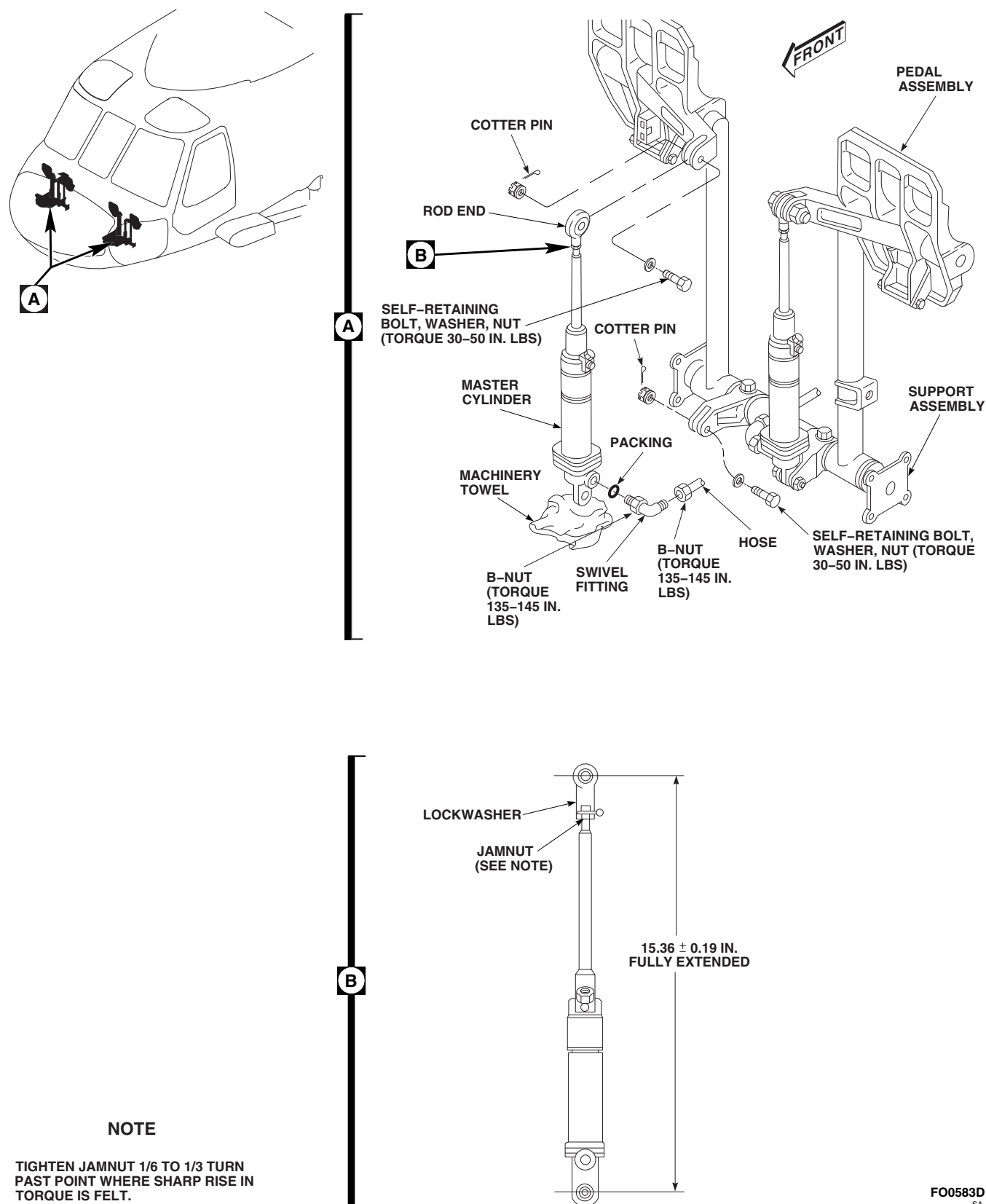


FIGURE 3-4-15-1. REPLACE MASTER BRAKE CYLINDER

3-4-15.1.2 Install.

- a. If removed, lubricate packing with hydraulic fluid, Item 154, Appendix D. Install packing on swivel fitting (Figure 3-4-15-1, Detail A).
- b. If removed, install swivel fitting in lower part of master cylinder. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**
- c. Fully extend master cylinder piston. Measure distance between master cylinder attachment holes, center-to-center. **DISTANCE MUST BE 15.36 ± 0.19 -INCHES.** If proper dimension is not met, adjust rod end as follows (Figure 3-4-15-1, Detail B):
 - (1) Loosen jamnut on piston rod.
 - (2) Adjust rod end on piston rod as required to obtain proper dimension.
 - (3) **TIGHTEN JAMNUT ON PISTON ROD $\frac{1}{6}$ - $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT**
 - (4) Using lockwire, Item 194, Appendix D, lockwire jamnut to lockwasher.
- d. Position master cylinder on pedal assembly and support assembly and install self-retaining bolts as follows (Figure 3-4-15-1, Detail A):

WARNING

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Installation of self-retaining bolts securing upper and lower ends of master cylinder is the same, except as noted.

- (1) Select washer combination per Table 3-4-15-1 to provide correct grip length and

cotter pin position. Install washer combination under bolthead.

- (2) Install self-retaining bolt and washer combination.
- (3) Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup using table.
- (4) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to proof torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

- (5) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** If bolt rotates, adjust using washer combinations in Table 3-4-15-1 or use next longer or shorter bolt.
- (6) Apply torque stripe (PARA 2-6-5).
- e. Unplug and connect hose to swivel fitting. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**
- f. Bleed appropriate master cylinder (PARA 3-4-20).
- g. Make sure area is clean and free of foreign material before closing up.
- h. Install pilot's/copilot's access panels (PARA 2-4-43).
- i. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- j. Set parking brake.
- k. Perform an operational check of brake system (PARA 3-2-1).

Table 3-4-15-1. Washer Combinations

Combination	Quantity	Washer T h i c k n e s s (Inch)
A	2	0.016
B	3	0.016
C	1	0.063
D	1	0.016
	1	0.063
E	2	0.016
	1	0.063

3-4-16 MAIN LANDING GEAR WHEEL BRAKE LINES AND HOSES.

PARAGRAPH BREAKDOWN

	PAGE
3-4-16.1 Replace Main Landing Gear Wheel Brake Lines and Hoses	3-4-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Container, 1-Quart
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 2-4-43
- PARA 3-2-1
- PARA 3-4-20
- PARA 11-4-7

3-4-16.1 Replace Main Landing Gear Wheel Brake Lines and Hoses.

NOTE

The following procedures apply to all brake lines and hoses, except as noted.

3-4-16.1.1 Remove.

- Chock helicopter wheels.
- Release parking brake.
- Remove cockpit floor cover (PARA 2-4-43).
- Remove pilot’s/copilot’s yaw control pedal boots (PARA 11-4-7).

- Remove pilot’s/copilot’s access panels (PARA 2-4-43).
- Remove screws, washers, nuts, clamps, and spacers where required (Figure 3-4-16-1, Sheet 1, Detail B, Sheet 2, Detail C, and Sheet 3, Detail D, Detail E, and Detail F).
- Place a 1-quart container under nut at one end of line, tube, or hose being replaced. Disconnect and drain lines, tubes, or hoses.

NOTE

Install caps and plugs as line, tube, or hose, and fitting is removed.

- Disconnect line, tube, or hose from union or elbow and remove from helicopter.

- i. Remove nuts, washers, and unions or elbows.

3-4-16.1.2 Install.

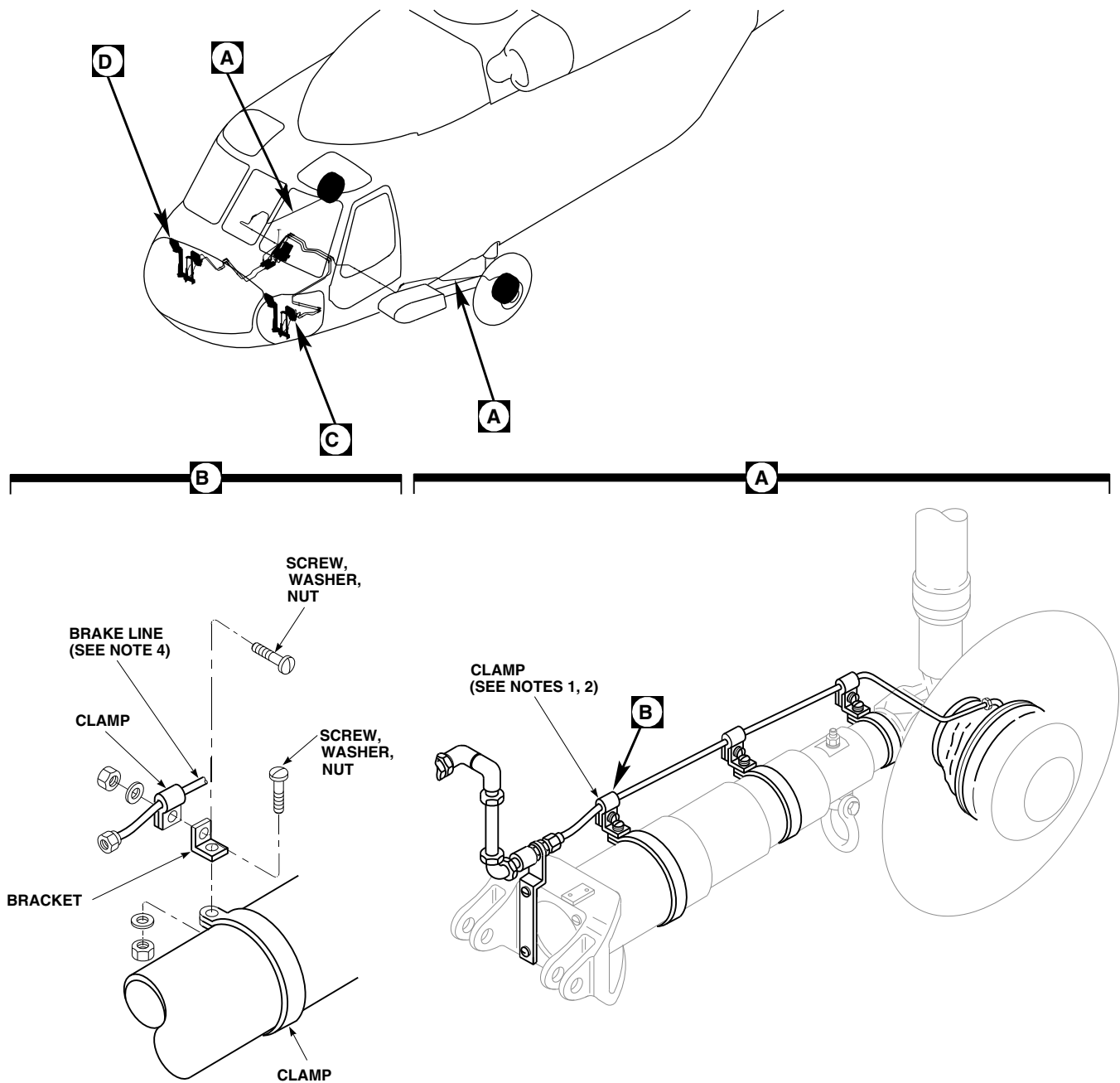
WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Remove caps and plugs as line, tube, or hose, and fitting is installed.

- a. Using clean, dry compressed air, blow line clear at installation.
- b. Install unions or bulkhead elbows using washers and nuts. **TORQUE NUT TO 85 - 105 INCH-POUNDS** (Figure 3-4-16-1, Sheet 2, Detail C, and Sheet 3, Detail D).
- c. Install packings lubricated with hydraulic fluid, Item 154, Appendix D, on parking brake valve and slave mixer valve unions. Install unions to valves. **TORQUE UNIONS TO 95 - 105 INCH-POUNDS** (Figure 3-4-16-1, Sheet 3, Detail D).
- d. Uncap and lubricate union or elbow threads with hydraulic fluid, Item 154, Appendix D (Figure 3-4-16-1, Sheet 2, Detail C, and Sheet 3, Detail D, Detail E, and Detail F).
- e. Install line or tube on swivel fittings. **TORQUE B-NUTS TO 120 - 135 INCH-POUNDS** (Figure 3-4-16-1, Sheet 1, Detail B, Sheet 2, Detail C, and Sheet 3, Detail D and Detail E).
- f. Install hose on elbows or swivel fittings. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS** (Figure 3-4-16-1, Sheet 2, Detail C, and Sheet 3, Detail D and Detail E).
- g. Install screws, washers, nuts, clamps, and spacers where required (Figure 3-4-16-1, Sheet 1, Detail B, Sheet 2, Detail C, and Sheet 3, Detail D, Detail E, and Detail F).
- h. Bleed brakes (PARA 3-4-20).
- i. Make sure area is clean and free of foreign material.
- j. Install pilot's/copilot's access panels (PARA 2-4-43).
- k. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- l. Install cockpit floor cover (PARA 2-4-43).
- m. Perform an operational check of brakes (PARA 3-2-1).
- n. Set parking brake.



NOTES

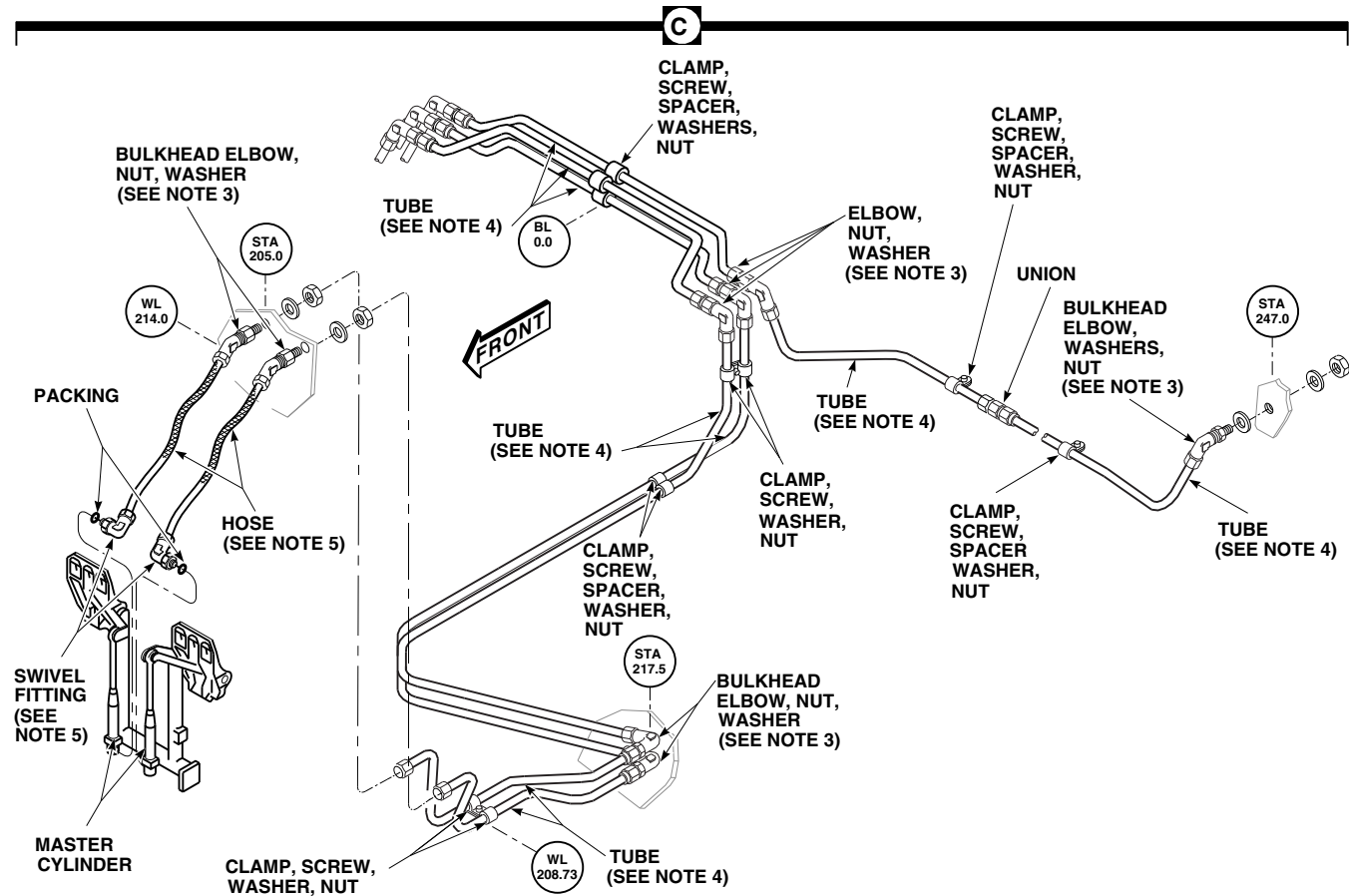
1. ONE CENTER CLAMP:
HELICOPTERS WITH DRAG BEAM
70250-12057-045, -046
2. THREE CLAMPS:
HELICOPTERS WITH DRAG BEAM
70250-12057-047, -048, -049

3. TORQUE BULKHEAD UNION AND
ELBOW NUTS TO 85-105 IN. LBS.
4. TORQUE TUBE B-NUTS TO 120-135 IN.
LBS.
5. TORQUE HOSE ASSEMBLY B-NUTS TO
135-145 IN. LBS.

6. TORQUE UNIONS 95-105 IN. LBS.

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**FIGURE 3-4-16-1. REPLACE MAIN LANDING GEAR WHEEL BRAKE LINES AND HOSES
(SHEET 1 OF 3)**

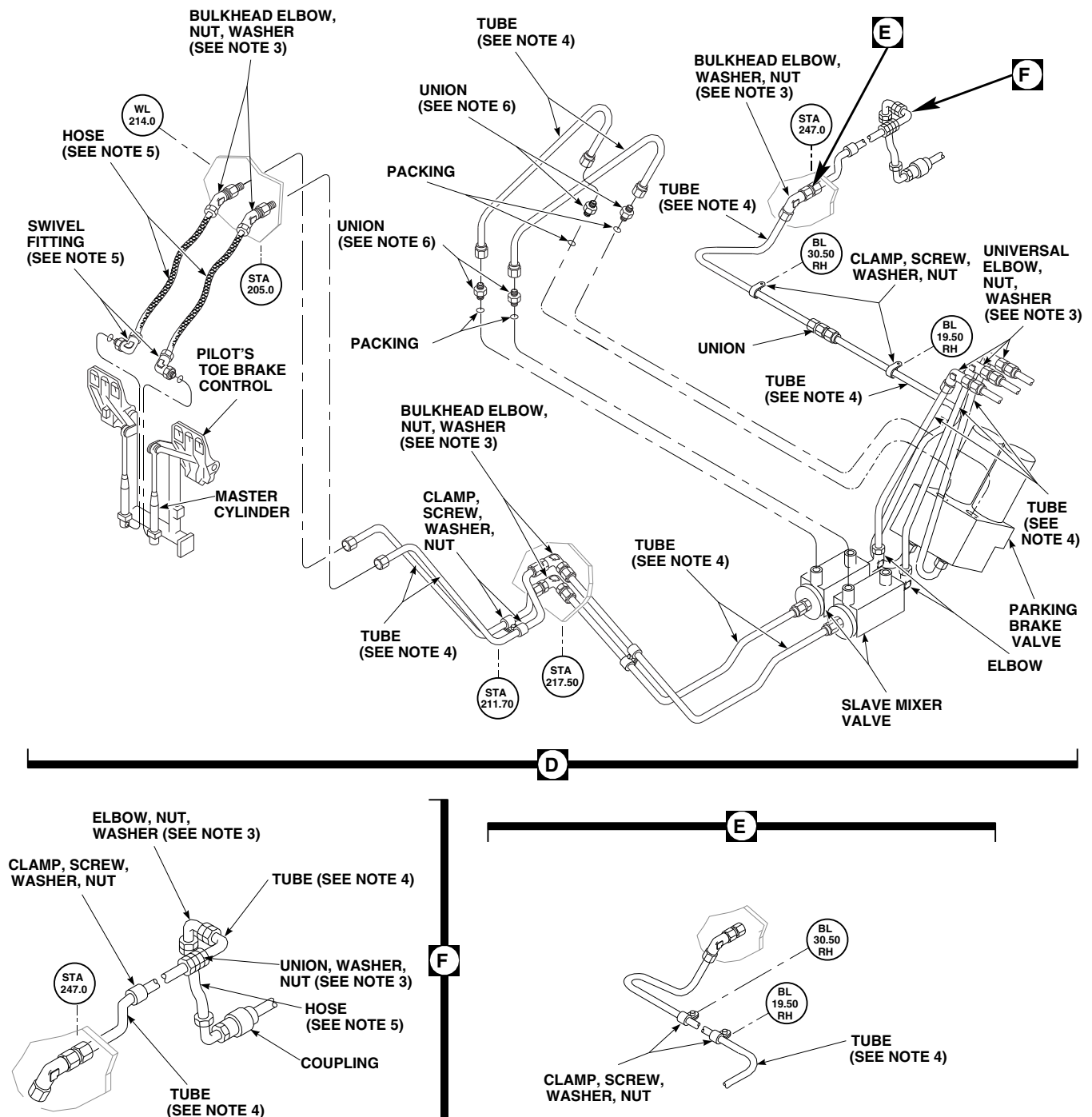


**FIGURE 3-4-16-1. REPLACE MAIN LANDING GEAR WHEEL BRAKE LINES AND HOSES
(SHEET 2 OF 3)**

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3-4-16-4 Change 7

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.



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FIGURE 3-4-16-1. REPLACE MAIN LANDING GEAR WHEEL BRAKE LINES AND HOSES (SHEET 3 OF 3)

Change 7

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3-4-17 PARKING BRAKE VALVE.

PARAGRAPH BREAKDOWN

	PAGE
3-4-17.1 Replace Parking Brake Valve	3-4-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-43
- PARA 3-2-1
- PARA 3-4-20

3-4-17.1 Replace Parking Brake Valve.

3-4-17.1.1. Remove.

- Chock helicopter wheels.
- Release parking brake.
- Turn off all electrical power (PARA 1-6-16).
- Remove cockpit floor cover (PARA 2-4-43).
- Remove cotter pin, pin, and washer securing rod to parking brake valve lever (Figure 3-4-17-1).
- Place machinery towel, Item 211, Appendix D, beneath parking brake valve.
- Disconnect tubes from inlet and outlet ports fittings/unions. Install protective plugs in tubes and cap ports.

- Remove bolts, washers, and nuts. Remove parking brake valve from support brackets.
- Remove screws securing switch to parking break valve. Do not disconnect wiring from switch.
- Remove inlet port unions and packings. Install protective plugs in inlet ports of parking valve. Discard packings.

3-4-17.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective caps and plugs from inlet ports and unions.
- Lubricate threads of inlet port unions and packing with hydraulic fluid, Item 154, Appendix D.

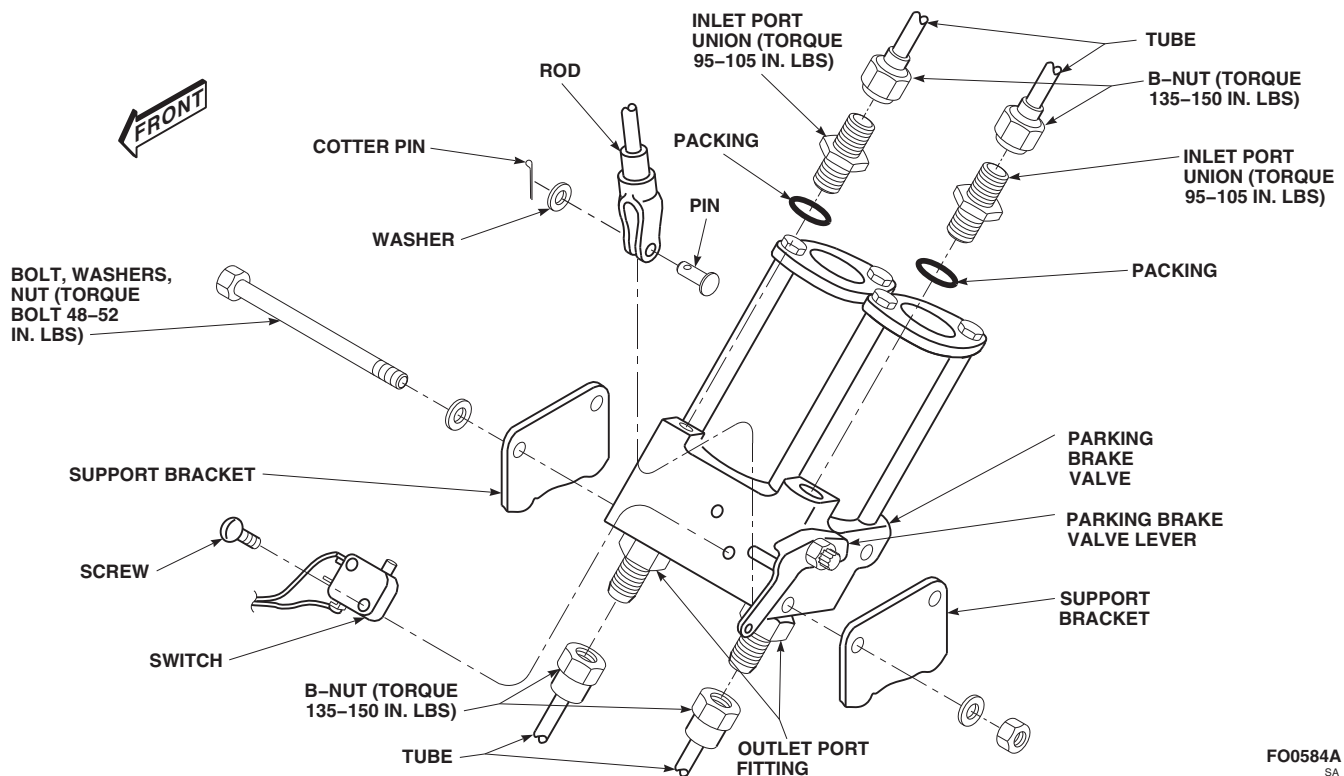
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FIGURE 3-4-17-1. REPLACE PARKING BRAKE VALVE

- d. Install inlet port unions and packings in inlet ports (Figure 3-4-17-1). **TORQUE UNIONS TO 95 - 105 INCH-POUNDS.**
- e. Install switch on parking valve with screws.
- f. Install parking brake valve to support brackets using bolts, washers, and nuts. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- g. Uncap outlet port fittings.
- h. Unplug and connect tubes to inlet port unions and outlet fittings. **WHILE HOLDING UNIONS, TORQUE B-NUTS TO 135 - 150 INCH-POUNDS.**
- i. Pin rod to parking brake valve lever. Install washer and cotter pin.
- j. Bleed pilot's brakes (PARA 3-4-20).
- k. Perform operational check of parking brake system (PARA 3-2-1).
- l. Make sure area is clean and free of all foreign matter.
- m. Install cockpit floor cover (PARA 2-4-43).
- n. Set parking brake.

3-4-18 PARKING BRAKE VALVE SWITCH.

PARAGRAPH BREAKDOWN

	PAGE
3-4-18.1 Replace Parking Brake Valve Switch	3-4-18-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-43
- PARA 3-2-1

3-4-18.1 Replace Parking Brake Valve Switch.

3-4-18.1.1 Remove.

- Chock helicopter wheels.
- Release parking brake.
- Turn off all electrical power (PARA 1-6-16).
- Remove cockpit floor cover (PARA 2-4-43).
- Remove screws and remove switch (Figure 3-4-18-1).
- Disconnect and tag wire from common terminal.
- Disconnect wire from normally closed terminal.

3-4-18.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Connect untagged wire to normally closed terminal (Figure 3-4-18-1).
- Connect tagged wire to common terminal and remove tag.
- Install switch on parking valve with screws.
- Make sure area is clean and free of all foreign material.
- Install cockpit floor cover (PARA 2-4-43).
- Perform operational check of parking brake system (PARA 3-2-1).
- Set parking brake.

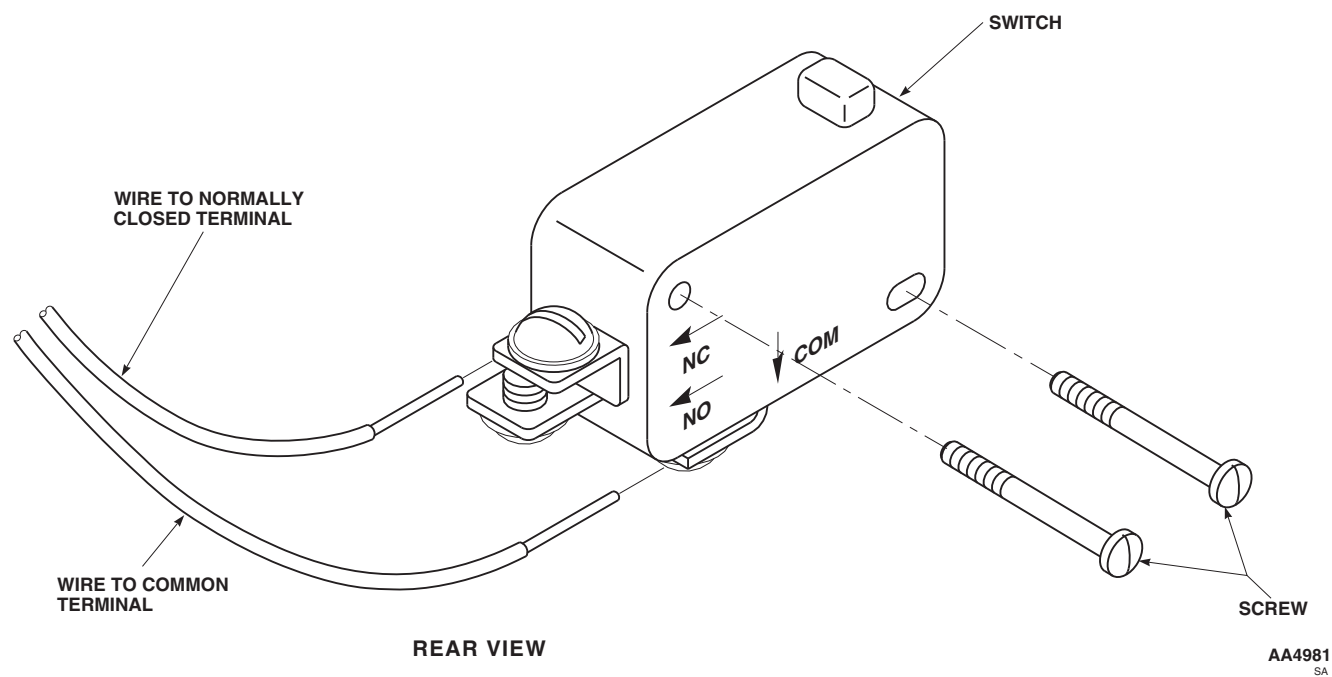


FIGURE 3-4-18-1. REPLACE PARKING BRAKE VALVE SWITCH

3-4-19 SLAVE MIXER VALVE.

PARAGRAPH BREAKDOWN

		PAGE
3-4-19.1	Replace Slave Mixer Valve	3-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Chocks
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Plugs and Caps

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-43
- PARA 3-2-1
- PARA 3-4-20

3-4-19.1 Replace Slave Mixer Valve.

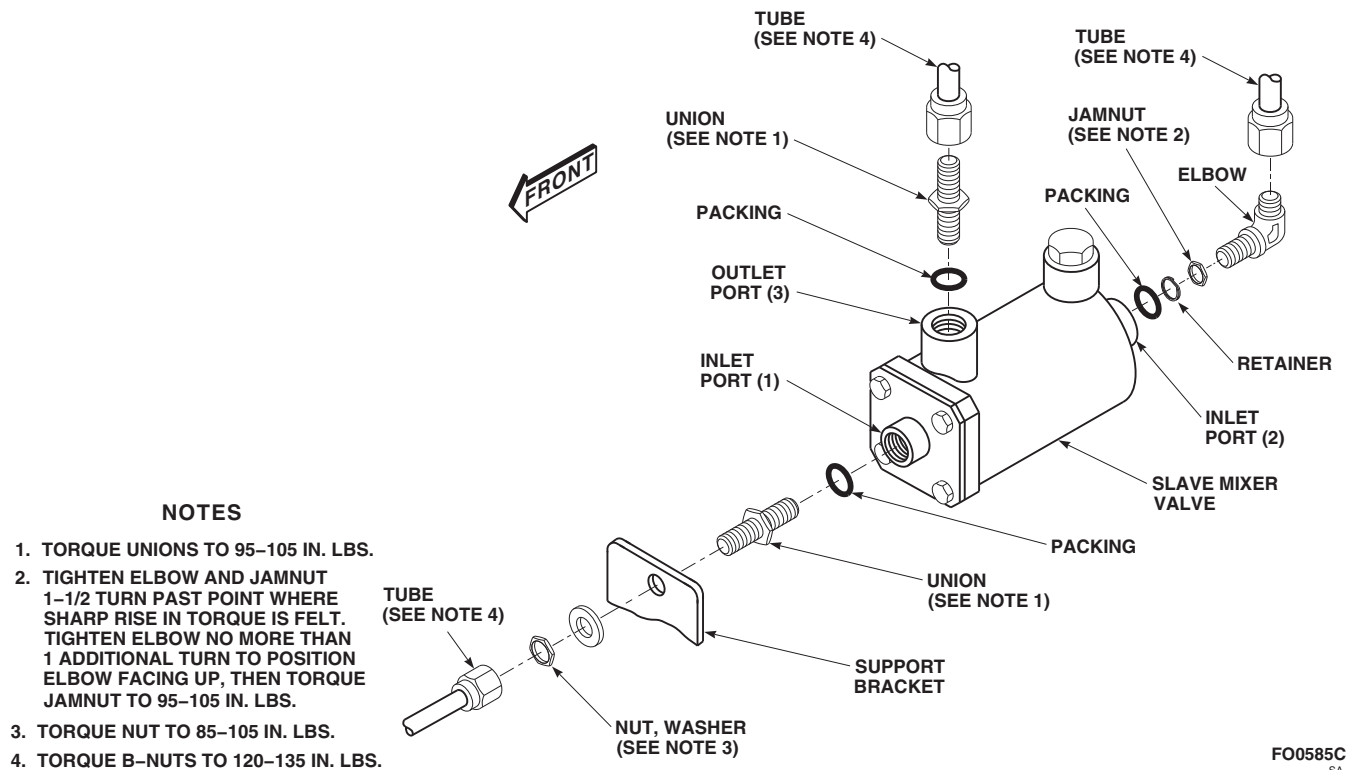
3-4-19.1.1. Remove.

- a. Chock helicopter wheels.
- b. Release parking brake handle.
- c. Turn off all electrical power (PARA 1-6-16).
- d. Remove cockpit floor cover (PARA 2-4-43).
- e. Place machinery towel, Item 211, Appendix D, beneath slave mixer valves.
- f. Disconnect tubes from unions and elbow (Figure 3-4-19-1). Install protective plugs in tubes and cap ports.
- g. Remove nut and washer, and remove slave mixer valve from support bracket.

- h. If valve is being replaced, remove unions, elbow, jamnut, packings, and retainer from slave mixer valve. Discard packings.
- i. Plug slave valve ports.

3-4-19.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective plugs or caps.
- c. Lubricate packings with hydraulic fluid, Item 154, Appendix D. Install packings on unions.
- d. Install unions in inlet port 1 and outlet port 3 of slave mixer valve (Figure 3-4-19-1). **TORQUE UNIONS TO 95 - 105 INCH-POUNDS.**
- e. Lubricate packing, retainer, and male threads of elbow with hydraulic fluid, Item 154, Appendix D.

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SA**FIGURE 3-4-19-1. REPLACE SLAVE MIXER VALVE**

- Install jamnut, retainer, and packing onto elbow. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of elbow.
- Install elbow into slave mixer valve until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN ELBOW AND JAMNUT 1-1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FACING UP.**
- PREVENT ELBOW FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
- Install slave mixer valve into support bracket with washer and nut. **TORQUE NUT TO 85 - 105 INCH-POUNDS.**
- Unplug and connect tubes to unions and elbow. **TORQUE B-NUTS TO 120 - 135 INCH-POUNDS.**
- Bleed pilot's and copilot's brakes (PARA 3-4-20).
- Make sure area is clean and free of all foreign matter.
- Install cockpit floor cover (PARA 2-4-43).
- Perform operational check of parking brake system (PARA 3-2-1).
- Set parking brake.

3-4-20 BLEED MAIN LANDING GEAR WHEEL BRAKE SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE
3-4-20.1 Bleed Main Landing Gear Wheel Brake System	3-4-20-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Container, 1-Gallon
- Container, 1-Quart
- Hydraulic Fluid Dispenser, AF5
- Hydraulic Servicing Unit
- Hypodermic Syringe
- Squirt Can

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 194, Appendix D
- Reducer, AN919-2
- Tubing, Nonmetallic, Item 344, Appendix D
- Union, AN815-3D

References

- Appendix D
- PARA 2-4-43
- PARA 3-2-1
- PARA 11-4-7

3-4-20.1 Bleed Main Landing Gear Wheel Brake System.

3-4-20.1.1. Prepare to Bleed.

NOTE

- Alternate procedures for bleeding main landing gear wheel brakes are given here. Method A uses a hydraulic fluid dispenser (PARA 3-4-20.1.2 and PARA 3-4-20.1.3). Method B (manual method) uses hypodermic syringe or squirt can to add hydraulic fluid (PARA 3-4-20.1.4 and PARA 3-4-20.1.5).

- Under emergency conditions, hydraulic fluid, Item 155, Appendix D, may be used to service main landing gear wheel brakes. However, reservice main landing gear wheel brakes with hydraulic fluid, Item 154, Appendix D, as soon as convenient.

- Chock helicopter wheels.
- Release parking brake.
- Remove pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- Remove pilot's/copilot's access panels (PARA 2-4-43).

3-4-20.1.2. Bleed Copilot's Brakes, Method A.

- Remove bleeder plug and packing from outer slave mixer valve (Figure 3-4-20-1, Sheet 1).

NOTE

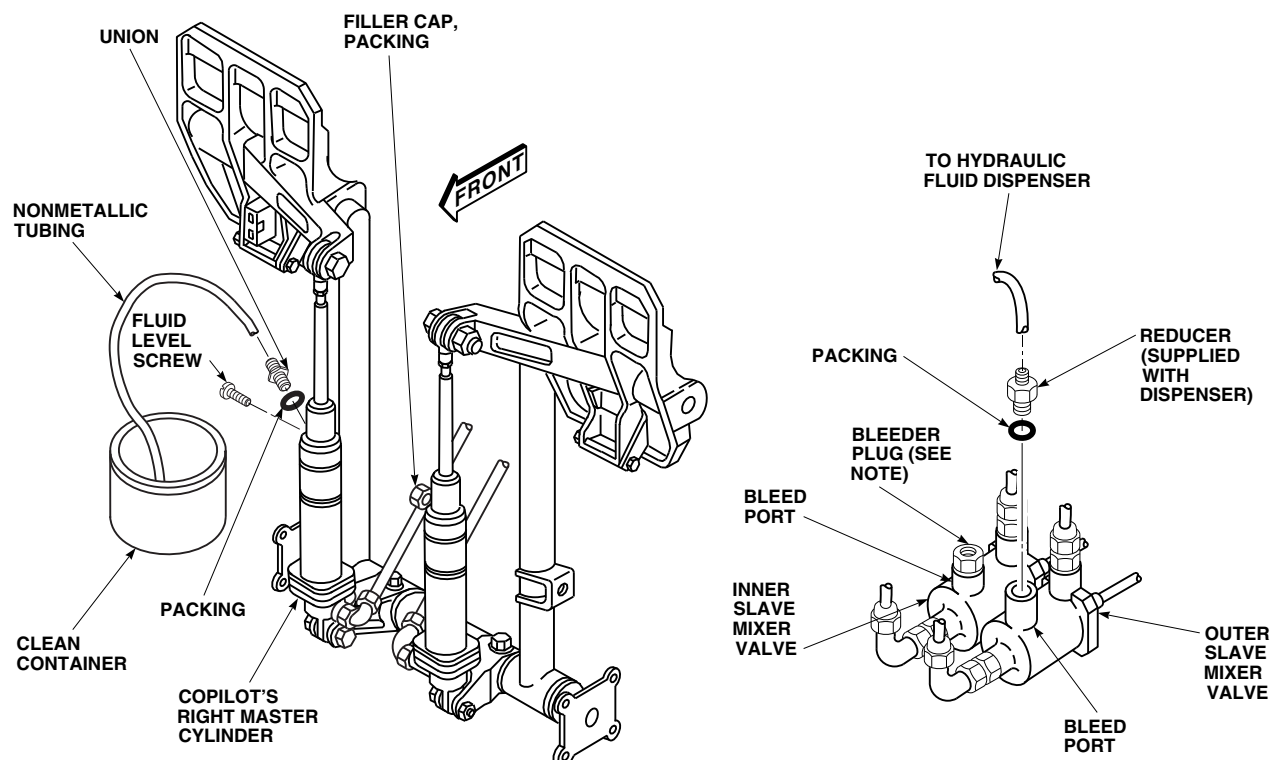
- Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.
- Reducer is normally supplied with fluid dispenser. Select a reducer with 7/16-20-inch thread to fit mixer valve.

- Install packing and reducer. Connect hydraulic fluid dispenser filled with hydraulic fluid, Item 154, Appendix D, to reducer.
- Remove filler cap from copilot's right master cylinder. Remove packing from cap.
- Install packing removed from filler cap on union. Install union and nonmetallic tubing, Item 344, Appendix D, in the master cylinder filler cap port.
- Place open end of tubing in a clean 1-quart container and make sure drain end of bleed tubing is completely submerged in clean hydraulic fluid.

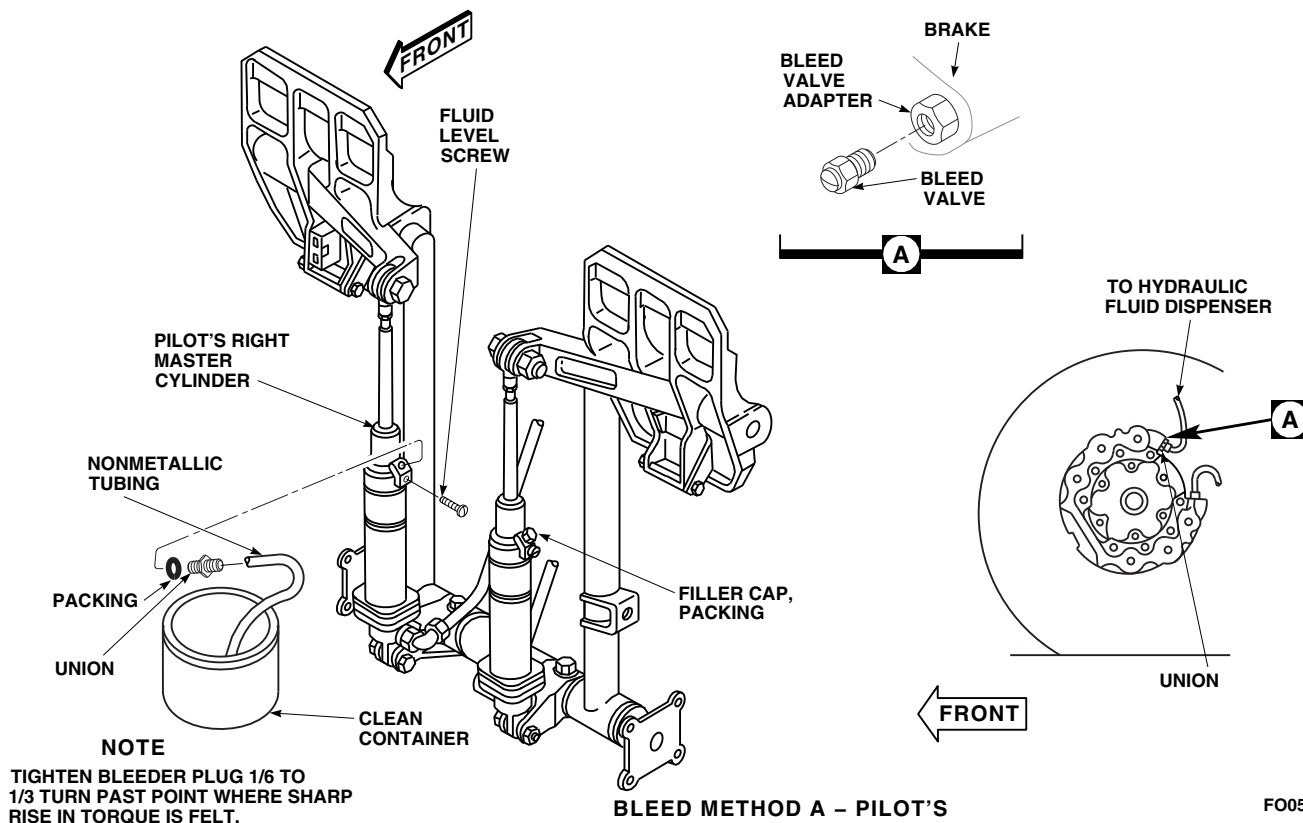
NOTE

During bleeding process, cycle parking brake ON-OFF several times to purge air from parking brake valve accumulators.

- While pumping copilot's right master cylinder, have an assistant operate hydraulic fluid dispenser to obtain about 0.2 gpm flow at 10 psi but not more than 15 psi. Stop pressure flow when air-free fluid flows in tubing, and cylinder is in brake-off position.
- Pinch off overflow line by kinking. Have assistant disconnect hydraulic fluid dispenser from outer slave mixer valve. Remove reducer. Install bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 194, Appendix D, bleeder plug.
- Remove nonmetallic tubing from master cylinder.
- Remove fluid level screw from master cylinder. Allow fluid to drain until level is even with bottom edge of screw hole.
- Install fluid level screw. Install filler cap and packing. Lockwire, Item 194, Appendix D, cap and screw together.



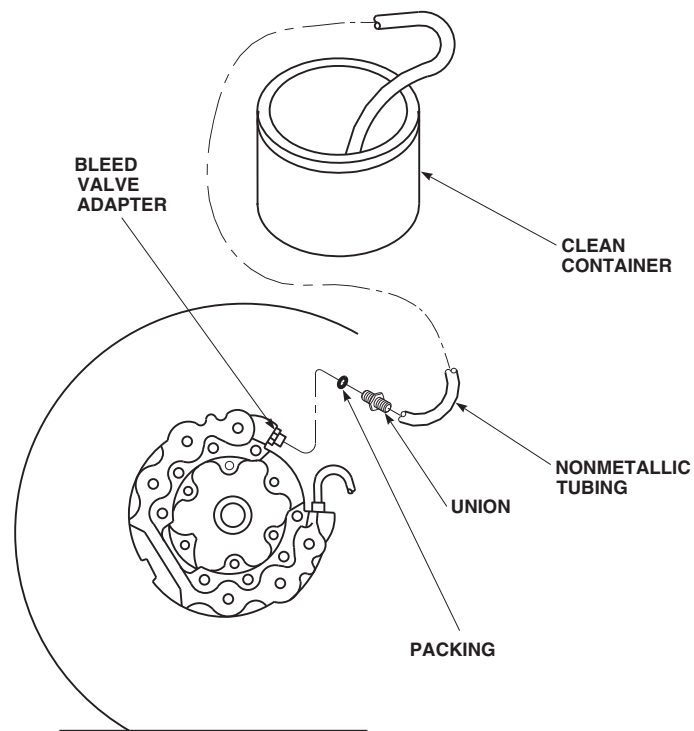
BLEED METHOD A - COPILOT'S



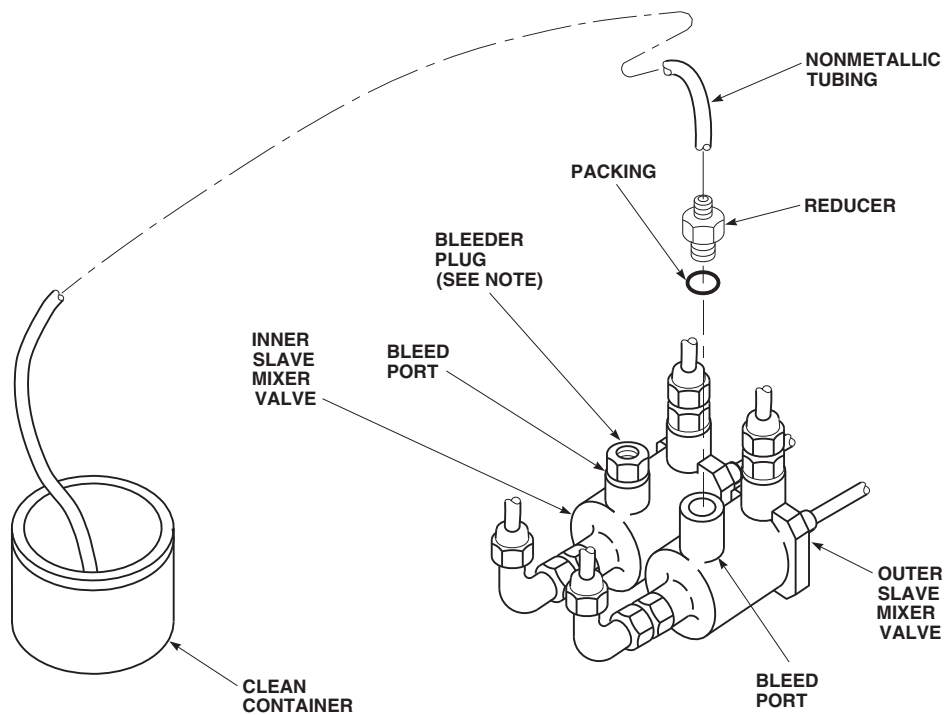
BLEED METHOD A - PILOT'S

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FIGURE 3-4-20-1. BLEED MAIN LANDING GEAR WHEEL BRAKE SYSTEM (SHEET 1 OF 2)



BLEED METHOD B - PILOT'S



BLEED METHOD B - COPILOT'S

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FIGURE 3-4-20-1. BLEED MAIN LANDING GEAR WHEEL BRAKE SYSTEM (SHEET 2 OF 2)

NOTE

Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.

- k. Remove bleeder plug and packing from inner slave mixer valve. Install packing and reducer. Connect hydraulic fluid dispenser to reducer.
- l. Bleed copilot's left master cylinder using same method as with right master cylinder.
- m. Make sure area is clean and free of foreign material before closing up.
- n. Install pilot's/copilot's access panels (PARA 2-4-43).
- o. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- p. Perform operational check of brakes (PARA 3-2-1).
- q. Set parking brake.

3-4-20.1.3. Bleed Pilot's Brakes, Method A.

- a. Remove bleed valve from bleed valve adapter on right wheel brake. Install union in bleed valve adapter (Figure 3-4-20-1, Sheet 1, Detail A).

NOTE

Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.

- b. Connect hydraulic servicing unit to union.
- c. Disconnect hose at drag beam quick-disconnect coupling.
- d. To remove air, bleed the brake and line using the hydraulic servicing unit.

- e. Connect hose at drag beam quick-disconnect coupling and proceed to bleed system.
- f. Remove filler cap on pilot's right master cylinder. Remove packing from cap (Figure 3-4-20-1, Sheet 1).
- g. Install packing removed from filler cap and install on union. Install union and nonmetallic tubing, Item 344, Appendix D, in the open port.
- h. Place open end of tubing in a clean 1-quart container and make sure drain end of bleed tubing is completely submerged in hydraulic fluid.

NOTE

During bleeding process, cycle parking brake ON-OFF several times to purge air from parking brake valve accumulators.

- i. Pump hydraulic servicing unit, keeping a constant pressure from dispenser of 0.2 gpm at 15 - 25 psi, but not over 30 psi, until air-free fluid flows in tubing, with cylinder in brake-off position. While pumping fluid dispenser, have an assistant push pilot's right brake pedal to allow pressure to build up, then slowly release pedal while holding fluid dispenser pressure. Continue this until fluid flows air-free in tubing.
- j. Disconnect hydraulic fluid dispenser from union installed in bleed valve adapter on brake.
- k. Install bleed valve in bleed valve adapter (Figure 3-4-20-1, Sheet 1, Detail A). Lockwire, Item 194, Appendix D, screw in bleed valve to brake.
- l. Remove nonmetallic tubing from master cylinder (Figure 3-4-20-1, Sheet 1).
- m. Remove fluid level screw from master cylinder. Allow fluid to drain until level is even with bottom edge of screwhole.
- n. Install fluid level screw. Install filler cap and packing. Lockwire, Item 194, Appendix D, cap and screw together.

- o. Bleed pilot's left master cylinder using same method as with right master cylinder.

NOTE

Hydraulic servicing unit is to be connected to left wheel brake.

- p. Disconnect hose at drag beam quick-disconnect coupling.
- q. To remove air, bleed the brake and line using the hydraulic servicing unit.
- r. Connect hose at drag beam quick-disconnect coupling and proceed to bleed system.
- s. Make sure area is clean and free of foreign materials before closing up.
- t. Install pilot's/copilot's access panels (PARA 2-4-43).
- u. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- v. Perform operational check of brakes (PARA 3-2-1).
- w. Set parking brake.
- g. Remove filler cap and fill master cylinder to top with hydraulic fluid, Item 154, Appendix D.
- h. Install filler cap and packing.
- i. Release right brake pedal pressure.
- j. Repeat steps c. through i. until no air bubbles appear in tubing.
- k. After pedal pressure is released, remove filler cap and fill right master cylinder with hydraulic fluid, Item 154, Appendix D, if required.
- l. Remove fluid level screw from right master cylinder and let fluid drain until level with bottom of screw hole.
- m. Install fluid level screw.
- n. Install filler cap and packing.
- o. Lockwire, Item 194, Appendix D, filler cap to fluid level screw.
- p. Remove nonmetallic tubing and reducer from outboard slave mixing valve.
- q. Install bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- r. Lockwire, Item 194, Appendix D, bleeder plug.
- s. Repeat steps a. through r. for bleeding inner slave mixer valve and copilot's left master cylinder.
- t. Make sure area is clean and free of foreign material before closing up.
- u. Install pilot's/copilot's access panels (PARA 2-4-43).
- v. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- w. Perform operational check of brakes (PARA 3-2-1).

3-4-20.1.4. Bleed Copilot's Brakes, Method B.

- a. Remove bleeder plug and packing from outer slave mixer valve (Figure 3-4-20-1, Sheet 2).
- b. Install packing and reducer. Install nonmetallic tubing, Item 344, Appendix D, and place open end in clean 1-gallon container.
- c. Remove filler cap from copilot's right master cylinder.
- d. Fill master cylinder to top with hydraulic fluid, Item 154, Appendix D, using hypodermic syringe or squirt can.
- e. Install filler cap and packing.
- f. Press right brake pedal until master cylinder piston bottoms out. Maintain pedal pressure.
- g. Remove filler cap and fill master cylinder to top with hydraulic fluid, Item 154, Appendix D.
- h. Install filler cap and packing.
- i. Release right brake pedal pressure.
- j. Repeat steps c. through i. until no air bubbles appear in tubing.
- k. After pedal pressure is released, remove filler cap and fill right master cylinder with hydraulic fluid, Item 154, Appendix D, if required.
- l. Remove fluid level screw from right master cylinder and let fluid drain until level with bottom of screw hole.
- m. Install fluid level screw.
- n. Install filler cap and packing.
- o. Lockwire, Item 194, Appendix D, filler cap to fluid level screw.
- p. Remove nonmetallic tubing and reducer from outboard slave mixing valve.
- q. Install bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- r. Lockwire, Item 194, Appendix D, bleeder plug.
- s. Repeat steps a. through r. for bleeding inner slave mixer valve and copilot's left master cylinder.
- t. Make sure area is clean and free of foreign material before closing up.
- u. Install pilot's/copilot's access panels (PARA 2-4-43).
- v. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- w. Perform operational check of brakes (PARA 3-2-1).

- x. Set parking brake.

3-4-20.1.5. Bleed Pilot's Brakes, Method B.

- a. Remove bleed valve from bleed valve adapter on right main wheel brake (Figure 3-4-20-1, Sheet 2).
- b. Install union and packing in bleed valve adapter.
- c. Install nonmetallic tubing, Item 344, Appendix D, and place open end in clean 1-gallon container.
- d. Remove filler cap from pilot's right master cylinder.
- e. Fill master cylinder to top with hydraulic fluid, Item 154, Appendix D, using hypodermic syringe or squirt can.
- f. Install filler cap and packing.
- g. Press right brake pedal until master cylinder piston bottoms out. Maintain pedal pressure.
- h. Remove filler cap and fill master cylinder to top with hydraulic fluid, Item 154, Appendix D.
- i. Install filler cap and packing.
- j. Release right brake pedal pressure.
- k. Repeat steps d. through j. until no air bubbles appear in tubing.
- l. After pedal pressure is released, remove filler cap and fill master cylinder with hydraulic fluid, Item 154, Appendix D, if required.

- m. Remove fluid level screw from master cylinder and let fluid drain until level with bottom of screw hole.
- n. Install fluid level screw.
- o. Install filler cap and packing.
- p. Lockwire, Item 194, Appendix D, filler cap to fluid level screw.
- q. Remove nonmetallic tubing and union from right wheel brake.
- r. Install bleed valve and packing in bleed valve adapter.
- s. Lockwire, Item 194, Appendix D, bleed valve screw to brake housing.
- t. Repeat procedures in steps a. through s. for bleeding pilot's left master cylinder and left wheel brake.
- u. Make sure area is clean and free of foreign material before closing up.
- v. Install pilot's/copilot's access panels (PARA 2-4-43).
- w. Install pilot's/copilot's yaw control pedal boots (PARA 11-4-7).
- x. Perform operational check of brakes (PARA 3-2-1).
- y. Set parking brake.

3-4-21 PARKING BRAKE HANDLE AND MECHANISM.

PARAGRAPH BREAKDOWN

		PAGE
3-4-21.1	Replace Parking Brake Handle and Mechanism, 70250-01002-011	3-4-21-1
3-4-21.2	Replace Parking Brake Handle and Mechanism, 70209-84101-013 and 70209-84101-014	3-4-21-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Chocks
- Feeler Gage

References

- PARA 1-6-16
- PARA 2-4-43

- PARA 2-4-44
- PARA 3-2-1
- TM 11-70-23

3-4-21.1 Replace Parking Brake Handle and Mechanism, 70250-01002-011.

3-4-21.1.1 Remove.

- Chock helicopter wheels.
- Release parking brake.
- Remove SAS/FPS or AFCC computer (TM 11-70-23).
- Remove cotter pin, pin, and washer securing brake handle rod end to bracket (Figure 3-4-21-1, Detail A).

- Remove screws and washers fastening brake handle housing to lower console. Remove brake handle from lower console.
- Remove cotter pins, pins, and washers securing rod to bracket and lever. Remove rod.

NOTE

- To aid in removing upper pin, lower console access cover, beneath pilot’s seat, may be removed.
- If necessary, remove access cover on side of lower console, beneath pilot’s seat, by removing screws.

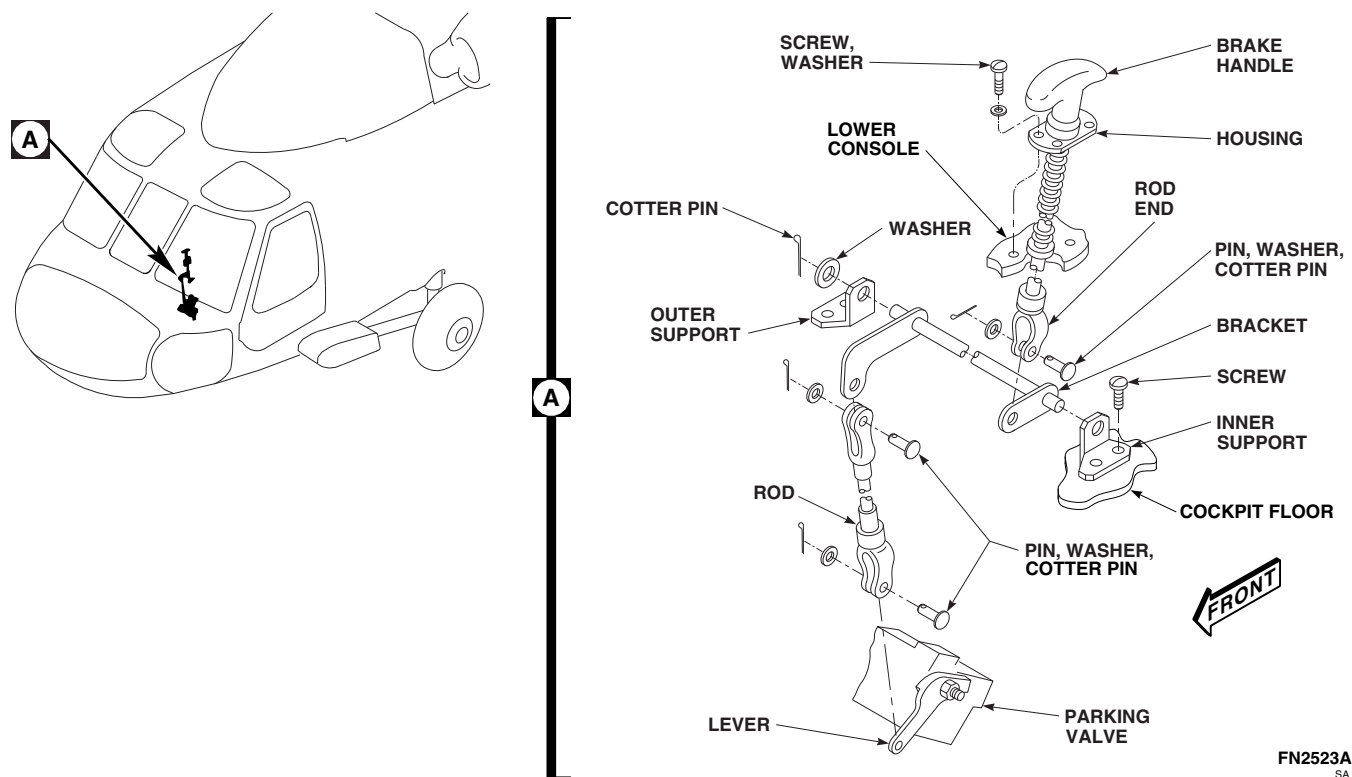
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FIGURE 3-4-21-1. REPLACE PARKING BRAKE HANDLE AND MECHANISM, 70250-01002-011

- h. Remove screws and inner support from cockpit floor.
- i. Remove cotter pin, pin, and washer securing bracket to outer support on cockpit floor. Remove bracket.
- f. Install SAS/FPS or AFCC computer (TM 11-70-23).
- g. Make sure area is clean and free of all foreign matter.
- h. If removed, install access cover on side of lower console, beneath pilot's seat, with screws.
- i. Perform operational check of parking brake system (PARA 3-2-1).
- j. Set parking brake.

3-4-21.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Place bracket in outer support on cockpit floor. Install inner support on bracket and secure support to cockpit floor with screws. Secure bracket to outer support with washer and cotter pin (Figure 3-4-21-1, Detail A).
- c. Secure rod to bracket and lever with pins, washers, and cotter pins.
- d. Install brake handle housing on lower console with screws and washers.
- e. Secure brake handle rod end to bracket with pin, washers, and cotter pin.

3-4-21.2 Replace Parking Brake Handle and Mechanism, 70209-84101-013 and 70209-84101-014.

3-4-21.2.1 Remove.

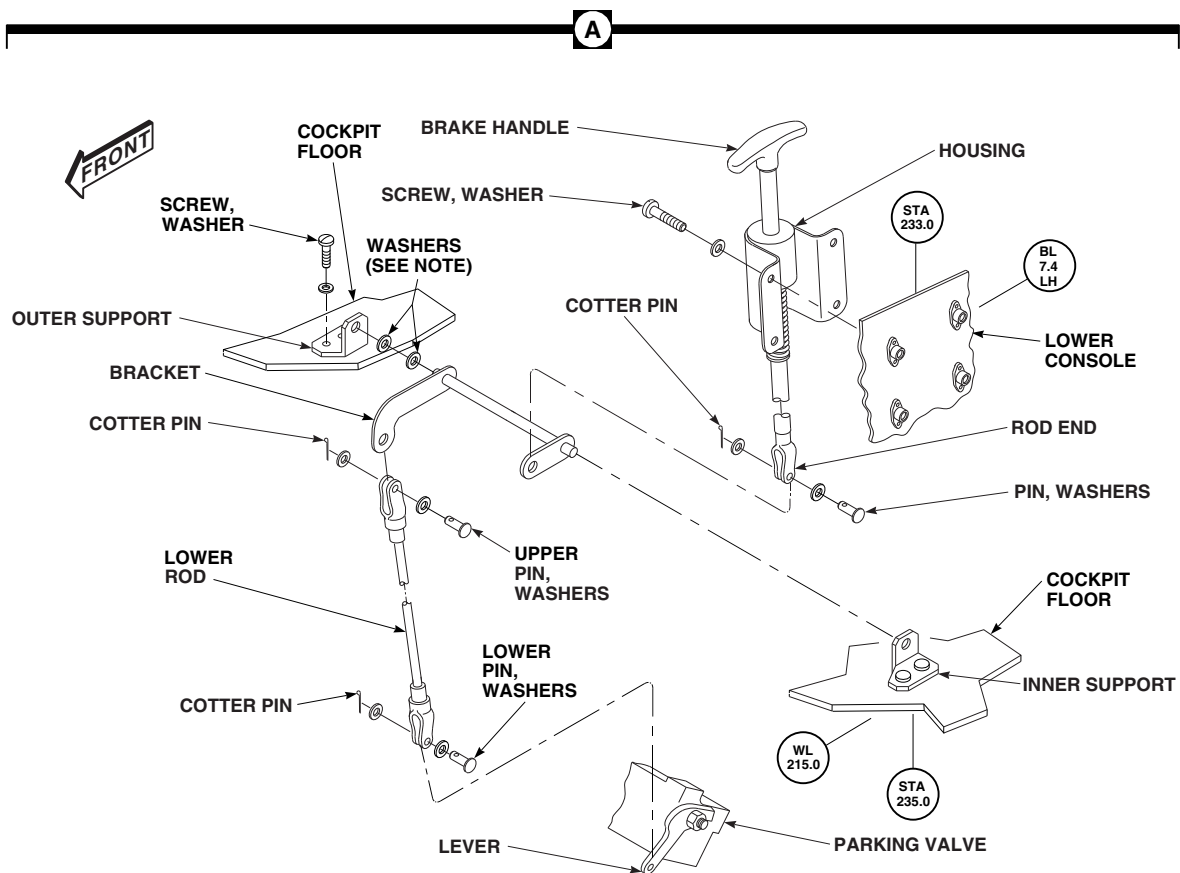
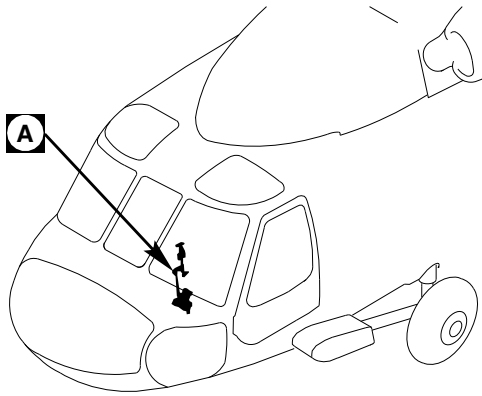
- a. Chock helicopter wheels.
- b. Release parking brake.
- c. Turn off all electrical power (PARA 1-6-16).

- d. If replacing lower rod, remove pilot's seat (PARA 2-4-44).
- e. Remove pilot's floor cover (PARA 2-4-43).
- f. Remove cotter pin, pin, and washer and remove rod end from bracket (Figure 3-4-21-2, Detail A).
- g. Remove screws and washers securing brake handle housing to inner side of lower console. Remove brake handle from lower console.
- h. Remove cotter pins, upper and lower pins, and washers securing lower rod to bracket and lever. Remove lower rod.
- i. Remove screws, washers, and outer support from cockpit floor.
- j. If installed, remove bracket and washers from inner support on cockpit floor.

3-4-21.2.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If removed, place bracket in inner support on cockpit floor. Install outer support on bracket and fasten to cockpit floor with screws and washers. Do not tighten screws now (Figure 3-4-21-2, Detail A).
- c. Secure lower rod to bracket and lever using upper and lower pins, washers, and cotter pins.

- d. Using feeler gage, check clearance as follows:
 - (1) Measure clearance between rod upper pin and cutout in floor. **MINIMUM CLEARANCE IS 0.030-INCH.**
 - (2) Measure gap between outer support and bracket arm. **MAXIMUM GAP IS 0.030-INCH.**
 - (3) Add washers between outer support and bracket as required to maintain minimum clearance between rod upper pin and cutout in floor and maximum gap between outer support and bracket arm.
 - (4) **TIGHTEN SCREWS SECURING OUTER SUPPORT TO FLOOR.**
- e. Install brake handle housing on inner side of lower console with screws and washers.
- f. Install rod end on bracket using pin, washer, and cotter pin.
- g. Make sure area is clean and free of all foreign material.
- h. Install pilot's floor cover (PARA 2-4-43).
- i. If removed, install pilot's seat (PARA 2-4-44).
- j. Perform operational check of parking brake system (PARA 3-2-1).
- k. Set parking brake.



NOTE

ADD WASHERS TO PROVIDE 0.030-IN. MINIMUM CLEARANCE BETWEEN ROD UPPER PIN AND CUT OUT IN FLOOR AND 0.030-IN. MAXIMUM GAP BETWEEN BRACKET AND OUTER SUPPORT.

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FIGURE 3-4-21-2. REPLACE PARKING BRAKE HANDLE AND MECHANISM 70209-84101-013 AND 70209-84101-014

3-4-21-4 Change 7

3-4-22 TAIL LANDING GEAR SHOCK STRUT.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-22.1	Replace Tail Landing Gear Shock Strut	3-4-22-2	3-4-22.2	Repair Tail Landing Gear Shock Strut	3-4-22-4

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beakers, 800 ml (2)
- Chocks
- Containers, 2-Gallon (2)
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 3, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Paint, Polyurethane, Item 222, Appendix D

- Petrolatum, Item 234, Appendix D
- Tail Landing Gear Strut Limiter, Locally-Made, Figure H-176, Appendix H
- Tubing, Nonmetallic, Item 344, Appendix D
- Union, AN815-4

References

- Appendix D
- Appendix H
- MIL-PRF-46170
- PARA 1-3-4
- PARA 1-6-3
- PARA 1-6-16
- PARA 2-4-97
- PARA 2-6-5
- PARA 3-4-23
- PARA 3-5-8

3-4-22.1 Replace Tail Landing Gear Shock Strut.

3-4-22.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Chock main landing gear wheels.
- c. Set parking brake.
- d. Jack tail landing gear (PARA 1-6-3).
- e. Remove screws and washers securing left and right tail cone access panels to tail cone (Figure 3-4-22-1, Detail A).
- f. Remove tail landing gear fairing (PARA 2-4-97).
- g. Use beakers to collect fluid from upper and lower stages, identify as upper or lower.
- h. Remove cap from upper and lower stage air valve (Figure 3-4-22-1, Detail C).
- i. Slip nonmetallic tubing, Item 344, Appendix D, over valve stem. Place free end of tubing in correct container to capture hydraulic fluid as nitrogen is released from shock strut chambers.

NOTE

A hissing sound will be heard from valve when nitrogen is being released.

- j. Turn outer hexnut counterclockwise to release nitrogen from upper/lower shock strut chamber.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- k. After all nitrogen pressure is released, turn inner hexnut counterclockwise to remove.

- l. Allow any remaining hydraulic fluid in upper/lower shock strut chambers to drain into containers.
- m. **IF DISCHARGE OF OIL FROM UPPER OR LOWER STAGE IS GREATER THAN 80-MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT.**
- n. **IF DISCHARGE OF OIL IS BETWEEN 40 AND 80-MILLILITERS (CUBIC CENTIMETERS), SERVICE STRUT (PARA 1-3-4). RE-INSPECT SHOCK STRUT AT 200 FLIGHT-HOURS.**
- o. **IF DISCHARGE OF OIL IS LESS THAN 40-MILLILITERS (CUBIC CENTIMETERS), SERVICE STRUT (PARA 1-3-4). RE-INSPECT SHOCK STRUT AT 500 FLIGHT-HOURS.**
- p. Install cap on both air valves.
- q. Remove nut and washer from bolt securing upper stage strut to fuselage support fitting. Do not remove bolt (Figure 3-4-22-1, Detail D).
- r. Remove nut, washers, and bolt securing lower stage to fitting on yoke.

NOTE

Strut weighs about 20 pounds. Support strut during removal.

- s. With assistant lifting up on shock strut, remove washer, limiter, and bolt attaching top of strut to fuselage fitting and remove strut from helicopter.

3-4-22.1.2 Inspect.

- a. Inspect upper and lower stage spherical bearing for maximum radial and axial play. **SHOCK STRUT FITTING ASSEMBLY SPHERICAL BEARING MAY HAVE UP TO 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY.** If out-of-tolerance, repair shock strut (PARA 3-5-8).
- b. Inspect upper and lower stage spherical bearing Teflon liner for disbonding. **NONE AL-**

LOWED. If disbonding is found, replace shock strut assembly.

- c. Inspect fuselage support fitting for chafing. **NONE ALLOWED.** Blend out chafing damage using abrasive cloth, Item 6, Appendix D. **DAMAGE MUST NOT EXCEED 0.045-INCH AFTER BLENDING.** If damage exceeds limit, replace fitting.
- d. Touch up protective coating (PARA 2-6-5).
- e. Inspect tail landing gear strut limiter for wear. If worn through, locally make tail landing gear strut limiter (Figure H-176, Appendix H).

3-4-22.1.3 Install.

- a. If replacement shock strut assembly is filled with hydraulic preservative fluid, MIL-PRF-46170, drain as follows:
 - (1) Remove cap from upper valve (Figure 3-4-22-1, Detail C).
 - (2) Release nitrogen pressure in shock strut by turning outer hexnut counterclockwise. A hissing sound can be heard when releasing nitrogen.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- (3) When valve stops hissing, turn inner hexnut counterclockwise and remove valve.
- (4) Remove upper bleeder plug from shock strut.
- (5) Install union with packing off bleeder plug, in bleeder port. Connect nonmetallic tubing, Item 344, Appendix D, to union. Place overflow line from bleeder port into 2-gallon container.

- (6) Install valve in shock strut upper stage (PARA 3-4-23).
- (7) Attach servicing chuck from bottled nitrogen, Item 221, Appendix D, to upper valve.
- (8) Lay shock strut on side with bleeder port facing down.

WARNING

Injury to personnel and damage to equipment will result if an unregulated nitrogen source is used. Apply pressure gradually from a regulated source of nitrogen.

NOTE

Do not pressurize strut with more than 100 psi pressure.

- (9) Increase nitrogen pressure until upper stage strut is fully extended.

NOTE

There are 39.54-inches between center of upper and lower attachment boltholes when both stages of shock strut are fully extended.

- (10) Disconnect nitrogen bottle from valve and continue draining shock strut until hydraulic fluid cannot be seen coming from bleeder port.
- (11) Remove union from bleeder port.
- (12) Install packing, lubricated with petrolatum, Item 234, Appendix D, on bleeder plug.
- (13) Screw bleeder plug into cylinder until contact is made with cylinder surface.
- (14) Using upper stage method, drain preservative fluid from lower stage of strut.

NOTE

Lower stage strut is identified by attachment point containing an air valve.

- b. Place shock strut lower stage between fitting ears on yoke (Figure 3-4-22-1, Detail D).
- c. Place limiter in fuselage support fitting clevis.

NOTE

Strut weighs about 20 pounds. Support strut during installation.

- d. Place lug of shock strut upper stage in clevis.
- e. Install bolt, washers, and nut attaching shock strut to fuselage support fitting. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin.
- f. With assistant raising up yoke, bolt shock strut lower stage to fitting on yoke with bolt, washers, and nut. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin.
- g. Service tail landing gear shock strut (PARA 1-3-4).
- h. After replacing shock strut with new strut, perform static leak inspection as follows:
 - (1) Wipe polished piston of shock strut dry using clean machinery towel, Item 211, Appendix D.
 - (2) Wait one hour and inspect polished piston for hydraulic fluid.
 - (3) If fluid leaks appear before one hour after servicing, replace strut.
 - (4) If no fluid is present, soak a clean machinery towel, Item 211, Appendix D, with hydraulic fluid, Item 154, Appendix D, and wipe all exposed polished portions of strut.

NOTE

Minor seepage in service is not considered cause for replacement.

- i. Lower helicopter and remove jack (PARA 1-6-3).
- j. Install tail landing gear fairing (PARA 2-4-97).
- k. Install left and right tail cone access covers on tail cone with screws and washers (Figure 3-4-22-1, Detail A).

3-4-22.2 Repair Tail Landing Gear Shock Strut.

- a. Blend out any minor scratches on housing with abrasive cloth, Item 3, Appendix D. **SCRATCHES AT RIGHT ANGLES TO CENTERLINE MUST NOT EXCEED 20 % OF CIRCUMFERENCE OF HOUSING. DEEPEST PART OF SCRATCH MUST NOT EXCEED 0.005-INCH AFTER BLENDING.**
- b. Blend out any small scratches or nicks on upper stage housing and lug with abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT EXCEED 0.010-INCH AFTER BLENDING.**
- c. Blend out any small scratches or nicks on lower stage housing with abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT EXCEED 0.010-INCH AFTER BLENDING.**
- d. No repair is allowed to upper or lower piston surface. **ALLOWABLE WEAR IS LIMITED TO 0.0015 INCH, OR SMALLEST DIAMETER OF 2.9935-INCHES.**
- e. Lower piston lug scratches or nicks may be blended out using abrasive cloth, Item 3, Appendix D. **DEEPEST PART OF REPAIRED AREA MUST NOT EXCEED 0.010-INCH AFTER BLENDING.**
- f. Clean any repaired area with dry-cleaning solvent, Item 124, Appendix D.
- g. Touch up using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 222, Appendix D (PARA 2-6-5).

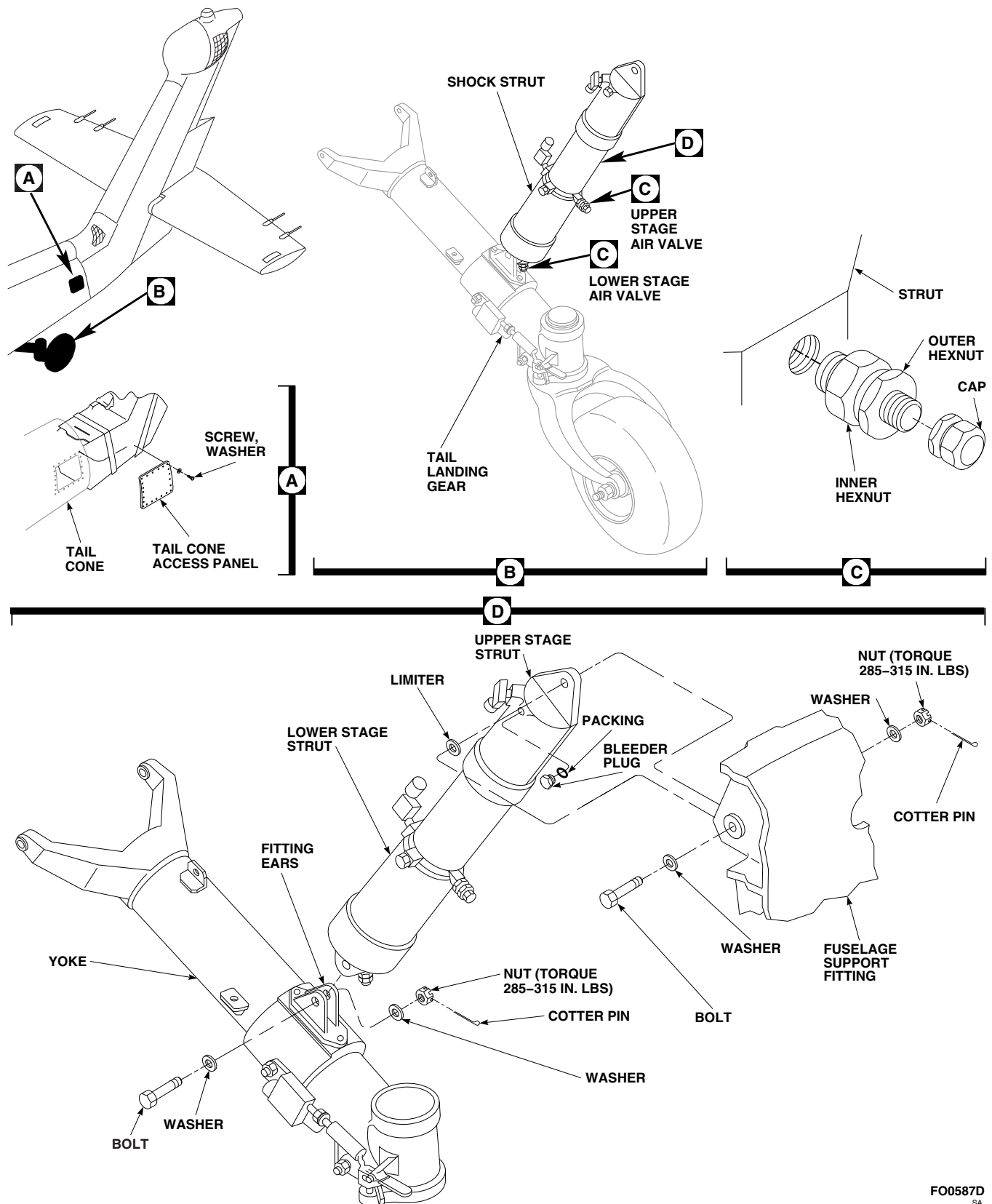


FIGURE 3-4-22-1. REPLACE TAIL LANDING GEAR SHOCK STRUT

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3-4-23 TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-23.1	Replace Tail Landing Gear Shock Strut Kneeling Valve	3-4-23-1	3-4-23.2	Repair Tail Landing Gear Shock Strut Kneeling Valve	3-4-23-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hydraulic Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-4
- PARA 2-4-97

3-4-23.1 Replace Tail Landing Gear Shock Strut Kneeling Valve.

NOTE

Replacement of either tail landing gear shock strut kneeling valve (upper or lower) is the same.

3-4-23.1.1 Remove.

- Remove screws and washers securing left and right tail cone access panels to tail cone (Figure 3-4-23-1, Detail A).
- Remove tail landing gear fairing (PARA 2-4-97).
- Deservice shock strut upper or lower stage as required (PARA 1-3-4).
- Loosen jamnut and remove kneeling valve from strut (Figure 3-4-23-1, Detail C).

- Remove packing, retainer, and jamnut from valve. Keep jamnut for installation of replacement valve. Discard packing and retainer.

3-4-23.1.2 Install.

- Using hydraulic fluid, Item 154, Appendix D, lubricate packing, retainer, and male threads of replacement kneeling valve (Figure 3-4-23-1, Detail C).
- Install jamnut, retainer, and packing onto kneeling valve. Turn jamnut and retainer down until packing is pushed firmly against lower threaded section of kneeling valve.
- Install kneeling valve in strut until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN KNEELING VALVE AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS**

FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN FOR UPPER STAGE OR UP FOR LOWER STAGE.

- d. Service stage of shock strut in which valve was replaced (PARA 1-3-4).
- e. Make sure area is clean and free of foreign material.
- f. Install tail landing gear fairing (PARA 2-4-97).
- g. Install left and right tail cone access covers on tail cone with screws and washers (Figure 3-4-23-1, Detail A).

3-4-23.2 Repair Tail Landing Gear Shock Strut Kneeling Valve.

3-4-23.2.1 Disassemble.

- a. Remove knob stop pin from valve body (Figure 3-4-23-2).
- b. Remove shaft by turning knob counterclockwise.
- c. Remove packing and retainers from shaft. Discard packing and retainers.
- d. Loosen jamnut and remove coupling half from valve body.
- e. Remove packing, retainer, and jamnut from coupling half. Discard packing and retainers.

3-4-23.2.2 Inspect.

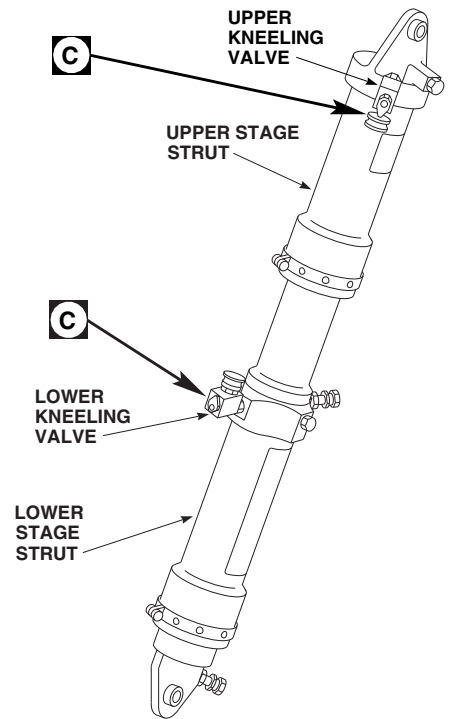
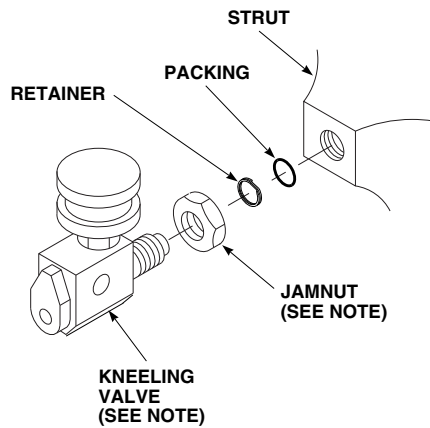
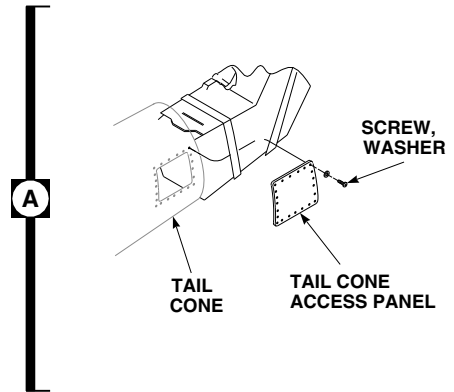
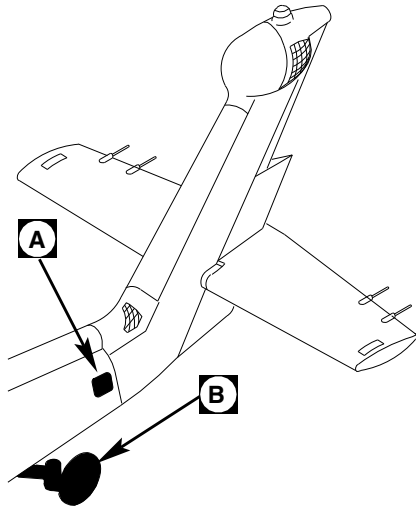
- a. Inspect shaft for corrosion or deformation. **NONE ALLOWED.** If corroded or deformed, replace shaft.
- b. Inspect shaft bore in valve body for corrosion or deformation. **NONE ALLOWED.** If corroded or deformed, replace valve body.

3-4-23.2.3 Assemble.

- a. Using hydraulic fluid, Item 154, Appendix D, lubricate packing, retainer, and male threads of coupling half (Figure 3-4-23-2).
- b. Install jamnut, retainer, and packing onto coupling half. Turn nut and retainer down until packing is pushed firmly against lower threaded section of coupling half.
- c. Install coupling half into valve body until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN COUPLING HALF AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- d. **PREVENT COUPLING HALF FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
- e. Using hydraulic fluid, Item 154, Appendix D, lubricate packing and retainers.
- f. Install retainer, packing, and second retainer on shaft.
- g. Insert shaft carefully into valve body.
- h. Turn knob clockwise to set specified clearance between knob and valve body. **CLEARANCE MUST RANGE FROM 0.060 TO 0.130-INCH.**
- i. Install knob stop pin in valve body.
- j. Using lockwire, Item 197, Appendix D, lockwire knob to hole in valve body.

3-4-23.2.4 Test.

Test for operation on helicopter. No bench test required.



NOTE
TIGHTEN KNEELING VALVE AND JAMNUT 1-1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN FOR UPPER STAGE OR UP FOR LOWER STAGE.

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FIGURE 3-4-23-1. REPLACE TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE

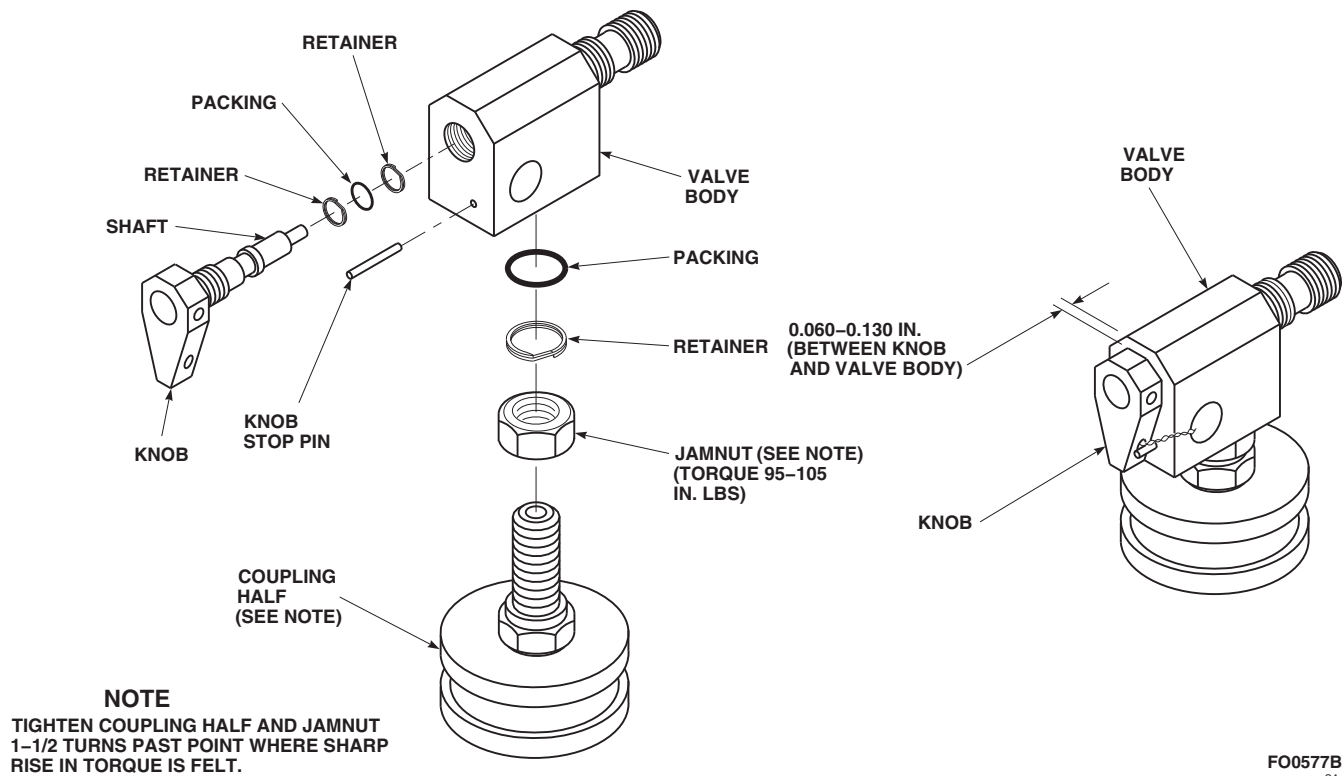


FIGURE 3-4-23-2. REPAIR TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE

3-4-24 TAIL LANDING GEAR SHOCK STRUT AIR VALVE.

PARAGRAPH BREAKDOWN

		PAGE
3-4-24.1	Replace Tail Landing Gear Shock Strut Air Valve	3-4-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-4
- PARA 2-4-97

3-4-24.1 Replace Tail Landing Gear Shock Strut Air Valve.

NOTE

Replacement of upper and lower tail landing gear shock strut air valves is the same.

3-4-24.1.1 Remove.

- Remove screws and washers and remove left and right access panels from tail cone.
- Remove tail landing gear fairing (PARA 2-4-97).
- Remove cap from air valve being replaced (Figure 3-4-24-1, Detail A).
- Turn outer hexnut counterclockwise until nitrogen hissing from valve is heard.



Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valves. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valves.

- When air valve stops hissing, turn inner hexnut counterclockwise and remove air valve.
- Check condition of packing. If packing is determined to be source of air leak, replace packing.
- If valve is determined to be malfunctioning, replace valve.

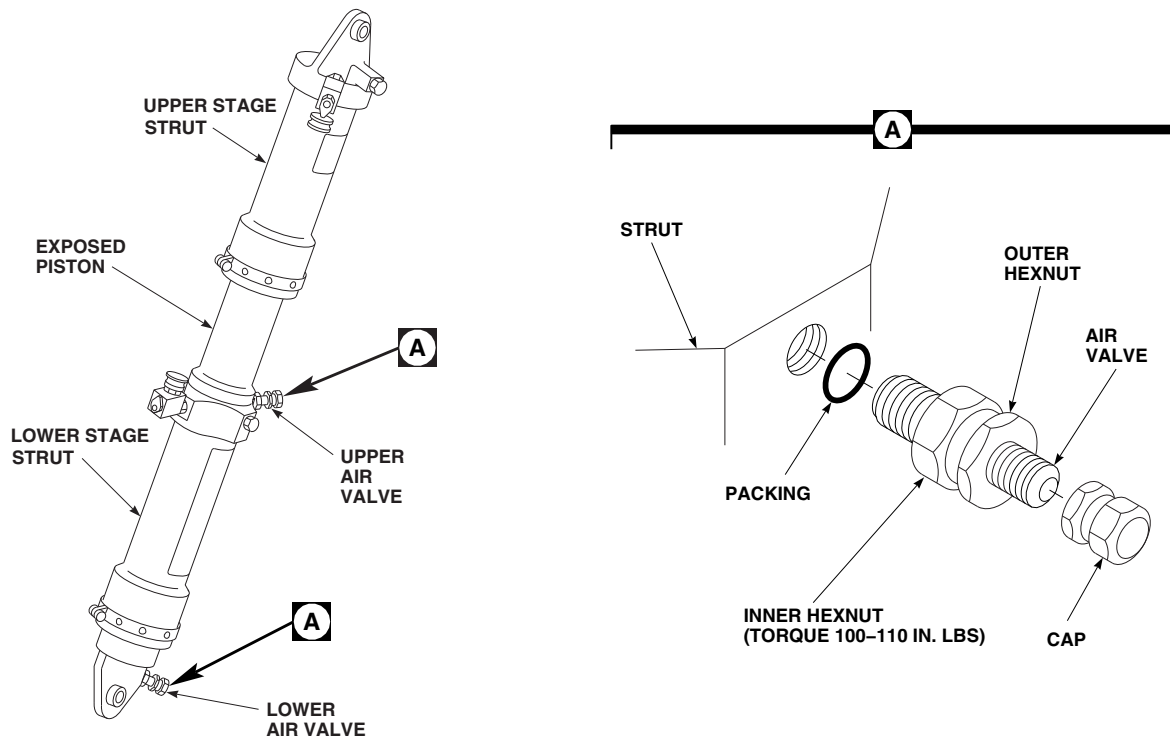
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FIGURE 3-4-24-1. REPLACE TAIL LANDING GEAR SHOCK STRUT AIR VALVE

3-4-24.1.2 Install.

NOTE

Replacement air valve includes new packing.

- a. Install replacement air valve in strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.** Lockwire, Item 197, Appendix D, inner hexnut to shock strut (Figure 3-4-24-1, Detail A).
- b. Service stage of shock strut from which valve was replaced (PARA 1-3-4).
- c. Make sure area is clean and free of foreign material before closing up.
- d. Install tail landing gear fairing (PARA 2-4-97).
- e. Using screws and washers, install left and right access panels on tail cone.

3-4-25 TAIL LANDING GEAR YOKE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-25.1	Replace Tail Landing Gear Yoke	3-4-25-2	3-4-25.5	Replace Teflon-Coated Washers	3-4-25-12
3-4-25.2	Adjust Tailwheel Locked/Unlocked Switches	3-4-25-7	3-4-25.6	Replace Teflon-Coated Sleeves	3-4-25-12
3-4-25.3	Replace Shock Strut Attachment Fitting	3-4-25-8			
3-4-25.4	Replace Tail Landing Gear Yoke Lockpin Bushing	3-4-25-11			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Brush
- Chocks
- Electrical Repairer’s Toolkit
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Multimeter
- Nonmetallic Drift, Locally-Made
- Nonmetallic Hammer
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs
- Wood Blocking (or suitable support)

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 12, Appendix D

- Abrasive Mat, Item 14, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Insulation Sleeving, Item 160A, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-3-4

- PARA 1-6-3
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-86
- PARA 2-4-87
- PARA 2-4-97
- PARA 2-6-5
- PARA 2-6-40
- PARA 3-2-2
- PARA 3-4-26
- PARA 3-4-27
- PARA 3-5-9

3-4-25.1 Replace Tail Landing Gear Yoke.

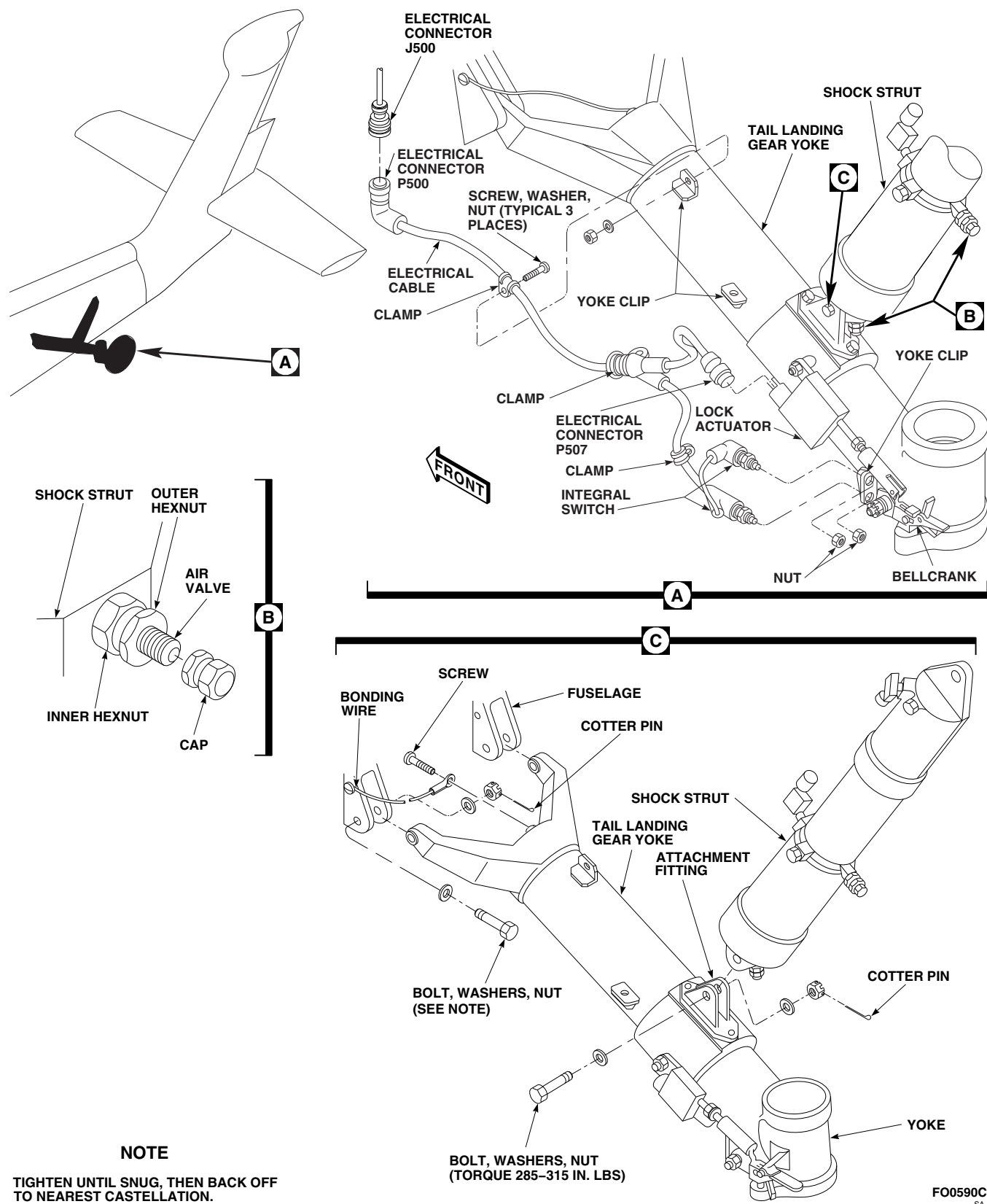
3-4-25.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Set parking brake.

WARNING

Injury to personnel and damage to equipment may occur if tailwheel lockpin is not engaged before jacking helicopter. Make sure to engage tailwheel lockpin.

- c. Engage tailwheel lockpin using either of the following methods:
 - (1) On tail landing gear, place tailwheel lockpin in ENGAGED position using manual operating lever (Figure 3-4-25-3, Detail A).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.
- d. Remove tail landing gear deflector and clamp (PARA 2-4-86 and PARA 2-4-87).
- e. Remove tail landing gear fairing (PARA 2-4-97).
- f. Jack tail landing gear (PARA 1-6-3).
- g. Remove caps from lower and upper stage air valves on shock strut (Figure 3-4-25-1, Detail B).
- h. Turn outer hexnuts counterclockwise until nitrogen hissing from valves is heard.
- i. When hissing stops, install cap.
- j. Remove tail landing gear fork (PARA 3-4-26).
- k. Remove electrical cable as follows:
 - (1) Disconnect integral switches by removing nuts on yoke clip at bellcrank (Figure 3-4-25-1, Detail A).
 - (2) Disconnect electrical connector, P507, from lock actuator.
 - (3) Disconnect electrical connector, P500, from J500.
 - (4) Install protective caps and plugs on electrical connectors.
 - (5) Remove screws, washers, and nuts securing electrical cable/clamps to yoke clips.
- l. Place wood blocking or other suitable support, under tail landing gear yoke.



NOTE

Rest yoke on support when disconnected from shock strut.

- m. Remove cotter pin, bolt, washers, and nut securing shock strut to attachment fitting on yoke (Figure 3-4-25-1, Detail C).
- n. Remove screw and bonding wire at yoke cover.
- o. Remove cotter pins, bolts, washers, and nuts securing yoke to fuselage. Remove yoke.
- p. Remove lock actuator and lock mechanism (PARA 3-4-27).
- q. Remove shock strut attachment fitting (PARA 3-4-25.3).

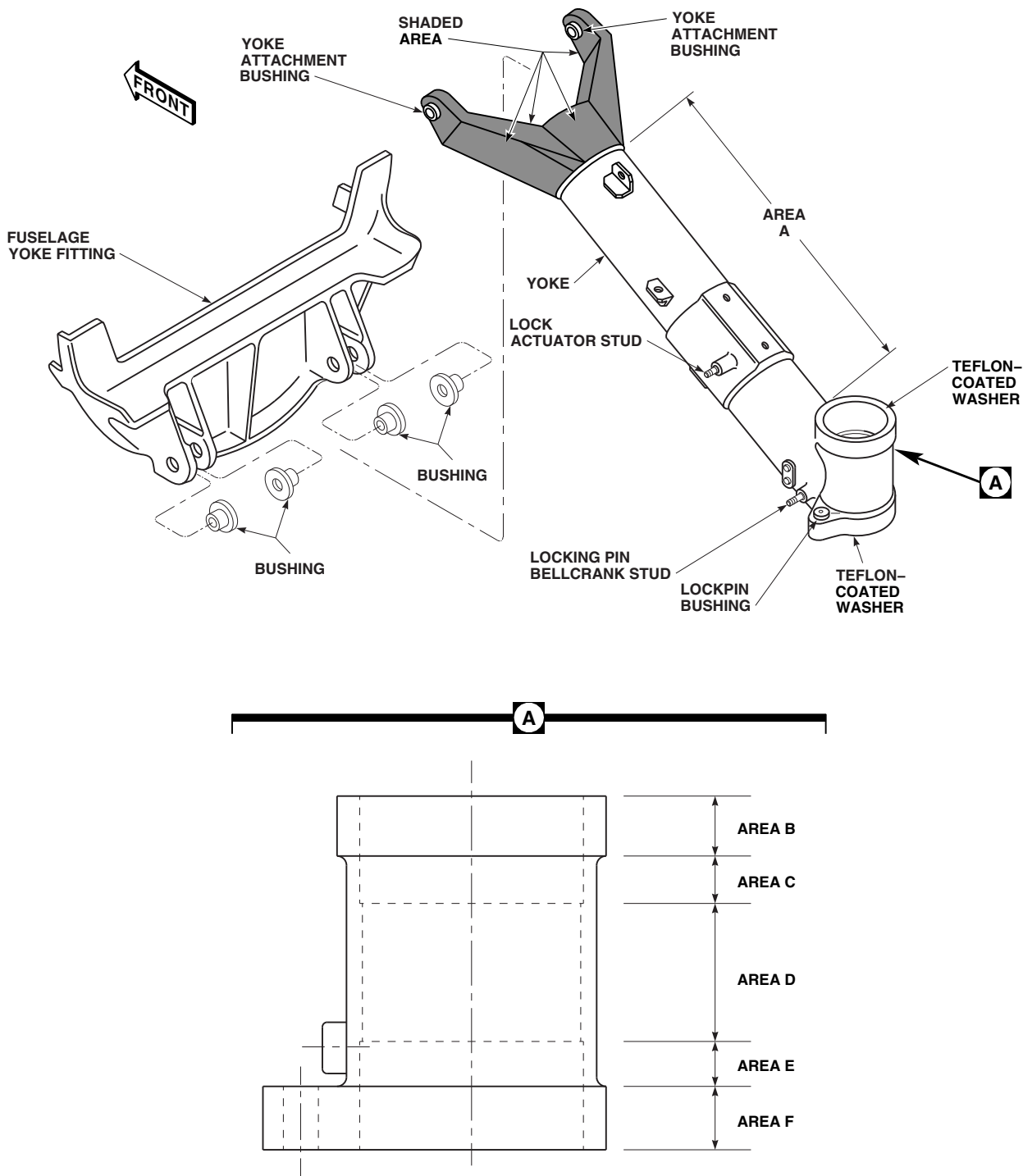
3-4-25.1.2 Inspect.

- a. Check Teflon-coated washers for disbonding (Figure 3-4-25-2). **NONE ALLOWED.** If disbonding is found, replace washer(s) (PARA 3-4-25.5).
- b. Check Teflon-coated sleeves and washers for wear. Replace Teflon-coated washers, and/or sleeves (PARA 3-4-25.5 and PARA 3-4-25.6) if:
 - (1) **WEAR SURFACE LOOKS LIKE SMEARED BRONZE WITH NO DARK SPOTS.**
 - (2) **20% OF SURFACE IS CORRODED OR SCORED.**
 - (3) **TEFLON IS WORN FROM MORE THAN 1/3 OF BEARING SURFACE.**
 - (4) **DISBONDING EXCEEDS 25% OF TOTAL BONDED SURFACE.**
 - (5) **THICKNESS OF TEFLON WASHERS, INCLUDING BOND LINE THICKNESS, IS LESS THAN 0.085-INCH.**
 - (6) **INSIDE DIAMETER OF TEFLON-COATED SLEEVE BEARING IS GREATER THAN 3.510-INCHES.**

- c. Check lock actuator and locking pin bellcrank studs for damage. Replace studs if:
 - (1) Threads are crossed, stripped, or flattened.
 - (2) Loose or bent.
 - (3) Flange of stud or lockring is above surface of yoke.
- d. Measure lockpin bushing inside diameter. **LARGEST INSIDE DIAMETER MUST NOT EXCEED 0.350-INCH.** If measurement exceeds limit, replace lockpin bushing (PARA 3-4-25.4).
- e. Measure yoke attachment bushings inside diameter. **LARGEST INSIDE DIAMETER MUST NOT EXCEED 0.6365-INCH, WITH NO MORE THAN 0.010-INCH OUT-OF-ROUND ALLOWED.** If beyond limits, replace yoke attachment bushings (PARA 3-5-9).
- f. Measure fuselage/yoke fitting bushings inside diameter. **INSIDE DIAMETER MUST NOT EXCEED 0.6365-INCH, WITH NO MORE THAN 0.010-INCH OUT-OF-ROUND ALLOWED.** If one is found out-of-tolerance, replace all four fuselage/yoke fitting bushings (PARA 3-5-9).
- g. Inspect and repair fuselage yoke fitting (PARA 2-6-40).

3-4-25.1.3 Repair.

- a. Repair shaded outline areas as follows (Figure 3-4-25-2):
 - (1) Using abrasive cloth, Item 12, Appendix D, blend out nicks, dents, scratches, and gouges. **YOKE ARM TOTAL CUMULATIVE REPAIR DEPTH (INCLUDING ANY PREVIOUS BLENDING) AT ANY ONE LOCATION MUST NOT BE GREATER THAN 0.0312-INCH. REPAIR MUST NOT EXCEED 0.0312-INCH MAXIMUM.**
 - (2) If damage exceeds limit after blending, replace yoke.



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FIGURE 3-4-25-2. INSPECT TAIL LANDING GEAR YOKE ASSEMBLY

b. Repair AREA A as follows:

- (1) **BLEND OUT NICKS, DENTS, SCRATCHES, OR GOUGES UP TO 0.010-INCH DEEP.**
- (2) Measure length of repair. **REPAIR AFTER BLENDING MUST NOT EXCEED 1.125-INCHES LONG.**
- (3) If damage exceeds limits after repair, replace yoke.

c. Repair AREA B (Figure 3-4-25-2, Detail A) as follows:

- (1) **ON OUTER SURFACE, BLEND UP TO 0.020-INCH DEEP (MINIMUM OUTSIDE DIAMETER IS 4.404-INCH). ON INNER SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM INSIDE DIAMETER IS 3.6972-INCH). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.3384-INCH. BLENDING MUST NOT EXCEED 1.375-INCH-LONG AT ANY CROSS SECTION.**
- (2) If damage exceeds limits after repair, replace yoke.

d. Repair AREA C as follows:

- (1) **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM OUTSIDE DIAMETER IS 4.00-INCH). ON INNER SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM INSIDE DIAMETER IS 3.6972-INCH). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.1204-INCH. BLENDING MUST NOT EXCEED 1.375-INCH-LONG AT ANY CROSS SECTION.**
- (2) If damage exceeds limits after repair, replace yoke.

e. Repair AREA D as follows:

- (1) **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM**

OUTSIDE DIAMETER IS 4.00-INCH). ON INNER SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM INSIDE DIAMETER IS 3.6972-INCH). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.179-INCH. BLENDING MUST NOT EXCEED 1.375-INCH-LONG AT ANY CROSS SECTION.

- (2) If damage exceeds limits after repair, replace yoke.

f. Repair AREA E as follows:

- (1) **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM OUTSIDE DIAMETER IS 4.00-INCH). ON INNER SURFACE, BLEND UP TO 0.030-INCH DEEP (MAXIMUM INSIDE DIAMETER IS 3.7172-INCH). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.1204-INCH. BLENDING MUST NOT EXCEED 1.375-INCH-LONG AT ANY CROSS SECTION.**
- (2) If damage exceeds limits after repair, replace yoke.

g. Repair AREA F as follows:

- (1) **ON OUTER SURFACE, BLEND UP TO 0.030-INCH DEEP (MINIMUM OUTSIDE RADIUS IS 2.182-INCH). ON INNER SURFACE, BLEND UP TO 0.030-INCH DEEP (MAXIMUM INSIDE DIAMETER IS 3.7172-INCH). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.3048-INCH. BLENDING MUST NOT EXCEED 1.375-INCH-LONG AT ANY CROSS SECTION.**
- (2) If damage exceeds limits after repair, replace yoke.

h. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace yoke.

- i. Using corrosion resistant coating, Item 118, Appendix D, touch up reworked aluminum surfaces. Apply primer and paint (PARA 2-6-5).

3-4-25.1.4 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install shock strut attachment fitting (PARA 3-4-25.3).
- c. Install lock actuator and lock mechanism (PARA 3-4-27).
- d. Position and support tail landing gear yoke for attachment to helicopter.
- e. Install yoke to fuselage with bolts, washers, and nuts. **TIGHTEN NUTS UNTIL SNUG, THEN BACK OFF TO NEAREST CASTELLATION.** Install cotter pins (Figure 3-4-25-1, Detail C).
- f. Check lateral movement of yoke assembly. **LATERAL MOVEMENT OF YOKE MUST NOT EXCEED 0.047-INCH.** If lateral movement exceeds limit, replace bushings (PARA 3-5-9).
- g. Secure bonding wire to yoke cover with screw.
- h. Position and support yoke for attachment to shock strut.
- i. Connect shock strut to attachment fitting with bolt, washers, and nut. Make sure thin washer is under bolthead and thick washer is under nut. **TORQUE NUTS TO 285 - 315 INCH-POUNDS.** Install cotter pin.
- j. Install electrical cable as follows:
 - (1) Connect integral switches into yoke clip at bellcrank with nuts (Figure 3-4-25-1, Detail A).
 - (2) Remove caps and plugs from electrical connectors.
 - (3) Inspect and treat electrical connectors (PARA 1-7-20).

- (4) Connect electrical connector, P507, to lock actuator.
- (5) Adjust tailwheel locked/unlocked switches (PARA 3-4-25.2).
- (6) Remove caps and plugs from electrical connectors.
- (7) Inspect and treat electrical connectors (PARA 1-7-20).
- (8) Connect electrical connector, P500 to J500.
- (9) Secure electrical cable/clamps to yoke clips with screws, washers, and nuts.
- (10) Apply sealing compound, Item 292, Appendix D, to all pockets, joints, and entrants on yoke cable clips.
- k. Install tail landing gear fork (PARA 3-4-26).
 - l. Lower tail landing gear (PARA 1-6-3).
- m. Service shock strut upper and lower stages (PARA 1-3-4).
- n. Install tail landing gear deflector and clamp (PARA 2-4-86 and PARA 2-4-87).
- o. Install tail landing gear fairing (PARA 2-4-97).
- p. Perform operational check of tailwheel lock (PARA 3-2-2).

3-4-25.2 Adjust Tailwheel Locked/Unlocked Switches.

3-4-25.2.1 Adjust.

- a. Park helicopter on level surface.
- b. Turn off all electrical power (PARA 1-6-16).
- c. Disconnect electrical connector, P500 from J500 (Figure 3-4-25-3, Detail A).
- d. Manually unlock tailwheel (PARA 1-6-3).
- e. Connect multimeter to pins 4 and 6 of electrical connector, P500.

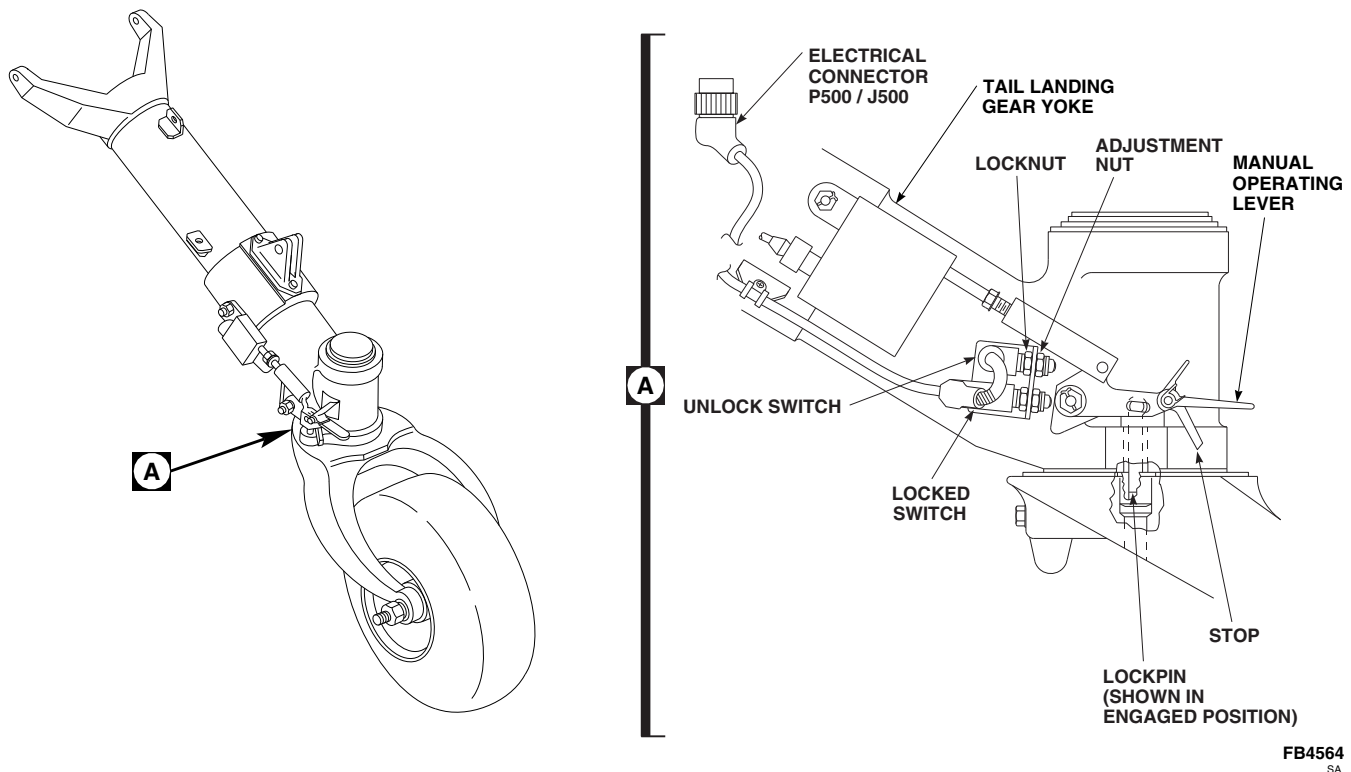
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FIGURE 3-4-25-3. ADJUST TAILWHEEL LOCKED/UNLOCKED SWITCHES

- f. Adjust unlocked switch until multimeter needle just shows continuity. Turn adjustment nut clockwise 1 full additional turn.
- g. **TIGHTEN LOCKNUT.** Using lockwire, Item 196, Appendix D, lockwire locknut.
- h. Disconnect multimeter.
- i. Manually lock tailwheel.
- j. Connect multimeter to pins 1 and 3 of electrical connector, P500.
- k. Adjust locked switch until multimeter needle just shows continuity. Turn adjustment nut clockwise $3 \frac{2}{3}$ - 4 additional turns.
- l. **TIGHTEN LOCKNUT.** Using lockwire, Item 196, Appendix D, lockwire locknut.
- m. Disconnect multimeter.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect electrical connector, P500 to J500.
- p. Perform operational check of tailwheel lock system (PARA 3-2-2).

3-4-25.3 Replace Shock Strut Attachment Fitting.

3-4-25.3.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Set parking brake.

WARNING

Injury to personnel and damage to equipment may occur if tailwheel lockpin is not engaged before jacking helicopter. Make sure to engage tailwheel lockpin.

- c. Engage tailwheel lockpin using either of the following methods:

- (1) On tail landing gear, place tailwheel lock-pin in ENGAGED position using manual operating lever (Figure 3-4-25-3, Detail A).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.
- d. Chock main landing gear wheels.
 - e. Remove tail landing gear fairing (PARA 2-4-97).
 - f. Jack tail landing gear (PARA 1-6-3).
 - g. Disconnect shock strut from tail landing gear yoke by removing cotter pin, bolt, washers, and nut securing shock strut to shock strut attachment fitting on yoke (Figure 3-4-25-1, Detail C).



Damage to shock strut attachment fitting will occur if shim plates are removed. Shim plates are bonded to fitting. Do not remove shim plates from fitting.

- h. Remove bolts, thin washers, and shim washers. Remove fitting, with shim plates, from yoke (Figure 3-4-25-4, Detail B).

3-4-25.3.2 Install.

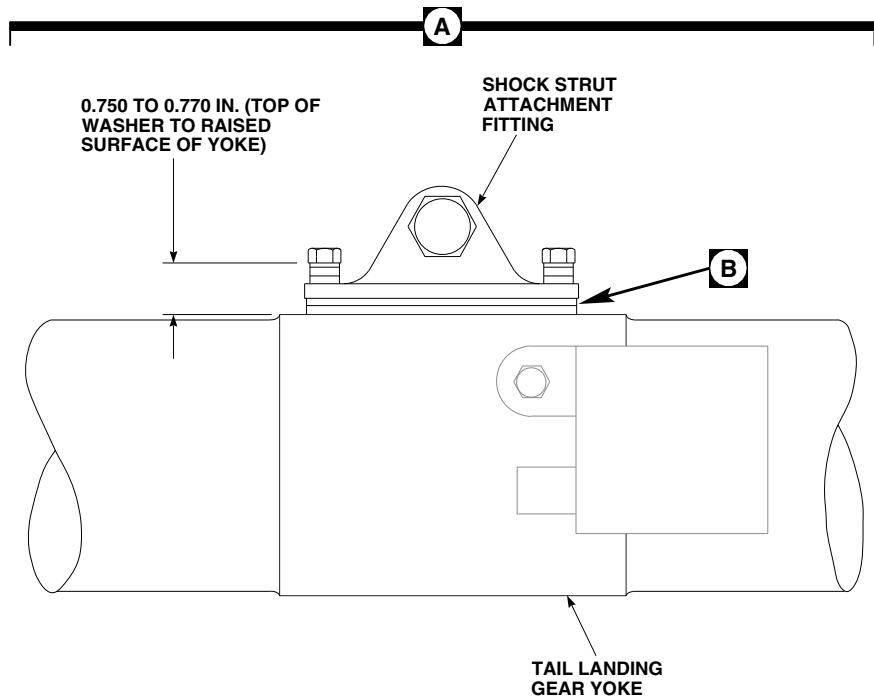
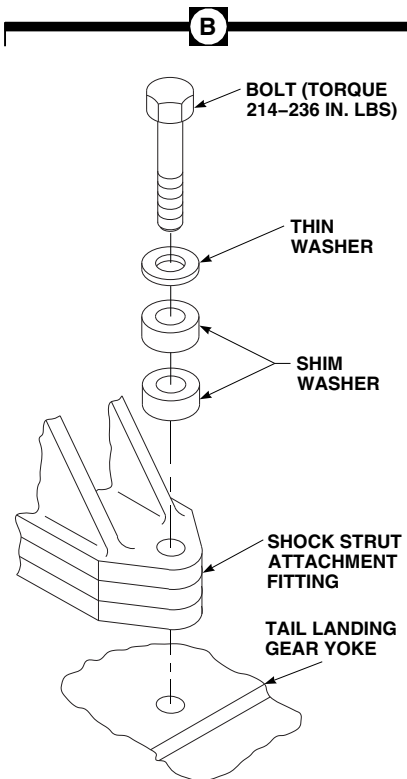
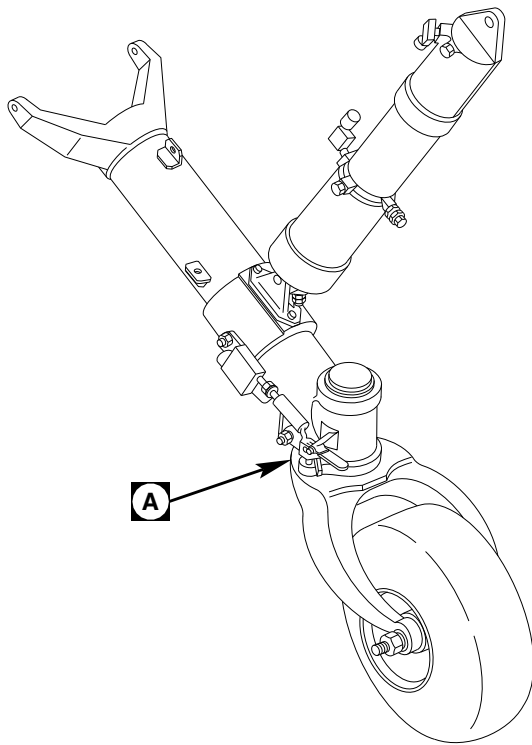
- a. Position shock strut attachment fitting and shims on tail landing gear yoke. Insert bolts with thin washer under bolthead and two 1/4-inch-thick shim washers (Figure 3-4-25-4, Detail A and Detail B).
- b. If fitting is being replaced, perform the following:
 - (1) Release nitrogen from upper and lower stages of shock strut (PARA 3-4-25.1).
 - (2) Fully compress upper and lower stages of shock strut.

- (3) Loosely attach fitting to shock strut, and peel shims on fitting to maintain centerline of tailwheel axle at WL 206.0. Use centerline of yoke to fuselage attachment boltholes at WL 210.5 as reference point to maintain axle at WL 206.0 (Figure 3-4-25-1, Detail C).
- (4) Install fitting on yoke, insert bolts with thin washer under bolthead and two 1/4-inch-thick shim washers (Figure 3-4-25-4, Detail B).



Damage to threaded inserts may occur, if bolts are not shimmed properly. Shim bolts to maintain distance of 0.750 - 0.770-inch above fitting.

- c. Using feeler gage, measure distance from top of washer to flat yoke mounting surfaces (Figure 3-4-25-4, Detail A). **DISTANCE MUST MEASURE 0.750 - 0.770-INCH.** Peel shim washer, as necessary, to obtain correct distance.
- d. Install bolts, thin washers, and shim washers in fitting. **TORQUE BOLTS TO 214 - 236 INCH-POUNDS** (Figure 3-4-25-4, Detail B).
- e. Using lockwire, Item 197, Appendix D, lockwire bolts.
- f. Slide insulation sleeving, Item 160A, Appendix D, over lockwire between bolts to prevent chafing.
- g. Connect fitting on yoke to shock strut with bolt, washers, and nut. Make sure thin washer is under bolthead and thick washer is under nut. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin (Figure 3-4-25-1, Detail C).
- h. Lower tail landing gear (PARA 1-6-3).
- i. Service shock strut upper and lower stages (PARA 1-3-4).



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FIGURE 3-4-25-4. REPLACE SHOCK STRUT ATTACHMENT FITTING

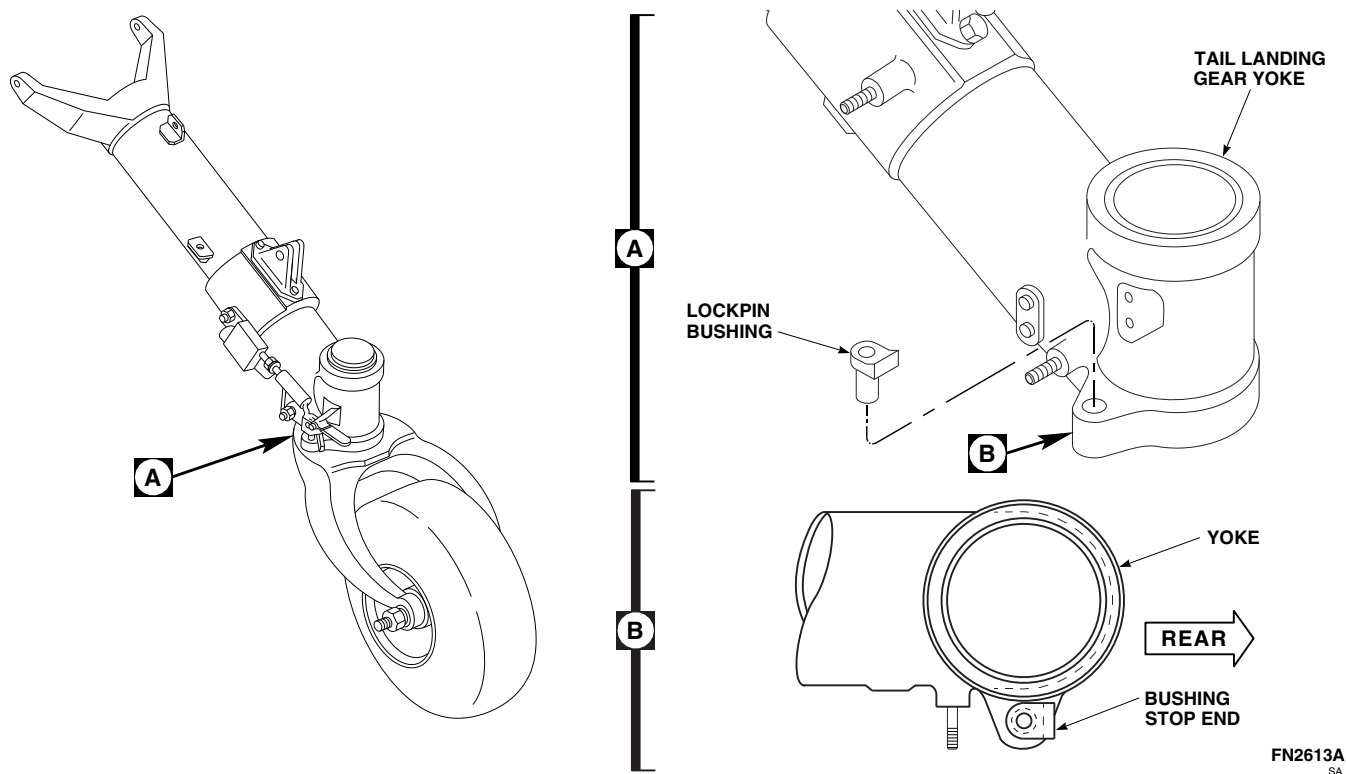


FIGURE 3-4-25-5. REPLACE TAIL LANDING GEAR YOKE LOCKPIN BUSHING

- j. Install tail landing gear fairing (PARA 2-4-97).

3-4-25.4 Replace Tail Landing Gear Yoke Lockpin Bushing.

3-4-25.4.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using nonmetallic scraper and methyl ethyl ketone, Item 217, Appendix D, remove old sealing compound from bushing.
- c. Using nonmetallic drift and nonmetallic hammer, drive lockpin bushing from tail landing gear yoke bore (Figure 3-4-25-5, Detail A).
- d. Using methyl ethyl ketone, Item 217, Appendix D, and abrasive mat, Item 14, Appendix D, remove old sealing compound from bushing bore. Using cheesecloth, Item 90, Appendix D, clean bore and exterior surfaces.

3-4-25.4.2 Inspect.

- a. Visually inspect lockpin bushing bore for corrosion. **NONE ALLOWED.** If corrosion is found, perform the following:
 - (1) Using abrasive cloth, Item 7, Appendix D, remove corrosion and blend out pits.
 - (2) Using feeler gage, measure inside diameter of lockpin bushing bore. **BORE MUST BE 0.5625 - 0.5631-INCH.** If not within limit, replace tail landing gear yoke (PARA 3-4-25.1).

3-4-25.4.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using methyl ethyl ketone, Item 217, Appendix D, and clean cheesecloth, Item 90, Appendix D, wipe new bushing clean. Wipe dry.

- c. Using sealing compound, Item 292, Appendix D, apply coat to bushing outside surfaces.
- d. Line up new lockpin bushing over bore on tail landing gear yoke with square raised portion facing rear (Figure 3-4-25-5, Detail B).
- e. Press new lockpin bushing into yoke while sealing compound is wet. Check that bushing shoulder is bottomed on yoke (Figure 3-4-25-5, Detail A).
- f. Wipe extra sealing compound out of bushing bore.

3-4-25.5 Replace Teflon-Coated Washers.

NOTE

Replacement of both Teflon-coated washers is the same.

3-4-25.5.1 Remove.

- a. Remove tail landing gear yoke (PARA 3-4-25.1.1).

CAUTION

Damage to yoke will occur if metallic drift is used to remove washers. Metal will scratch yoke. Use nonmetallic drift to remove washers.

- b. Using nonmetallic drift, remove Teflon-coated washer(s) from yoke (Figure 3-4-25-6, Detail A).
- c. Using nonmetallic scraper, remove sealing compound from yoke.
- d. If necessary, remove any remaining sealing compound using clean cheesecloth, Item 90, Appendix D dampened with dry-cleaning solvent, Item 124, Appendix D.

3-4-25.5.2 Inspect/Repair.

- a. Inspect area of tail landing gear yoke where Teflon-coated washers will be bonded for cor-

rosion, nicks, dents, scratches, and gouges. **NONE ALLOWED.** If necessary, blend out corrosion, nicks, dents, scratches, or gouges up to 0.010-inch deep on upper and lower surfaces.

- b. Measure length of repair. **REPAIR AFTER BLENDING MUST NOT EXCEED 1.125-INCHES LONG OR 25 % OF THE SURFACE AREA FOR CORROSION.** If damage exceeds limits after repair, replace tail landing gear yoke.

3-4-25.5.3 Install.

- a. Using brush, apply uniform coat, approximately 0.010-inch thick, of sealing compound, Item 292, Appendix D, to metallic surface of Teflon-coated washer(s).
- b. Install washer(s) on tail landing gear yoke. Apply pressure to Teflon surface of washer(s). Make sure that sealing compound has squeezed out uniformly around entire circumference of washer(s) (Figure 3-4-25-6, Detail A and Detail B). No sealing compound permissible on outer flat surface of washer(s).
- c. If a continuous squeeze out is not achieved, remove washer(s) and repeat step a. and step b.
- d. If necessary, make sealing compound flush with Teflon surface of washer(s) using nonmetallic scraper.
- e. Allow sealing compound to cure for a minimum of 72 hours.
- f. Install tail landing gear yoke (PARA 3-4-25.1.4).

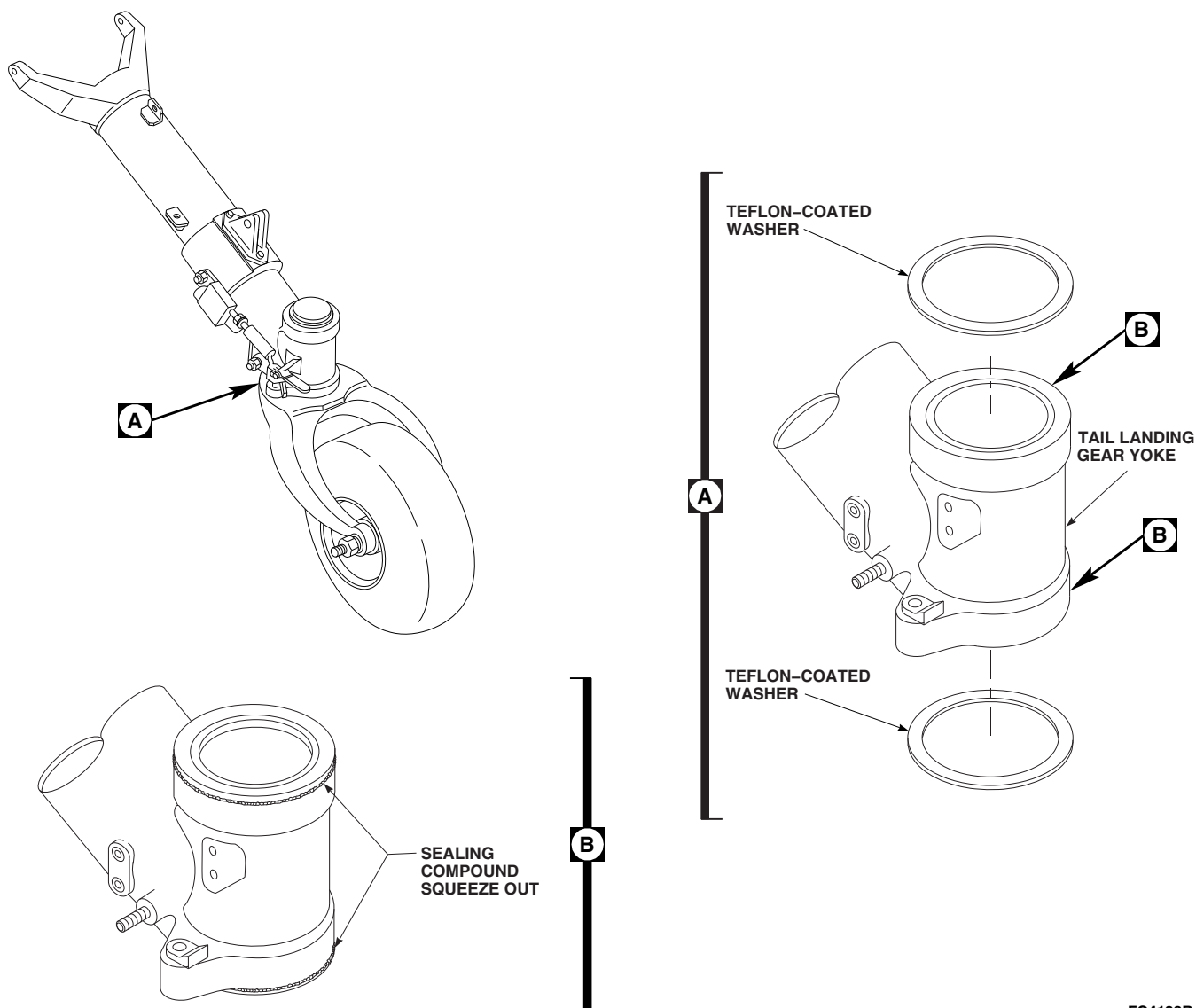
3-4-25.6 Replace Teflon-Coated Sleeves.

NOTE

Replacement of both Teflon-coated sleeves is the same.

3-4-25.6.1 Remove.

- a. Remove tail landing gear yoke (PARA 3-4-25.1.1).



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FIGURE 3-4-25-6. REPLACE TEFLON-COATED WASHERS

CAUTION

Damage to yoke will occur if metallic drift is used to remove sleeves. Metal will scratch yoke. Use nonmetallic drift to remove sleeves.

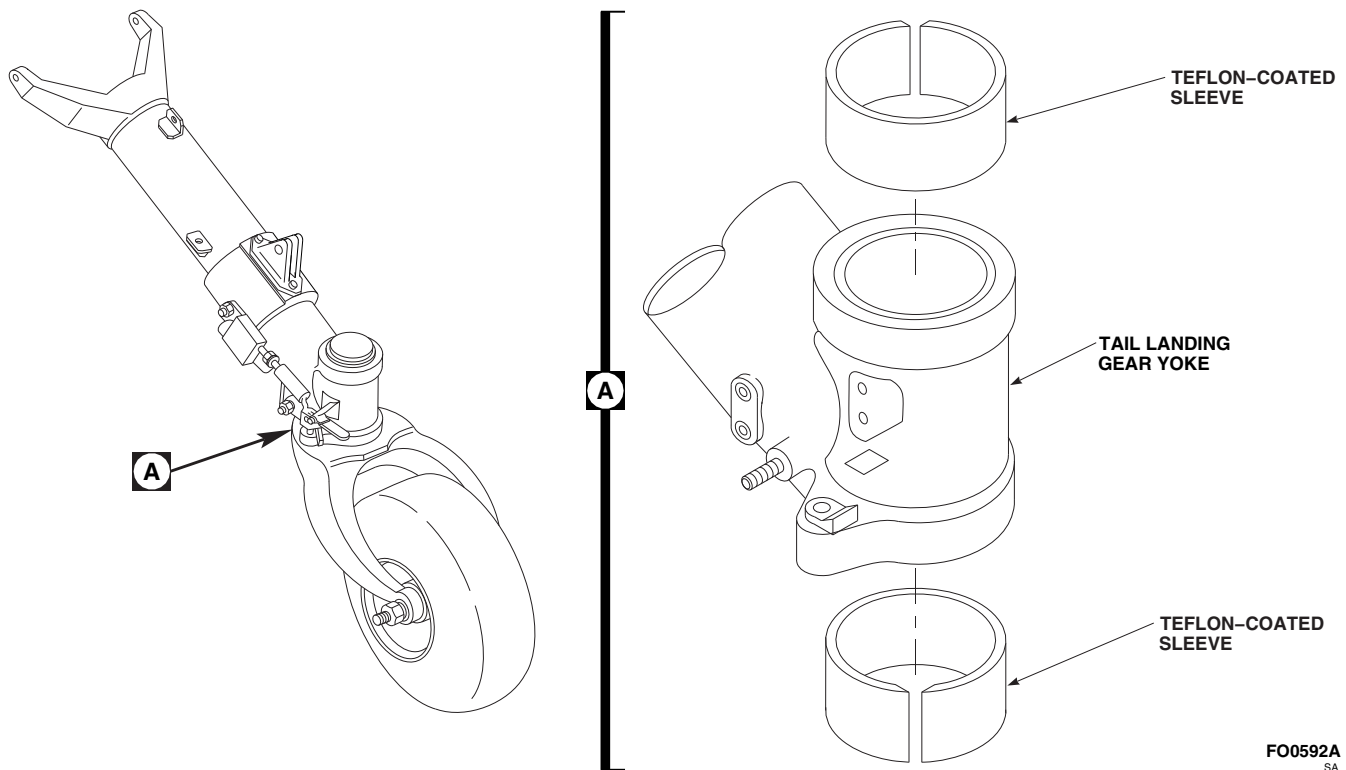
- b. Using nonmetallic drift, remove Teflon-coated sleeves (Figure 3-4-25-7, Detail A).
- c. Using clean cheesecloth, Item 90, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D, clean yoke bore. Remove all grime.

3-4-25.6.2 Inspect.

Inspect bore of tail landing gear yoke for corrosion. **NONE ALLOWED.** If corrosion is found, repair yoke bore (PARA 3-4-25.1.3).

3-4-25.6.3 Install.

- a. Using clean cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, wipe bore of tail landing gear yoke and sleeve mating surfaces. Wipe dry with clean cheesecloth. Continue cleaning until no soil shows on clean cloth. Do not allow solvent to dry on parts. Wipe parts dry.

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- b. Using epoxy primer coating kit, Item 135, Appendix D, apply coat of epoxy primer to outside diameter of Teflon-coated sleeve and yoke bore.
- c. Hand-press sleeve into yoke while primer is wet. Check that sleeve is bottomed in yoke bore (Figure 3-4-25-7, Detail A).
- d. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe primer off Teflon-coated surfaces.
- e. Install tail landing gear yoke (PARA 3-4-25.1.4).

3-4-26 TAIL LANDING GEAR FORK.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-26.1	Replace Tail Landing Gear Fork	3-4-26-2	3-4-26.3	Replace Teflon-Coated Wear Plate	3-4-26-8
3-4-26.2	Replace Lockpin Bushing and Jackpad	3-4-26-6	3-4-26.4	Replace Axle Bushings	3-4-26-9

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Brush
- Chocks
- Fluorescent Penetrant Inspection Kit
- Hand Reamer Set, Adjustable
- Heat Gun
- Medium Cut File
- Nonmetallic Drift, Locally-Made
- Nonmetallic Scraper, Locally-Made
- Steering Bar, 70700-20374-041
- Tail Landing Gear Lockpin Bushing Drift, Locally-Made, Figure H-204, Appendix H
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 8, Appendix D

- Abrasive Wheel, Item 22, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 110, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 178, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tail Landing Gear Oversized Axle Bushing, Locally-Made, Figure H-210, Appendix H

References

- Appendix D
- Appendix H

- ASTM E1417
- PARA 1-6-3
- PARA 1-6-16
- PARA 2-6-5
- PARA 3-2-2
- PARA 3-4-28
- PARA 3-5-10

3-4-26.1 Replace Tail Landing Gear Fork.

3-4-26.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Set parking brake.
- c. Jack tail landing gear (PARA 1-6-3).
- d. Remove tail landing gear wheel and tire (PARA 3-4-28).
- e. Disengage tailwheel lockpin using either of the following methods:
 - (1) On tail landing gear, place tailwheel lockpin in DISENGAGED position using manual operating lever.
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to release tailwheel lock. Place BATT switch OFF.
- f. If installed, remove insulation sleeving from yoke (Figure 3-4-26-4).

CAUTION

Damage to equipment will occur if fork is not supported prior to removing lockring. Fork will drop from yoke. Support fork prior to removing lockring.

- g. Remove lockring and fork from yoke (Figure 3-4-26-2).

3-4-26.1.2 Inspect/Repair.

NOTE

Corrosion is indicated by red film or pitting.

- a. Visually inspect lockring for corrosion.
- b. Remove corrosion and blend out pits, using abrasive cloth, Item 7, Appendix D. **LOCKRING MUST NOT BE LESS THAN 0.108-INCH THICK OR LESS THAN 0.264-INCH WIDE AFTER REPAIR.** If lockring is not within limits, replace lockring.
- c. Clean lockring and fork shaft using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry using clean machinery towel.
- d. Inspect Teflon-coated wear plate as follows:
 - (1) Measure wear from AREA H of Teflon-coated wear plate to AREA G of wear plate (Figure 3-4-26-3). **MAXIMUM WEAR MUST NOT EXCEED 0.010-INCH.**
 - (2) Check Teflon-coated wear plate AREA G and AREA H for scoring and corrosion. **SCORING OR CORROSION MUST NOT EXCEED 20 % OF TEFLON SURFACE.**

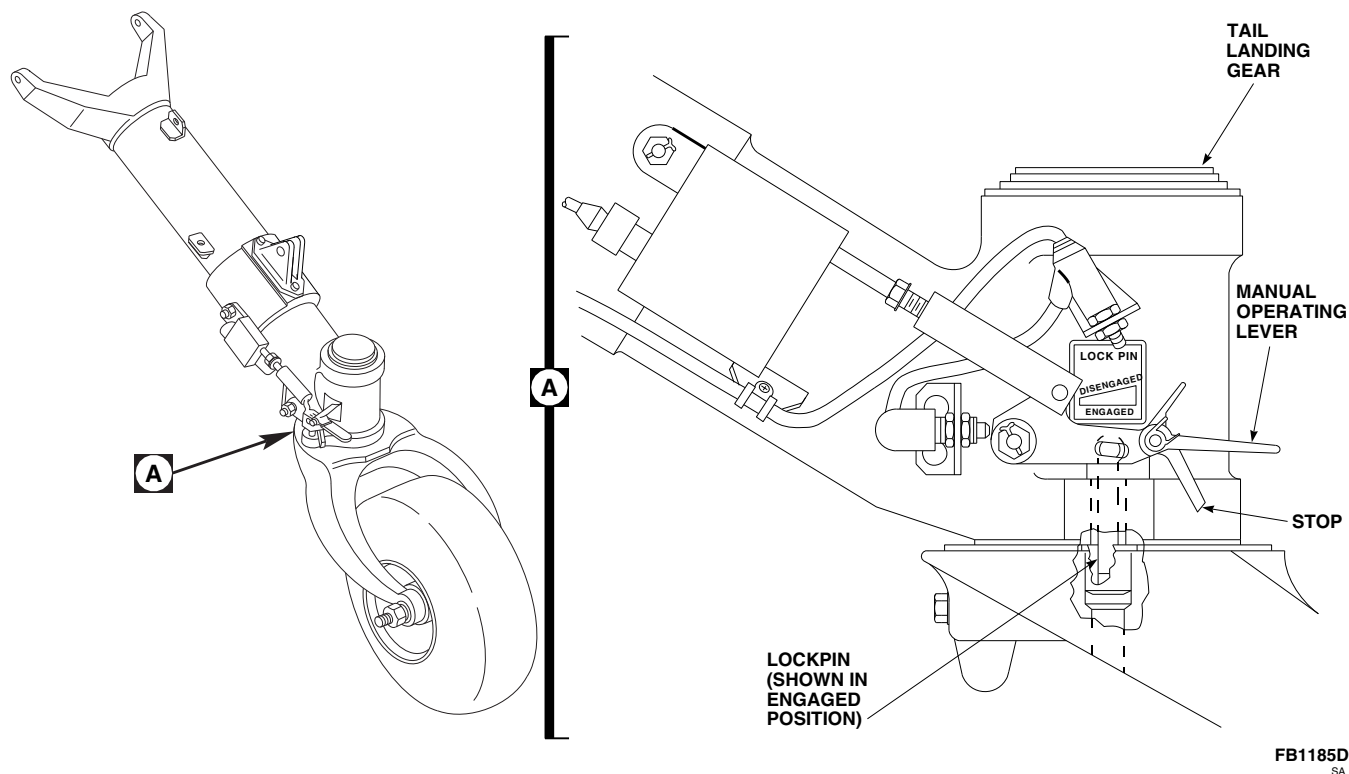


FIGURE 3-4-26-1. ENGAGE/DISENGAGE TAILWHEEL LOCKPIN

NOTE

- Absence of Teflon is acceptable if surface is sprayed with corrosion preventive compound, Item 110, Appendix D, every 30 days to prevent corrosion.
 - Disbonding, up to a total peripheral length of 50% with no indication of corrosion or wear plate movement must be repaired up to next 30-hour/42-day inspection. Inspect for corrosion on a daily basis until wear plate is replaced.
- (3) Check Teflon-coated wear plate for disbonding. **DISBONDING OF TEFLON WEAR PLATE MUST NOT EXCEED A TOTAL OF 50 % OF TOTAL PERIPHERAL LENGTH. NO CORROSION OR MOVEMENT OF PLATE IS ALLOWED.**
 - (4) Check difference between thickness at inner diameter (AREA G) and outer diameter (AREA H). **DIFFERENCE MUST NOT EXCEED 0.010-INCH.**
 - (5) If damage exceeds limits, replace Teflon-coated wear plate (PARA 3-4-26.3).
- e. Inspect aluminum/bronze wear plate as follows:
- (1) Measure wear from AREA H of aluminum/bronze wear plate to AREA G of wear plate. **MAXIMUM WEAR MUST NOT EXCEED 0.040-INCH.**
 - (2) Check aluminum/bronze wear plate AREA G and AREA H for scoring and corrosion. **SCORING OR CORROSION MUST NOT EXCEED 20 % OF AREA G AND 30 % OF AREA H ON WEAR PLATE SURFACE.**
 - (3) Check aluminum/bronze wear plate for disbonding. **DISBONDING OF ALUMINUM/BRONZE WEAR PLATE MUST NOT EXCEED A TOTAL OF 50 % OF TOTAL PERIPHERAL LENGTH. NO CORROSION OR MOVEMENT OF PLATE IS ALLOWED.**

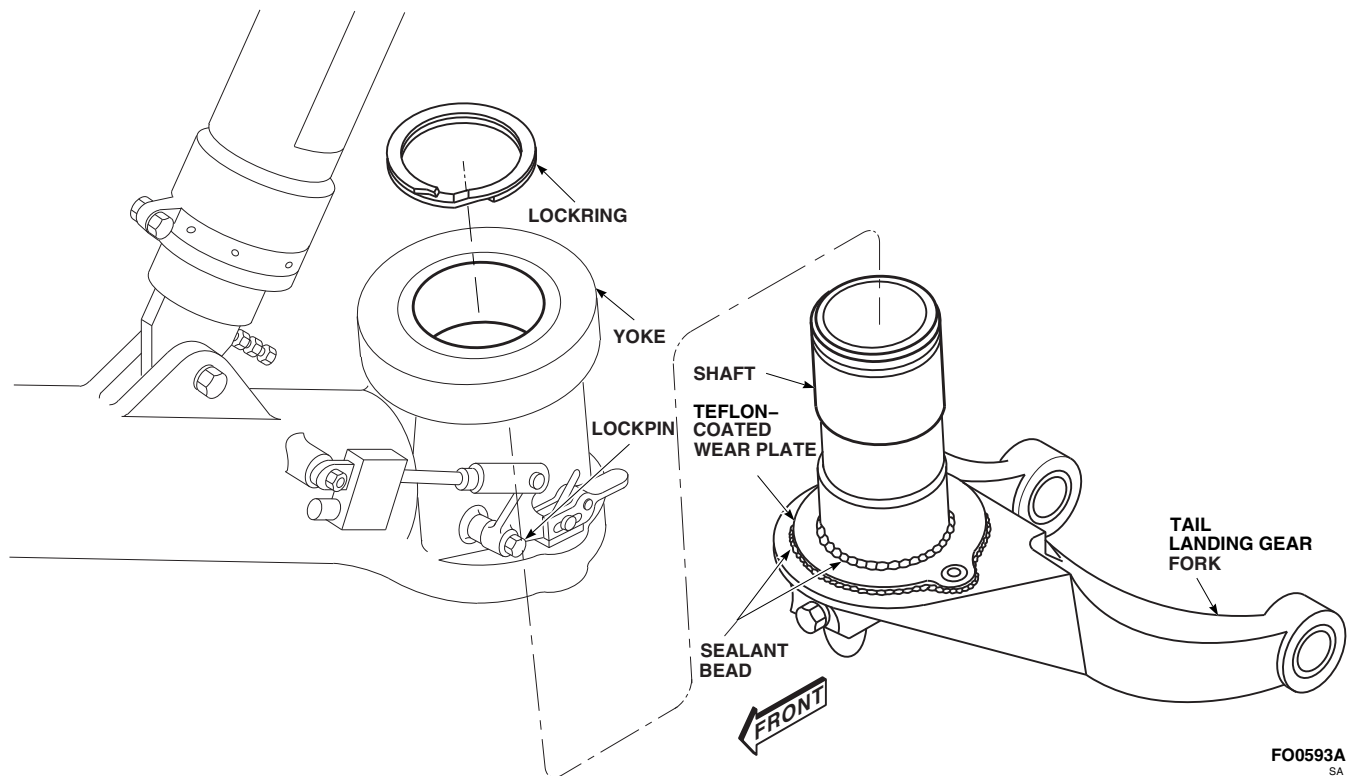


FIGURE 3-4-26-2. REPLACE TAIL LANDING GEAR FORK

- (4) If damage exceeds limits, replace aluminum/bronze wear plate (PARA 3-4-26.3).
- f. Inspect chrome-plating in AREA F as follows:
 - (1) Inspect chrome-plating for evidence of exposed base metal. **NO EXPOSED BASE METAL ALLOWED.** If exposed base metal is found, contact Sikorsky Aircraft ILS.
 - (2) Inspect for damage extending below locking groove. **NONE ALLOWED.** If damage extends below locking groove, contact Sikorsky Aircraft ILS.
- g. Measure locking groove for wear (Figure 3-4-26-3, Detail A). **GROOVE WIDTH MUST NOT EXCEED 0.126-INCH.** If groove width exceeds limit, replace fork.
- h. Measure axle bushings inside diameter (Figure 3-4-26-3). **LARGEST INSIDE DIAMETER MUST NOT EXCEED 1.508-INCHES.** If measurement exceeds limit, replace axle bushings (PARA 3-4-26.4).
- i. Measure lockpin bushing inside diameter. **LARGEST INSIDE DIAMETER MUST NOT EXCEED 0.362-INCH.** If measurement exceeds limit, replace axle bushing (PARA 3-4-26.2).

NOTE

In the following steps, perform all blending with an abrasive wheel, Item 22, Appendix D, abrasive cloth, Item 8, Appendix D, or a medium cut file, as necessary.

- j. Repair AREA A as follows:
 - (1) **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.062-INCH DEEP.** If damage exceeds limit after blending, replace fork.
 - (2) Measure length of repair. Measure length of four sides of fork leg at repair. **LENGTH OF REPAIR IN DIRECTION AROUND FORK LEG MUST NOT EXCEED 10 % OF TOTAL LENGTH AROUND LEG.** If damage exceeds limit after repairs, replace fork.

- k. Repair AREA B as follows:
 - (1) **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.031-INCH DEEP AND 1/4-INCH IN DIAMETER.**
 - (2) If damage exceeds limits after repair, replace fork.
- l. Repair AREA C as follows:
 - (1) **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.094-INCH DEEP AND 1/2-INCH IN DIAMETER.**
 - (2) If damage exceeds limits after repair, replace fork.
- m. Repair AREA D as follows:
 - (1) **CHECK CHROME-PLATING ON JOURNALS FOR CHIPPING, FLAKING, SCORING, OR WEAR THROUGH TO BASE METAL.** If damage is found, replace fork.
 - (2) Measure diameters of journals. **DIAMETERS MUST NOT BE LESS THAN 3.497-INCHES.** If journals measure below limit, replace fork.
- n. Repair AREA E as follows:
 - (1) **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.010-INCH DEEP.**
 - (2) Measure length of repair. **REPAIR AFTER BLENDING, MUST NOT EXCEED 1.062-INCHES LONG.** If measurement exceeds limits, replace fork.
- o. Repair AREA F as follows:
 - (1) **CHECK CHROME-PLATING ABOVE LOCKRING GROOVE FOR CHIPPING, FLAKING, SCORING, OR WEAR THROUGH TO BASE METAL.**
 - (2) If damage is within limits, repair as follows:
 - (a) Blend damaged chrome-plating smooth. **NO BLENDING OF BASE METAL ALLOWED.**
 - (b) Apply corrosion resistant coating, Item 118, Appendix D, to reworked areas. Touch up paint as required (PARA 2-6-5).
 - (3) If damage exceeds limits, contact Sikorsky Aircraft ILS with condition of component.
 - p. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace fork.
 - q. Apply corrosion resistant coating, Item 118, Appendix D, to reworked areas. Touch up paint as required (PARA 2-6-5).

3-4-26.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Apply bead of sealing compound, Item 292, Appendix D, around fork shaft between fork and wear plate and outer edge of wear plate (Figure 3-4-26-2). Using nonmetallic scraper, make sealing compound flush with top surface of wear plate.
- c. Apply corrosion preventive compound, Item 114, Appendix D, to entire fork shaft and lockring.
- d. Insert fork into yoke and install lockring on fork. Apply corrosion preventive compound, Item 114, Appendix D, to exposed shaft and lockring.
- e. Cut a piece of heat-shrinkable insulation sleeving, Item 178, Appendix D, to a length of 2.5-inches.
- f. Install insulation sleeving over top of yoke assembly until it extends 0.125-inch below shoulder of yoke (Figure 3-4-26-4).
- g. Shrink insulation sleeving around body of yoke and over top of fork using heat gun.

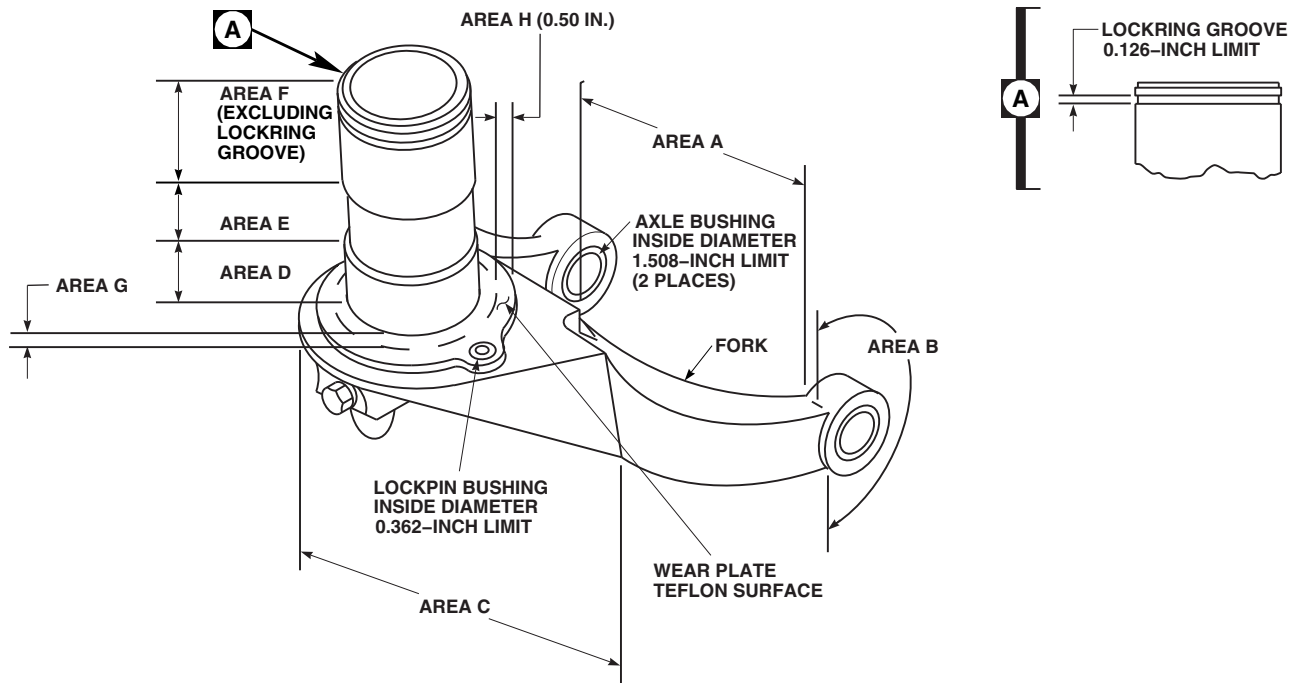
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FIGURE 3-4-26-3. REPAIR TAIL LANDING GEAR FORK

- h. Place fork in trail position.
- i. Engage tailwheel lockpin using either of the following methods:
 - (1) On tail landing gear, place tailwheel lockpin in ENGAGED position using manual operating lever (Figure 3-4-26-1, Detail A).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.
- j. Install tail landing gear wheel and tire (PARA 3-4-28).
- k. Lower tail landing gear and remove jack (PARA 1-6-3).

3-4-26.2 Replace Lockpin Bushing and Jackpad.

3-4-26.2.1 Remove.

- a. Locally make tail landing gear lockpin bushing drift (Figure H-204, Appendix H).
- b. Install steering bar.
- c. Disengage tailwheel lockpin using either of the following methods:
 - (1) On tail landing gear, place tailwheel lockpin in DISENGAGED position using manual operating lever (Figure 3-4-26-1, Detail A).
 - (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to release tailwheel lock. Place BATT switch OFF.

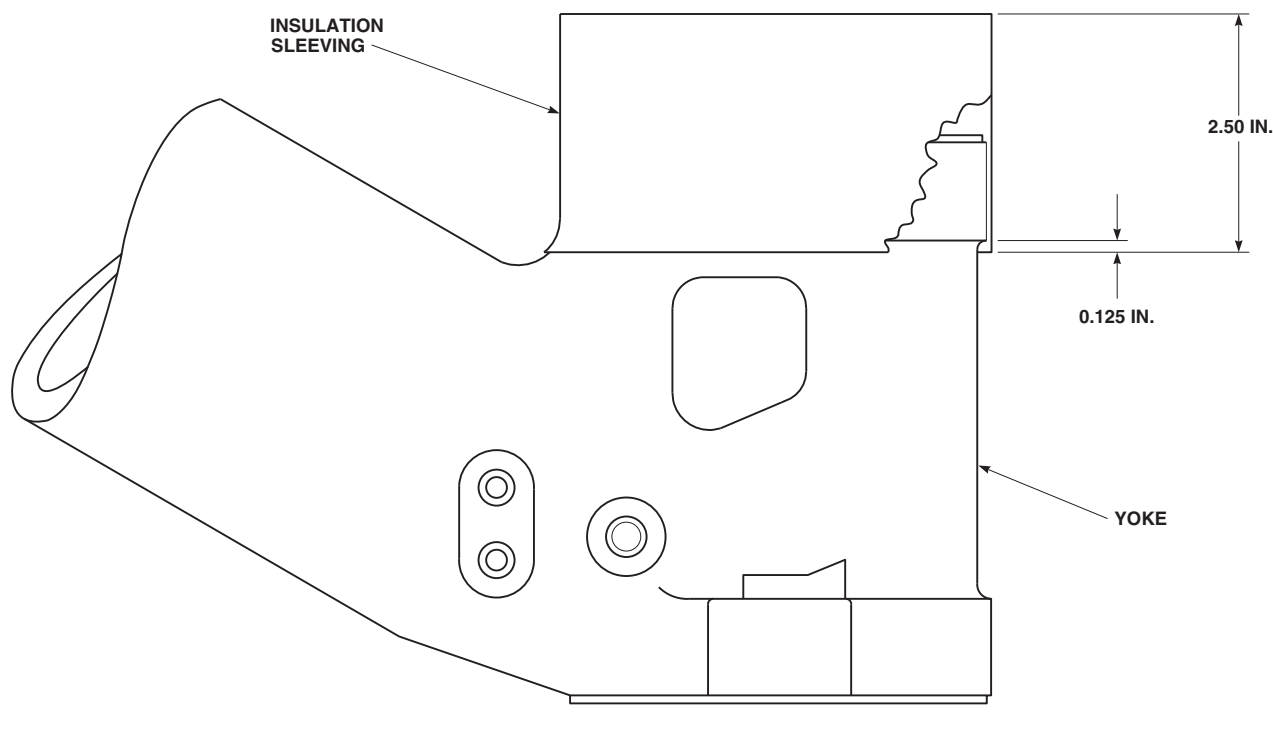


FIGURE 3-4-26-4. REPLACE TAIL LANDING GEAR FORK INSULATION SLEEVING

- d. Have assistant operate brakes.
- e. Position chocks 8 - 12-inches away from front and rear of main gear tires.
- f. Release parking brake.
- g. Have assistant push on right side of tail cone while turning tailwheel 20° - 30° to right so lockpin bushing is moved clear of yoke-lockpin boss.
- h. Chock main wheels. Set parking brake.
- i. Working from underside of fork, press or drive lockpin bushing from fork, using locally-made tail landing gear lockpin bushing drift (Figure 3-4-26-5).
- j. If removing jackpad, remove bolt and washer and remove from fork.

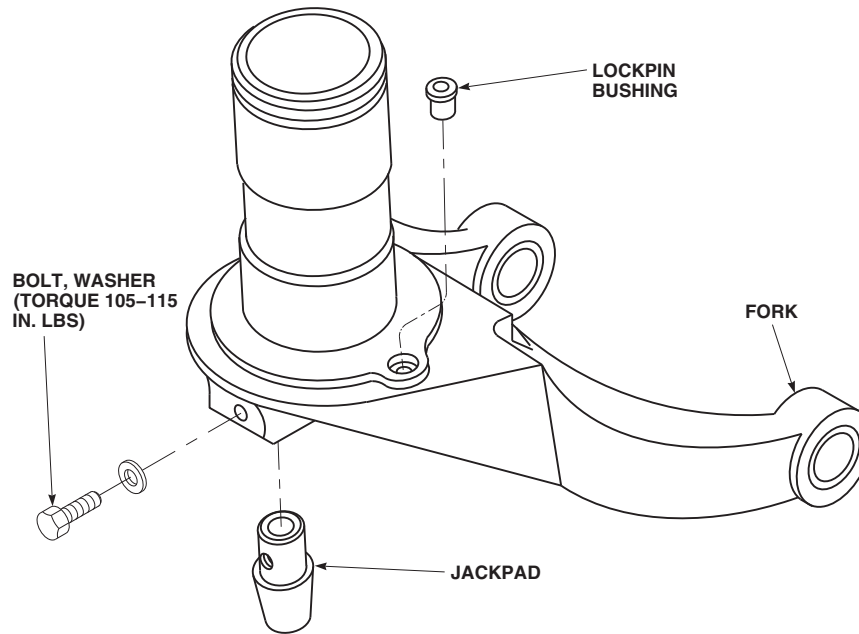
3-4-26.2.2 Inspect.

Check lockpin bushing hole in fork for wear and elongation. **NONE ALLOWED.** If worn or

elongated, repair lockpin bushing hole (PARA 3-5-10).

3-4-26.2.3 Install.

- a. Using light coat of sealing compound, Item 292, Appendix D, apply around new bushing.
- b. Press lockpin bushing into fork (Figure 3-4-26-5).
- c. If jackpad was removed, install it in fork and line up hole in jackpad with hole in fork.
- d. Install bolt and washer. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolt to fork.
- e. Have assistant operate brakes.
- f. Release parking brake.
- g. Move chocks 8 - 12 inches away from front and rear of main gear tires.
- h. Have assistant push on left side of tail cone while turning tailwheel to trail position.

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SA**FIGURE 3-4-26-5. REPLACE LOCKPIN BUSHING AND JACKPAD**

- i. Engage tailwheel lockpin using either of the following methods:
- b. Remove lockpin bushing (PARA 3-4-26.2).

- (1) On tail landing gear, place tailwheel lockpin in ENGAGED position using manual operating lever (Figure 3-4-26-1, Detail A).
- (2) In cockpit on upper console, place BATT switch ON. On miscellaneous switch panel, push TAILWHEEL switch to engage tailwheel lock. Place BATT switch OFF.

- j. Lock and unlock tailwheel a few times, to check operation of lockpin.
- k. Remove steering bar.
- l. Perform an operational check of tail wheel locking system (PARA 3-2-2).

CAUTION

Damage to fork will occur if metallic drift is used to remove Teflon-coated wear plate. Metal will scratch fork. Use nonmetallic drift to remove wear plate.

- c. Using nonmetallic drift, remove wear plate from fork (Figure 3-4-26-6).
- d. Using nonmetallic scraper, remove sealing compound from fork.
- e. If necessary, remove any remaining sealing compound with clean cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

3-4-26.3 Replace Teflon-Coated Wear Plate.**3-4-26.3.1 Remove.**

- a. Remove tail landing gear fork (PARA 3-4-26.1).

3-4-26.3.2 Install.

- a. Using brush, apply uniform coat of sealing compound, Item 292, Appendix D, approximately 0.010-inch thick, to Teflon-coated

wear plate (side with chamfer on inside diameter).

- b. Install wear plate on fork with sealing compound-coated surface down (Figure 3-4-26-6). Apply pressure to Teflon surface of wear plate. Make sure sealing compound has squeezed out uniformly around entire circumference of wear plate.
- c. If continuous squeeze out is not achieved, remove wear plate and repeat step a. and step b.
- d. Install lockpin bushing (PARA 3-4-26.2).
- e. Apply bead of sealing compound, Item 292, Appendix D, around fork shaft between fork and wear plate and outer edge of wear plate.
- f. If necessary, make sealing compound flush with Teflon surface of wear plate using nonmetallic scraper. Allow sealing compound to cure for a minimum of 72 hours.
- g. Install tail landing gear fork (PARA 3-4-26.1).

3-4-26.4 Replace Axle Bushings.

3-4-26.4.1 Remove.

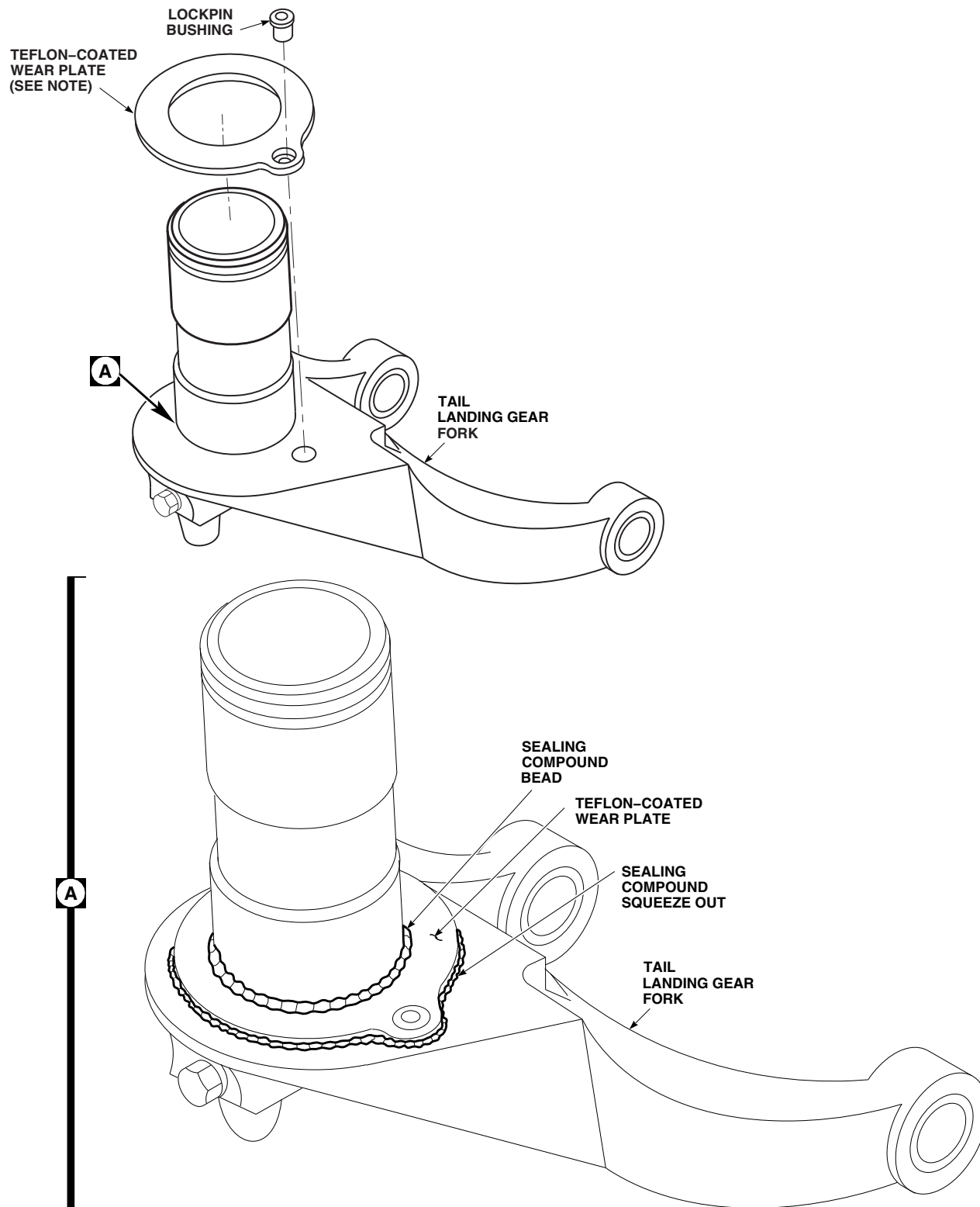
Place fork in arbor press and carefully remove bushings (Figure 3-4-26-7).

3-4-26.4.2 Inspect.

Inspect fork inside diameter for damage. **NICKS AND SCRATCHES UP TO 0.031-INCH DEEP MAY BE REPAIRED BY REAMING. LUG MAY BE REAMED UP TO 1.6875-INCHES IN DIAMETER.**

3-4-26.4.3 Install.

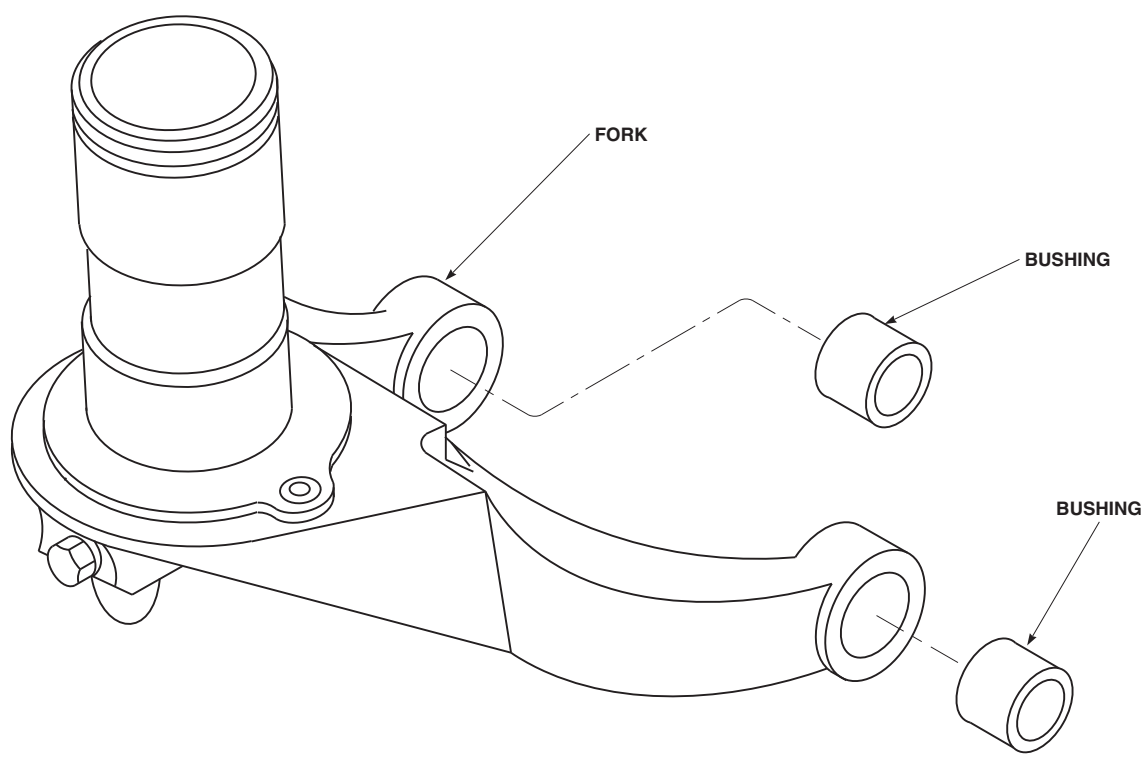
- a. Locally-made tail landing gear oversized axle bushing (Figure H-210, Appendix H) must be used for reworked lugs. Interference fit between bushing and lug must be 0.001 - 0.0035-inch.
- b. To ease installation, chill bushing in dry ice, Item 125, Appendix D.
- c. Remove bushing from dry ice. Using epoxy primer coating kit, Item 135, Appendix D, coat bushing.
- d. While primer is still wet, install bushing with arbor press (Figure 3-4-26-7). Allow bushings to return to room temperature.
- e. Using hand reamer set, line-ream both bushings to 1.502 - 1.503-inches.
- f. Using machinery towel, Item 211, Appendix D, clean inside diameter of bushings.

**NOTE**

INSTALL WITH CHAMFERED INSIDE
DIAMETER SIDE DOWN.

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FIGURE 3-4-26-6. REPLACE WEAR PLATE



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FIGURE 3-4-26-7. REPLACE AXLE BUSHINGS

3-4-27 TAIL LANDING GEAR LOCK MECHANISM.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-27.1	Replace Lock Actuator	3-4-27-1	3-4-27.2	Replace Lock Mechanism	3-4-27-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Tail Landing Gear Lock Mechanism Bellcrank Bushing Drift, Locally-Made, Figure H-203, Appendix H
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 110, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appendix H
- PARA 1-6-16
- PARA 1-7-20
- PARA 3-2-2

3-4-27.1 Replace Lock Actuator.

3-4-27.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P507, from lock actuator (Figure 3-4-27-1).
- Install protective caps and plugs on electrical connectors.
- Remove cotter pin, washer, and clevis pin from housing.
- Remove nut and washers, and remove lock actuator from yoke.

- Loosen jamnut on lock actuator shaft and unscrew adapter, with housing, from shaft (Figure 3-4-27-1, Detail C).
- Remove lockwasher and jamnut from lock actuator shaft. Keep lockwasher and jamnut for reinstallation.

3-4-27.1.2. Disassemble.

- Remove bushing from housing (Figure 3-4-27-1, Detail A).
- Slide out adapter, with parts attached, from housing.

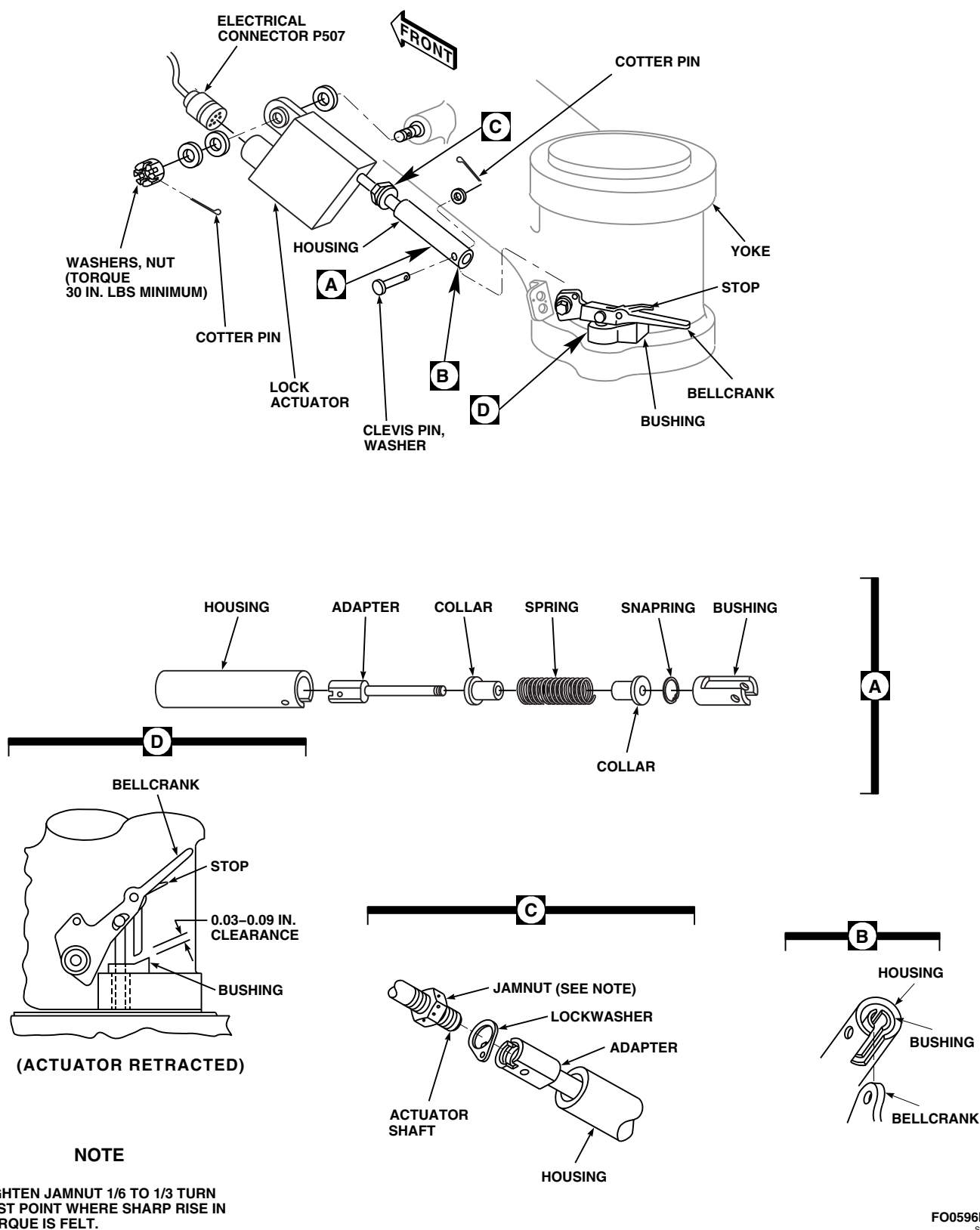


FIGURE 3-4-27-1. REPLACE LOCK ACTUATOR

3-4-27-2

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing collar. Spring is under pressure. Take care when removing collar.

- c. Remove snapping from adapter groove while restraining collar, to avoid sudden release of spring pressure.
- d. Remove two collars and spring from adapter.
- e. Clean all parts with dry-cleaning solvent, Item 124, Appendix D.

3-4-27.1.3. Inspect.

- a. Inspect parts for corrosion, damage, cracks, or deformation. **NONE ALLOWED.**
- b. Replace damaged parts.

3-4-27.1.4. Assemble.

- a. Apply corrosion preventive compound, Item 110, Appendix D, to spring and to inside surfaces of housing.
- b. Install collars and spring on adapter (Figure 3-4-27-1, Detail A).
- c. Compress spring by squeezing collars and install snapping in adapter groove.
- d. Install adapter, with parts attached, in housing.
- e. Coat inside and outside of bushing with corrosion preventive compound, Item 110, Appendix D, and install in housing.

3-4-27.1.5. Install.

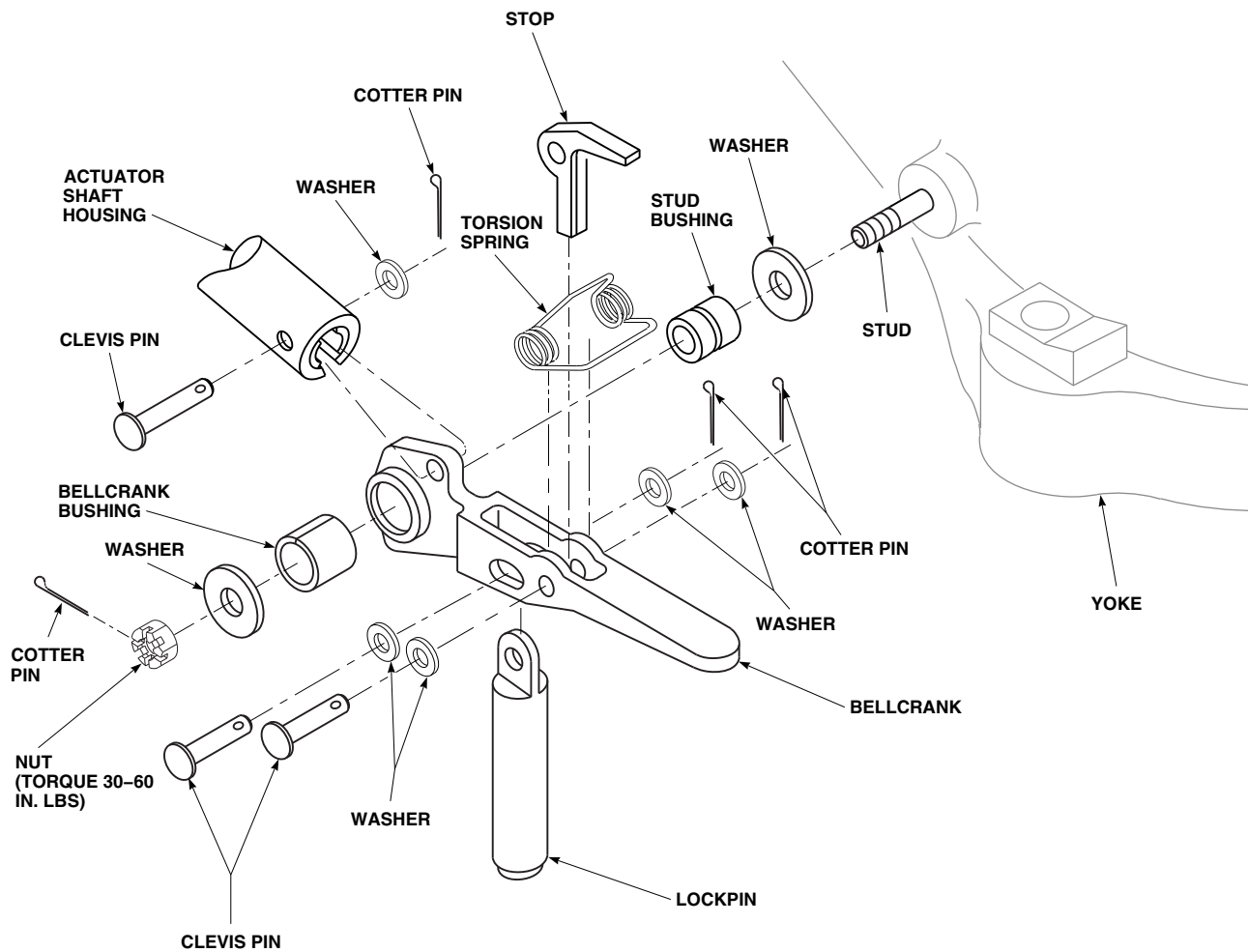
- a. Turn off all electrical power (PARA 1-6-16).
- b. Thread jamnut, lockwasher, and adapter with housing on lock actuator shaft (Figure 3-4-27-1, Detail C).

- c. Slide housing and bushing onto bellcrank (Figure 3-4-27-1, Detail B). Install clevis pin, washer, and cotter pin (Figure 3-4-27-1).
- d. Install lock actuator on yoke with washers and nut. Install cotter pin.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P507, to lock actuator.
- h. Adjust throw of lock actuator as follows:
 - (1) Turn on electrical power and retract lock actuator.
 - (2) **ADJUST ADAPTER ON LOCK ACTUATOR SHAFT, SO THAT WHEN STOP ON BELL CRANK IS IN ENGAGED POSITION ON TOP OF BUSHING ON YOKE, THERE IS A 0.03 - 0.09-INCH CLEARANCE BETWEEN STOP AND BUSHING** (Figure 3-4-27-1, Detail D).
 - (3) **TIGHTEN JAMNUT AGAINST LOCK-WASHER AND ADAPTER. TIGHTEN 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 197, Appendix D, jamnut to lockwasher (Figure 3-4-27-1, Detail C).
 - (4) Squeeze and release stop to make sure it clears bushing on yoke. If stop hangs up on bushing, readjust adapter.
- i. Perform an operational check of tailwheel lock system (PARA 3-2-2).

3-4-27.2 Replace Lock Mechanism.

3-4-27.2.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove clevis pin and washer from actuator shaft housing and disconnect housing from bellcrank (Figure 3-4-27-2).

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SA**FIGURE 3-4-27-2. REPLACE LOCK MECHANISM**

- c. Unlock tail wheel.
- d. Remove clevis pin, washers, stop, and torsion spring from bellcrank.
- e. Remove clevis pin and washers from bellcrank and lockpin.
- f. Remove nut, washers, and stud bushing from stud.
- g. Remove bellcrank from yoke.

3-4-27.2.2. Inspect/Repair.

- a. Check stud bushing, lockpin, bellcrank, bellcrank bushing, clevis pins, stop, and torsion spring for cracks, wear, and corrosion. Repair or replace worn or corroded parts (Figure 3-4-27-2).
- b. Locally make tail landing gear lock mechanism bellcrank bushing drift (Figure H-203, Appendix H).

- c. Remove worn bellcrank bushing using tail landing gear lock mechanism bellcrank bushing drift.
- d. Wipe replacement bushing and bellcrank mating surfaces clean using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
- e. Wet bushing and bellcrank mating surface using epoxy primer coating kit, Item 135, Appendix D.
- f. Press bushing into bellcrank while primer is wet. Wipe primer out of bushing bore.

3-4-27.2.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install lockpin in yoke (Figure 3-4-27-2).
- c. Install washer (with Teflon face outboard) on stud.

- d. Install bellcrank as follows:
 - (1) Install stud bushing and bellcrank on stud.
 - (2) Install washer (with Teflon face inboard) and nut.
 - (3) **TORQUE NUT TO 30 - 60 INCH-POUNDS.**
 - (4) Install cotter pin.
- e. Fasten lockpin to bellcrank with clevis pin, washers, and cotter pin.
- f. Install stop and torsion spring in bellcrank and secure with clevis pin, washers, and cotter pin.
- g. Install lock actuator shaft housing on bellcrank and secure with clevis pin, washer, and cotter pin.
- h. Perform an operational check of lock mechanism system (PARA 3-2-2).

3-4-28 TAIL LANDING GEAR WHEEL AND TIRE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-4-28.1	Replace Tail Landing Gear Axle, Wheel, and Tire	3-4-28-2	3-4-28.2	Replace Tail Landing Gear Tire/Tube	3-4-28-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Aircraft Tire Bead Breaker
- Aircraft Tire Inflation Cage
- Medium Cut File
- Protection, Eye
- Tire Inflator Kit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 11, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Antiseize Compound, Item 76, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 149, Appendix D
- Grease, Item 151, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lubricant, Inner Tube, Item 200, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-3
- PARA 3-3-7
- PARA 3-5-11

3-4-28.1 Replace Tail Landing Gear Axle, Wheel, and Tire.

3-4-28.1.1 Remove.

- a. Jack tail landing gear (PARA 1-6-3).
- b. Set parking brake. Leave brake on until wheel is replaced.
- c. Deflate tire as follows:

WARNING

To prevent injury to personnel, make sure air pressure is completely released from tire before removing tire from wheel. Wheel could explode. Do not remove valve core from valve stem until tire is completely deflated.

- (1) Remove valve cap from valve stem (Figure 3-4-28-1, Detail B).
- (2) Completely deflate tire, then remove valve core from valve stem.
- (3) Reinstall valve core.
- (4) Attach deflation tag.

CAUTION

Damage to spacers, bearing seals, and bearings will result if dropped when removing axle. Spacers, bearing seals, and bearings will drop free when axle is removed. Catch bearings as axle is removed to avoid damaging cones.

NOTE

If wheel is being replaced, keep bearings and bearing seals with wheel and send to intermediate level for repair.

- d. Straighten tab on lockwasher. Remove nut, lockwasher, axle, wheel and tire, spacers, bear-

ing seals, and bearings from fork (Figure 3-4-28-1, Detail A).

3-4-28.1.2 Inspect.

WARNING

Injury to personnel and damage to equipment will result if tire pressure exceeds 20 psi prior to installation. Make sure tire pressure is 20 psi or less prior to installation.

- a. Inspect wheel for obvious damage.
- b. Check bearing cups in wheel halves for looseness, scratches, pitting, corrosion, and signs of overheating. **NONE ALLOWED.** If damaged, replace cup (PARA 3-5-11).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Do not allow bearings to spin while drying.

- c. Using dry-cleaning solvent, Item 124, Appendix D, clean bearings. If available, dry bearings with compressed air, or allow to air-dry.
- d. Inspect bearings for scratches, water stains, heat discoloration, worn rollers, or with a

cracked or distorted bearing cage. **NONE ALLOWED.** If any of these conditions are found, replace bearing.

- e. If bearing is determined serviceable, repack it using grease, Item 149, Appendix D.
- f. Inspect axle for nicks, scratches, and gouges. Using abrasive wheel, Item 22, Appendix D, abrasive cloth, Item 11, Appendix D, or a medium cut file, blend out nicks, scratches, or gouges on axle. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.005-INCH AFTER BLENDING. SCRATCHES AT RIGHT ANGLES TO CENTERLINE MUST NOT BE MORE THAN 1-INCH IN LENGTH.** If damage exceeds limits, replace axle.

NOTE

Clamp-up forces will increase if length of spacer is less than 2.064-inches.

- g. Measure length of spacer. **SPACER MUST NOT BE LESS THAN 2.064-INCHES.** If length of spacer is less than limit, replace spacer.

3-4-28.1.3 Install.

- a. Using grease, Item 149, Appendix D, lubricate bearing cups in wheel, and bearing seals.
- b. Carefully install bearings and bearing seals in wheel (Figure 3-4-28-1, Detail A).
- c. Using grease, Item 149, Appendix D, lubricate axle.
- d. Position wheel and tire between fork ears while making sure bearings and seals do not fall out of wheel.
- e. Position both spacers between fork ear and bearing seal, slide axle through fork ear, first spacer, wheel, second spacer and fork ear.

NOTE

- Reuse of lockwasher is permitted for not more than three times, provided it is not cracked, corroded, or discolored, and new lockwasher is not available.

- If lockwasher is reused, bend must not overlap or cross a previous bend.
- f. Inspect lockwasher for cracks, corrosion, discoloration, or three bends in tang. **NONE ALLOWED.** If damage or three bends are found, replace lockwasher.
- g. Install lockwasher and nut on axle. **SPIN WHEEL AND TIGHTEN NUT UNTIL WHEEL CAN NO LONGER BE ROTATED BY HAND.** Loosen nut until wheel rotates freely with no binding. Bend lockwasher over flat on nut. **NO ENDPLAY ALLOWED IN WHEEL ASSEMBLY.**

WARNING

- To prevent injury to personnel or damage to equipment, use a tire inflator kit containing a pressure relief valve or similar device. Before inflating tire, check tire inflator to make sure that nitrogen source is vented at 145 psi maximum.
- If required equipment is inoperative or unavailable, tire inflation must not be attempted.
- To prevent injury to personnel or damage to equipment, do not inflate tire above its rated inflation pressure.
- To prevent injury to personnel, do not use an unregulated nitrogen pressure source to service tire.
- To prevent injury to personnel, never stand facing sidewall of tire being serviced. Stand facing tire tread.
- h. Using regulated source of nitrogen, Item 221, Appendix D, connected to tire inflator, **INFLATE TIRE TO 115 - 125 PSI.**
- i. After a 5 - 10 minute waiting period, check wheel for leakage as follows:
 - (1) Apply leak detection fluid, Item 189, Appendix D, around tire beads and to valve

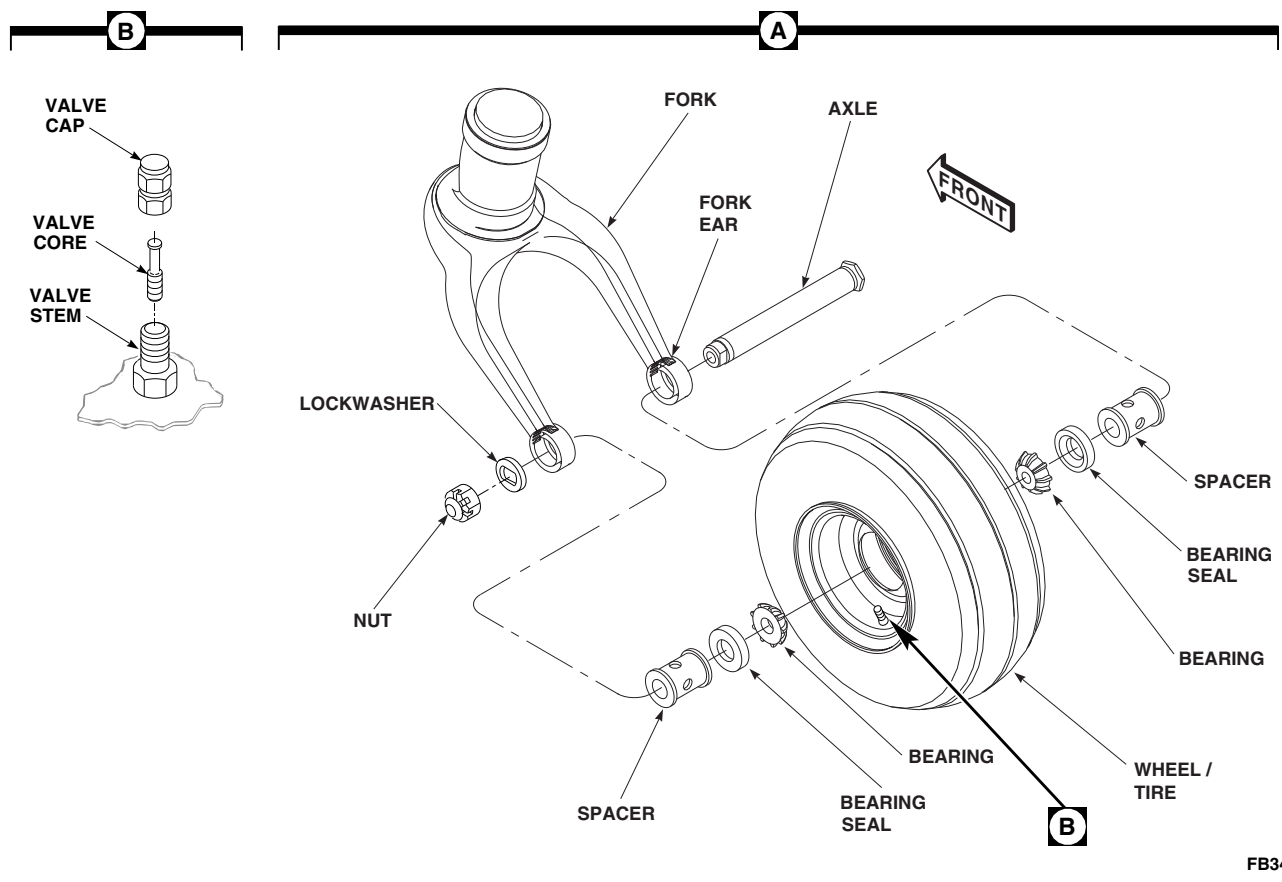
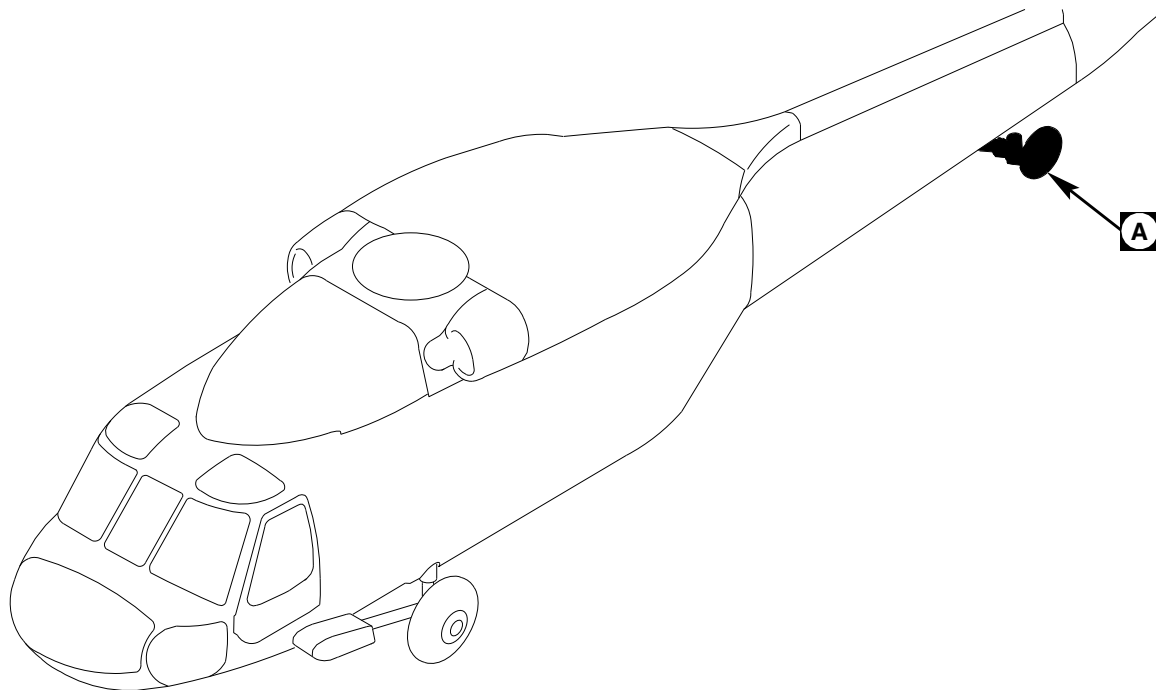

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FIGURE 3-4-28-1. REPLACE TAIL LANDING GEAR AXLE, WHEEL, AND TIRE

stem area (Figure 3-4-28-1, Detail A and Detail B).

- (2) If bubbles appear, deflate, disassemble, and correct at point of leakage (PARA 3-4-28.2).
- j. Install valve cap on valve stem (Figure 3-4-28-1, Detail B).
- k. Lower tail landing gear (PARA 1-6-3).
- l. Using sealing compound, Item 292, Appendix D, seal around nut only.

3-4-28.2 Replace Tail Landing Gear Tire/Tube.

3-4-28.2.1 Remove.

- a. If not already done, completely deflate tire and remove valve core as follows:

WARNING

To prevent injury to personnel, make sure air pressure is completely released from tire before removing tire from wheel. Wheel could explode. Do not remove valve core from valve stem until tire is completely deflated.

- (1) Remove valve cap from valve stem (Figure 3-4-28-2, Detail B).
- (2) Completely deflate tire, then remove valve core from valve stem.
- (3) Reinstall valve core.
- (4) Attach deflation tag.

CAUTION

- Use aircraft tire bead breaker on clean flat surface to disassemble and assemble wheel. Use care not to damage protective finish.

- Do not pry between tire bead and wheel flange with sharp tools.

- b. Using aircraft tire bead breaker, break tire beads away from wheel flanges by applying even pressure to tire sides.

CAUTION

Damage to wheel, wheel bolts, and nuts will result if impact or power wrenches are used to remove wheel bolt nuts. Do not use impact or power wrenches to remove wheel bolt nuts.

- c. Remove bolts, countersunk washers, and nuts. Separate wheel halves (Figure 3-4-28-2, Detail A).

NOTE

Replacement wheel may be either part number, 5004016-1 or 040-22800. Only part number, 040-22800, has packing on inner wheel half. Packing used on wheel, 040-22800, is not required when installing tube and tire. Packing may be discarded at the discretion of the helicopter operator.

- d. If wheel, 040-22800, is present, remove and discard packing on inner wheel half.
- e. Remove tire from wheel half.
- f. Remove tube from tire.

WARNING

Injury to personnel or damage to equipment may occur if wheel halves are not inspected whenever wheel is disassembled. Inspect wheel halves every time wheel is disassembled.

- g. Inspect and repair wheel (PARA 3-5-11).

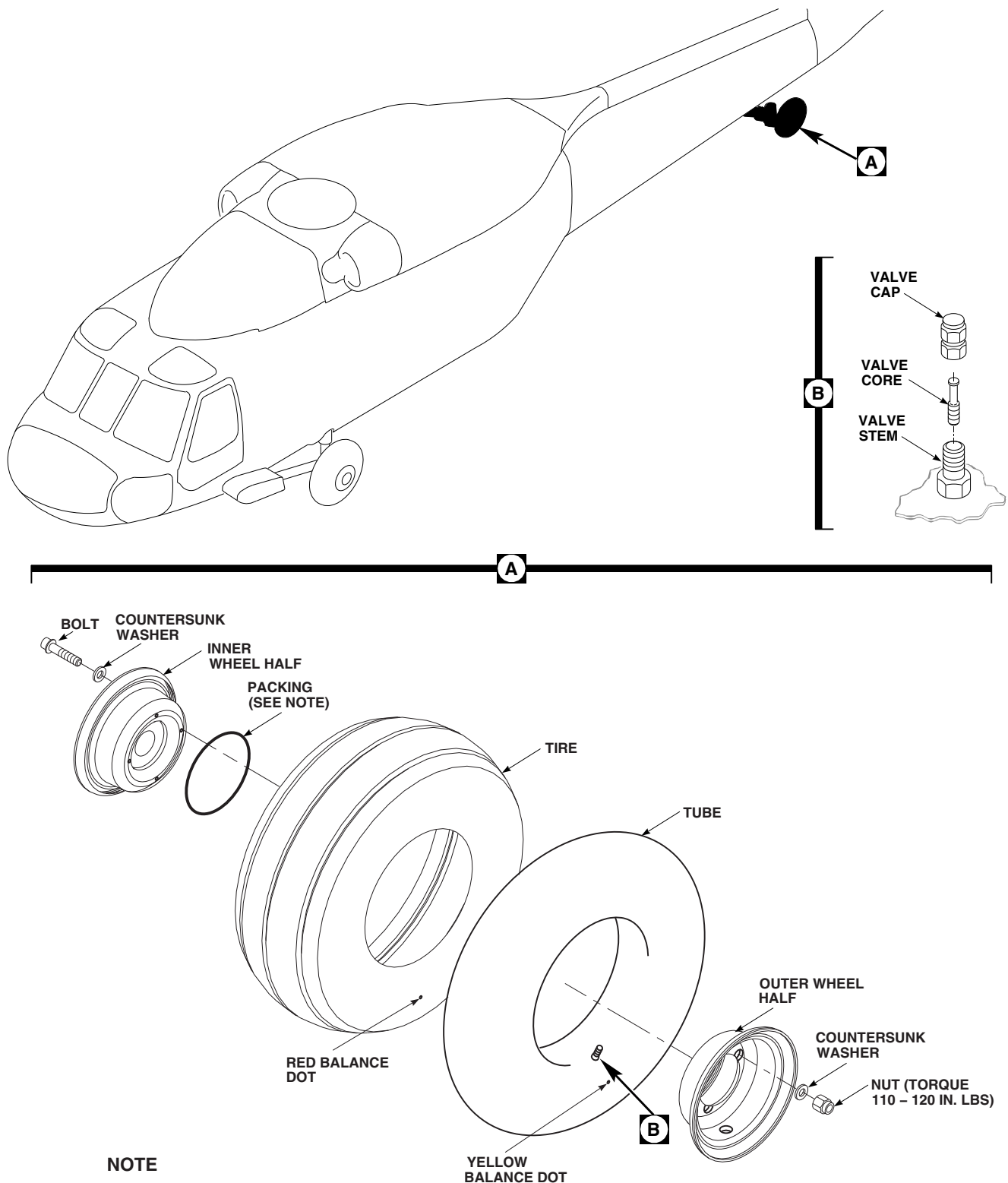
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FIGURE 3-4-28-2. REPLACE TAIL LANDING GEAR TIRE/TUBE

3-4-28.2.2 Inspect.

- a. Inspect tire for damage (PARA 3-3-7). **NONE ALLOWED.** If damaged, replace tire.
- b. Inspect new tire beads for damage. **NONE ALLOWED.** If damaged, replace tire.
- c. Inspect bolts for thread damage and cracks under boltheads. **NONE ALLOWED.** If damaged, replace bolt.
- d. To inspect nuts for damage, thread bolts into nuts by hand. If end of bolt is allowed to thread beyond other side of nut, replace nut.

3-4-28.2.3 Install.

- a. Before inserting tube in tire, dust tire generously with inner tube lubricant, Item 200, Appendix D. Using machinery towel, Item 211, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, clean wheel flanges and tire beads and remove any lubricant that settles on beads.
- b. Install valve core in valve stem.
- c. Line up yellow balance dot of tube with red balance dot of tire. Insert tube in tire (Figure 3-4-28-2, Detail A).
- d. Using regulated source of nitrogen, Item 221, Appendix D, connected to tire inflator, inflate tire slowly until beads start to spread. Probe between tube and tire with fingers, to release possible trapped air.

NOTE

Packing installed on inner wheel half is not required when installing tube and tire. Packing may be discarded at the discretion of the helicopter operator.

- e. On wheel, 040-22800, if packing will be used, lubricate packing using grease, Item 151, Appendix D, and install packing on inner wheel half.
- f. Carefully place tire on inner wheel half.

- g. Place other wheel half in tire and line up bolt-holes in both wheel halves.
- h. Apply antiseize compound, Item 76, Appendix D, on bolt and nut threads, face of nuts, under boltheads, and both sides of washers.

CAUTION

Damage to wheel, bolts, and nuts will result if impact or power wrenches are used to remove nuts. Do not use impact or power wrenches to tighten or torque bolts or nuts.

- i. Install bolts, countersunk washers, and nuts securing wheel halves together, with boltheads on side of wheel opposite valve stem. Install countersunk washers so countersunk sides are toward boltheads and nut faces, respectively. **TIGHTEN NUTS IN A CRISSCROSS PATTERN IN EVEN INCREMENTS OF 30 INCH-POUNDS, UNTIL A FINAL TORQUE OF 110 - 120 INCH-POUNDS IS OBTAINED.**

WARNING

- To prevent injury to personnel or damage to equipment, use a tire inflator kit containing a pressure relief valve or similar device. Before inflating tire, check tire inflator to make sure that nitrogen source is vented at 145 psi maximum.
- If required equipment is inoperative or unavailable, tire inflation must not be attempted.
- To prevent injury to personnel, or damage to equipment, do not inflate tire beyond its rated inflation pressure.
- To prevent injury to personnel, do not use an unregulated nitrogen pressure source.

- To prevent injury to personnel, never stand facing sidewall of tire being serviced. Stand facing tire tread.
 - To prevent injury to personnel, and damage to equipment, use aircraft tire inflation cage when inflating tire.
- j. Place wheel in aircraft tire inflation cage.
- k. Using regulated source of nitrogen, Item 221, Appendix D, connected to tire inflator, inflate tire with enough pressure to seat tire beads. **DO NOT EXCEED 110 PSI.**
- l. **ONCE TIRE IS CORRECTLY SEATED, INFLATE IT TO 115 - 125 PSI.**
- m. After a 5 - 10 minute waiting period, check wheel for leakage as follows:
- (1) Apply leak detection fluid, Item 189, Appendix D, to valve stem area and around tire bends (Figure 3-4-28-1, Detail A and Detail B).
 - (2) If bubbles appear, deflate, disassemble, and correct at point of leakage (PARA 3-4-28.2).
- n. **REDUCE TIRE PRESSURE TO 20 PSI.** Tag wheel, stating psi in tire. Tire will be inflated to its rated inflation pressure after wheel is installed on helicopter.
- o. Install valve cap on valve stem (Figure 3-4-28-2, Detail B).
- p. Remove wheel from tire inflation cage.
- q. Inspect and install tail landing gear axle, wheel, and tire (PARA 3-4-28.1.2 and PARA 3-4-28.1.3).

SECTION 3-5**MAINTENANCE PROCEDURES (BENCH)****SECTION OVERVIEW**

PARAGRAPH	TITLE
3-5-1	Main Landing Gear Shock Strut (BENCH)
3-5-2	Main Landing Gear Shock Strut Lower Stage Spherical Bearing (BENCH)
3-5-3	Main Landing Gear Wheel (BENCH)
3-5-4	Main Landing Gear Wheel Brake (BENCH)
3-5-5	Master Brake Cylinder (BENCH)
3-5-6	Parking Brake Valve (BENCH)
3-5-7	Slave Mixer Valve (BENCH)
3-5-8	Tail Landing Gear Shock Strut (BENCH)
3-5-9	Tail Landing Gear Yoke (BENCH)
3-5-10	Tail Landing Gear Fork (BENCH)
3-5-11	Tail Landing Gear Wheel (BENCH)

3-5-1 MAIN LANDING GEAR SHOCK STRUT (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
3-5-1.1	Repair Main Landing Gear Shock Strut (BENCH)	3-5-1-2

INITIAL SETUP

Tools

- Air Pump, Manual
- Aircraft Mechanic’s Toolkit
- Arbor Press
- Container, 1-Gallon
- Drift
- Fluorescent Penetrant Inspection Kit
- Hydraulic Repairer’s Toolkit
- Hydraulic Shutoff Valve
- Hydraulic Test Stand
- Magnetic Particle Inspection Kit
- Orifice Assembly Gage, Locally-Made, Figure H-267, Appendix H
- Pail, Utility
- Pencil
- Pick, Plastic or Wood, Locally-Made
- Protection, Eye
- Puller Kit, Mechanical
- Rawhide Mallet

- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Lockwire, Item 199, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Petrolatum, Item 234, Appendix D
- Screw, MS35275-247
- Tags

References

- Appendix D
- Appendix H
- ASTM E1417
- ASTM E1444

- PARA 1-3-2
- PARA 3-4-8
- PARA 3-4-9
- PARA 3-5-2

3-5-1.1 Repair Main Landing Gear Shock Strut (BENCH).

3-5-1.1.1 Disassemble.

- Clean strut exterior using dry-cleaning solvent, Item 124, Appendix D, and machinery towels, Item 211, Appendix D.

NOTE

Work area and tools must be clean. Use a surface covered with clean machinery towels, Item 211, Appendix D, for disassembled parts.

- Remove upper and lower stage air valve caps. Loosen outer hexnut and depressurize upper and lower stage struts. Leave valves open (Figure 3-5-1-1, Sheet 1, Detail A).

WARNING

Injury to personnel and damage to equipment will occur if shock strut is disassembled without removing upper and lower bleeder plugs. Residual pressure will cause shock strut piston to blow out during disassembly. Make sure hydraulic bleeder plugs are slowly removed to release possible residual pressure resulting from leakage from air side to hydraulic side of shock strut.

- Remove upper and lower stage bleeder plugs. Discard packings (Figure 3-5-1-1, Sheet 1). Drain hydraulic fluid from strut into 1-gallon container. Discard used fluid.
- Remove upper and lower kneeling valves (PARA 3-4-8).
- Lay strut down. Remove screws, washers, and band. Remove six pins using screw, MS35275-

247, and mechanical puller kit (Figure 3-5-1-1, Sheet 1, Detail B).

NOTE

Split bearing halves may fall free as piston clears cylinder.

- Separate piston from cylinder. Catch split bearing halves as piston is removed from cylinder.
- Remove six pins, retainer, lower orifice assembly, dampener, and bearing (Figure 3-5-1-1, Sheet 1, Detail C).

NOTE

Disassembly of lower orifice assembly is not required for regular maintenance. Disassemble only if an overhaul is required or if orifice appears damaged or discrepant. An identification tag is necessary only if orifice assembly is removed without disassembly.

- If necessary, disassemble lower orifice assembly as follows (Figure 3-5-1-1, Sheet 3, Detail F):
 - Using arbor press and suitable drift, compress spring and tension spring washers, and back nut out of lower orifice assembly.
 - Release pressure applied by arbor press and remove nut and support.
 - Remove washer, spring, and tension spring washers.
 - Remove orifice and poppet from lower orifice assembly.

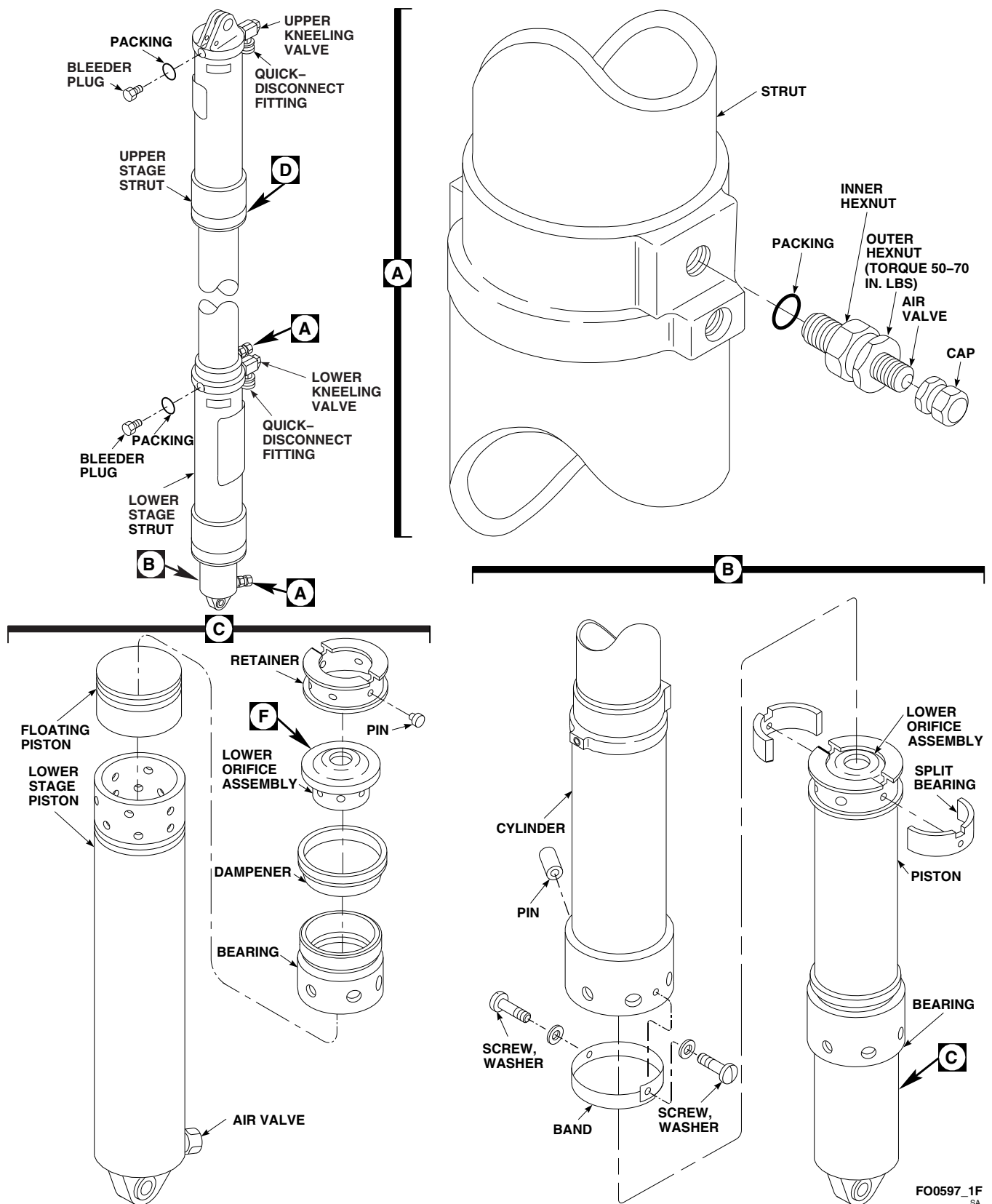


FIGURE 3-5-1-1. DISASSEMBLE (BENCH) (SHEET 1 OF 3)

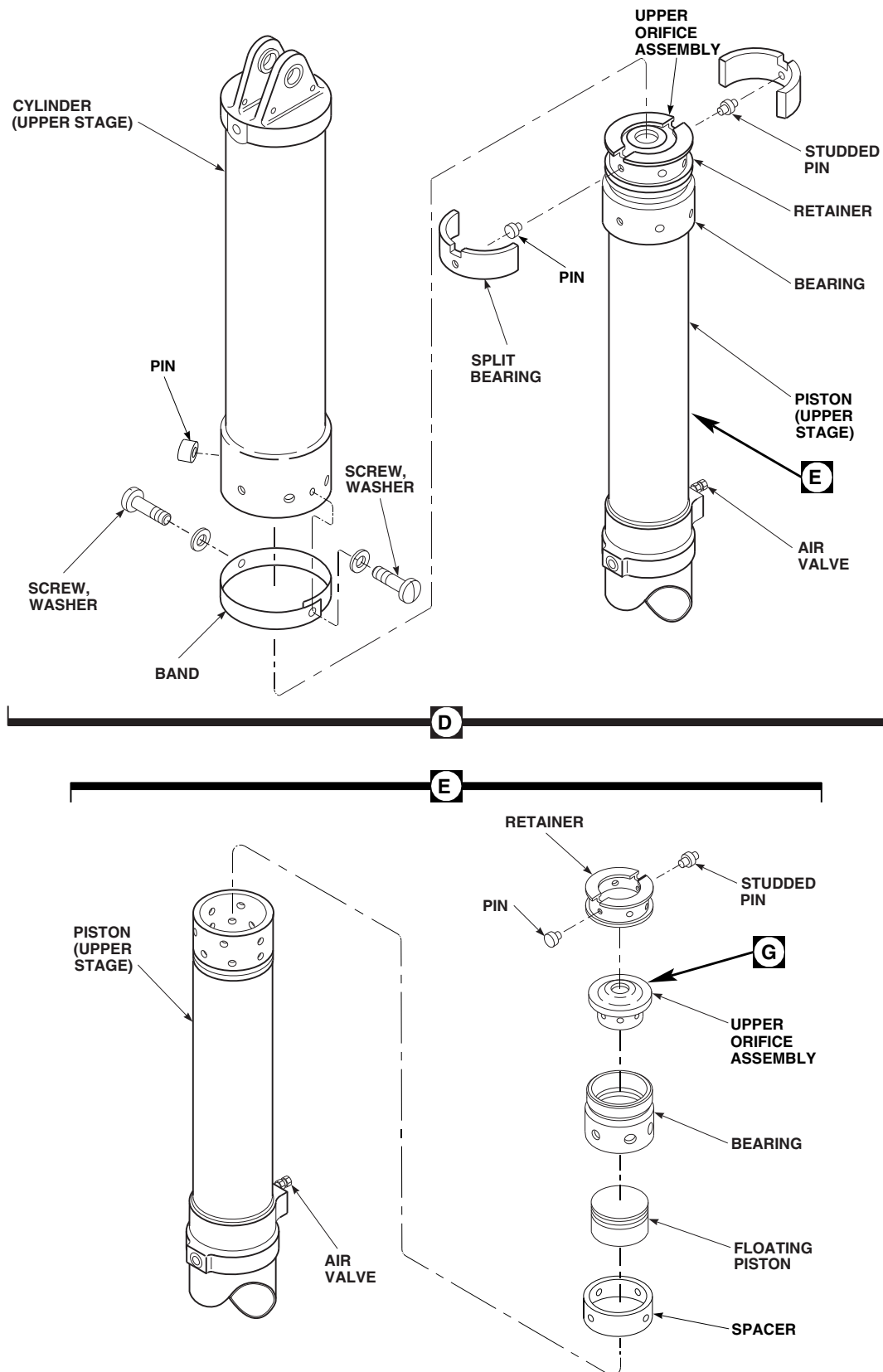
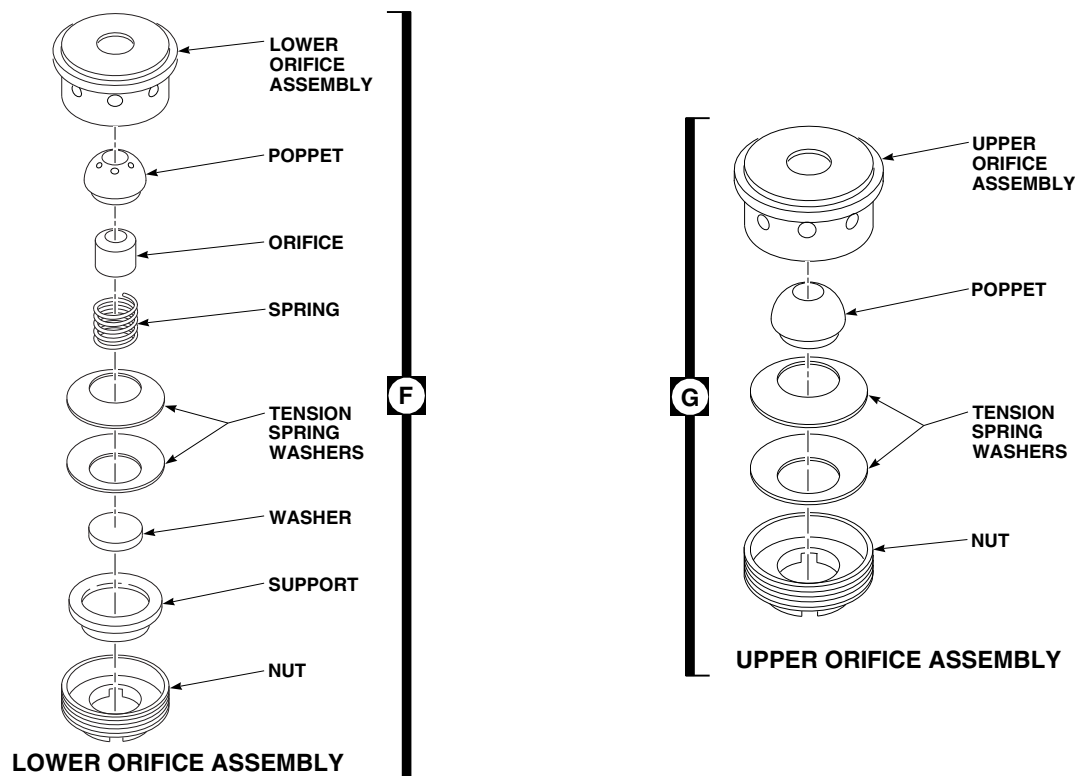


FIGURE 3-5-1.1. DISASSEMBLE (BENCH) (SHEET 2 OF 3)

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**FIGURE 3-5-1-1. DISASSEMBLE UPPER AND LOWER ORIFICE ASSEMBLY (BENCH)
(SHEET 3 OF 3)**

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate air pump at low pressure when removing floating piston.

- i. Attach manual air pump to lower air valve. Apply 2 - 5 psi pressure. Catch floating piston and spacer in a utility pail filled with machinery towels, Item 211, Appendix D. Remove manual air pump.
- j. Remove lower stage air valve (Figure 3-5-1-1, Sheet 1, Detail A). Discard packing.
- k. Remove screws, washers, and band (Figure 3-5-1-1, Sheet 2, Detail D).
- l. Remove six pins using screw, MS35275-247, and mechanical puller kit.

NOTE

Split bearing halves may fall free as piston clears cylinder.

- m. Separate upper stage piston from upper stage cylinder. Catch split bearing halves as piston is removed from cylinder.
- n. Remove six pins, retainer, upper orifice assembly, and bearing (Figure 3-5-1-1, Sheet 2, Detail E).

NOTE

Disassembly of upper orifice assembly is not required for regular maintenance. Disassemble only if an overhaul is required or if orifice appears damaged or discrepant. An identification tag is necessary only if orifice assembly is removed without disassembly.

- o. If necessary, disassemble upper orifice assembly as follows (Figure 3-5-1-1, Sheet 3, Detail G):

- (1) Using arbor press and suitable drift, compress tension spring washers, and back nut out of upper orifice assembly.
- (2) Release pressure applied by arbor press and remove nut, tension spring washers, and poppet from upper orifice assembly.

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate air pump at low pressure when removing floating piston.

- p. Attach manual air pump to upper air stage valve. Apply 2 - 5 psi pressure. Catch floating piston in a utility pail filled with machinery towels, Item 211, Appendix D.
- q. Remove upper stage air valve (Figure 3-5-1-1, Sheet 1, Detail A). Discard packing.
- r. Using plastic or wooden pick, remove all packings, seals, and scrapers. Discard packings, seals, and scrapers.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- s. Clean parts with dry-cleaning solvent, Item 124, Appendix D. Blow dry with dry compressed air.

3-5-1.1.2 Inspect.

- a. Inspect strut parts for cracks, scoring, scratches, and corrosion. **NONE ALLOWED.** Use

metal flaw detection procedures as necessary to verify suspected cracks or fractures (ASTM E1444). If damaged, replace parts as necessary.

- b. Inspect upper cylinder mounting lug bushings for wear. **BUSHING INSIDE DIAMETER MUST BE 0.995 - 1.001-INCHES. DISTANCE BETWEEN BUSHINGS MUST BE 1.002 - 1.006-INCHES** (Figure 3-5-1-2, Detail A). If worn bushings are found, replace cylinders.

- c. Inspect piston assembly chrome-plated surface for surface blemishes or discolorations. Buff out surface blemishes or discolorations using abrasive cloth, Item 10, Appendix D. **PISTON OUTSIDE DIAMETER MUST NOT BE LESS THAN 3.495-INCHES AFTER BUFFING.** Slight discoloration after buffing is acceptable providing minimum dimension and surface finish is maintained. If limits are exceeded, replace piston.

- d. Inspect piston and cylinder chrome-plated surfaces for wear. Take measurements along several points, 90° apart as follows:

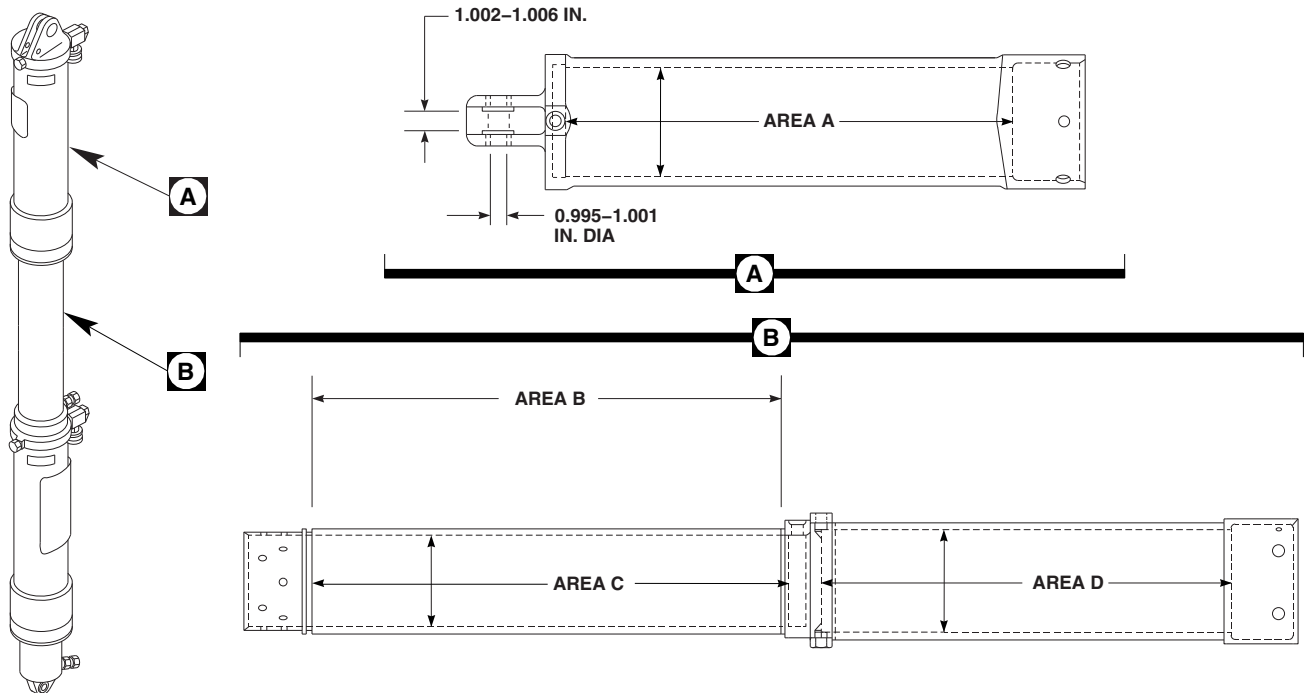
- (1) **AREA A - UPPER STAGE CYLINDER INSIDE DIAMETER MUST NOT EXCEED 4.003-INCHES.** If out-of-tolerance, replace cylinder.

- (2) **AREA B - UPPER STAGE PISTON OUTSIDE DIAMETER MUST NOT BE LESS THAN 3.495-INCHES** (Figure 3-5-1-2, Detail B). If out-of-tolerance, replace piston/cylinder assembly.

- (3) **AREA C - UPPER STAGE PISTON BORE INSIDE DIAMETER MUST NOT EXCEED 3.336-INCHES.** If out-of-tolerance, replace piston/cylinder assembly.

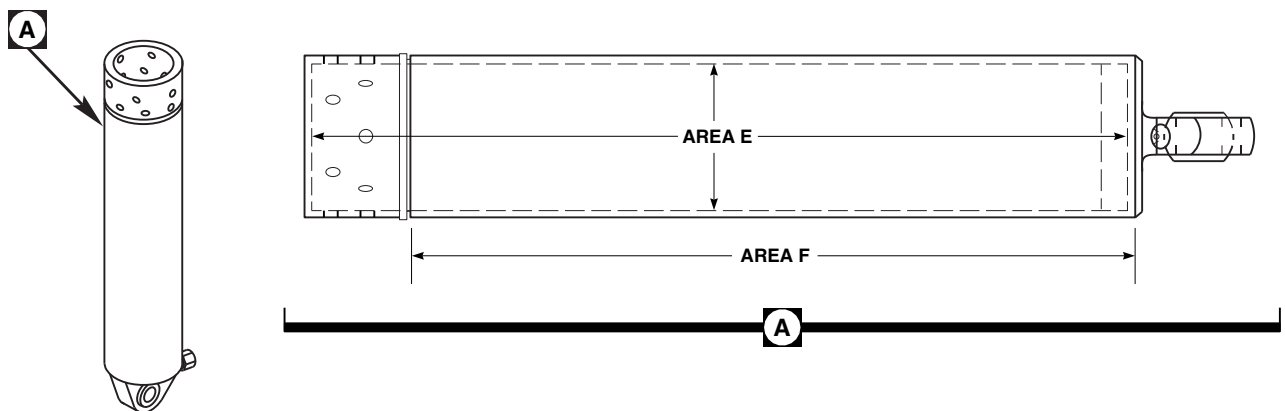
- (4) **AREA D - LOWER STAGE CYLINDER INSIDE DIAMETER MUST NOT EXCEED 4.002-INCHES.** If out-of-tolerance, replace cylinder assembly.

- (5) **AREA E - LOWER STAGE PISTON BORE INSIDE DIAMETER MUST NOT EXCEED 3.336-INCHES** (Figure



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FIGURE 3-5-1-2. INSPECT UPPER STAGE (BENCH)



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FIGURE 3-5-1-3. INSPECT LOWER STAGE (BENCH)

3-5-1-3, Detail A). If out-of-tolerance, replace piston/cylinder assembly.

- (6) **AREA F - LOWER STAGE PISTON OUTSIDE DIAMETER MUST NOT BE LESS THAN 3.495-INCHES.** If out-of-tolerance, replace piston.

- e. Inspect split bearing halves for wear at several points around circumference. **SPLIT BEAR-**

ING THICKNESS MUST NOT BE LESS THAN 0.092-INCH (Figure 3-5-1-4, Detail A). If out-of-tolerance, replace bearings.

- f. Inspect retainer for wear at several points around outside diameter. **ON STRUT, 70250-12051-042, UPPER AND LOWER STAGE RETAINERS MUST NOT BE LESS THAN 3.993-INCHES. ON STRUTS, 70250-12051-**

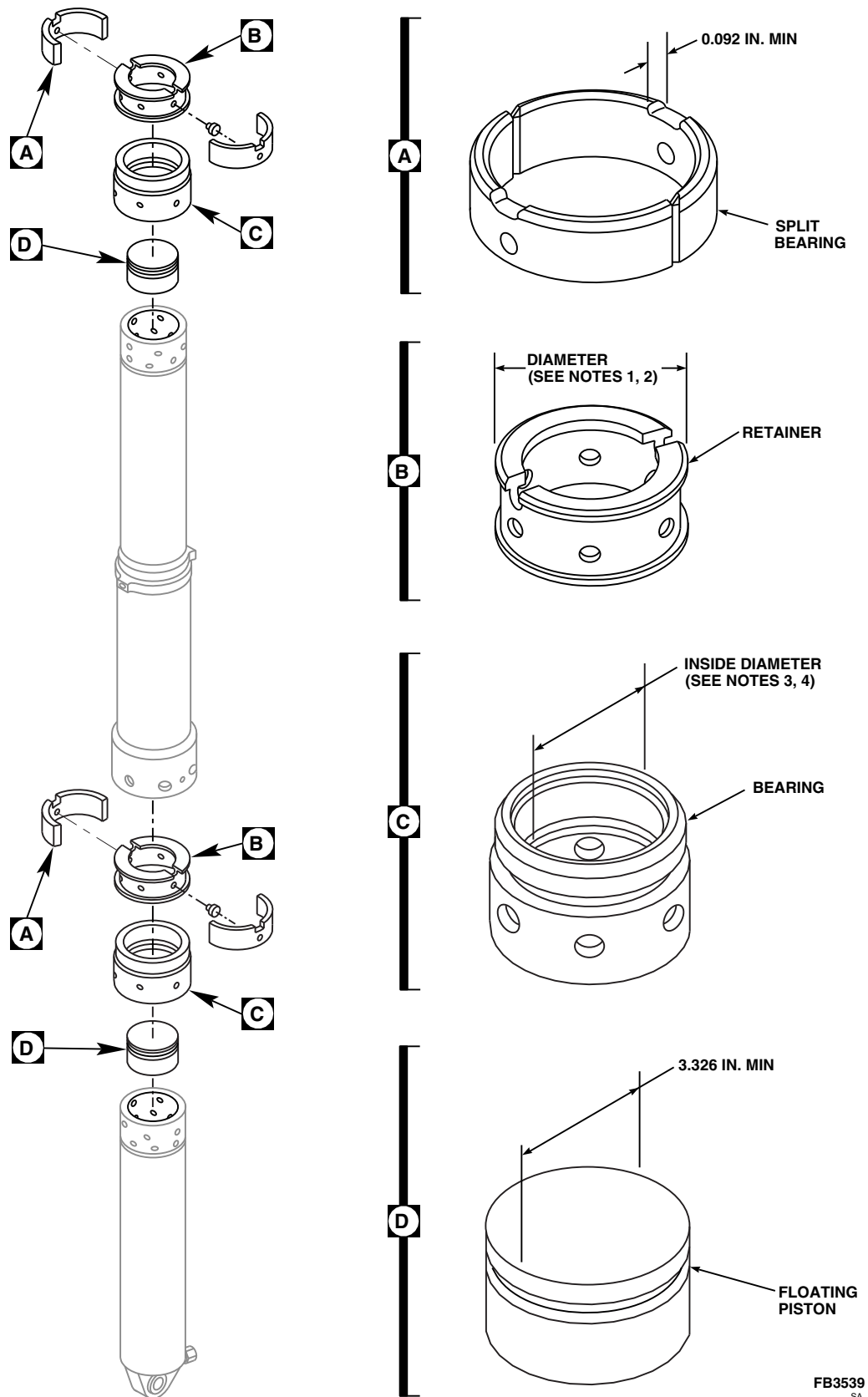
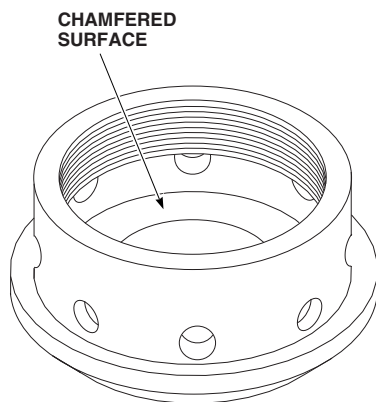
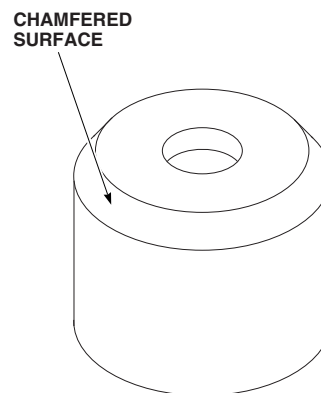


FIGURE 3-5-1-4. INSPECT UPPER AND LOWER STAGE COMPONENTS (BENCH)



**UPPER/LOWER
ORIFICE ASSEMBLY**
70250-12020-102



ORIFICE
70250-12046-103

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FIGURE 3-5-1-5. INSPECT ORIFICE ASSEMBLY (BENCH)

043 THROUGH 70250-12051-045, UPPER AND LOWER STAGE RETAINERS MUST NOT BE LESS THAN 3.973-INCHES (Figure 3-5-1-4, Detail B). If out-of-tolerance, replace retainer.

- g. Inspect bearings for wear at several points around inside diameter. **ON STRUT, 70250-12051-042, UPPER AND LOWER STAGE BEARING INSIDE DIAMETER MUST NOT EXCEED .503-INCHES. ON STRUTS, 70250-12051-043 THROUGH 70250-12051-045, UPPER AND LOWER STAGE BEARING INSIDE DIAMETER MUST NOT EXCEED 3.506-INCHES** (Figure 3-5-1-4, Detail C). If out-of-tolerance, replace bearings.
- h. Inspect floating pistons for wear at several points around outside diameter. **UPPER AND LOWER STAGE FLOATING PISTONS MUST NOT BE LESS THAN 3.326-INCHES** (Figure 3-5-1-4, Detail D). If out-of-tolerance, replace floating pistons.
- i. Inspect lower stage piston bearing for maximum radial play. **LOWER STAGE PISTON**

BEARING MAY HAVE UP TO 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY. If out-of-tolerance, replace bearing (PARA 3-5-2).

- j. Inspect Teflon liner for disbonding. **NONE ALLOWED.** If liner is disbonded, replace bearing (PARA 3-5-2).
- k. Inspect upper orifice assembly components as follows:
 - (1) Inspect nut, 70250-12029-101, as follows:
 - (a) Visually inspect nut for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace nut.
 - (b) Visually inspect nut for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are crossed, stripped, or flattened, replace nut.
 - (c) Using magnetic particle inspection kit, inspect nut for fractures or

cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace nut.

- (2) Inspect upper orifice assembly, 70250-12020-102, as follows (Figure 3-5-1-5):

- (a) Visually inspect upper orifice assembly for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace upper orifice assembly.
- (b) Visually inspect chamfered surface of upper orifice assembly for nicks, burrs, or scratches. **NONE ALLOWED.** If nick, burr, or scratch is found, replace upper orifice assembly.
- (c) Visually inspect upper orifice assembly for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are crossed, stripped, or flattened, replace upper orifice assembly.
- (d) Measure diameter of each hole in upper orifice assembly. **DIAMETER MUST BE 0.365 - 0.385-INCH.**
- (e) Using magnetic particle inspection kit, inspect upper orifice assembly for fractures or cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace upper orifice assembly.

- (3) Inspect poppet, 70250-12030-102, as follows:

- (a) Visually inspect poppet for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace poppet.
- (b) Visually inspect poppet for nicks, burrs, or scratches. **NONE ALLOWED.** If nick, burr, or scratch is found, replace poppet.

- (c) Using magnetic particle inspection kit, inspect poppet for fractures or cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace poppet.

1. Inspect lower orifice assembly components as follows:

- (1) Inspect nut, 70250-12029-101, as follows:

- (a) Visually inspect nut for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace nut.
- (b) Visually inspect nut for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are crossed, stripped, or flattened, replace nut.
- (c) Using magnetic particle inspection kit, inspect nut for fractures or cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace nut.

- (2) Inspect support, 70250-12045-103, as follows:

- (a) Visually inspect support for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace support.
- (b) Using fluorescent penetrant inspection kit, inspect support for fractures or cracks (ASTM E1417). **NONE ALLOWED.** If crack or fracture is found, replace support.

- (3) Inspect washer, BS2-1-3.00, as follows:

- (a) Visually inspect washer for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace washer.

- (4) Inspect orifice, 70250-12046-103, as follows:

- (a) Visually inspect orifice for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace orifice.
- (b) Visually inspect chamfered surface of orifice for nicks, burrs, or scratches. **NONE ALLOWED.** If nick, burr, or scratch is found, replace orifice.
- (c) Using fluorescent penetrant inspection kit, inspect orifice for fractures or cracks (ASTM E1417). **NONE ALLOWED.** If crack or fracture is found, replace orifice.
- (d) Measure inside diameter of hole in orifice. **DIAMETER MUST BE 0.325 - 0.329-INCH.**
- (5) Inspect poppet, 70250-12030-107, as follows:
 - (a) Visually inspect poppet for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace poppet.
 - (b) Visually inspect spherical radius surface of poppet for nicks, burrs, or scratches. **NONE ALLOWED.** If nick, burr, or scratch is found, replace poppet.
 - (c) Using magnetic particle inspection kit, inspect poppet for fractures or cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace poppet.
 - (d) Measure inside diameter of poppet. **DIAMETER MUST BE 0.873 - 0.877-INCH.**
 - (e) Visually inspect seating surface of poppet for nicks, burrs, or scratches.
- (6) Inspect lower orifice assembly, 70250-12020-102, as follows:
 - (a) Visually inspect lower orifice assembly for burrs and corrosion. **BURRS AND CORROSION MUST NOT EXCEED 0.005-INCH.** If burrs and corrosion are beyond limit, replace lower orifice assembly.
 - (b) Visually inspect chamfered surface of lower orifice assembly for nicks, burrs, or scratches. **NONE ALLOWED.** If nick, burr, or scratch is found, replace lower orifice assembly.
 - (c) Visually inspect lower orifice assembly for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are crossed, stripped, or flattened, replace lower orifice assembly.
 - (d) Measure diameter of each hole in lower orifice assembly. **DIAMETER MUST BE 0.365 - 0.385-INCH.**
 - (e) Using magnetic particle inspection kit, inspect lower orifice assembly for fractures or cracks (ASTM E1444). **NONE ALLOWED.** If crack or fracture is found, replace lower orifice assembly.

3-5-1.1.3 Assemble.

- a. Make sure all parts are clean.

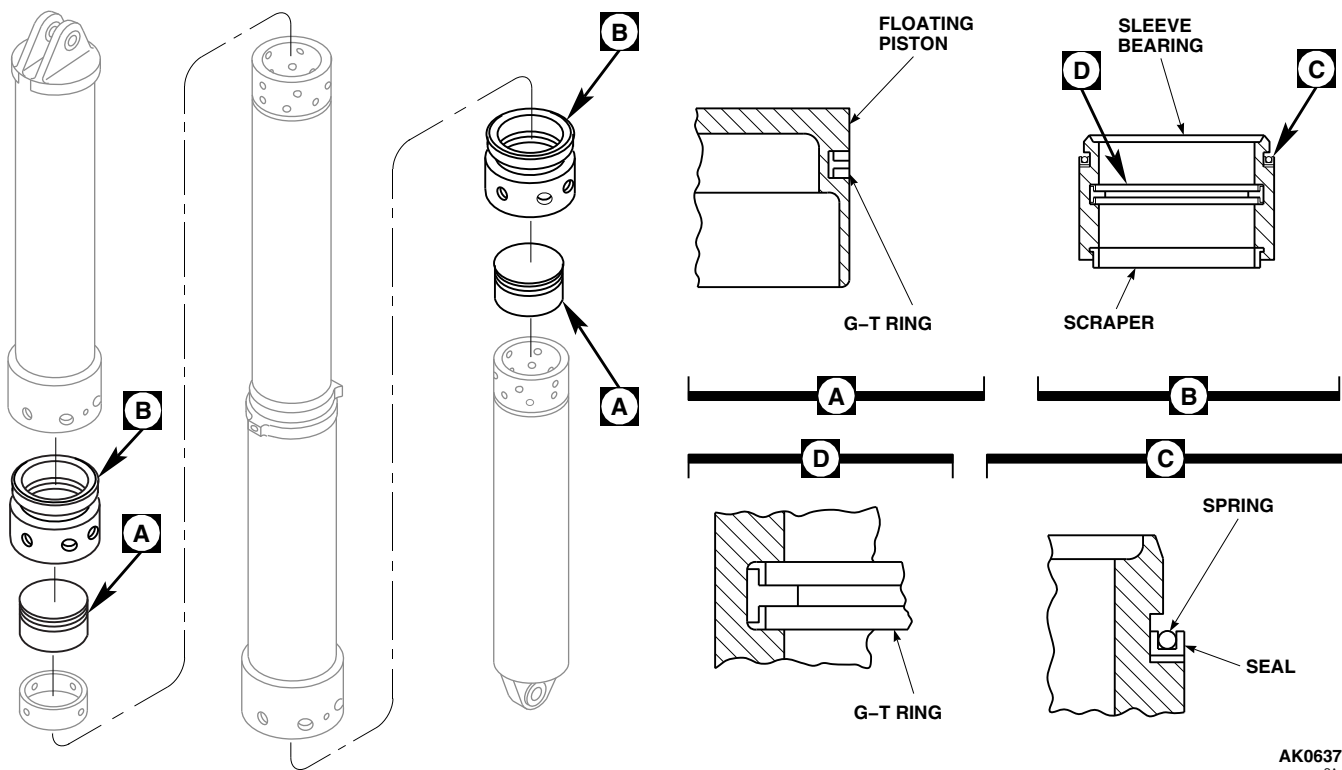
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FIGURE 3-5-1-6. INSTALL FLOATING PISTON/SLEEVE BEARING (BENCH)

NOTE

- If strut will be stored, use hydraulic fluid, Item 155, Appendix D, as an assembly and servicing fluid.
 - If strut is scheduled for immediate return to service, use hydraulic fluid, Item 154, Appendix D, as an assembly and servicing fluid.
- b. Soak packings, retainers, G-T rings, and seals in hydraulic fluid, Item 154 or Item 155, Appendix D, before installation.
 - c. Coat all chrome-plated sliding and threaded parts with hydraulic fluid, Item 154 or Item 155, Appendix D, before assembly.
 - d. Install G-T rings on upper and lower stage floating pistons, with radius of two Teflon rings facing center packing (Figure 3-5-1-6, Detail A).
 - e. Install G-T rings inside upper and lower bearings (Figure 3-5-1-6, Detail D).
 - f. With springs facing tops of sleeves, install seals on outside diameter of upper and lower bearings (Figure 3-5-1-6, Detail C).
 - g. Install scrapers, inside lower edge of bearings (Figure 3-5-1-6, Detail B).
 - h. With holes closest to spacer edge toward bottom, install spacer in upper stage piston. Line up one spacer hole with air valve port (Figure 3-5-1-7, Sheet 1, Detail A and Detail B).
 - i. Install floating pistons, open ends down, in upper and lower stage piston bores. Push floating pistons to bottom of bore (Figure 3-5-1-7, Sheet 1, Detail A and Figure 3-5-1-8, Detail A).
 - j. Slip bearings, scraper ends first, over upper and lower stage pistons.

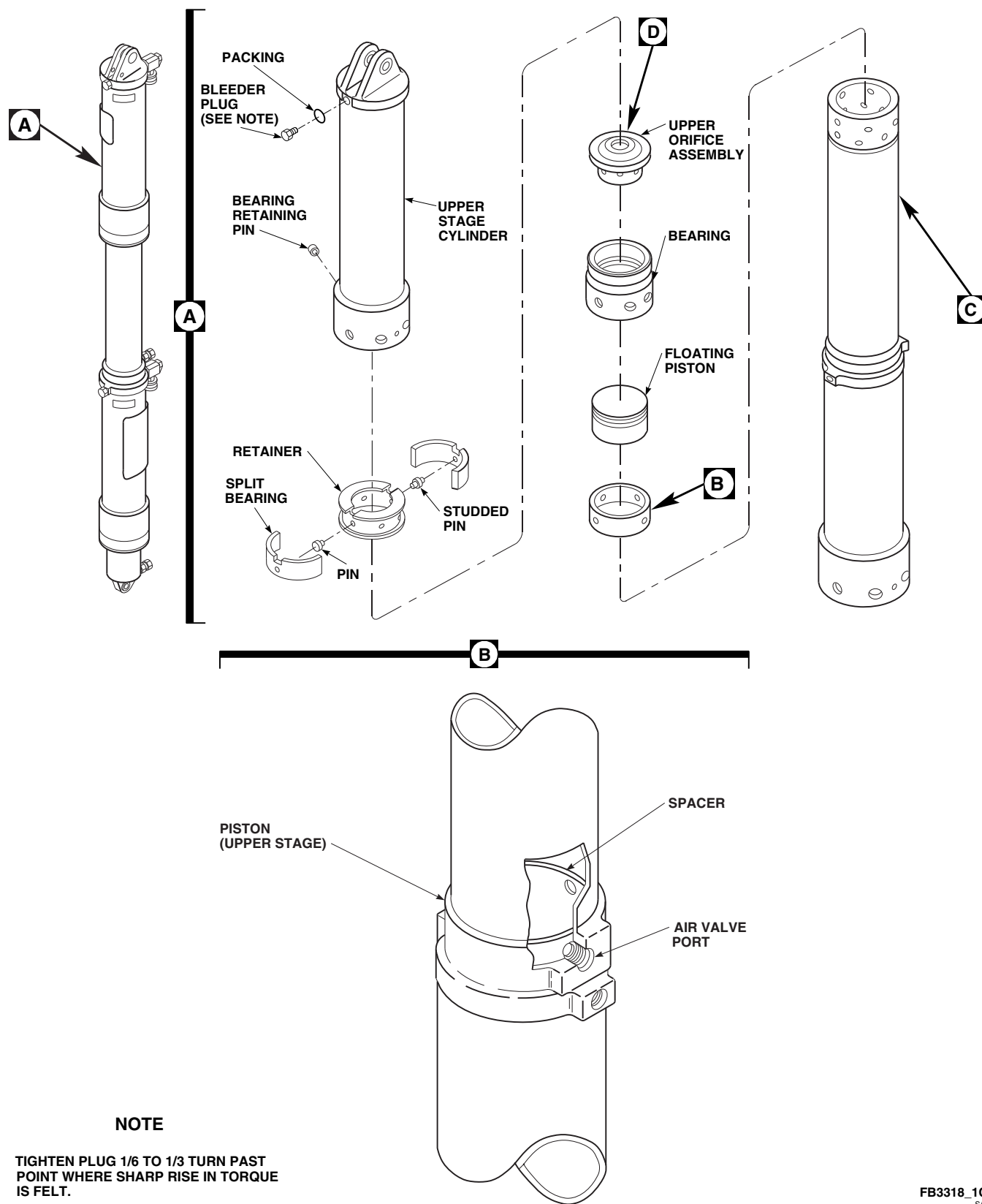


FIGURE 3-5-1-7. ASSEMBLE UPPER STAGE (BENCH) (SHEET 1 OF 2)

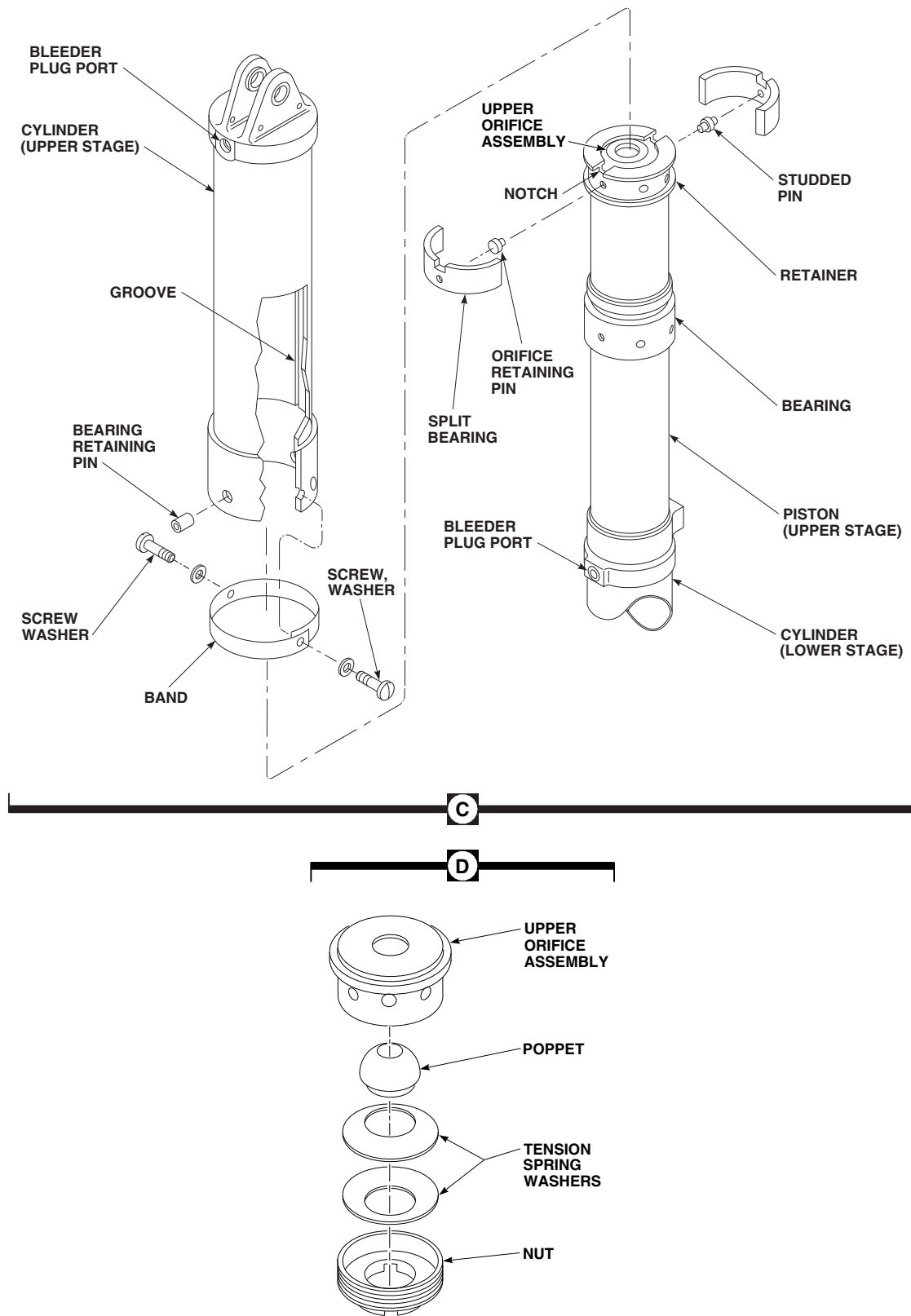


FIGURE 3-5-1-7. ASSEMBLE UPPER STAGE (BENCH) (SHEET 2 OF 2)

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NOTE

Upper orifice assembly has a single hole in center of poppet valve and no coil spring.

- k. Assemble upper orifice assembly as follows (Figure 3-5-1-7, Sheet 2, Detail D and Figure 3-5-1-9):

- (1) Insert poppet into upper orifice assembly.
- (2) Locally make orifice assembly gage (Figure H-267, Appendix H).
- (3) Install orifice assembly gage on poppet.
- (4) Using hydraulic fluid, Item 154, Appendix D, coat threads of nut.
- (5) Thread nut into upper orifice assembly. Tighten nut until it bottoms on orifice assembly gage.
- (6) Using pencil, make index marks on nut for depth and radial location inside upper orifice assembly.
- (7) Remove nut and orifice assembly gage from upper orifice assembly.
- (8) Install tension spring washers in upper orifice assembly and secure with nut.
- (9) Using arbor press and suitable drift, compress tension spring washers.
- (10) Thread nut into upper orifice assembly and line up index marks noted in step k. (6). Lining up index marks will allow tension spring washers to compress to a height of 0.594 ± 0.010 -inches.
- (11) Using lockwire, Item 196, Appendix D, line up slot in nut with upper orifice assembly and lockwire hole.
- (12) Remove arbor press and drift.
- (13) Using lockwire, Item 196, Appendix D, lockwire nut to upper orifice assembly

internally. Make sure both lockwire and lockwire tail do not protrude past end of upper orifice assembly.

- l. Insert upper stage orifice assembly in upper stage piston (Figure 3-5-1-7, Sheet 1, Detail A).
- m. Install retainer in upper stage piston. Line up slots with kneeling and air valve ports. Line up orifice retaining pin through retainer and piston wall in line with retainer notch, kneeling, and air valve ports (Figure 3-5-1-7, Sheet 2, Detail C).

NOTE

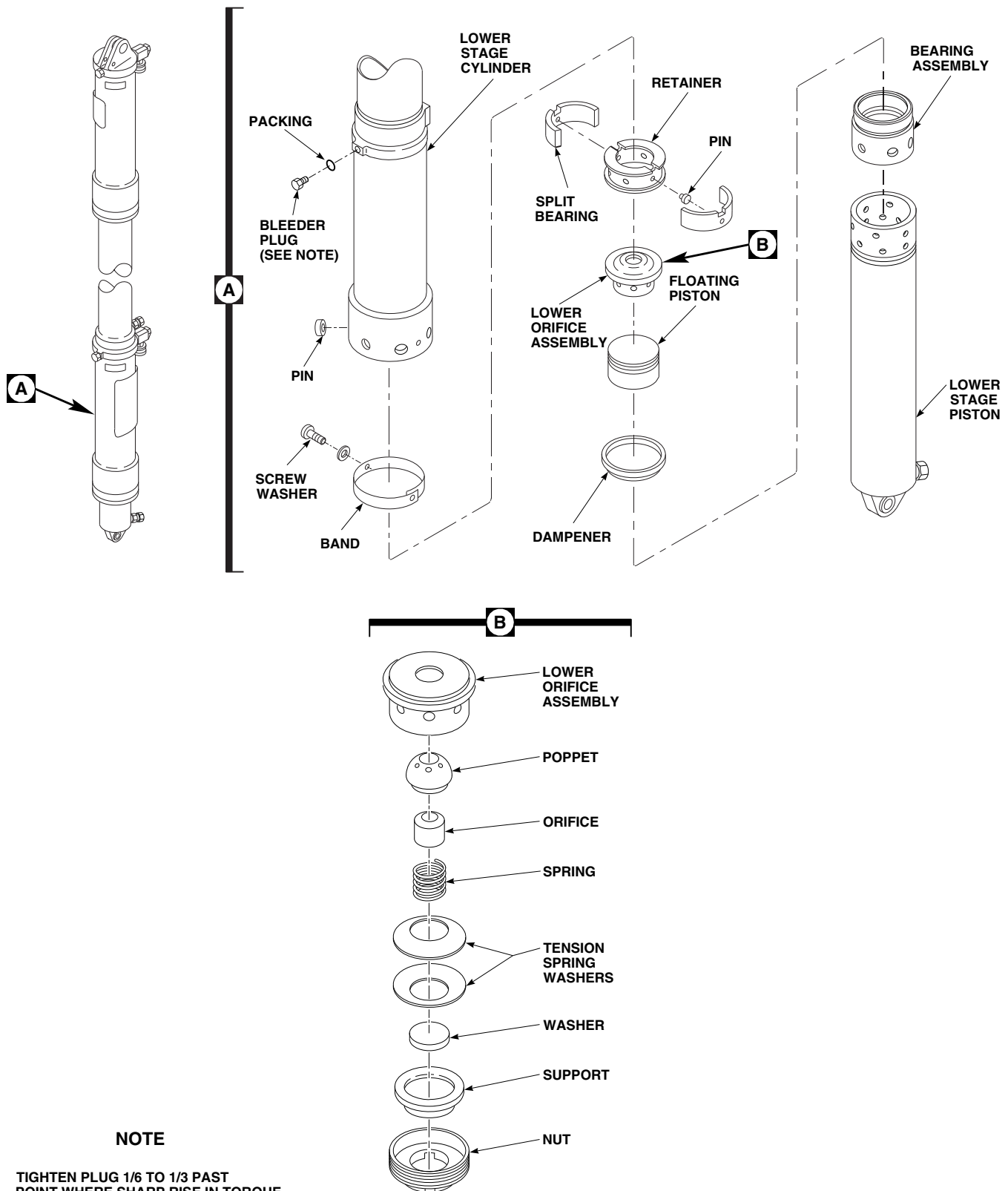
Installation of studded pin in any other pin hole will cause misalignment of strut stages.

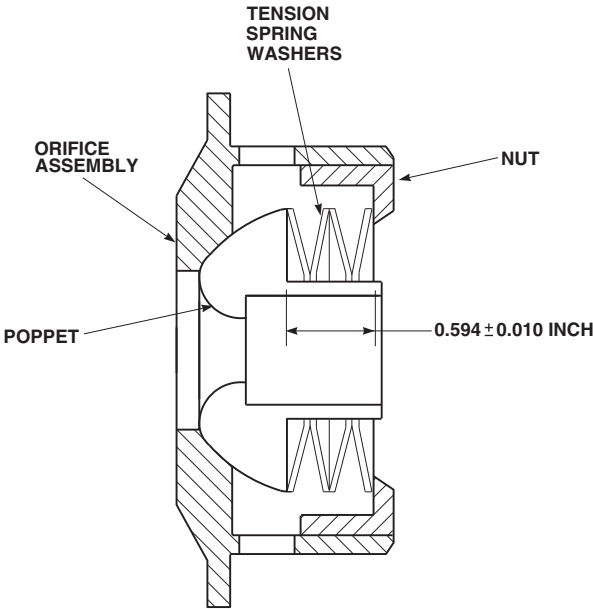
- n. Install five remaining orifice retaining pins.
- o. Install split bearing halves on retainer. Line up upper stage cylinder groove with studded orifice retaining pin. Slide upper stage cylinder onto upper stage piston.
- p. Slide bearing into upper stage cylinder. Line up bearing retaining pin holes and install six pins. Seat retaining pins with rawhide mallet as required.
- q. Install band over bearing retaining pins with screws and washers. Using lockwire, Item 196, Appendix D, lockwire screws.
- r. With dampener flange facing open end of piston, install dampener on lower stage piston (Figure 3-5-1-8, Detail A).
- s. Slide bearing assembly, scraper end first, onto lower stage piston.

NOTE

Lower orifice assembly has more than one hole in poppet valve and has a coil spring installed.

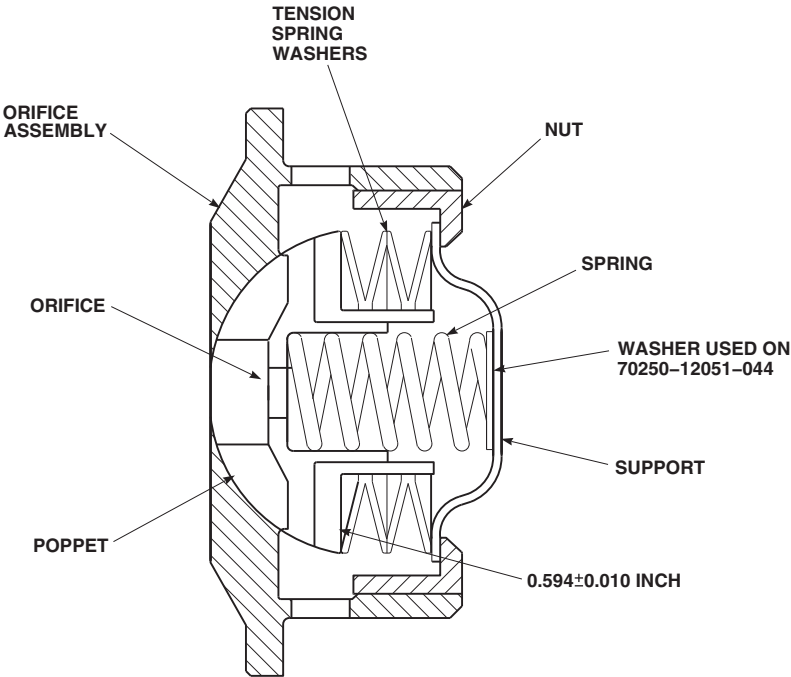
- t. Assemble lower orifice assembly as follows (Figure 3-5-1-8, Detail B and Figure 3-5-1-10):

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FIGURE 3-5-1-9. ASSEMBLE UPPER ORIFICE ASSEMBLY (BENCH)



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FIGURE 3-5-1-10. ASSEMBLE LOWER ORIFICE ASSEMBLY (BENCH)

- (1) Insert orifice and tension spring washers in poppet and install poppet into lower orifice assembly.
 - (2) Locally make orifice assembly gage (Figure H-267, Appendix H).
 - (3) Install orifice assembly gage on poppet.
 - (4) Using hydraulic fluid, Item 154, Appendix D, coat threads of nut. Insert support in orifice and thread nut into lower orifice assembly.
 - (5) Tighten nut until it bottoms on orifice assembly gage.
 - (6) Using pencil, make index marks on nut for depth and radial location inside lower orifice assembly.
 - (7) Remove nut, support, and orifice assembly gage from lower orifice assembly. Install tension spring washers and washer.
 - (8) Using arbor press and suitable drift, compress spring and tension spring washers.
 - (9) Thread nut into lower orifice assembly and line up index marks noted in step t. (6). Lining up index marks will allow tension spring washers to compress to a height of 0.594 ± 0.010 -inches.
 - (10) Using lockwire, Item 196, Appendix D, line up slot in nut with lower orifice assembly and lockwire hole.
 - (11) Remove arbor press and drift.
 - (12) Using lockwire, Item 196, Appendix D, lockwire nut to lower orifice assembly internally. Make sure both lockwire and lockwire tail do not protrude past end of lower orifice assembly.
- u. Insert lower stage orifice assembly into lower stage piston.
 - v. Install retainer. Line up holes and install six orifice retaining pins.
 - w. Install split bearing halves on retainer. Slide piston into lower stage cylinder.
 - x. Slide bearing into lower stage cylinder. Line up bearing retaining pin holes. Install six pins. Seat retaining pins with rawhide mallet as required.
 - y. Install band over bearing pins with screws and washers. Using lockwire, Item 196, Appendix D, lockwire screws.
 - z. Test and service strut (PARA 3-5-1.1.4 and PARA 1-3-2).

3-5-1.1.4 Test.

- a. Install upper and lower kneeling valves (PARA 3-4-8).
- b. Remove upper stage coupling dust cover. Install coupling half and drain hose on kneeling valve.
- c. Support strut vertical. Route drain hose into a 1-gallon container.



Damage to strut will result if clearance does not exist prior to testing and servicing. Testing and servicing will extend strut to its maximum length of 72.94-inches measured between center of upper and lower attachment boltholes. Make sure enough clearance exists above strut.

- d. Connect a shutoff valve and pressure line from hydraulic test stand to upper stage bleeder port. Start test stand. Open pressure line valve.
- e. Apply 100 psi hydraulic pressure, at minimum flow, to fill upper stage strut and to bottom lower stage piston. Slowly close kneeling valve after fluid has flowed from drain line for 20 - 30 seconds.
- f. Reduce hydraulic pressure to 0 psi.

- g. Apply minimum test stand pressure and flow to strut. Open kneeling valve and maintain fluid flow until a steady air-free stream is obtained.
- h. Close kneeling valve. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi.
- i. Allow a few minutes for trapped air to rise to top of strut. Open kneeling valve and fully compress strut. Close kneeling valve.
- j. Apply 100 psi hydraulic pressure, at minimum flow, to fill upper stage strut.
- k. Reduce hydraulic pressure and flow to minimum. Open kneeling valve. Check draining fluid for air.
- l. Repeat preceding steps until hydraulic fluid is completely air-free.
- m. Close pressure line shutoff valve. Compress upper stage strut, if extended.
- n. Close kneeling valve. Reduce test stand pressure to 0 psi.
- o. Disconnect hydraulic line at bleeder port.
- p. Lubricate packing with petrolatum, Item 234, Appendix D.
- q. Install packing on bleeder plug. Install bleeder plug in upper stage bleeder port. **TIGHTEN BLEEDER PLUG $\frac{1}{8}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 3-5-1-7, Sheet 1, Detail A).
- r. Disconnect drain line from upper kneeling valve. Install valve dust cap (Figure 3-5-1-1, Sheet 1). Using lockwire, Item 199, Appendix D, lockwire kneeling valve knob.
- s. Connect drain line to lower kneeling valve.
- t. Connect pressure line to lower stage bleeder port.
- u. Bleed lower stage strut, using same steps given for upper stage.
- v. Disconnect pressure line to lower stage bleeder port.
- w. Lubricate packing with petrolatum, Item 234, Appendix D.
- x. Install packing on bleeder plug. Install bleeder plug in lower stage bleeder port. **TIGHTEN BLEEDER PLUG $\frac{1}{8}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 3-5-1-8, Detail A).
- y. Disconnect drain line from lower kneeling valve. Install valve dust cap (Figure 3-5-1-1, Sheet 1). Using lockwire, Item 199, Appendix D, lockwire kneeling valve knob.
- z. Install upper and lower stage air valves (PARA 3-4-9).
- aa. Connect pressure regulated nitrogen, Item 221, Appendix D, to upper stage air valve (Figure 3-5-1-1, Sheet 1, Detail A). Slowly pressurize upper stage to 740 psi. Close air valve.
- ab. Connect nitrogen to lower stage air valve and pressurize lower stage to 210 psi. Close air valve. **TORQUE UPPER AND LOWER AIR VALVE OUTER HEXNUTS TO 50 - 70 INCH-POUNDS.**
- ac. Using leak detection fluid, Item 189, Appendix D, inspect upper and lower stage air valves for leakage. **NONE ALLOWED.** If leakage is found, replace air valve (PARA 3-4-9).
- ad. Check upper and lower stage struts for evidence of nitrogen and or hydraulic fluid leakage. **NONE ALLOWED.** If leakage is found, repair or replace strut.
- ae. Install cap on air valves. Using lockwire, Item 197, Appendix D, lockwire lower hexnuts.
- af. Using lockwire, Item 197, Appendix D, lockwire upper and lower stage bleed plugs.

3-5-2 MAIN LANDING GEAR SHOCK STRUT LOWER STAGE SPHERICAL BEARING (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
3-5-2.1	Replace Main Landing Gear Shock Strut Lower Stage Spherical Bearing (BENCH)	3-5-2-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Anvil
- Arbor Press
- Clamp
- Drill Press
- Driver
- Fly Cutter
- Nonmetallic Scraper, Locally-Made
- Spinning Tool

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- ASTM E1444
- PARA 2-6-5
- PARA 3-4-7

3-5-2.1 Replace Main Landing Gear Shock Strut Lower Stage Spherical Bearing (BENCH).

3-5-2.1.1 Remove.

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

- a. Clamp fitting to baseplate of drill press. Using fly cutter, machine off bearing outer race swaged over chamfer on fitting (Figure 3-5-2-1, Detail B).
- b. Press out bearing using arbor press with anvil and driver.

3-5-2.1.2 Inspect/Repair.

- a. Clean primer residue from bore and chamfer, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.
- b. Check bearing bore lug for galling or scoring. **NO MORE THAN 0.010-INCH ALLOWED IF 25% OF CONTACT AREA IS AFFECTED.** If necessary, blend out galling or scoring in bearing bore lug using abrasive cloth, Item 10, Appendix D.
- c. Magnetic particle inspect bearing bore lug (ASTM E1444). Replace main landing gear shock strut if cracks are found (PARA 3-4-7).

3-5-2.1.3 Install.

- a. Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.
- b. Apply a light coat of primer, Item 257, Appendix D, to inside of fitting bore and outside surface of outer race of bearing.

- c. Press bearing into fitting, using arbor press with anvil and driver (Figure 3-5-2-1, Detail B).
- d. Clean primer, if any, from bearing V-grooves, using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
- e. Fix fitting with bearing on drill press base plate, with proper spinning tool and anvil.

CAUTION

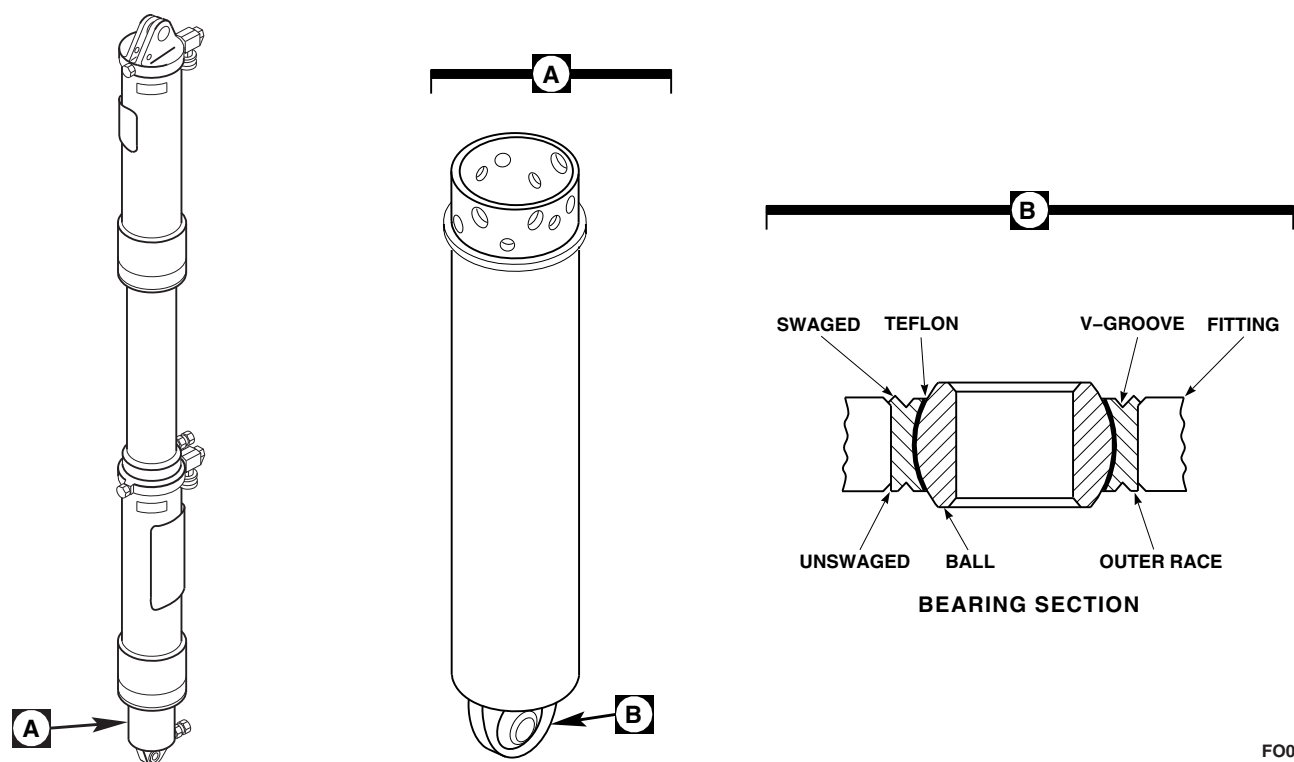
Do not allow petrolatum on the ball of bearing, or on exposed edge of fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

- f. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 234, Appendix D.

CAUTION

Do not allow spinning tool to make contact with ball side of groove as binding may result.

- g. Swage V-groove lip over chamfer in fitting bore.
- h. Turn fitting over and repeat swaging process on other side.
- i. Clean petrolatum from V-groove with cheesecloth, Item 90, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D.
- j. Touch up bearing surfaces with corrosion preventive compound, Item 113, Appendix D.
- k. Touch up paint (PARA 2-6-5).



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FIGURE 3-5-2-1. REPLACE MAIN LANDING GEAR SHOCK STRUT LOWER STAGE SPHERICAL BEARING (BENCH)

3-5-3 MAIN LANDING GEAR WHEEL (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-3.1	Repair Main Landing Gear Wheel (BENCH)	3-5-3-2	3-5-3.2	Replace Bearing Cups (BENCH)	3-5-3-6

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage, 0.002-inch
- Fluorescent Penetrant Inspection Kit
- Mallet
- Nonmetallic Scraper, Locally-Made
- Oven
- Phenolic Drift Pin

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry Ice, Item 125, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 222A, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 3-4-10
- PARA 3-4-11
- PARA 3-4-12
- PARA 3-4-13

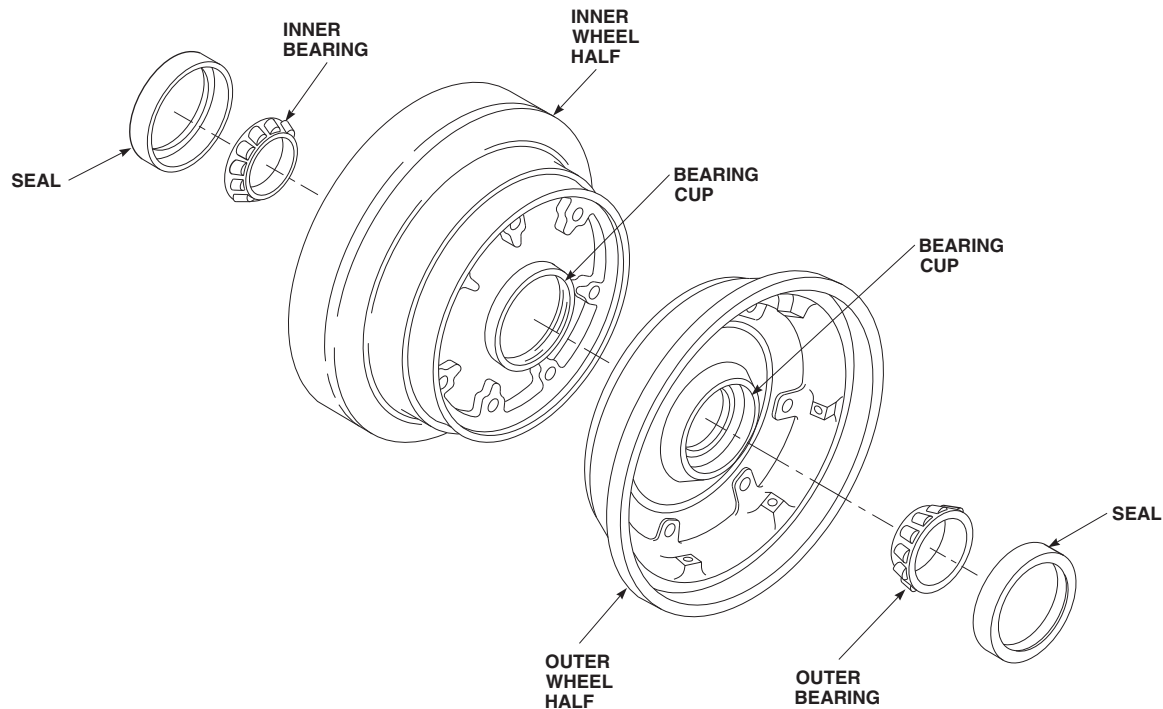
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FIGURE 3-5-3-1. REPAIR MAIN LANDING GEAR WHEEL (BENCH) (SHEET 1 OF 3)

3-5-3.1 Repair Main Landing Gear Wheel (BENCH).**WARNING**

Injury to personnel and damage to equipment may occur if wheel halves are not inspected whenever wheel is disassembled. Make sure wheel halves are inspected whenever wheel is disassembled.

- a. Remove main landing gear tire (PARA 3-4-13).
- b. Inspect seals and bearings (PARA 3-4-13).

NOTE

New bearings packaged in preservative do not need degreasing before being packed with grease.

- c. Inspect bearing cups in wheel halves for looseness, scratches, pitting, corrosion, and evidence

of overheating (Figure 3-5-3-1, Sheet 1). **NONE ALLOWED.** If cups are damaged, replace cups (PARA 3-5-3.2).

WARNING

Injury to personnel or damage to equipment may occur if hardware is reused on wheel, 5003889-2 or 5003889-3. Replace all bolts and nuts on wheel, 5003889-2 or 5003889-3.

- d. Inspect valve for stripped or crossed threads. **NONE ALLOWED.** If damaged, replace valve (PARA 3-4-11).
- e. Inspect instruction plate on outer wheel half for security and readability. **INSTRUCTION PLATE MUST BE SECURE AND LEGIBLE.** If loose or illegible, replace instruction plate.
- f. Remove main landing gear wheel disc drive keys (PARA 3-4-12).

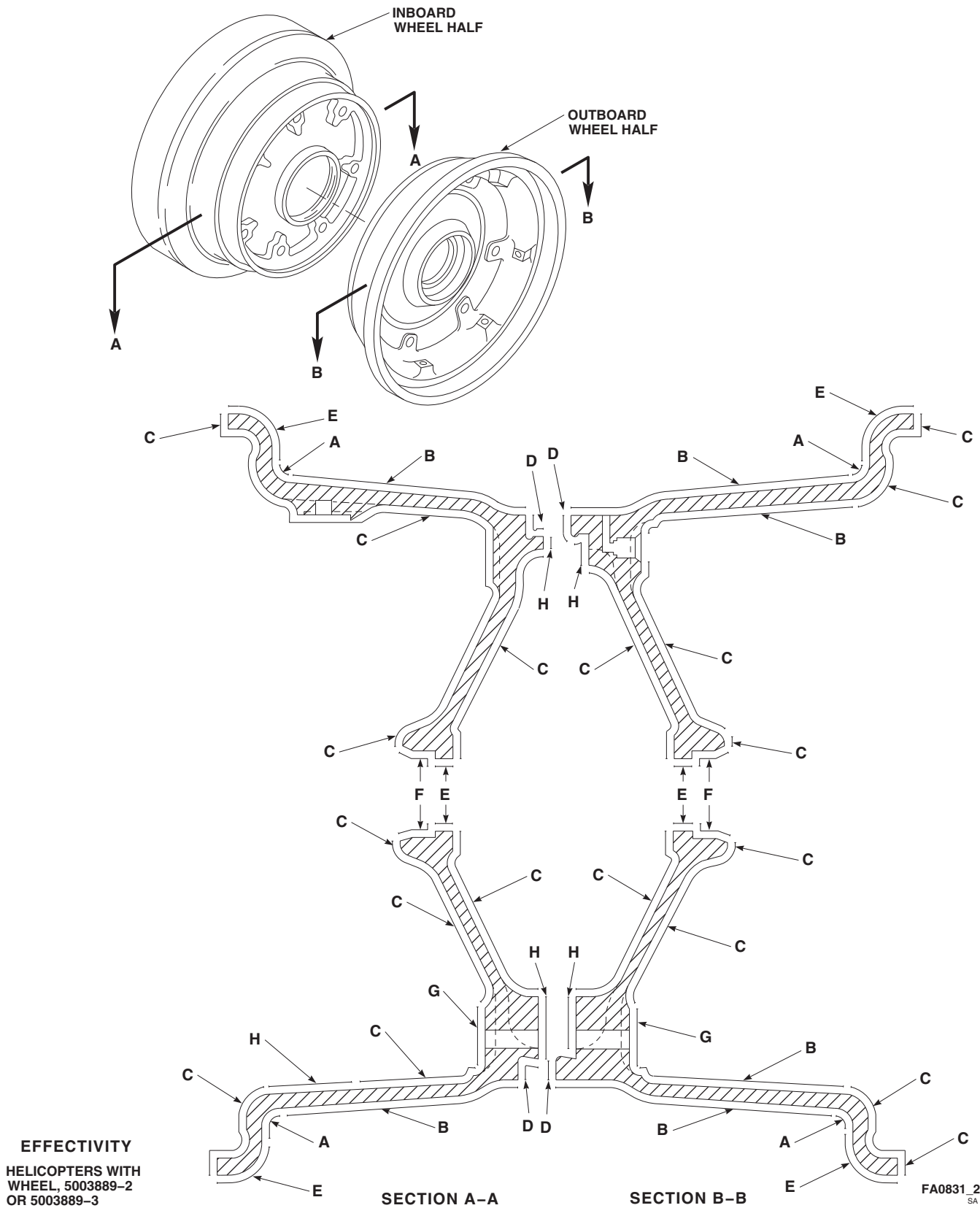


FIGURE 3-5-3-1. REPAIR MAIN LANDING GEAR WHEEL (BENCH) (SHEET 2 OF 3)

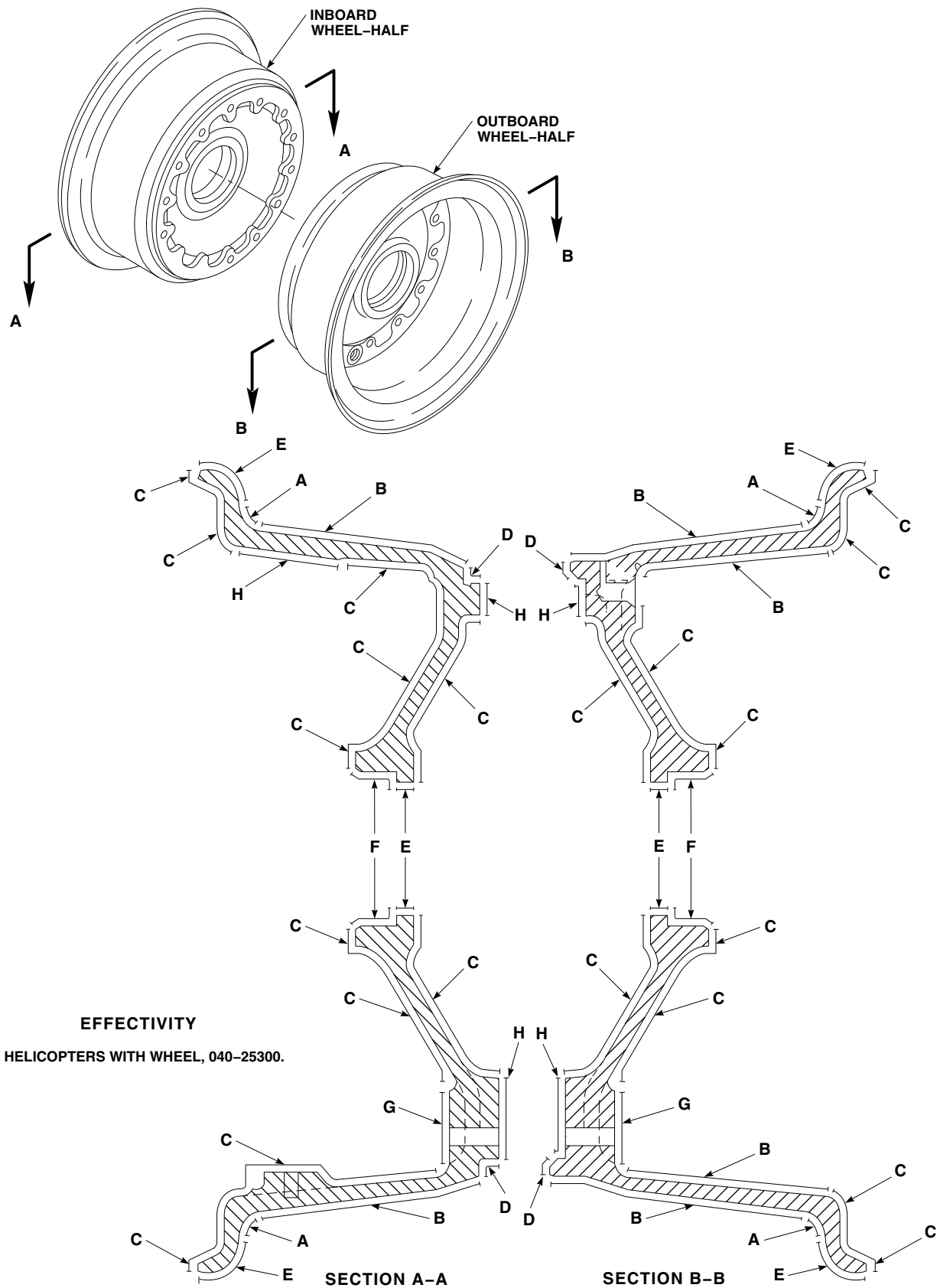


FIGURE 3-5-3-1. REPAIR MAIN LANDING GEAR WHEEL (BENCH) (SHEET 3 OF 3)

- g. Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect wheel halves as follows:

NOTE

Fluorescent penetrant inspection is not required on wheel, 040-25300.

- (1) Using paint remover, Item 230, Appendix D, and nonmetallic scraper, remove paint from wheel halves.
- (2) Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, remove paint residue.
- (3) Using clean machinery towel, Item 211, Appendix D, wipe dry.
- (4) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect wheel, 5003889-2 or 5003889-3, for cracks and structural damage. Pay particular attention to bead seat, bolt boss, male/female mating surfaces, and valve hole areas. **NO CRACKS ALLOWED.** If cracks are found, replace wheel.

- h. Inspect wheel halves (Figure 3-5-3-1, Sheet 2 and Sheet 3) for scratches, nicks, and pits. If damage is found, repair as follows:

NOTE

Using abrasive cloth, Item 8, Appendix D, and abrasive cloth, Item 10, Appendix D, blend and polish wheel halves.

- (1) Repair AREA A as follows:
 - (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.015-INCH DEEP AND 1/2-INCH-LONG.**
 - (b) If damage or repairs exceed limits, replace main landing gear wheel (PARA 3-4-10).

- (2) Repair AREA B as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.020-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**
- (b) **DAMAGE MUST NOT BE DIRECTLY OPPOSITE OTHER DAMAGE OR A PREVIOUSLY REPAIRED AREA.**
- (c) If damage or repairs exceed limits, replace main landing gear wheel (PARA 3-4-10).

- (3) Repair AREA C as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.020-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**
- (b) If damage or repairs exceed limits, replace main landing gear wheel (PARA 3-4-10).

- (4) Repair AREA D as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP.**
- (b) **REPAIR AFTER BLENDING AND POLISHING, MUST NOT INTERFERE WITH SEALING QUALITIES IN MALE/FEMALE MATING SURFACE AREAS.**
- (c) If damage or repairs exceed limits, or if sealing qualities in male/female mating surface area is not maintained, replace main landing gear wheel (PARA 3-4-10).

- (5) Repair AREA E as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.030-INCH DEEP, 1/2 SQUARE INCH IN AREA, IN A MAXIMUM OF 2 PLACES.**

- (b) If damage or repairs exceed limits, replace main landing gear wheel (PARA 3-4-10).

(6) Repair AREA F as follows:

- (a) Remove bearing cups (PARA 3-5-3.2.1).
- (b) **BLEND AND POLISH SCRATCHES OR CORROSION.**
- (c) **REPAIR AFTER BLENDING AND POLISHING MUST NOT AFFECT BEARING CUP RETENTION.**
- (d) If damage or repairs affect bearing cup retention, replace main landing gear wheel (PARA 3-4-10).

- (e) Install bearing cups (PARA 3-5-3.2.2).

(7) Repair AREA G as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP ON FACE OF EACH BOLT BOSS.**
- (b) If damage or repairs exceed limit, replace main landing gear wheel (PARA 3-4-10).

(8) Repair AREA H as follows:

- (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP AND 1/2-SQUARE INCH ON EACH INTERFACE BOSS.**
- (b) If damage or repairs exceed limits, replace main landing gear wheel (PARA 3-4-10).

- i. Using corrosion resistant coating, Item 118, Appendix D, touch up any blended areas.
- j. Using epoxy primer coating kit, Item 135, Appendix D, prime wheel.

NOTE

Male/female mating surfaces, and bolt bosses need primer only.

- k. Paint wheel with polyurethane paint, Item 222A, Appendix D (PARA 2-6-5).
- l. Install main landing gear wheel disc drive keys (PARA 3-4-12).
- m. Install main landing gear tire (PARA 3-4-13).

3-5-3.2 Replace Bearing Cups (BENCH).

3-5-3.2.1 Remove.

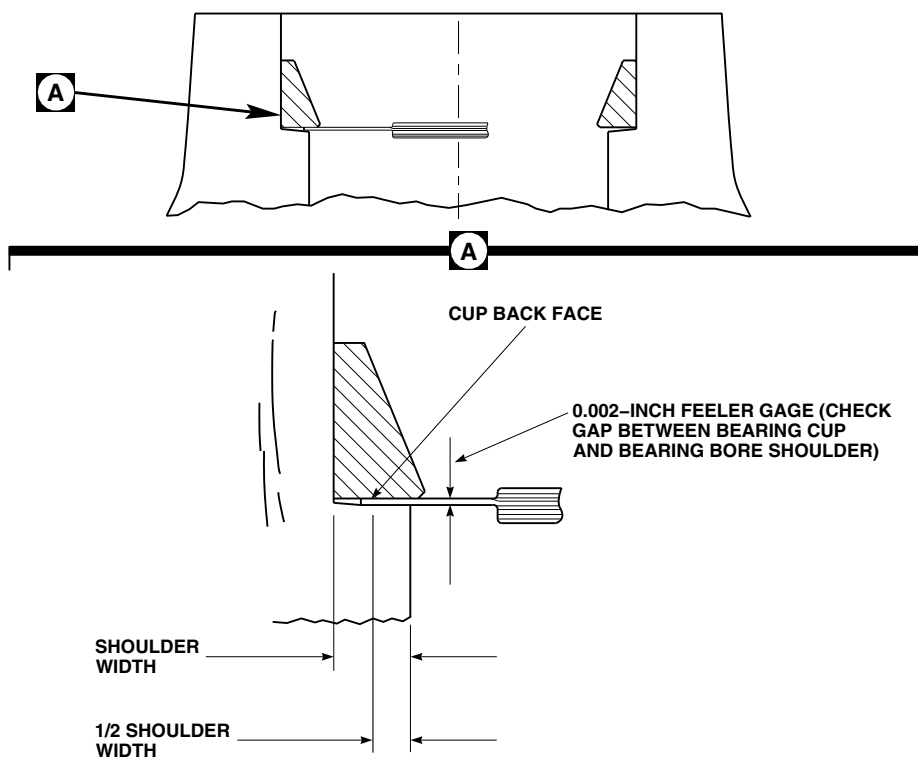
CAUTION

Damage to wheel half will result if overheated. Overheating will effect service life of wheel. Do not overheat wheel half or heat longer than required.

- a. Heat wheel half in an oven not over 250°F (121°C), or place in boiling water for 30 minutes.

CAUTION

- Be careful when removing bearing cup to avoid raising burrs in hub bore.
- Support wheel half on hub and not on flange.
- b. Remove wheel half from oven and immediately remove cup from bore (Figure 3-5-3-2, Detail A). After heating wheel, bearing cup should be loose enough to fall out of bore when wheel is turned upside down. If cup does not fall out, tap cup from bore evenly using phenolic drift pin and mallet.
- c. Carefully remove any primer or other foreign matter from hub bore. Make sure bearing bore and shoulder are free of burrs.



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FIGURE 3-5-3-2. REPLACE BEARING CUPS (BENCH)

3-5-3.2.2 Install.

- a. Place replacement bearing cup in container of dry ice, Item 125, Appendix D.

install in hub bore while primer is wet. Use phenolic drift pin and mallet to tap cup into bore. Make sure cup is against shoulder and avoid cocking cup during installation.

CAUTION

Damage to wheel half will result if overheated. Overheating will affect service life of wheel. Do not overheat wheel half or heat longer than required.

- b. Heat wheel half in an oven not over 250°F (121°C), or place in boiling water for 30 minutes.
- c. Remove wheel half from heat source.
- d. Remove bearing cup from dry ice. Make sure cup is thoroughly dry.
- e. Using primer, Item 257, Appendix D, coat wheel contacting surface of bearing cup, and

CAUTION

Damage to wheel half will result if supported on flange. Support wheel half on hub and not on flange.

- f. Make sure cup is correctly seated by inspecting for any gap between cup back face and shoulder in wheel hub. Use 0.002-inch feeler gage and try to insert it between cup and shoulder. **A 0.002-INCH-MAXIMUM GAP BETWEEN BACK FACE AND SHOULDER IS ALLOWED, PROVIDED THAT FEELER GAGE CANNOT BE INSERTED MORE THAN ONE-HALF THE WIDTH OF SHOULDER AT ANY LOCATION AROUND BACK FACE OF CUP** (Figure 3-5-3-2, Detail A).

- g. Allow wheel to cool to room temperature. Apply light coat of grease, Item 149, Appendix D, to bearing cup to prevent rust and corrosion.

3-5-4 MAIN LANDING GEAR WHEEL BRAKE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
3-5-4.1	Main Landing Gear Wheel Brake Repair (BENCH)	3-5-4-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Protection, Eye
- Test Stand, Hydraulic
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Alcohol, Denatured, Item 70, Appendix D

- Antiseize Compound, Item 76, Appendix D
- Brush, Acid, Item 84, Appendix D
- Brush, Paint, Item 86, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Lacquer, Item 188, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

3-5-4.1 Main Landing Gear Wheel Brake Repair (BENCH).

3-5-4.1.1 Disassemble.

- a. Place brake assembly housing on a clean flat surface.

CAUTION

Damage to brake will result if impact or power wrenches are used to remove bolts or nuts. Do not use impact or power wrenches to remove bolts or nuts.

- b. Remove eight self-locking nuts. Do not remove bolts now (Figure 3-5-4-1).
- c. Remove backing plate with six wear pads attached.
- d. Remove six wear pads from backing plate.
- e. Remove rotating disc (outboard).
- f. Remove stationary disc.
- g. Remove rotating disc (inboard).
- h. Remove torque tube from housing.
- i. Remove eight bolts and countersunk washers.
- j. Remove six wear pads and six piston insulators from housing.
- k. Remove six brake pistons from housing (Figure 3-5-4-1, Detail A).
- l. Remove six packing assemblies from housing.
- m. Remove bleeder screw and washer.
- n. Remove bleeder valve.
- o. Remove bleeder valve adapter.
- p. Remove packing from housing. Discard packing.
- q. Remove protective plug, or hydraulic fitting, from inlet in housing.
- r. Remove packing from bushing. Discard packing.
- s. Remove inlet bushing.
- t. Remove packing. Discard packing.
- u. Remove two wear pin nuts (Figure 3-5-4-1, Detail B).
- v. Remove two wear pin indicators and helical springs from housing.

3-5-4.1.2 Clean.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- a. Using compressed air and acid brush, Item 84, Appendix D, clean two rotating and one stationary disc.
- b. Using dry-cleaning solvent, Item 124, Appendix D, clean all other metal parts. If required, use paint brush, Item 86, Appendix D.
- c. Using denatured alcohol, Item 70, Appendix D, clean all piston insulators. Dry with machinery towel, Item 211, Appendix D.

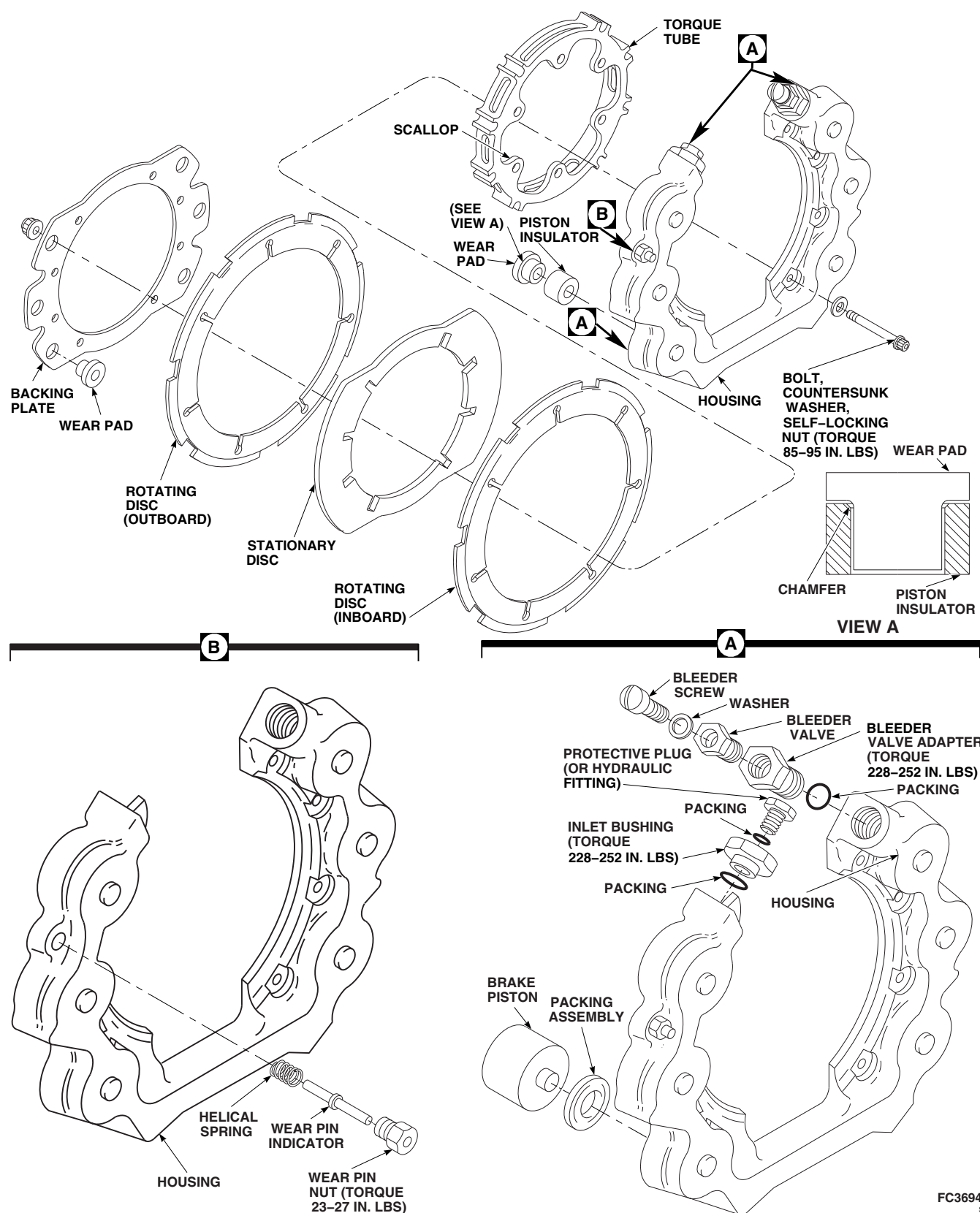


FIGURE 3-5-4-1. REPAIR MAIN LANDING GEAR WHEEL BRAKE (BENCH)

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- d. Using filtered and dried compressed air, dry all metal parts thoroughly after cleaning.

3-5-4.1.3 Inspect.

- a. Remove all foreign material from cavities, threaded holes, pockets, or other entrapment areas in metal parts.
- b. Visually inspect all parts for cracks, wear, structural damage, corrosion, and damaged threads. If cracks, wear, corrosion, structural, or thread damage is evident, replace part as necessary.
- c. Inspect backing plate for cracks around wear pad and mounting boltholes. **NONE ALLOWED.** If crack is found or plate is worn, replace backing plate.
- d. Inspect backing plate for dishing and distortion. **WARPING MUST NOT BE 0.015-INCH OR MORE.** If warping exceeds limit, replace backing plate.
- e. Inspect bolt for bends, cracks, or thread damage. **NONE ALLOWED.** If damaged, replace bolt.
- f. Inspect rotating discs for corrosion at relief slot terminal holes. Using abrasive cloth, Item 7, Appendix D, or abrasive wheel, Item 22, Appendix D, remove corrosion. **CORROSION REMOVAL MUST NOT EXCEED 0.015-INCH FOR RELIEF SLOTS AND 0.281-INCH-DIAMETER FOR TERMINAL HOLES.**

- g. Inspect rotating discs specifically for cracks around six relief slot terminal holes and seven drive slot radii. **NONE ALLOWED.** If crack is found, replace discs.
- h. Inspect rotating discs specifically for wear. **REPLACE DISC WHEN THICKNESS ACROSS UNCHIPPED MIX MEASURES 1/8-INCH OR LESS, OR IF MIX IS WORN TO 0.010 INCH-THICK AT ANY POINT ON EITHER FACE OF DISC. AS MUCH AS 20% OF MIX MAY BE LOST THROUGH CHIPPING IF MIX LOSS IS SCATTERED. NO MORE THAN 25% LOSS OF MIX IS ALLOWED ON ANY SEGMENT.**
- i. Inspect rotating discs for shrinkage and distortion. **REPLACE DISCS THAT ARE DISHED 0.015-INCH OR MORE, OR ARE 1/8-INCH OUT-OF-ROUND, OR HAVE SHRUNK TO AN INSIDE DIAMETER OF 6.940-INCHES OR LESS AT ANY POINT.**
- j. Inspect rotating disc drive slots for wear and battering. **REPLACE DISC IF ONE OR MORE TANGS ARE BENT, OR IF ANY SLOT IS 0.780-INCH OR MORE IN WIDTH.**
- k. Inspect stationary disc specifically for cracks around key slot corners. **NONE ALLOWED.** If crack is found, replace discs.
- l. Inspect stationary disc for thickness. **IF WORN TO 0.140-INCH OR LESS IN THICKNESS AT ANY POINT ON DISC, REPLACE DISC.**
- m. Inspect stationary disc key slots for wear and battering. **IF WIDTH OF ANY KEY SLOT IS LESS THAN 0.515-INCH OR GREATER THAN 0.560-INCH, REPLACE DISC.**
- n. Inspect stationary disc for shrinkage and distortion. **IF DISHED 0.015-INCH OR MORE, OR IS 1/8-INCH OUT-OF-ROUND, OR HAS SHRUNK TO AN INSIDE DIAMETER OF 6.320-INCHES OR LESS WHEN MEASURED BETWEEN ENDS OF KEY SLOT TANGS, REPLACE DISC.**

- o. Inspect inside diameter of stationary disc for size. If disc cannot be installed on torque tube without interference, replace stationary disc.
- p. Inspect torque tube for cracks. **NONE ALLOWED.** If crack is found, replace tube.
- q. Inspect width of torque tube key. **IF KEY WIDTH IS 0.470-INCH OR LESS AT ANY POINT ALONG LENGTH OF ANY KEY, REPLACE TUBE.**
- r. Inspect pistons for wear as follows:
 - (1) Measure diameter of each piston at three places around it. **IF DIAMETER MEASURES 1.240-INCHES OR LESS AT ANY POINT. REPLACE ANY PISTONS WITH BURRS, SCRATCHES, OR NICKS MORE THAN 0.003-INCH-DEEP ON SEAL CONTACTING SURFACES, REPLACE PISTON.**
 - (2) Measure surface of piston that contacts insulator. **IF WEAR EXCEEDS 0.015-INCH, REPLACE PISTON.**
- s. Inspect piston insulators for deterioration or wear. **IF INSULATORS WORN TO 0.635-INCH OR LESS IN LENGTH, OR WITH BLISTERS AND RAISED AREAS, OR WITH END FACES NOT PARALLEL, REPLACE PISTON.**
- t. Using paint remover, Item 230, Appendix D, and nonmetallic scraper, remove paint from housing.
- u. Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, remove paint residue.
- v. Using clean machinery towel, Item 211, Appendix D, wipe dry.
- w. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect housing subassembly for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace housing.
- x. Using 1-inch masking tape, Item 212, Appendix D, mask piston bores, fitting bosses, and wear pin holes.
- y. Using epoxy primer coating kit, Item 135, Appendix D, prime housing with one coat of primer.
- z. Using lacquer, Item 188, Appendix D, paint housing (PARA 2-6-5).
- aa. Inspect thread size and condition in housing inlet and bleeder ports. If threads are damaged, replace housing.

CAUTION

Damage to brake will result if new wear pads are mixed with used pads. If any wear pad on piston or on backing plate is worn beyond limits, replace that set of six wear pads.

- ab. Inspect thickness of backplate wear pads. **IF ONE OR MORE ARE WORN TO 0.380-INCH-THICK OR LESS, REPLACE ALL SIX BACKPLATE WEAR PADS.**
- ac. Inspect thickness of piston wear pads. **IF ANY ONE PAD IS WORN TO 0.380-INCH-THICK OR LESS, REPLACE ALL SIX PADS.**
- ad. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect bolts specifically for cracks at radius under bolthead and within threads next to bolt shank (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bolts.

3-5-4.1.4 Assemble.

CAUTION

- Do not stretch or twist packings during assembly.
 - Do not use impact or power wrenches to install or tighten bolts or nuts.
 - Assemble brake on clean, flat surface, free of foreign objects.
- a. Install two helical springs, wear indicator pins, and wear pin nuts in housing (Figure 3-5-4-1, Detail B). **TORQUE WEAR PIN NUTS TO 23 - 27 INCH-POUNDS.**
 - b. Lubricate packing with hydraulic fluid, Item 154, Appendix D. Install packing and inlet bushing in inlet port of brake housing. **TORQUE INLET BUSHING TO 228 - 252 INCH-POUNDS** (Figure 3-5-4-1, Detail A).
 - c. Lubricate packing with hydraulic fluid, Item 154, Appendix D. Install packing and protective plug (or hydraulic fitting) in inlet bushing.
 - d. Lubricate packing with hydraulic fluid, Item 154, Appendix D. Install packing and bleeder valve adapter in bleeder port of brake housing. **TORQUE BLEEDER VALVE ADAPTER TO 228 - 252 INCH-POUNDS.**
 - e. Install bleeder valve in bleeder valve adapter.
 - f. Install washer and bleeder screw.
 - g. Lubricate, with hydraulic fluid, Item 154, Appendix D, six packing assemblies, and six housing cavities in areas which contact piston surfaces. Install packing assemblies in piston cavities of housing.
 - h. Lubricate six brake pistons with hydraulic fluid, Item 154, Appendix D. Install six pistons in housing cavities, without damaging them or moving packings.
 - i. Using finger pressure, bottom pistons in housing.
 - j. Using antiseize compound, Item 76, Appendix D, coat threads of eight bolts and self-locking nuts, and bearing surfaces of boltheads, washers, and self-locking nuts (Figure 3-5-4-1).
 - k. Put a countersunk washer under each bolthead (countersunk side faces bolthead) and insert bolt through housing.
 - l. Place brake housing, with bolts in place, on a clean, flat surface, with pistons up.
 - m. Place torque tube over bolts. Make sure torque tube scallops face housing.
 - n. Insert wear pad into chamfered end of piston insulator (Figure 3-5-4-1, View A).
 - o. Install six piston insulators and six wear pads in housing (Figure 3-5-4-1).
 - p. Install inboard rotating disc, over torque tube, add stationary disc and outboard rotating disc, over tube.
 - q. Install backplate and six wear pads, with flat face down, against torque tube, with bolts extending through backplate.
 - r. Install self-locking nuts on eight bolts. **TIGHTEN EACH NUT A LITTLE AT A TIME UNTIL SNUG. TORQUE NUTS TO 85 - 95 INCH-POUNDS.**

3-5-4.1.5 Test.

- a. Connect repaired brake assembly in vise to hydraulic test stand with hydraulic fluid, Item 154, Appendix D, able to apply 800 psi hydraulic pressure to brake.
- b. Apply pressure and bleed brake.
- c. Apply 794 psi pressure, then release pressure. Repeat several times. **ROTATING DISCS SHOULD BE FREE TO ROTATE AS EVIDENCE OF PROPER BRAKE RELEASE FUNCTION.**
- d. Apply 5.5 psi pressure to brake. **BRAKE CLEARANCE MUST BE ZERO AND ROTATING DISCS MUST BE MOVABLE BY HAND.**

- e. Apply 794 psi pressure. Maintain steady pressure for 15 - 30 seconds and check for leaks.

NONE ALLOWED. If leaks are evident, replace brake.

3-5-5 MASTER BRAKE CYLINDER (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-5.1	Repair Master Brake Cylinder (BENCH)	3-5-5-1	3-5-5.2	Test Master Brake Cylinder (BENCH)	3-5-5-5

INITIAL SETUP

Tools

- Filter
- Hand Pump
- Hydraulic Pressure Gage, 0 - 100 psi
- Hydraulic Pressure Gage, 0 - 200 psi
- Hydraulic Pressure Gage, 0 - 3000 psi
- Hydraulic Repairer’s Toolkit
- Hydraulic Test Stand
- Protection, Eye and Skin

- Test Fixture, T-23555-1
- Test Fixture, T-3430
- Valve, Hydraulic Shutoff (4)

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 193, Appendix D
- Lockwire, Item 194, Appendix D

References

- Appendix D

3-5-5.1 Repair Master Brake Cylinder (BENCH).

3-5-5.1.1 Disassemble.

- Clean cylinder exterior surfaces with dry-cleaning solvent, Item 124, Appendix D.
- Remove cap screws and end cap (Figure 3-5-5-1).
- Drain hydraulic fluid.
- Remove piston metering valve assembly and packing from cylinder body. Discard packing.
- Remove filler cap and gasket (Figure 3-5-5-1). Discard gasket.
- Remove fluid level screw and Dyna seal.
- Remove rod end, lockwasher, jamnut, and packing. Discard packing.
- Remove piston rod from cylinder body.
- Disassemble piston metering valve assembly as follows (Figure 3-5-5-1, Detail A):
 - Compress return spring to expose spring clip and retaining pin.
 - Remove spring clip, retaining pin, piston retainer, spring retainer, and seat.
 - Remove packing from seat. Discard packing.

- (4) Remove piston head, return spring, and guide.
- (5) Remove packing from piston head. Discard packing.
- (6) Remove relief cap, poppet spring, nylon lock, poppet, and pressure sensing piston from piston assembly.
- (7) Remove packing from pressure sensing piston. Discard packing.

3-5-5.1.2 Clean.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

Wash all parts in clean dry-cleaning solvent, Item 124, Appendix D. Blow parts dry with filtered dry compressed air.

3-5-5.1.3 Inspect.

- a. Check all chrome-plated wear surfaces for scratches, dents, gouges, corrosion, or pitting. **NONE ALLOWED.** If damaged, replace parts.
- b. Check threads for damage. **NONE ALLOWED.** If damaged, repair or replace threaded parts.

3-5-5.1.4 Assemble.

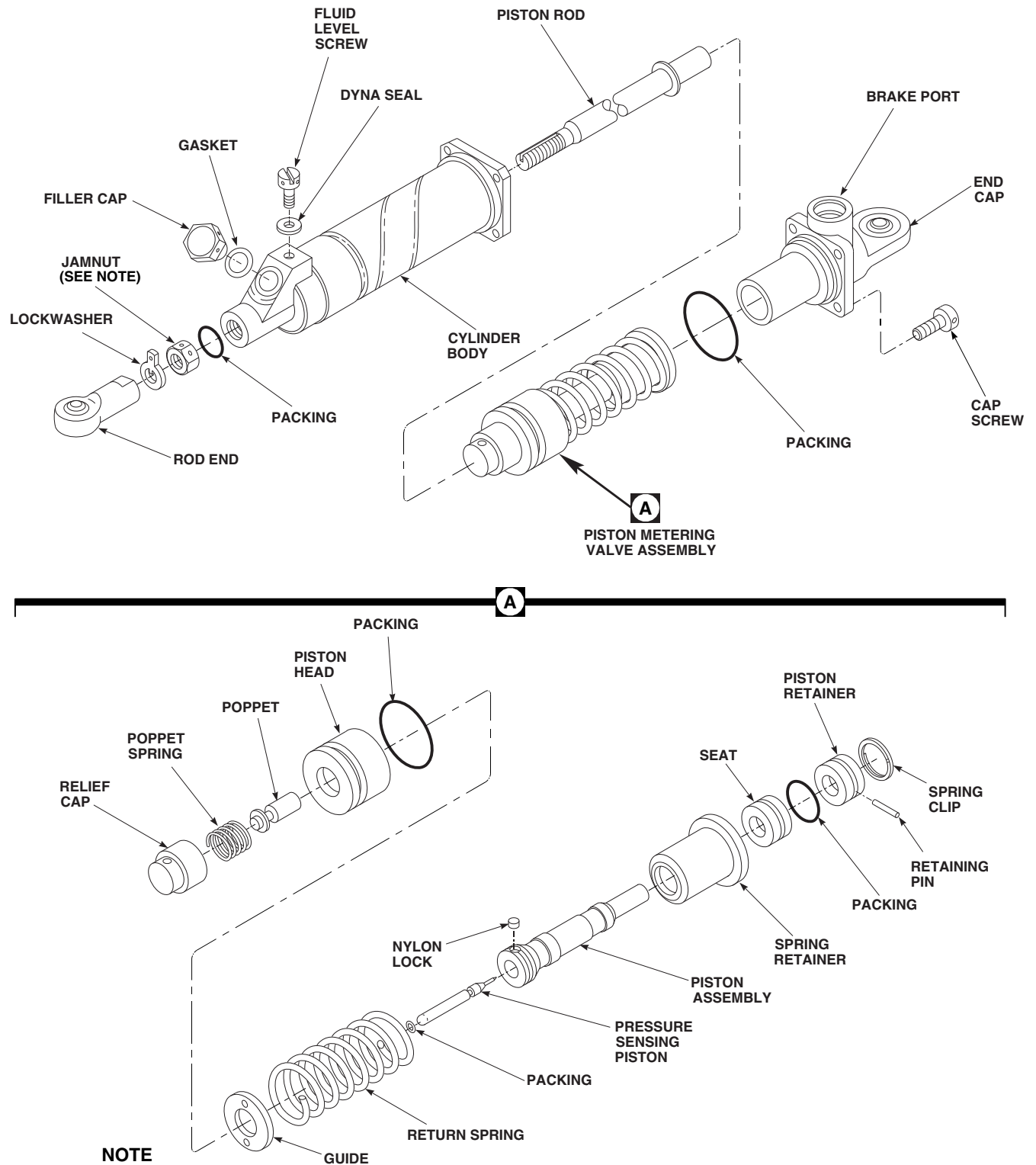
- a. Immerse new packings and parts in hydraulic fluid, Item 154, Appendix D, before assembly.
- b. Assemble piston metering valve assembly as follows (Figure 3-5-5-1, Detail A):

- (1) Install packing on piston head groove.
- (2) Install packing on seat groove.
- (3) Install packing on pressure sensing piston.
- (4) Install pressure sensing piston, poppet, and nylon lock in piston assembly.
- (5) Position poppet spring in relief cap. Screw cap and spring onto piston assembly until bottomed.
- (6) Install piston head on piston assembly with open end away from relief cap.
- (7) Position guide and return spring inside piston head.
- (8) Stack seat and piston retainer in spring retainer.
- (9) Position spring retainer inside return spring. Compress spring. Line up retaining pin holes. Install retaining pin through piston retainer and piston assembly. Install spring clip over retaining pin.

CAUTION

Injury to personnel will result if care is not taken when testing piston metering valve assembly. Hydraulic fluid is under pressure. Wear protective goggles and clothing when testing piston metering valve assembly.

- c. Perform piston metering valve assembly operational test as follows:
 - (1) Install appropriate fitting in test fixture, T-3430 (Figure 3-5-5-2).
 - (2) Connect piston metering valve assembly in test fixture. Connect hydraulic test stand to test fixture port.
 - (3) Using hand pump, slowly increase inlet port pressure to the fixture past the crack-



TIGHTEN JAMNUT 1/6 TO 1/3 TURN
PAST POINT WHERE SHARP RISE IN
TORQUE IS FELT.

FO0600A
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FIGURE 3-5-5-1. REPAIR MASTER BRAKE CYLINDER (BENCH)

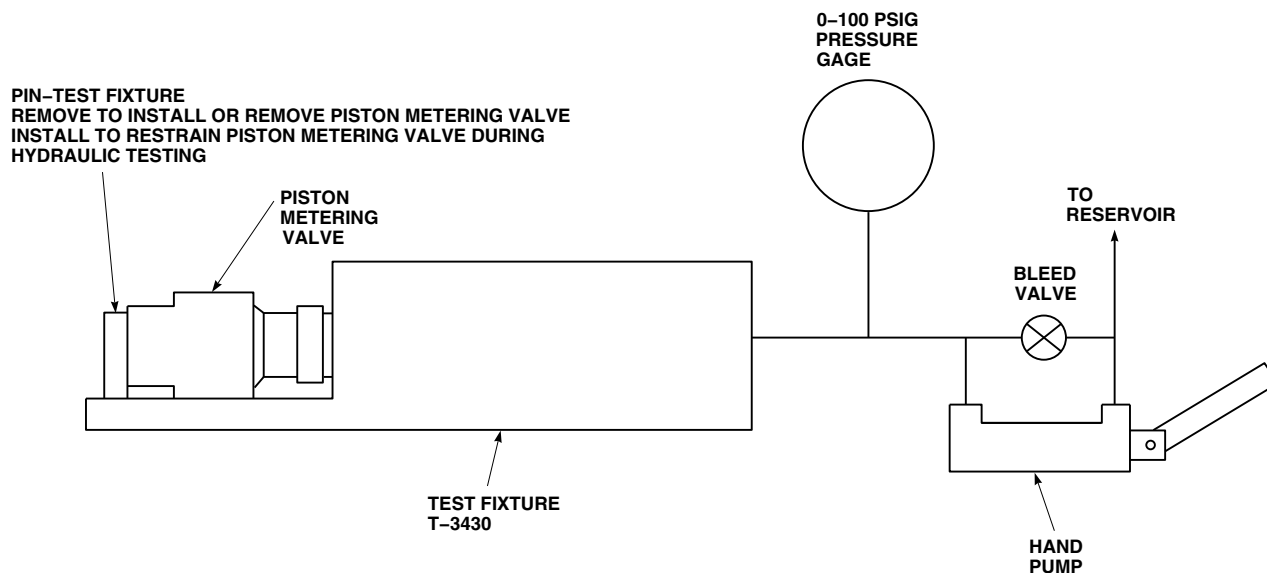
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FIGURE 3-5-5-2. TEST PISTON METERING VALVE ASSEMBLY (BENCH)

ing point of the piston metering valve allowing hydraulic fluid to flow from the exposed cap. Reduce inlet port pressure to zero.

- (4) Starting at 0 psi inlet port pressure, slowly increase pressure to the fixture until the piston valve cracks. **CRACKING PRESSURE SHALL BE 65 - 85 PSI.**
- (5) If pressure is not within range, adjust cracking pressure by removing piston metering valve assembly from test fixture and either tighten cap to increase cracking pressure or loosen cap to decrease cracking pressure. Repeat step c. (2) through step c. (4).
- (6) Apply inlet port pressure to the fixture to the cracking point of the piston metering valve assembly. Block off inlet port pressure and observe for one minute. **GAGE PRESSURE AT THE INLET PORT SHALL NOT DECREASE BELOW 55**

PSI MINIMUM. Repair or replace piston metering valve if pressure decrease is below limits.

- (7) Repeat step c. (2) through step c. (6) to verify proper operation of the piston metering valve assembly.
- d. Install packing in cylinder body piston rod bore (Figure 3-5-5-1).
- e. Position piston rod into cylinder body. Install jamnut, lockwasher, and rod end on piston rod. **DO NOT TIGHTEN ROD END.**
- f. Install piston metering valve assembly in cylinder body.
- g. Install packing on end cap groove.
- h. Install end cap to cylinder body with cap screws. Using lockwire, Item 193, Appendix D, lockwire screws.
- i. Fill cylinder with hydraulic fluid, Item 154, Appendix D.

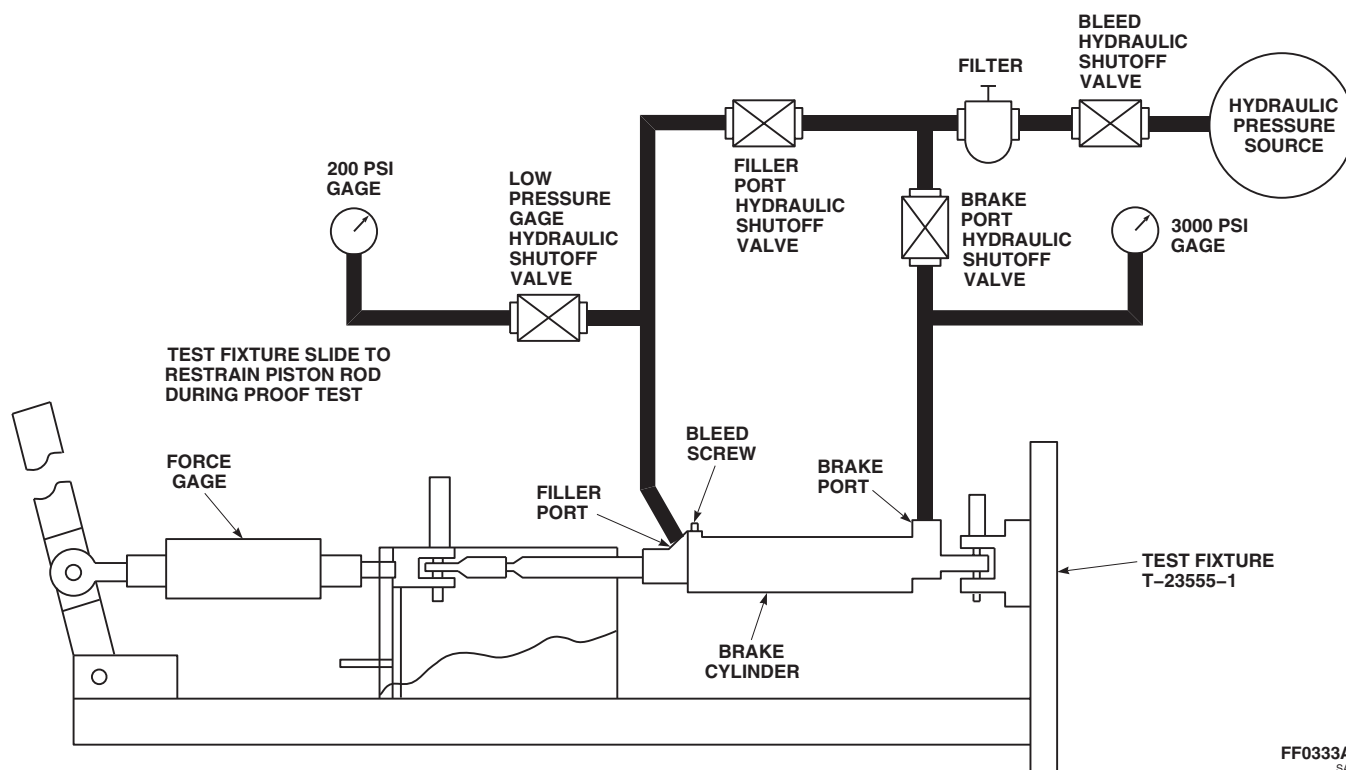


FIGURE 3-5-5-3. TEST MASTER BRAKE CYLINDER (BENCH)

- j. Install Dyna seal on fluid level screw.
- k. Install new gasket on filler cap.
- l. Install fluid level screw and filler cap on cylinder. Secure cap and screw with lockwire, Item 194, Appendix D.
- m. Fully extend master cylinder piston. **ADJUST ROD END ON PISTON ROD TO OBTAIN LENGTH OF 15.36 ± 0.19 -INCHES BETWEEN MASTER CYLINDER ATTACHMENT HOLES, CENTER-TO-CENTER.**
- n. **TIGHTEN JAMNUT ON PISTON ROD $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Secure jamnut to lockwasher with lockwire, Item 194, Appendix D.
- o. Test master brake cylinder (PARA 3-5-5.2).

3-5-5.2 Test Master Brake Cylinder (BENCH).

CAUTION

Injury to personnel will result if care is not taken when testing master brake cylinder. Hydraulic fluid is under pressure. Wear protective goggles and clothing when testing master brake cylinder.

- a. Install master brake cylinder in test fixture, T-23555-1. Adjust test fixture to compress master cylinder piston rod $\frac{1}{8}$ -inch (Figure 3-5-5-3).
- b. Attach hydraulic lines to FILLER and BRAKE ports.
- c. Open BRAKE, FILLER, and BLEEDER line shutoff valves.

- d. Pump hydraulic fluid through lines until air is removed. Refill hydraulic reservoir as required.
- e. Close FILLER port shutoff valve. Open BRAKE port and bleeder line shutoff valves.
- f. Close bleed line shutoff valves.
- g. Close low pressure gage hydraulic shutoff valve.
- h. Apply 2250 psi to BRAKE port. Hold pressure for five minutes. Check exterior for leakage. **NONE ALLOWED.** If leakage is evident, repair or replace master brake cylinder.
- i. Open BRAKE port shutoff valve and reduce pressure to 0 psi. **THE PISTON ROD SHALL RETRACT RAPIDLY AND SMOOTHLY AND THE PRESSURE SHALL IMMEDIATELY DROP TO 0 PSI.**
- j. Close BRAKE port shutoff valve.
- k. Open low pressure gage hydraulic shutoff valve.
- l. Open FILLER port shutoff valve. Apply 150 psi to FILLER port. Hold pressure for five minutes. Check cylinder for external leakage. **NONE ALLOWED.** Reduce pressure to 0 psi and then close FILLER port shutoff valve.

NOTE

Staging pressure readings must be taken from the FILLER port gage.

- m. Open BRAKE port shutoff valve. Apply gradually increasing force to BRAKE port until a change in brake pressure occurs (staging pres-

sure). **PRESSURE SHALL BE 65 - 85 PSI. PRESSURE FLUCTUATIONS FROM 0 PSI TO STAGING PRESSURE SHALL NOT EXCEED 6 PSI.**

- n. Open FILLER port shutoff valve and apply 10 psi to piston rod. Close FILLER port shutoff valve and check pressure over 2 minute period for any drop in pressure. Pressure drop indicates internal leakage. **NO PRESSURE DROP ALLOWED.** If leakage is evident, repair or replace master brake cylinder.

NOTE

Pressure readings must be taken from the FILLER port gage.

- o. Open BRAKE port shutoff valve and apply 65 - 85 psi to BRAKE port. Close BRAKE port shutoff valve and check pressure over two minute period for any drop in pressure. Pressure drop indicates internal leakage. **NO PRESSURE DROP ALLOWED.** If leakage is evident, repair or replace master brake cylinder.
- p. Increase piston rod force to 265 - 285 psi and observe brake pressure. Make sure that piston rod has not bottomed-out after two minute waiting period. **BRAKE PRESSURE SHALL BE 800 PSI MINIMUM AT 265 - 285 LBS ROD FORCE. THE PISTON SHALL NOT HAVE BOTTOMED-OUT AT THE END OF A TWO MINUTE PERIOD.**
- q. Manually actuate piston rod as fast as possible without taking pressure gage needle over 1100 psi. **PRESSURE SHALL BE 850 PSI MINIMUM AT END OF STROKE.**
- r. Remove brake cylinder from test fixture.

3-5-6 PARKING BRAKE VALVE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-6.1	Repair Parking Brake Valve (BENCH)	3-5-6-1	3-5-6.2	Test Parking Brake Valve (BENCH)	3-5-6-6

INITIAL SETUP

Tools

- Container, 2-Gallon
- Discharge Manifold, Locally-Made
- Filter, Hydraulic Fluid
- Fluorescent Penetrant Inspection Kit
- Gage Tester
- Hydraulic Pressure Gage, 0 - 3000 psi (2)
- Hydraulic Repairer’s Toolkit
- Pressure Manifold, Locally-Made
- Protection, Eye
- Spring Scale
- Stopwatch

- Torque Wrench, 30 - 150 in. lbs
- Valve, Hydraulic Shutoff (4)
- Valve Test Fixture, Locally-Made, Figure H-202, Appendix H

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 202, Appendix D
- Petrolatum, Item 234, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417

3-5-6.1 Repair Parking Brake Valve (BENCH).

3-5-6.1.1 Disassemble.

WARNING

Injury to personnel, or damage to components will result if caution is not

used when removing can covers. Covers compress powerful springs. Use caution when removing can covers.

- Remove bolts and can covers (Figure 3-5-6-1, Sheet 2, Detail D).
- Remove accumulator cans and accumulator springs.
- Remove cylinders.

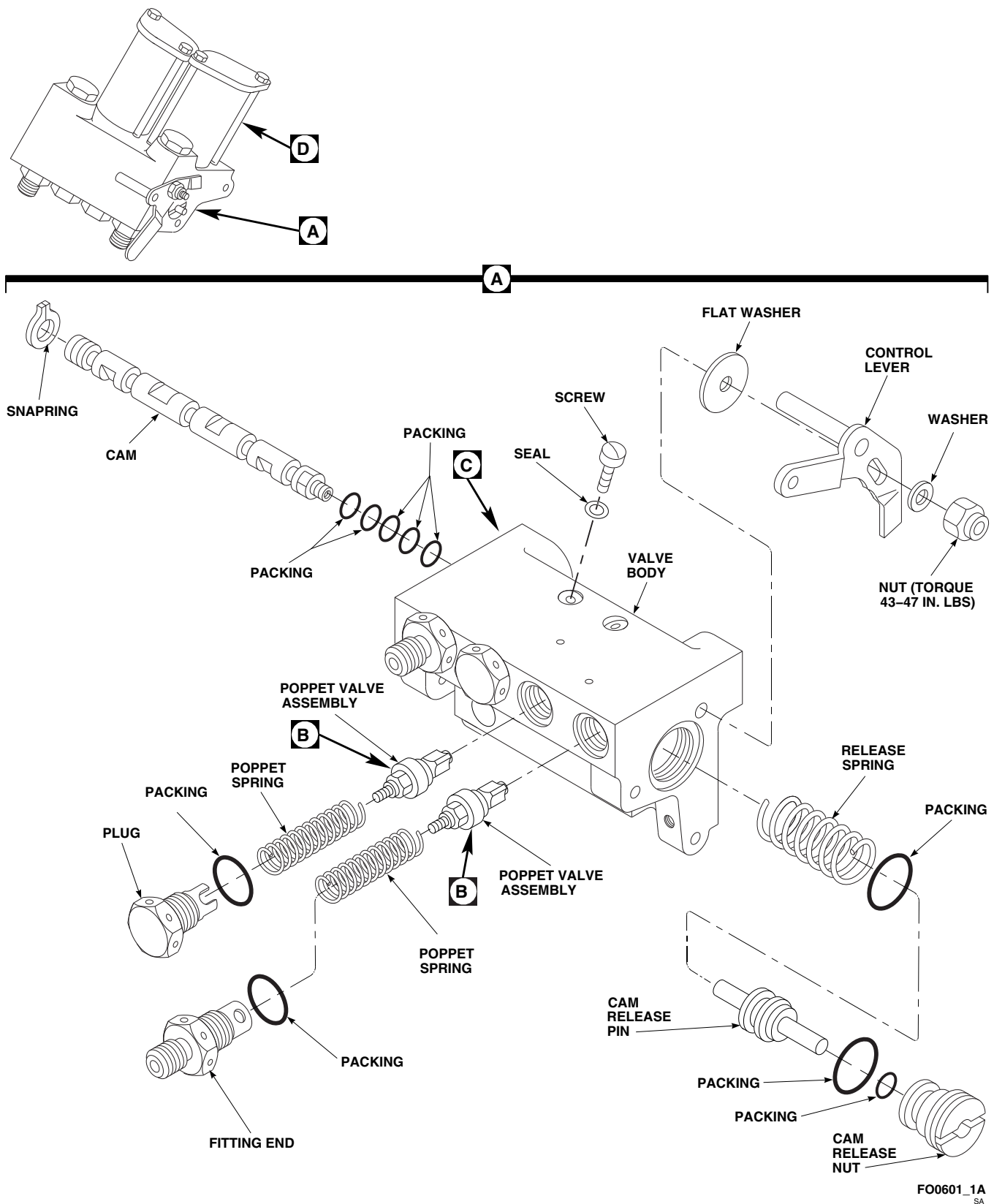


FIGURE 3-5-6-1. REPAIR PARKING BRAKE VALVE (BENCH) (SHEET 1 OF 2)

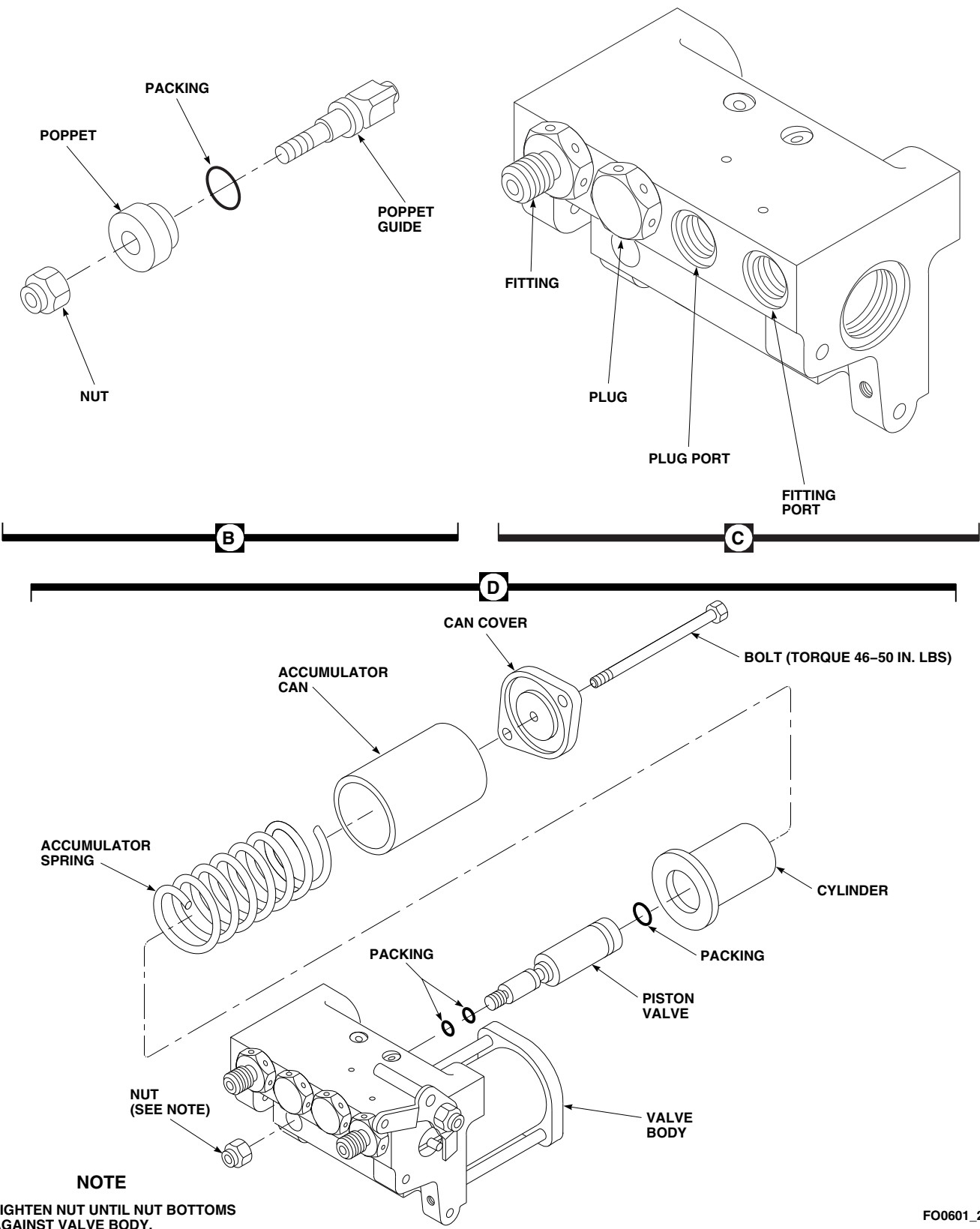


FIGURE 3-5-6-1. REPAIR PARKING BRAKE VALVE (BENCH) (SHEET 2 OF 2)

- d. Remove nuts from ends of piston valves. Remove piston valves from valve body.
- e. Remove packings from piston valves. Discard packings.
- f. Remove nut and washers and remove control lever from cam (Figure 3-5-6-1, Sheet 1, Detail A).
- g. Remove fitting ends and plugs from valve body.
- h. Remove packings from fitting ends and plugs. Discard packings.
- i. Remove poppet springs from poppet valve assembly. Discard springs.
- j. Remove poppet valve assemblies from valve body.
- k. Remove snapping and remove cam from valve body.
- l. Remove packings from cam. Discard packings.

NOTE

Cam release pin nut is spring-loaded.

- m. Remove cam release nut from valve body.
- n. Remove packings from cam release nut. Discard packings.
- o. Remove cam release pin from valve body.
- p. Remove packing from cam release pin. Discard packing.
- q. Remove release spring from valve body. Discard spring.
- r. Remove screws and seals from valve body.
- s. Remove poppet valve nuts, poppets, and packings from poppet guides (Figure 3-5-6-1, Sheet 2, Detail B). Discard packings.

3-5-6.1.2 Clean.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to valve parts will result if dirty solvent is used for cleaning. Make sure clean solvent is used.

Using dry-cleaning solvent, Item 124, Appendix D, clean valve body and components. Use filtered, dry compressed air to dry valve body and parts. Make sure valve passages are dry and clean.

3-5-6.1.3 Inspect.

- a. Inspect valve parts for scratches, wear, cracks, and corrosion. **NONE ALLOWED.** If corrosion, crack, or scratch is found, replace part as necessary.
- b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect valve body and cam release pin for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace valve body or cam release pin.
- c. Inspect accumulator cans for dents interfering with spring action. **NONE ALLOWED.** If dent interferes with spring action, replace accumulator cans.
- d. Inspect accumulator cylinder bores for foreign matter, corrosion, or scoring. **NONE ALLOWED.** If damaged, replace accumulator cylinders.

- e. Inspect threads for damage. **NONE ALLOWED.** If damaged, replace parts as necessary.
- f. Replace all packings and springs.

3-5-6.1.4 Assemble.

- a. Immerse all packings in hydraulic fluid, Item 154, Appendix D, before installing.
- b. Using hydraulic fluid, Item 154, Appendix D, or petrolatum, Item 234, Appendix D, lubricate threads.
- c. Place seals on screws and install screws in valve body. Using lockwire, Item 197, Appendix D, lockwire screws (Figure 3-5-6-1, Sheet 1, Detail A).
- d. Assemble packings, poppets, and nuts onto poppet guides. **TIGHTEN NUTS SO PACKINGS PROTRUDE 0.010-INCH** (Figure 3-5-6-1, Sheet 2, Detail B).
- e. Using new poppet springs, install poppet valve assemblies in valve body (Figure 3-5-6-1, Sheet 1, Detail A).
- f. Install packings on plugs.

NOTE

Plugs are installed in center two valve body ports (Figure 3-5-6-1, Sheet 2, Detail C).

- g. Compress spring on poppet valve assemblies and install plugs. Lockwire, Item 197, Appendix D (Figure 3-5-6-1, Sheet 1, Detail A).
- h. Install packings on fitting ends.

NOTE

Fitting ends are installed in outboard two valve body ports (Figure 3-5-6-1, Sheet 2, Detail C).

- i. Compress spring on poppet valve assemblies and install fitting ends. Lockwire, Item 197, Appendix D (Figure 3-5-6-1, Sheet 1, Detail A).

- j. Install new release spring in valve body.
- k. Install packing on cam release pin.
- l. Install cam release pin in valve body.
- m. Install packing on outer groove and packing on inside groove of cam release nut.
- n. Install cam release nut in valve body. **TIGHTEN SO THAT NUT IS FLUSH TO 0.010-INCH BELOW VALVE BODY SURFACE.**
- o. Install packings on cam.
- p. Using hydraulic fluid, Item 154, Appendix D, lubricate cam. While observing orientation of cam as shown, install in valve body.
- q. Install 1-inch-diameter flat washer over cam threaded end, followed by control lever, washer and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
- r. Install packings on small end of piston valves (Figure 3-5-6-1, Sheet 2, Detail D).
- s. Install packings on large end of piston valves.
- t. Using hydraulic fluid, Item 154, Appendix D, lubricate piston valves and install in valve body.
- u. Using lubricating oil, Item 202, Appendix D, lubricate threads of piston valve nuts and install in valve body. **TIGHTEN PISTON VALVE NUTS UNTIL NUTS BOTTOM AGAINST VALVE BODY. DO NOT OVERTIGHTEN.**
- v. Install cylinders over piston valves.
- w. Test parking brake valve (PARA 3-5-6.2).
- x. Install accumulator springs over cylinders.
- y. Install accumulator cans over accumulator springs.
- z. Install can covers and bolts. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts.

3-5-6.2 Test Parking Brake Valve (BENCH).**3-5-6.2.1 Test.****NOTE**

Assemble valve to point described in PARA 3-5-6.1. Before completing assembly, test valve seals for fluid leakage.

- a. Locally make valve test fixture (Figure H-202, Appendix H).
- b. Install parking brake valve in locally-made test fixture.
- c. Make hydraulic test pressure and discharge manifolds (Figure 3-5-6-2).
- d. Using hydraulic fluid, Item 154, Appendix D, service gage tester.
- e. Install hydraulic fluid filter on outlet port of gage tester.
- f. Attach pressure manifold to filter outlet port. Open both pressure manifold shutoff valves.
- g. Pump gage tester until a steady stream of air free hydraulic fluid flows from hose.
- h. Attach pressure manifold to parking brake valve cylinder ports.
- i. Attach discharge manifold to parking brake valve wheel brake ports. Route hose end into a 2-gallon container
- j. Using new hydraulic fluid, Item 154, Appendix D, refill gage tester reservoir as required.
- k. Put control level in BRAKE-ON position.
- l. Open all four manifold shutoff valves.
- m. Pump all remaining air out of hoses and valves.
- n. Close discharge manifold shutoff valves.
- o. Raise pressure to 1,940 - 2,140 psi. Using stopwatch, hold pressure for two minutes.
- p. Inspect brake valve for external leakage or distortion. **NONE ALLOWED.** Repair or replace valve parts as required (PARA 3-5-6.1).
- q. Open discharge manifold valves. Reduce pressure to zero.
- r. Put control lever in BRAKE-OFF position.
- s. Close discharge manifold shutoff valves.
- t. Apply 1,000 psi at cylinder ports. Move valve control lever to BRAKE-ON position.
- u. Open gage tester return valve and reduce pressure to zero at brake valve cylinder ports.
- v. Inspect locking pin extension. Failure of locking pin to extend may be caused by binding of release pin in locking pin and piston bore. Inspect and repair as necessary (PARA 3-5-6.1).
- w. Using stopwatch, check for valve pressure loss over five minutes. Any pressure loss indicates brake poppet valve or accumulator packing seal leakage. **NONE ALLOWED.** Inspect and repair as necessary (PARA 3-5-6.1).
- x. Close gage tester return valve.
- y. Close pressure manifold shutoff valves.
- z. Using gage tester pump, apply maximum of 100 - 130 psi.
- aa. Attach spring scale to end of control lever. Apply 7 1/2-pounds tension while pulling lever toward BRAKE-OFF position.
- ab. Open left pressure manifold shutoff valve. **CONTROL LEVER LOCKING PIN MUST RETRACT WITHOUT HESITATION** allowing control lever to be rotated toward BRAKE-OFF position. Inspect and repair faulty pin retraction or binding cam lever action (PARA 3-5-6.1).
- ac. Open right hand pressure manifold shutoff valve.
- ad. Apply 556 psi to valve cylinder ports.

- ae. Close discharge manifold shutoff valves.
- af. Put control lever in BRAKE-ON position to energize accumulators.
- ag. Open gage tester return valve to relieve pressure at valve cylinder ports.
- ah. Note time. Leave valve in brake-on position for eight hours. **INDICATED PRESSURE MUST BE 556 - 334 PSI AT END OF EIGHT**

HOURS. A pressure drop below 334 psi indicates unacceptable leakage at poppet valves and accumulators. Disassemble, inspect, and repair as necessary (PARA 3-5-6.1).

- ai. Retest parking brake valve using preceding test sequence. **FAILURE TO MEET ANY ONE OF THE TEST REQUIREMENTS MUST BE CAUSE FOR REJECTION.**

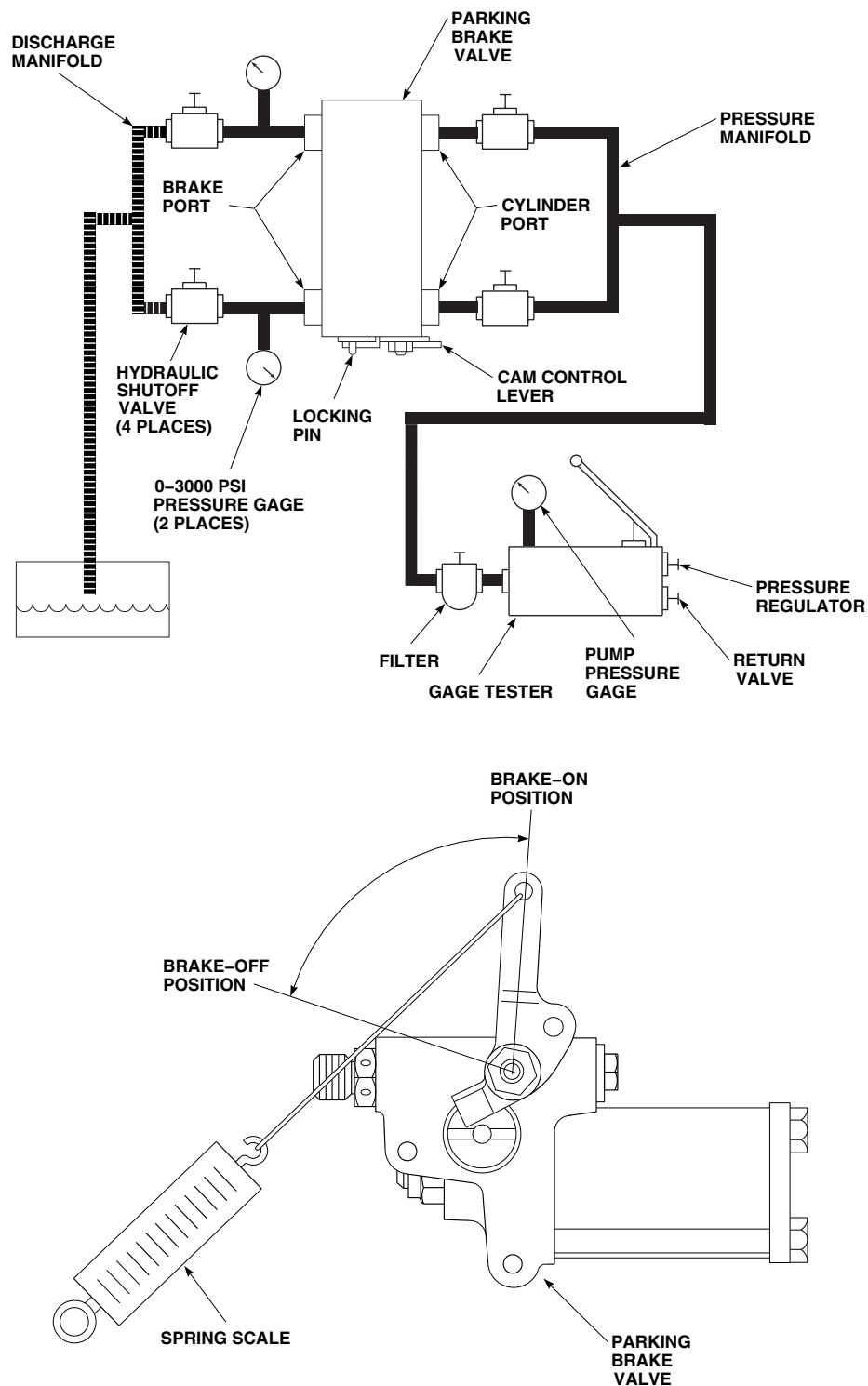
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FIGURE 3-5-6-2. TEST PARKING BRAKE VALVE (BENCH)

3-5-7 SLAVE MIXER VALVE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-7.1	Repair Slave Mixer Valve (BENCH)	3-5-7-1	3-5-7.2	Test Slave Mixer Valve (BENCH)	3-5-7-3

INITIAL SETUP

Tools

- Bleeder Valve
- Dial Indicator
- Flow Meter
- Fluorescent Penetrant Inspection Kit
- Hydraulic Pressure Gage, 0 - 200 psi (2)
- Hydraulic Pressure Gage, 0 - 3000 psi (2)
- Hydraulic Repairer’s Toolkit
- Hydraulic Test Stand

- Needle Valve (2)
- Protection, Eye and Skin
- Vise, Padded

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Lockwire, Item 193, Appendix D

References

- Appendix D
- ASTM E1417

3-5-7.1 Repair Slave Mixer Valve (BENCH).

3-5-7.1.1 Disassemble.

WARNING

Injury to personnel and damage to equipment will occur if care is not taken when removing slave mixer valve cap. Cap is spring loaded. Take care when removing cap.

- Remove screws and cap from valve body (Figure 3-5-7-1).

- Remove packing retainer, packing, and spring. Discard packing.
- Remove piston assembly from valve body.
- Remove valve cap, poppet, and poppet spring from piston (Figure 3-5-7-1, Detail A).
- Remove nylon lock and packing from piston. Discard packing.
- Remove plug and packing from valve body. Discard packing (Figure 3-5-7-1).

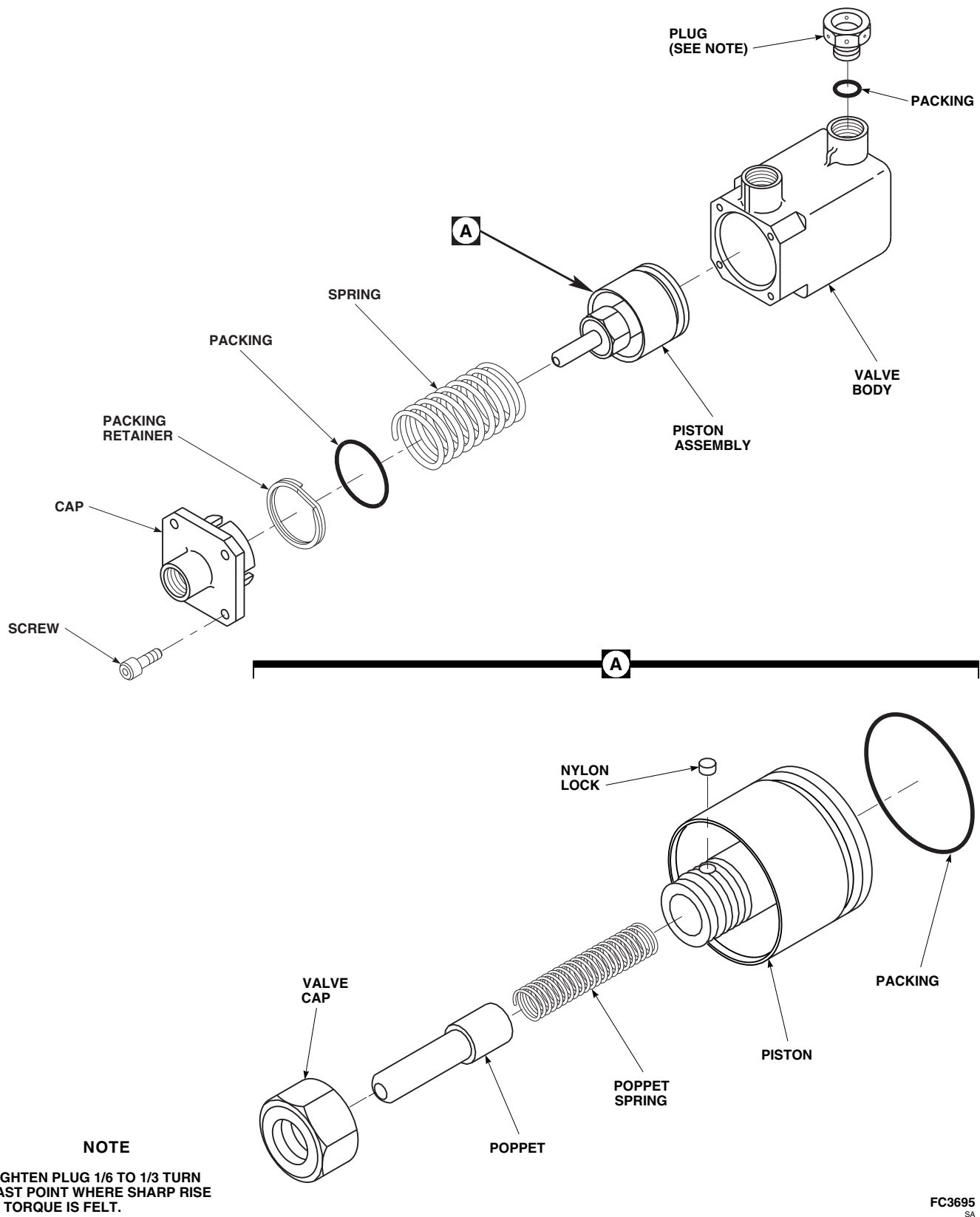


FIGURE 3-5-7-1. REPAIR SLAVE MIXER VALVE (BENCH)

3-5-7.1.2 Clean.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

Clean parts using dry-cleaning solvent, Item 124, Appendix D. Use filtered, dry compressed air to dry parts.

3-5-7.1.3 Inspect.

- Inspect valve parts for scratches, wear, cracks, and corrosion. **NONE ALLOWED.** If corrosion, crack or scratch is found, replace parts as necessary.
- Using fluorescent penetrant inspection kit (ASTM E1417), inspect valve body and piston assembly for cracks. **NONE ALLOWED.** If crack is found, replace valve body or piston assembly.
- Inspect valve body bore for corrosion, scoring, scratches, or wear. **NONE ALLOWED.** If corrosion, scoring, scratch or wear is found, replace valve body.
- Inspect threads for damage. **NONE ALLOWED.** If damaged, replace parts as necessary.

3-5-7.1.4 Assemble.

- Lubricate all packings using hydraulic fluid, Item 154, Appendix D.
- Install poppet spring and poppet into piston and secure with valve cap and nylon lock (Figure 3-5-7-1, Detail A).

- Install packing on piston. Install piston assembly into valve body (Figure 3-5-7-1).
- Install packing retainer and packing on cap.
- Install spring and cap into valve body. Secure cap with screws. Using lockwire, Item 193, Appendix D, lockwire screws.
- Test slave mixer valve (PARA 3-5-7.2).
- Install plug and packing into valve body. **TIGHTEN PLUG 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 193, Appendix D, lockwire plug.

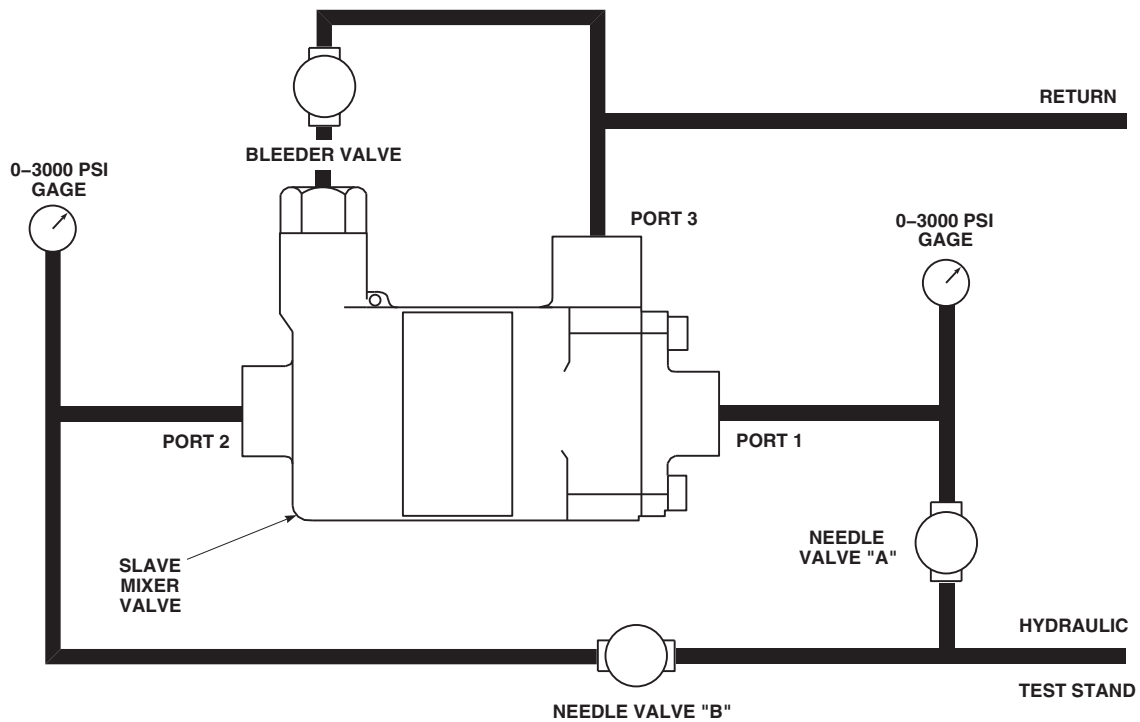
3-5-7.2 Test Slave Mixer Valve (BENCH).

WARNING

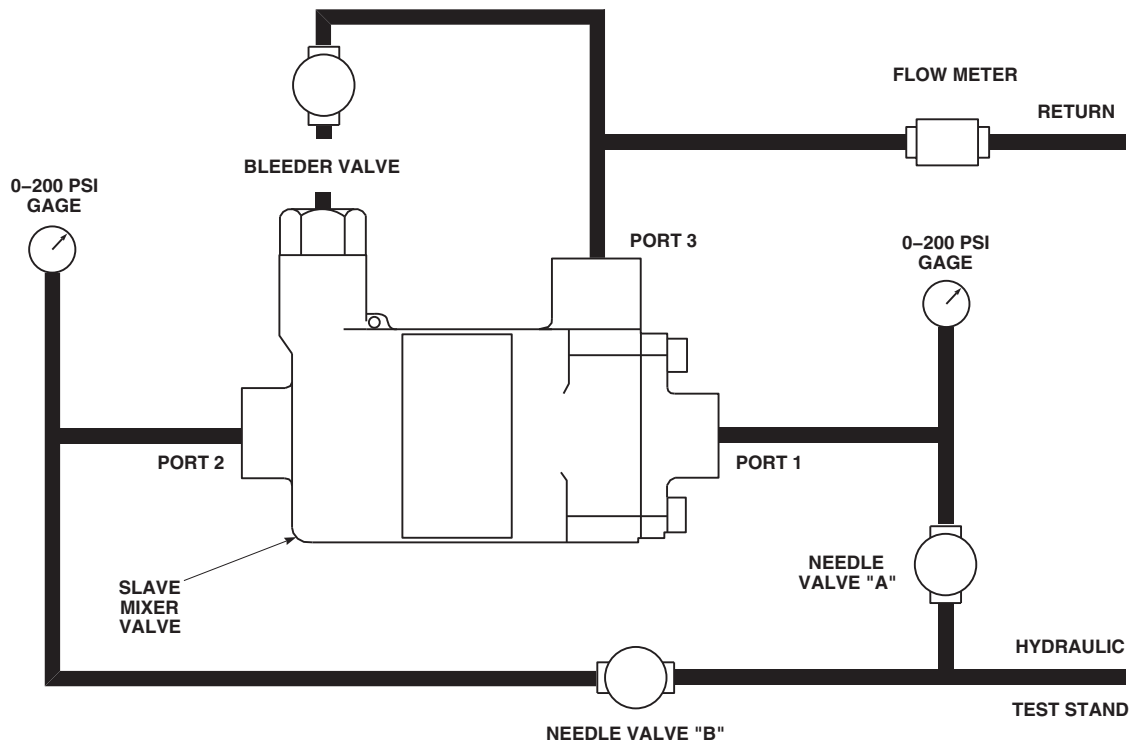
Injury to personnel will result if care is not taken when testing slave mixer valve. Hydraulic fluid is under high pressure. Wear protective goggles and clothing when testing slave mixer valve.

3-5-7.2.1 Proof Pressure Test.

- Set up slave mixer valve for test (Figure 3-5-7-2, Sheet 1).
- Close needle valve B and open needle valve A.
- Slowly apply hydraulic pressure to port 1 until air-free flow is observed from port 3. Close valve A.
- Cap port 3 and adjust valve A to apply 2040 psi to port 1. Hold pressure for two minutes.
- Check slave mixer valve for deformation, external leakage, and internal leakage from port 2. **NONE ALLOWED.** If deformation or leakage is found, repair or replace slave mixer valve.
- Close valve A. Reduce hydraulic pressure to 100 psi.



PROOF PRESSURE TEST



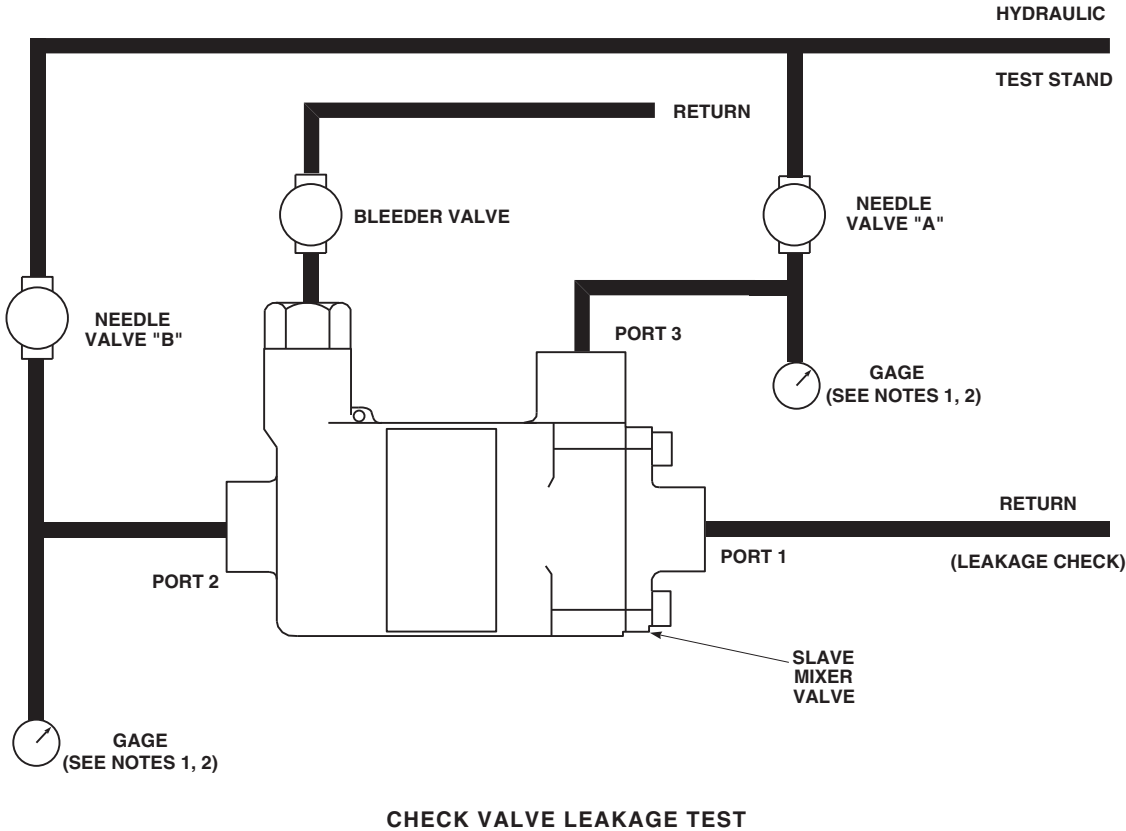
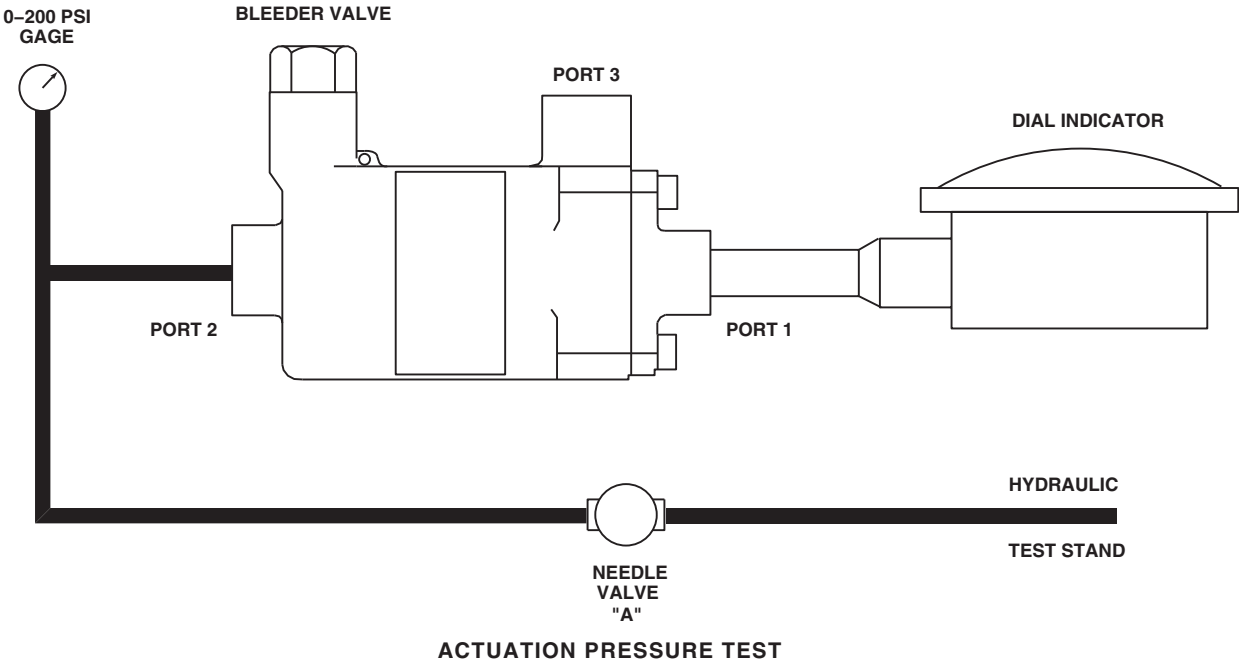
NOTES

1. LOW PRESSURE LEAKAGE TEST USE 0-200 PSI GAGES.
2. HIGH PRESSURE LEAKAGE TEST USE 0-3000 PSI GAGES.

CHECK VALVE RELIEF PRESSURE TEST

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FIGURE 3-5-7-2. TEST SLAVE MIXER VALVE (BENCH) (SHEET 1 OF 2)



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FIGURE 3-5-7-2. TEST SLAVE MIXER VALVE (BENCH) (SHEET 2 OF 2)

- g. Open bleeder valve and valve B. Apply low pressure to port 2 to purge air from slave mixer valve.
- h. Close bleeder valve. Reduce test stand pressure to 0 psi. Open valve A and return line valve to vent ports 1 and 3. Close valve A and return line valve.
- i. Adjust valve B to apply 2040 psi to port 2. Hold pressure for two minutes.
- j. Check slave mixer valve for external or internal leaks from ports 1 and 3. **NO LEAKAGE OR DEFORMATION ALLOWED.** If leakage or deformation is found, repair or replace slave mixer valve.
- k. Reduce test stand pressure to 0 psi and close valve B.

3-5-7.2.2 Check Valve Relief Pressure Test.

- a. Replace 0 - 3000 psi gages with 0 - 200 psi gages (Figure 3-5-7-2, Sheet 1).
- b. Install flow meter along return line from port 3.
- c. Close valve A. Open valve B and apply 100 psi to port 2. Open valve A and slowly apply pressure to port 1 until a flow of 20 cc/min is recorded out port 3.
- d. Record pressure at port 1. **PORT 1 PRESSURE MUST NOT EXCEED 30 PSI.** If port 1 pressure exceeds limit, repair or replace slave mixer valve.

3-5-7.2.3 Check Valve Leakage Test.

- a. Perform low pressure leakage test as follows:
 - (1) Set up slave mixer valve for test (Figure 3-5-7-2, Sheet 2). Use 0 - 200 psi gages.
 - (2) Adjust valve B to apply 40 - 45 psi to port 2. Adjust valve A to apply 20 psi to port 3. Close valve A.

- (3) Hold pressure for two minutes and check port 1 for leakage. **PORT 1 LEAKAGE MUST NOT EXCEED 1 DROP PER MINUTE.** If leakage exceeds limit, repair or replace slave mixer valve.

- b. Perform high pressure leakage test as follows:

- (1) Replace 0 - 200 psi gages with 0 - 3000 psi gages.
- (2) Adjust valve B to apply 520 - 525 psi to port 2. Adjust valve A to apply 500 psi to port 3. Close valve A.
- (3) Hold pressure for two minutes and check port 1 for leakage. **PORT 1 LEAKAGE MUST NOT EXCEED 1 DROP PER MINUTE.** If leakage exceeds limit, repair or replace slave mixer valve.

- (4) Adjust valve B to apply 1025 - 1030 psi to port 2. Adjust valve A to apply 1000 psi to port 3. Close valve A.

- (5) Hold pressure for two minutes and check port 1 for leakage. **PORT 1 LEAKAGE MUST NOT EXCEED 1 DROP PER MINUTE.** If leakage exceeds limit, repair or replace slave mixer valve.

3-5-7.2.4 Actuation Pressure Test.

- a. Set up slave mixer valve for test in a padded vise (Figure 3-5-7-2, Sheet 2).
- b. Install plug and packing (PARA 3-5-7.1).
- c. Insert dial indicator into port 1 until contact is made with poppet and reading is noted on indicator.
- d. Adjust valve A until dial indicator movement is noted. **PRESSURE MUST NOT EXCEED 12 PSI.** If pressure exceeds limit, repair or replace slave mixer valve.

3-5-8 TAIL LANDING GEAR SHOCK STRUT (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
3-5-8.1	Repair Tail Landing Gear Shock Strut (BENCH)	3-5-8-2
3-5-8.2	Replace Tail Landing Gear Shock Strut Spherical Bearings (BENCH)	3-5-8-13

INITIAL SETUP

Tools

- Air Pump, Manual
- Aircraft Mechanic’s Toolkit
- Anvil
- Arbor Press
- Clamp
- Container, 1-Gallon
- Drain Hose
- Drill Press
- Driver
- Fly Cutter
- Hydraulic Repairer’s Toolkit
- Hydraulic Shutoff Valve
- Hydraulic Test Stand
- Magnetic Particle Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Pail, Utility

- Pick, Plastic or Wood, Locally-Made
- Pilot
- Protection, Eye
- Puller Kit, Mechanical
- Rawhide Mallet
- Spinning Tool
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 110, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Lockwire, Item 199, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Nitrogen, Item 221, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 257, Appendix D
- Screw, MS35275-247
- PARA 2-6-5
- PARA 3-4-22
- PARA 3-4-23
- PARA 3-4-24

References

- Appendix D
- ASTM E1444

3-5-8.1 Repair Tail Landing Gear Shock Strut (BENCH).

3-5-8.1.1 Disassemble.

NOTE

If replacing lower stage piston assembly, 70250-13167-041 (part of shock strut, 70250-13101-043), replace shock strut with shock strut, 70250-13101-044.

- Using dry-cleaning solvent, Item 124, Appendix D, and machinery towels, Item 211, Appendix D, clean strut exterior.

NOTE

Work area and tools must be clean. Use a surface covered with clean machinery towels, Item 211, Appendix D, for disassembled parts.

- Remove upper and lower stage air valve caps. Loosen outer hexnut and depressurize upper and lower stage struts. Leave valves open (Figure 3-5-8-1, Detail B).

WARNING

Injury to personnel and damage to equipment will occur if shock strut is

disassembled without removing upper and lower bleeder plugs. Residual pressure will cause shock strut piston to blow out during disassembly. Make sure hydraulic bleeder plugs are slowly removed to release possible residual pressure resulting from leakage from air side to hydraulic side of shock strut.

- Remove upper and lower stage bleeder plugs. Discard packings. Drain hydraulic fluid from strut into 1-gallon container. Discard used fluid (Figure 3-5-8-1, Detail A).
- Remove upper and lower kneeling valves (PARA 3-4-23).
- Remove screws, washers, and band from piston and cylinder assembly. Using screw, MS35275-247, and mechanical puller kit, remove pins.

NOTE

Split bearing halves may fall free as piston clears cylinder.

- Separate lower stage piston assembly from piston and cylinder assembly. Catch split bearing halves as piston is removed from cylinder.
- Remove pins, retainer, orifice assembly, spacer, dampener, and bearing from lower stage piston assembly. Do not disassemble orifice assembly.

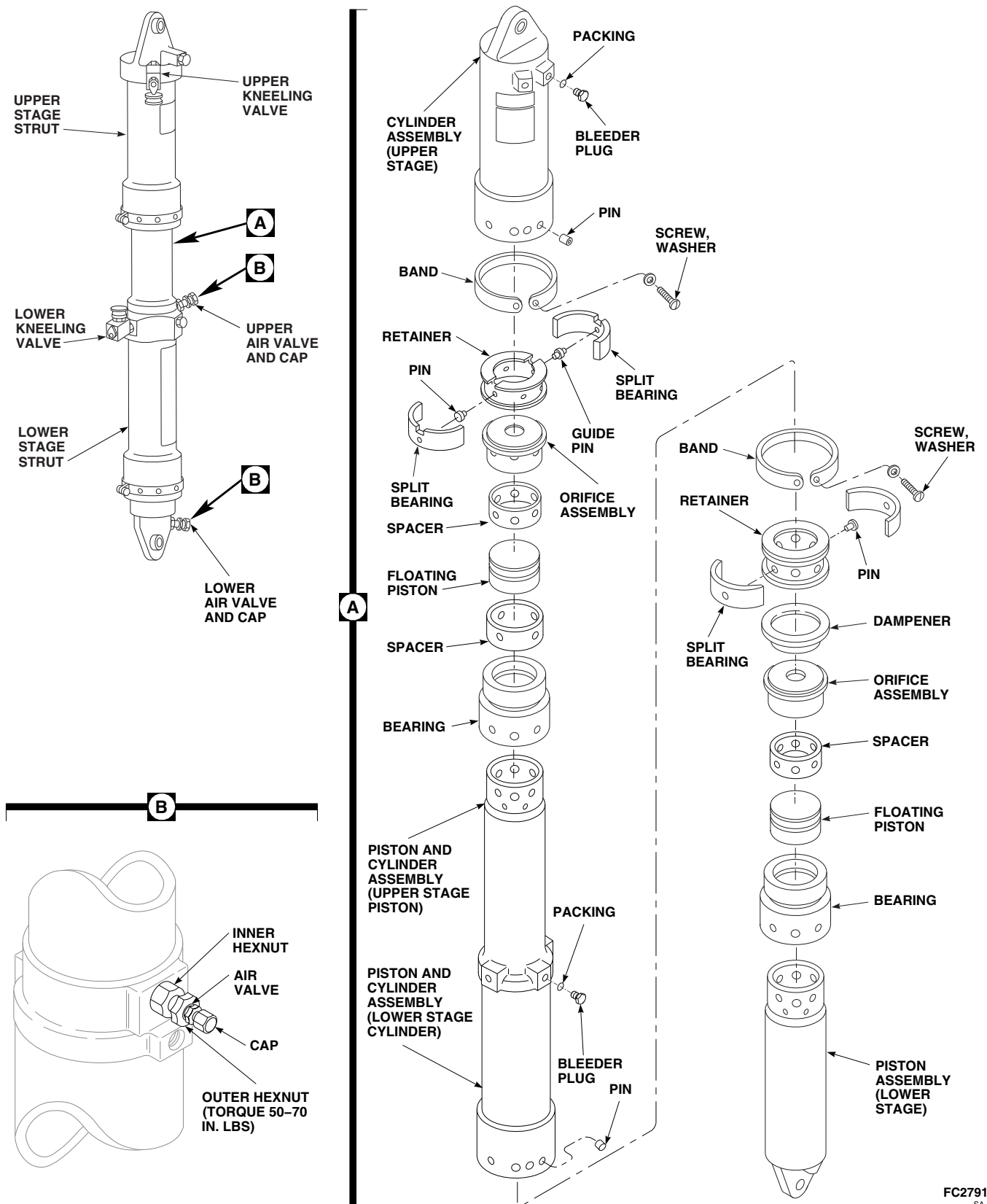


FIGURE 3-5-8-1. DISASSEMBLE TAIL LANDING GEAR SHOCK STRUT (BENCH)

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate air pump at low pressure when removing floating piston.

- h. Attach manual air pump to lower air valve. Apply 2 - 5 psi pressure. Catch floating piston in a utility pail filled with machinery towels, Item 211, Appendix D. Remove manual air pump.
- i. Remove lower stage air valve (PARA 3-4-24).
- j. Remove screws, washers, and band from piston and cylinder assembly.
- k. Using screw, MS35275-247, and mechanical puller kit, remove pins.

NOTE

Split bearing halves may fall free as piston clears cylinder.

- l. Separate piston and cylinder assembly from upper stage cylinder assembly. Catch split bearing halves as piston is removed from cylinder.
- m. Remove pins, retainer, orifice assembly, spacer, and bearing. Do not disassemble orifice.

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate air pump at low pressure when removing floating piston.

- n. Attach manual air pump to upper air valve. Apply 2 - 5 psi pressure. Catch floating piston in a utility pail filled with machinery towels, Item 211, Appendix D. Remove manual air pump.

- o. Remove upper air valve (PARA 3-4-24).
- p. Using plastic or wooden pick, remove packings, seals, and scrapers. Discard used packings, seals, and scrapers.

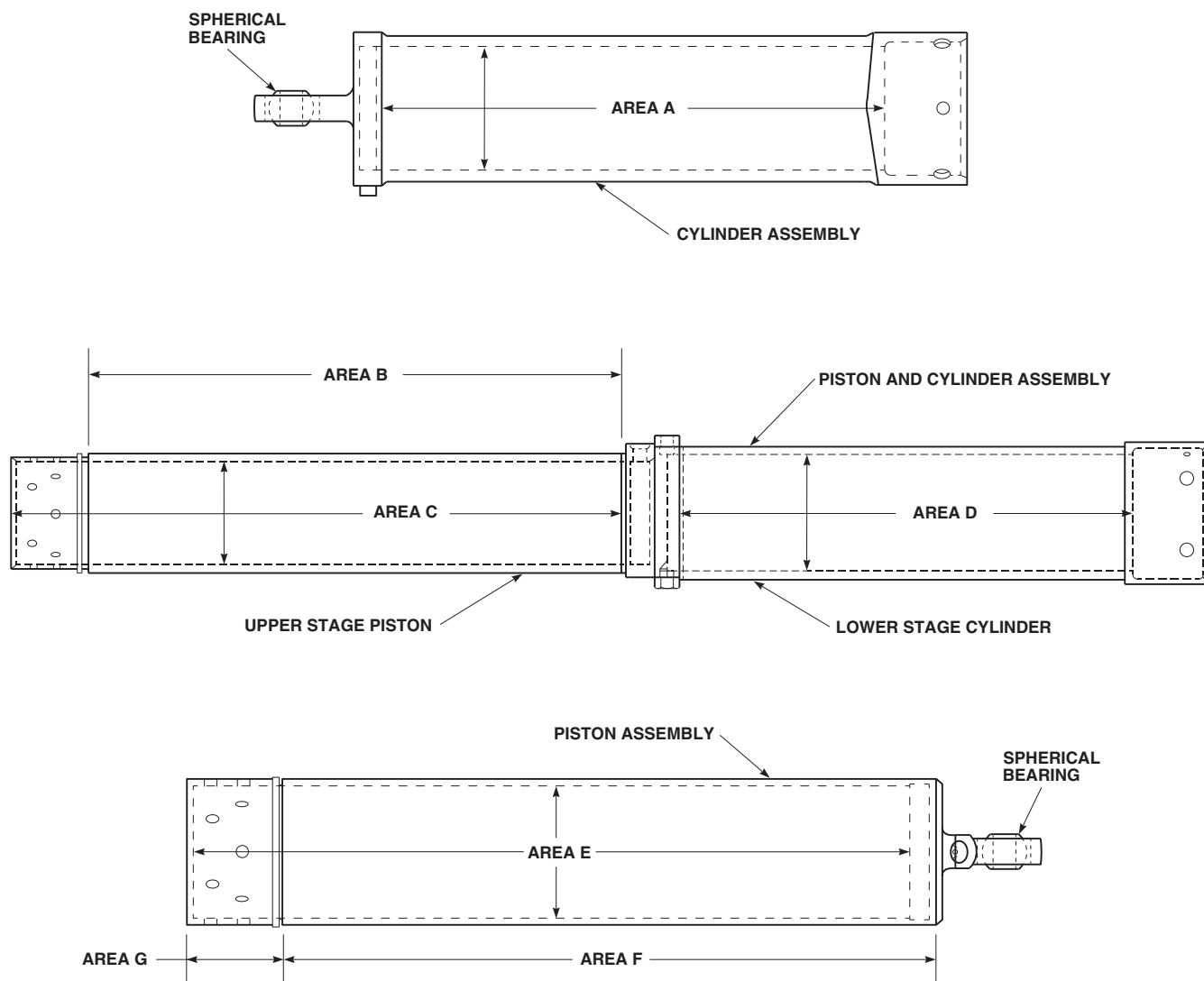
WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- q. Clean parts with dry-cleaning solvent, Item 124, Appendix D. Blow dry with dry compressed air.

3-5-8.1.2 Inspect.

- a. Inspect strut parts for cracks, scoring, scratches, and corrosion. Use metal flaw detection procedures as necessary to verify suspected cracks or fractures (ASTM E1444). **NO SCORING, CRACKS, OR CORROSION ALLOWED.** If damage is found, replace part as necessary.
- b. Inspect piston assembly, piston and cylinder assembly, and cylinder assembly chrome-plated surfaces for wear (Figure 3-5-8-2). Take measurements along several points, 90° apart as follows:
 - (1) **AREA A - CYLINDER INSIDE DIAMETER MUST NOT EXCEED 3.503-INCHES.** If out-of-tolerance, replace cylinder assembly.
 - (2) **AREA B - PISTON OUTSIDE DIAMETER MUST NOT BE LESS THAN 2.9935 -INCHES.** If out-of-tolerance, replace piston and cylinder assembly.
 - (3) **AREA C - PISTON BORE INSIDE DIAMETER MUST NOT EXCEED 2.813-**



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FIGURE 3-5-8-2. INSPECT CYLINDER, PISTON AND CYLINDER, AND PISTON ASSEMBLIES (BENCH)

INCHES. If out-of-tolerance, replace piston and cylinder assembly.

- (4) **AREA D - CYLINDER INSIDE DIAMETER MUST NOT EXCEED 3.503-INCHES.** If out-of-tolerance, replace piston and cylinder assembly.
- (5) **AREA E - PISTON BORE INSIDE DIAMETER MUST NOT EXCEED 2.812-INCHES.** If out-of-tolerance, replace piston assembly.

- (6) **AREA F - LOWER STAGE PISTON OUTSIDE DIAMETER MUST NOT BE LESS THAN 2.995-INCHES.** If out-of-tolerance, replace piston.

- (7) **AREA G - LOWER STAGE PISTON PIN ORIFICE BORE DIAMETER MUST NOT EXCEED 0.4385-INCHES.** If out-of-tolerance, replace piston.

- c. Inspect split bearing halves for wear at several points around circumference. **SPLIT BEARING THICKNESS MUST NOT BE LESS**

THAN 0.063-INCHES (Figure 3-5-8-3, Detail A). If out-of-tolerance, replace bearings.

d. Inspect retainers for wear at several points around outside diameter. **RETAINER MUST NOT BE LESS THAN 3.486-INCHES** (Figure 3-5-8-3, Detail B). If out-of-tolerance, replace retainer.

e. Inspect bearings for wear at several points around inside diameter. **BEARING INSIDE DIAMETER MUST NOT EXCEED 3.0035-INCHES** (Figure 3-5-8-3, Detail C). If out-of-tolerance, replace bearings.

f. Inspect floating pistons for wear at several points around outside diameter. **FLOATING PISTON MUST NOT BE LESS THAN 2.8035-INCHES** (Figure 3-5-8-3, Detail D). If out-of-tolerance, replace floating pistons.

g. Inspect piston assembly and cylinder assembly spherical bearings for radial and axial play. **BEARINGS MAY HAVE 0.010-INCHES MAXIMUM RADIAL PLAY AND 0.030-INCHES MAXIMUM AXIAL PLAY.** If out-of-tolerance, replace bearing (PARA 3-5-8.2).

h. Inspect Teflon liners for disbonding. **NONE ALLOWED.** If liner is disbonded, replace bearing (PARA 3-5-8.2).

3-5-8.1.3 Assemble.

a. Make sure all parts are clean.

NOTE

- If strut is to be stored, use hydraulic fluid, Item 155, Appendix D, as an assembly and servicing fluid.
- If strut is scheduled for immediate return to service, use hydraulic fluid, Item 154, Appendix D, as an assembly and servicing fluid.
- If replacing lower stage piston assembly, 70250-13167-041 (part of shock strut, 70250-13101-043), replace shock strut with shock strut, 70250-13101-044.

b. Soak packings, retainers, G-T rings, and seals in hydraulic fluid, Item 154 or Item 155, Appendix D, before assembly.

c. Coat chrome-plated sliding and threaded parts with hydraulic fluid, Item 154 or Item 155, Appendix D, before assembly.

d. Install G-T rings on upper and lower stage floating pistons, with radius of Teflon rings facing center packing (Figure 3-5-8-4, Detail A).

e. Install G-T rings inside upper and lower bearings (Figure 3-5-8-4, Detail D).

f. Install seals on outside diameter of upper and lower bearings, with springs facing tops of sleeves (Figure 3-5-8-4, Detail C).

g. Install scrapers, inside lower edge of sleeve bearings (Figure 3-5-8-4, Detail B).

h. Install spacer in upper stage piston with holes closest to spacer edge toward bottom. Line up one spacer hole with air valve port (Figure 3-5-8-5, Sheet 1, Detail B).

i. Install floating pistons, open ends down, in upper and lower stage piston bores. Push pistons to bottom of bore (Figure 3-5-8-5, Sheet 1, Detail A, and Figure 3-5-8-6, Detail A).

j. Slip bearing assemblies, scraper ends first, over upper and lower stage pistons.

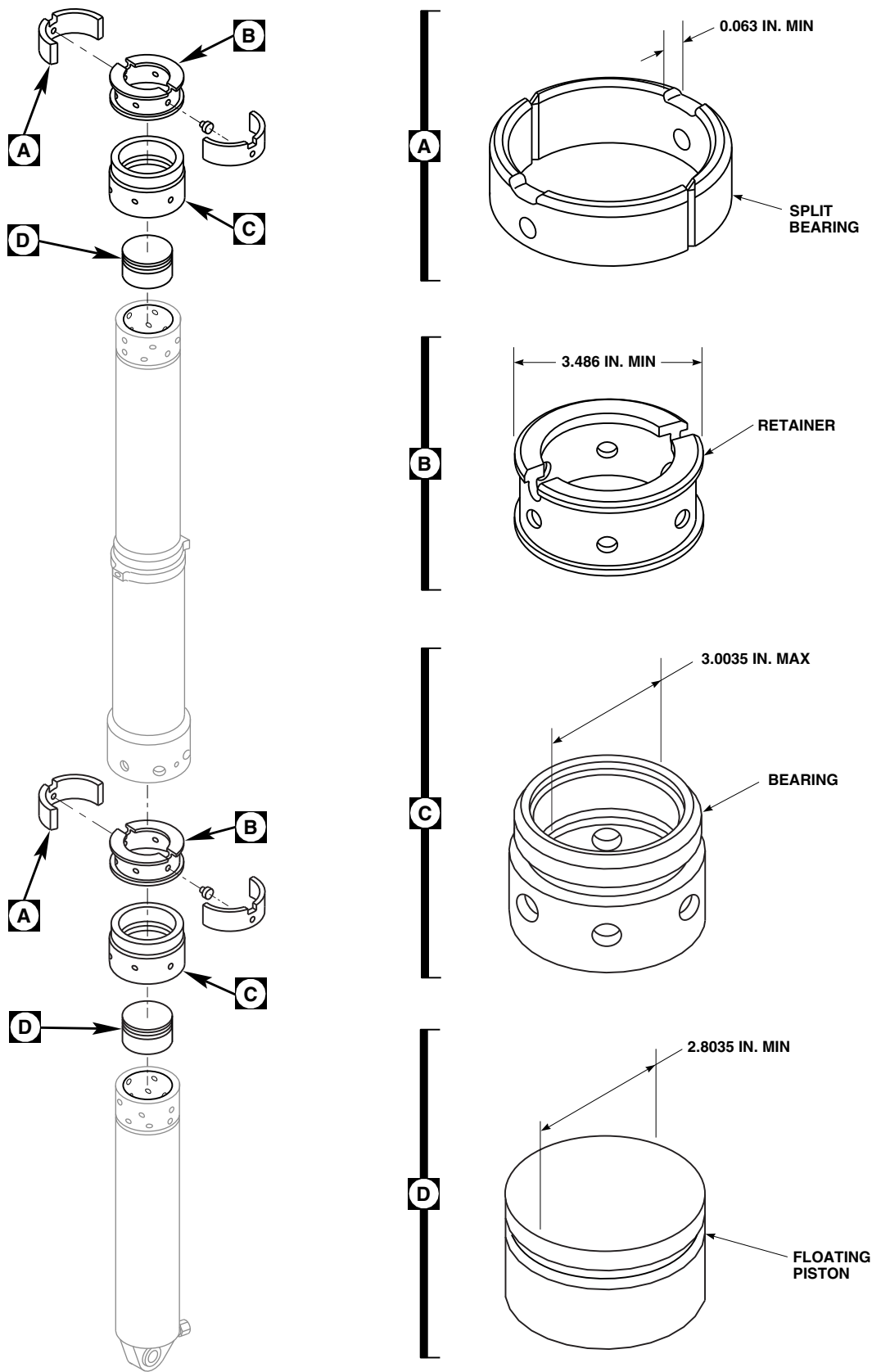
NOTE

Do not allow spacer to fall into piston assembly bore.

k. Install spacer in upper stage piston. Line up holes in spacer with holes at top of upper stage piston.

l. Insert upper stage orifice assembly in piston (Figure 3-5-8-5, Sheet 1, Detail A).

m. Install upper stage retainer. Line up slots with kneeling and air valve ports. Line up guide pin through retainer and piston wall in line with cylinder assembly notch, kneeling, and air valve ports (Figure 3-5-8-5, Sheet 2, Detail C).



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FIGURE 3-5-8-3. INSPECT BEARINGS, RETAINER, AND FLOATING PISTON (BENCH)

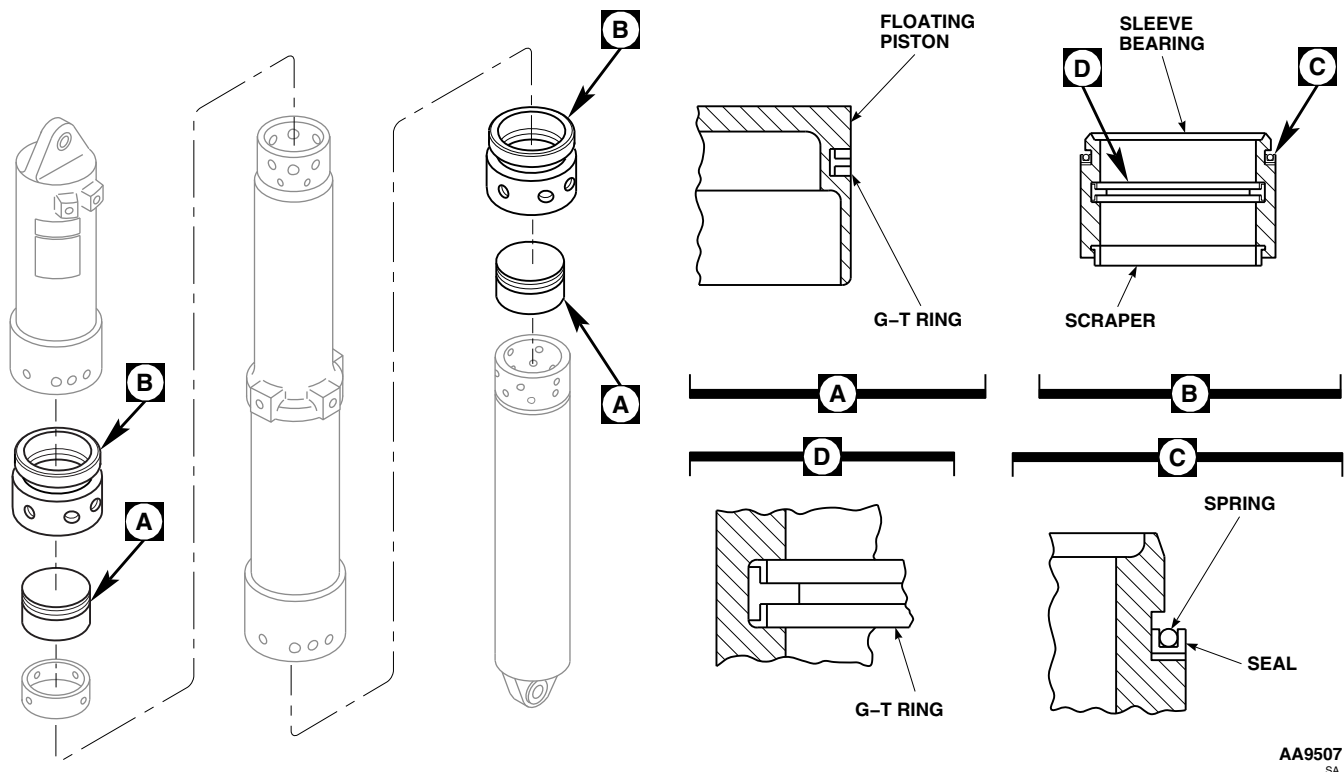
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FIGURE 3-5-8-4. ASSEMBLE FLOATING PISTON/SLEEVE BEARING (BENCH)

NOTE

Installation of guide pin in any other pin hole will prevent further strut assembly.

- n. Install remaining orifice assembly retaining pins.
- o. Install split bearing halves on retainer. Line up upper stage cylinder groove with studded pin. Slide cylinder onto piston.
- p. Slide bearing into cylinder. Line up bearing retaining pin holes and install pins. Seat retaining pins with rawhide mallet as required.
- q. Install band over bearing pins and secure with screws and washers. Using lockwire, Item 196, Appendix D, lockwire screws.
- r. Slide bearing assembly, scraper end first, onto lower piston (Figure 3-5-8-6, Detail A).
- s. Install dampener on piston assembly, with dampener flange facing open end of piston.

NOTE

Do not allow spacer to fall into piston assembly bore.

- t. Install spacer in lower stage piston. Line up holes in spacer with holes at top of lower stage piston.
- u. Insert lower stage orifice assembly into piston.
- v. Install retainer. Line up holes and install orifice assembly retaining pins.
- w. Install split bearing halves on retainer. Slide piston into cylinder.
- x. Slide bearing into cylinder. Line up bearing retaining pin holes. Install pins. Seat retaining pins with rawhide mallet as required.
- y. Install band over bearing pins and secure with screws and washers. Using lockwire, Item 196, Appendix D, lockwire screws.
- z. Test shock strut (PARA 3-5-8.1.4).

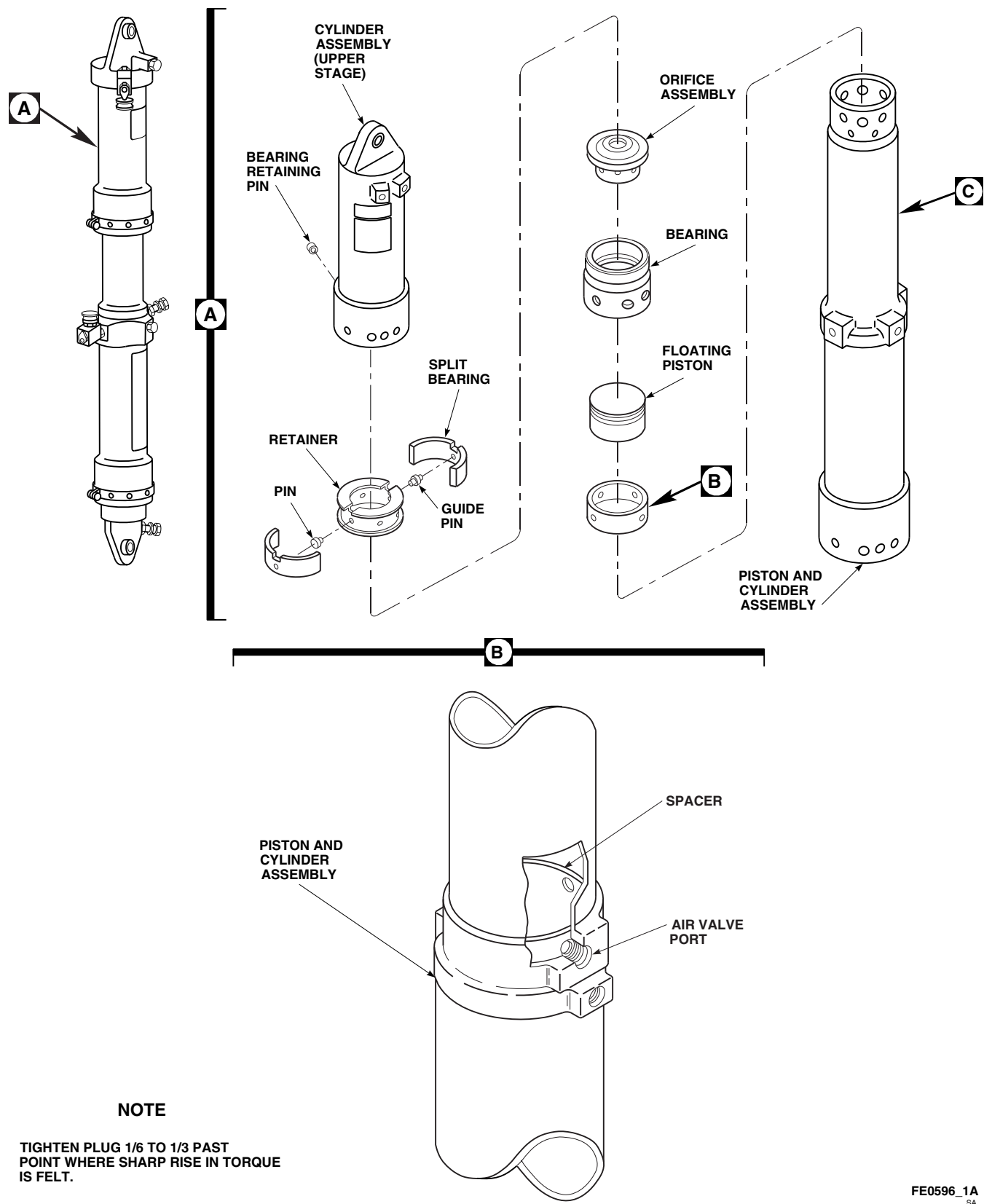


FIGURE 3-5-8-5. ASSEMBLE UPPER STAGE (BENCH) (SHEET 1 OF 2)

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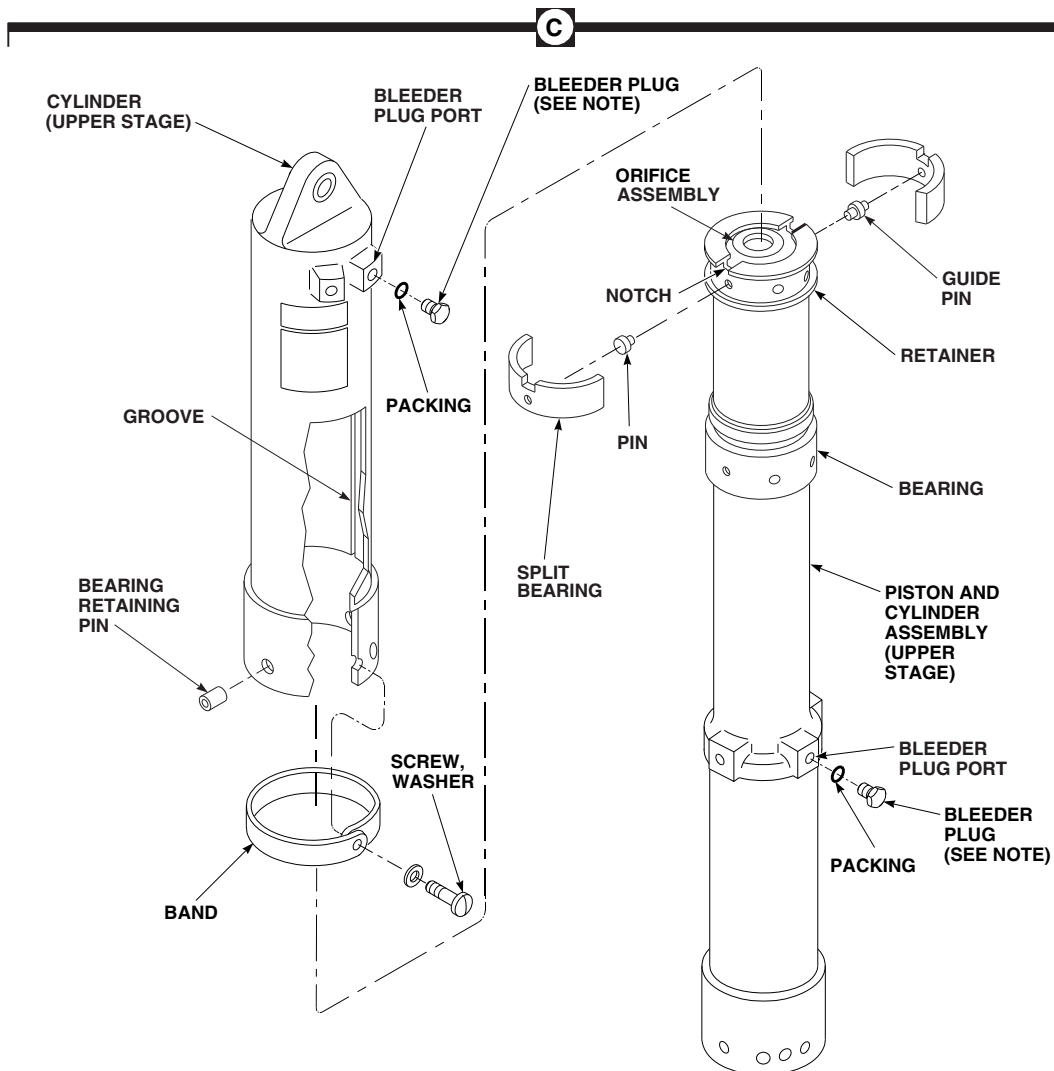
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FIGURE 3-5-8-5. ASSEMBLE UPPER STAGE (BENCH) (SHEET 2 OF 2)

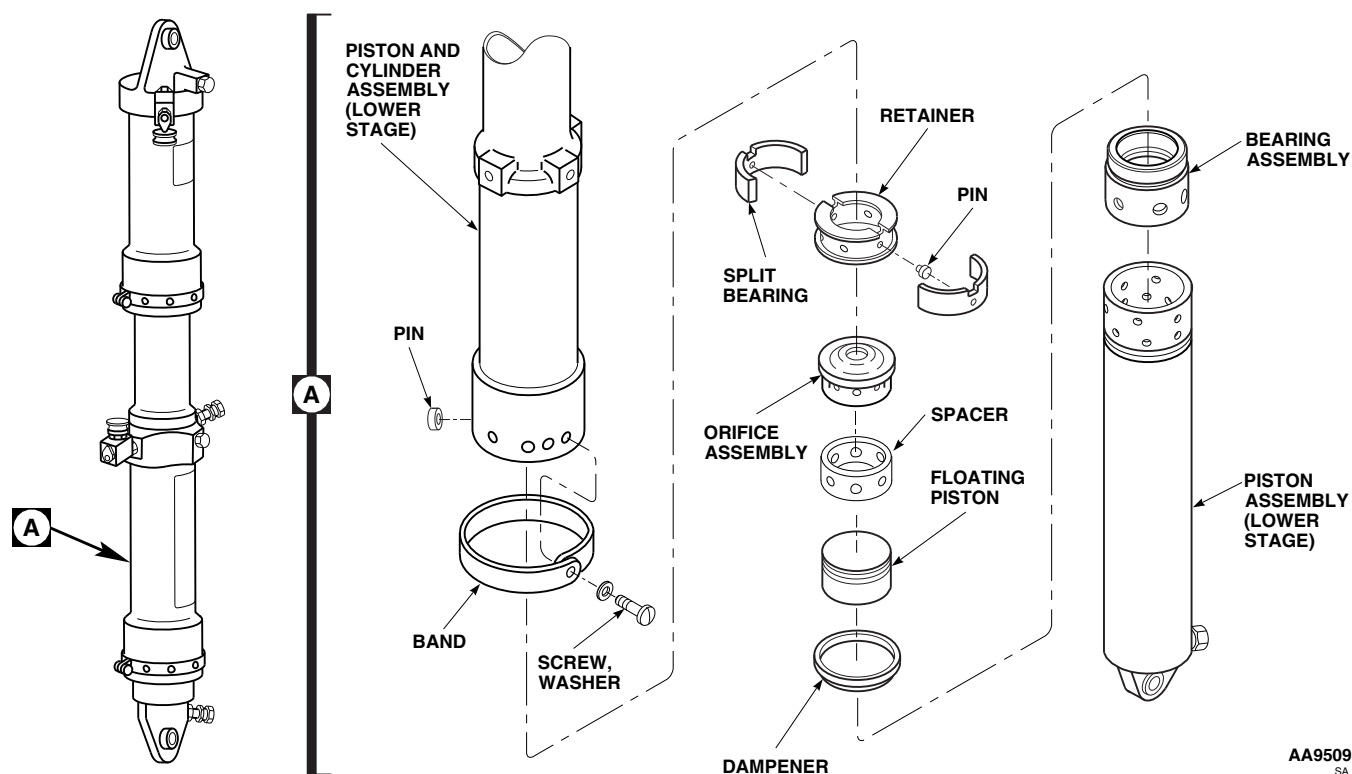


FIGURE 3-5-8-6. ASSEMBLE LOWER STAGE (BENCH)

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3-5-8.1.4 Test.

- a. Install upper and lower kneeling valves (PARA 3-4-23).
- b. Remove upper stage coupling dust cover. Install coupling half and drain hose on kneeling valve.
- c. Route drain hose into a 1-gallon container.

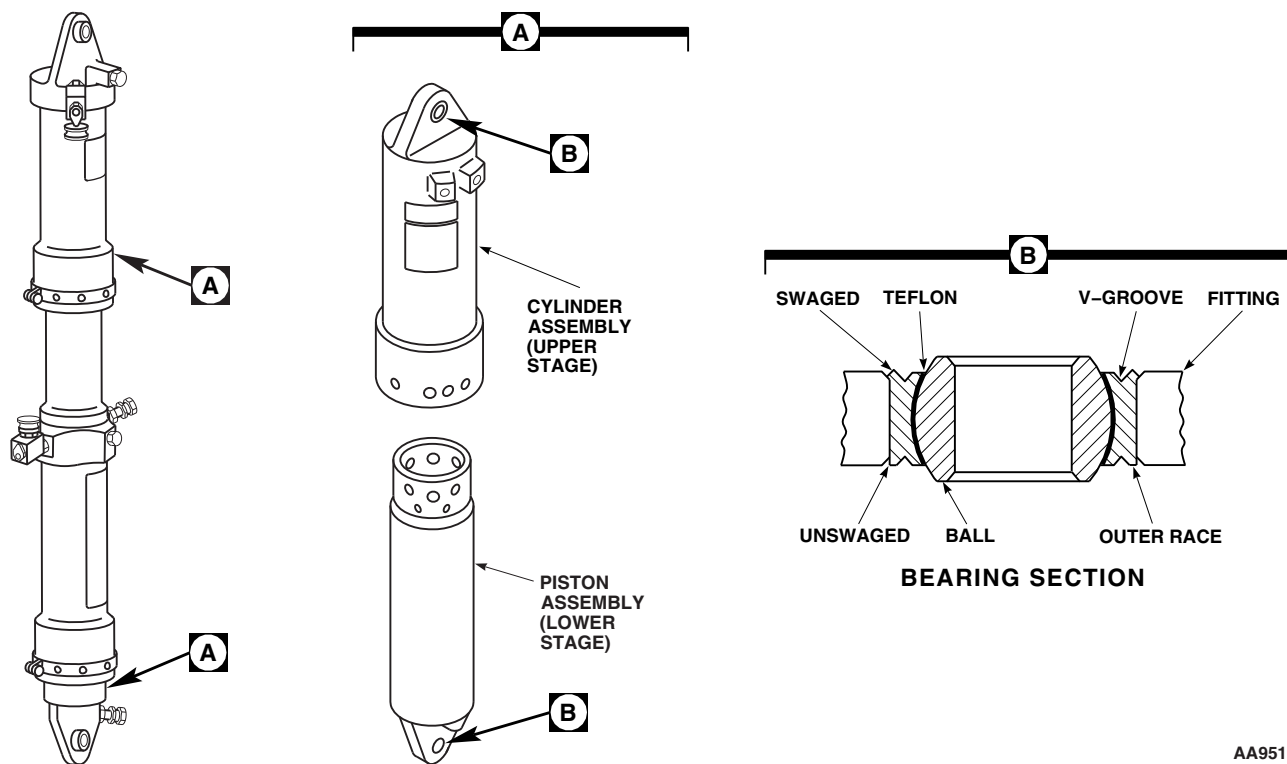
CAUTION

Damage to strut will result if clearance does not exist prior to testing and servicing. Testing and servicing will extend strut to its maximum length of 39.54-inches. Make sure enough clearance exists above strut.

- d. Connect hydraulic shutoff valve and pressure line from hydraulic test stand to upper stage bleeder port. Start test stand. Open pressure line valve.

- e. Slowly apply hydraulic pressure to a maximum of 100 psig, at minimum flow, to fill upper stage strut and to bottom floating piston. Slowly close kneeling valve after fluid has flowed from drain line for 20 - 30 seconds.
- f. Reduce hydraulic pressure to 0 psi.
- g. Apply minimum test stand pressure and flow to strut. Open kneeling valve and maintain fluid flow until a steady air-free stream is obtained.
- h. Close kneeling valve. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi.
- i. Allow a few minutes for trapped air to rise to top of strut. Open kneeling valve and slowly compress strut fully. Close kneeling valve.
- j. Slowly apply hydraulic pressure to a maximum of 100 psig, at minimum flow, to extend and fill upper stage strut.
- k. Reduce hydraulic pressure and flow to minimum. Open kneeling valve. Inspect draining fluid for air.

- l. Repeat step e. through step k. until hydraulic fluid is completely air-free.
- m. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi. If extended, compress upper stage strut.
- n. Close kneeling valve.
- o. Disconnect hydraulic line at bleeder port.
- p. Lubricate packing with petrolatum, Item 234, Appendix D.
- q. Install packing on bleeder plug. Install bleeder plug in upper stage bleeder port. **TIGHTEN BLEEDER PLUG $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- r. Disconnect drain line from upper kneeling valve. Install valve dust cap. Using lockwire, Item 199, Appendix D, lockwire kneeling valve knob.
- s. Connect drain line to lower kneeling valve.
- t. Connect pressure line to lower stage bleeder port.
- u. Using step b. through step t. for upper stage, bleed strut lower stage.
- v. Install upper and lower stage air valves (PARA 3-4-24).
- w. Connect pressure regulated nitrogen, Item 221, Appendix D, to upper stage air valve. Slowly pressurize upper stage to 900 pounds line pressure on shock strut, 70250-13101-041, or 1025 pounds line pressure on shock struts, 70250-13101-042 or 70250-13101-043. Close air valve, allow pressure to stabilize for 10 to 15 minutes, and check for leaks.
- x. Connect nitrogen to lower stage air valve and slowly pressurize lower stage to 275 pounds line pressure. Close air valve, allow pressure to stabilize for 10 to 15 minutes, and check for leaks. **TORQUE UPPER AND LOWER AIR VALVE OUTER HEXNUTS TO 50 - 70 INCH-POUNDS.**
- y. Using leak detection fluid, Item 189, Appendix D, check upper and lower stage air valves for leakage. **NONE ALLOWED.** If leakage is found, replace leaking air valve (PARA 3-4-24).
- z. Position strut assembly at a 45° angle and observe for a 24-hour period. **NO AIR LEAKAGE ALLOWED AFTER ONE HOUR FROM UPPER AND LOWER STAGES. MAXIMUM HYDRAULIC FLUID LEAKAGE AFTER 24 HOURS MUST NOT EXCEED 2 DROPS PER STAGE.** If limits are exceeded, repair or replace strut.
- aa. Remove air lines, open upper and lower air shutoff valves, and reduce pressure to 0 psi.
- ab. Make sure upper and lower kneeling valves are closed. If installed, remove coupling from upper and lower stage kneeling valves.
- ac. Connect hydraulic drain lines to upper and lower stage kneeling valves and vent lines to drain tray.
- ad. Open upper and lower stage kneeling valves and completely compress strut assembly. Close upper and lower stage kneeling valves.
- ae. Remove all hydraulic lines. Install coupling upper and lower stage kneeling valves. Install coupling dust cap.
- af. Install air valve dust caps. Using lockwire, Item 197, Appendix D, lockwire lower hex-nuts.
- ag. Using lockwire, Item 197, Appendix D, lockwire upper and lower stage bleed plugs.



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FIGURE 3-5-8-7. REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (BENCH)

3-5-8.2 Replace Tail Landing Gear Shock Strut Spherical Bearings (BENCH).

3-5-8.2.1 Remove.

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

- Clamp fitting to baseplate of drill press. Using fly cutter and pilot, machine off bearing outer race swaged over chamfer on fitting (Figure 3-5-8-7, Detail B).
- Using arbor press with anvil and driver, press out bearing.

3-5-8.2.2 Repair.

- Using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, clean primer residue from bore and chamfer.
- Inspect bearing bore lugs for galling or scoring. Using abrasive cloth, Item 10, Appendix D, blend out galling or scoring in bearing bore lugs. **GALLING OR SCORING NO DEEPER THAN 0.010-INCHES MAY BE BLENDED PROVIDED NOT MORE THAN 25% OF CONTACT AREA IS AFFECTED.**
- Magnetic particle inspect bearing bore lugs (ASTM E1444). If crack is found, replace tail landing gear shock strut (PARA 3-4-22).

3-5-8.2.3 Install.

- Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.

- b. Apply light coat of primer, Item 257, Appendix D, to inside of fitting bore and to outside surface of outer race of bearing.
- c. Using arbor press with anvil and driver, press bearing into fitting, (Figure 3-5-8-7, Detail B).
- d. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer, if any, from bearing V-grooves. Wipe dry.
- e. Fix fitting with bearing on drill press base plate, with proper spinning tool and anvil.

CAUTION

Do not allow petrolatum to get on the ball of bearing, or on the exposed edge of the fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

- f. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 234, Appendix D.

CAUTION

Do not allow spinning tool to make contact with ball side of groove as binding may result.

- g. Swage V-groove lip over chamfer in fitting bore.
- h. Turn fitting over and repeat swaging process on other side.
- i. Clean petrolatum from V-groove with cheesecloth, Item 90, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D.
- j. Touch up bearing surfaces with corrosion preventive compound, Item 110, Appendix D.
- k. Touch up paint (PARA 2-6-5).

3-5-9 TAIL LANDING GEAR YOKE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-9.1	Replace Yoke Attachment Bushings (BENCH)	3-5-9-2	3-5-9.3	Replace Inserts, Studs, and Lockrings (BENCH)	3-5-9-4
3-5-9.2	Replace Fuselage/Yoke Fitting Bushings (BENCH)	3-5-9-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press (or equivalent)
- Bushing Installation Draw Bolt, Locally-Made, consisting of:
 - Bolt, NAS6205-11
 - Nut, MS17826-5
 - Phenolic Washer
 - Washer, MS20002-5
 - Washer, MS20002-9
- Fluorescent Penetrant Inspection Kit
- Sand Bags (or similar protective surface)
- Slide Hammer
- Stud Removal/Insert Kit

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Mat, Item 14, Appendix D

- Brush, Paint, Item 86, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 248, Appendix D (or equivalent)

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 3-4-25

3-5-9.1 Replace Yoke Attachment Bushings (BENCH).

3-5-9.1.1 Remove.

- a. Place yoke assembly on sand bags or similar protective surface.
- b. Using slide hammer, remove yoke attachment bushings (Figure 3-5-9-1, Detail A). Discard old bushings.
- c. Using abrasive mat, Item 14, Appendix D, and methyl ethyl ketone, Item 217, Appendix D, remove old primer from yoke attachment bushing holes. Wipe clean and dry with cheesecloth, Item 90, Appendix D.

3-5-9.1.2 Install.

- a. Inspect yoke attachment bushing bores as follows:
 - (1) Inspect bores for scoring and corrosion. **DAMAGE TO BORES MUST NOT BE DEEPER THAN 0.010-INCH.** If damage to bores exceeds limit, replace yoke.
 - (2) Using abrasive cloth, Item 12, Appendix D, blend out any damage. **BLENDED AREA MUST NOT EXCEED 25% OF BORE AREA AFTER REPAIRS.** If blended area exceeds limit after repairs, replace yoke.
 - (3) Measure diameter of bores. **BORE DIAMETER MUST NOT EXCEED 0.8133-INCHES IN UNBLENDED AREAS.** If bore diameter exceeds limit in unblended areas, replace yoke.
 - (4) Using fluorescent penetrant inspection kit, inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace yoke (PARA 3-4-25).
 - (5) Using corrosion resistant coating, Item 118, Appendix D, touch up reworked surfaces.
- b. Using epoxy primer coating kit, Item 135, Appendix D, and paint brush, Item 86, Appendix

D, coat yoke attachment bushing bore and bushing outside surface.

- c. Using arbor press or equivalent, press yoke attachment bushing halves into bore while primer is wet (Figure 3-5-9-1, Detail A).
- d. Touch up paint as required (PARA 2-6-5).

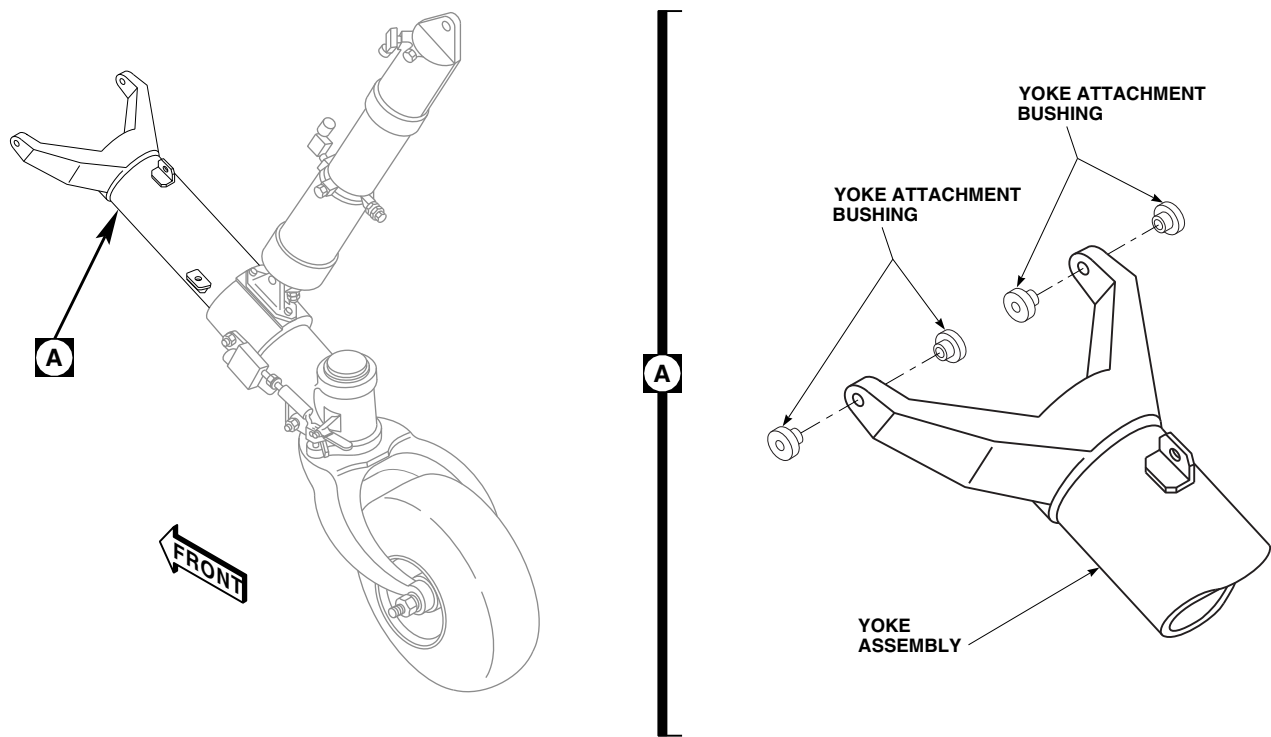
3-5-9.2 Replace Fuselage/Yoke Fitting Bushings (BENCH).

3-5-9.2.1 Remove.

- a. Using slide hammer, remove fuselage/yoke fitting bushing halves. Discard old bushings (Figure 3-5-9-2, Detail A).
- b. Using methyl ethyl ketone, Item 217, Appendix D, and abrasive mat, Item 14, Appendix D, remove old primer from fuselage fitting holes. Wipe clean and dry with cheesecloth, Item 90, Appendix D.

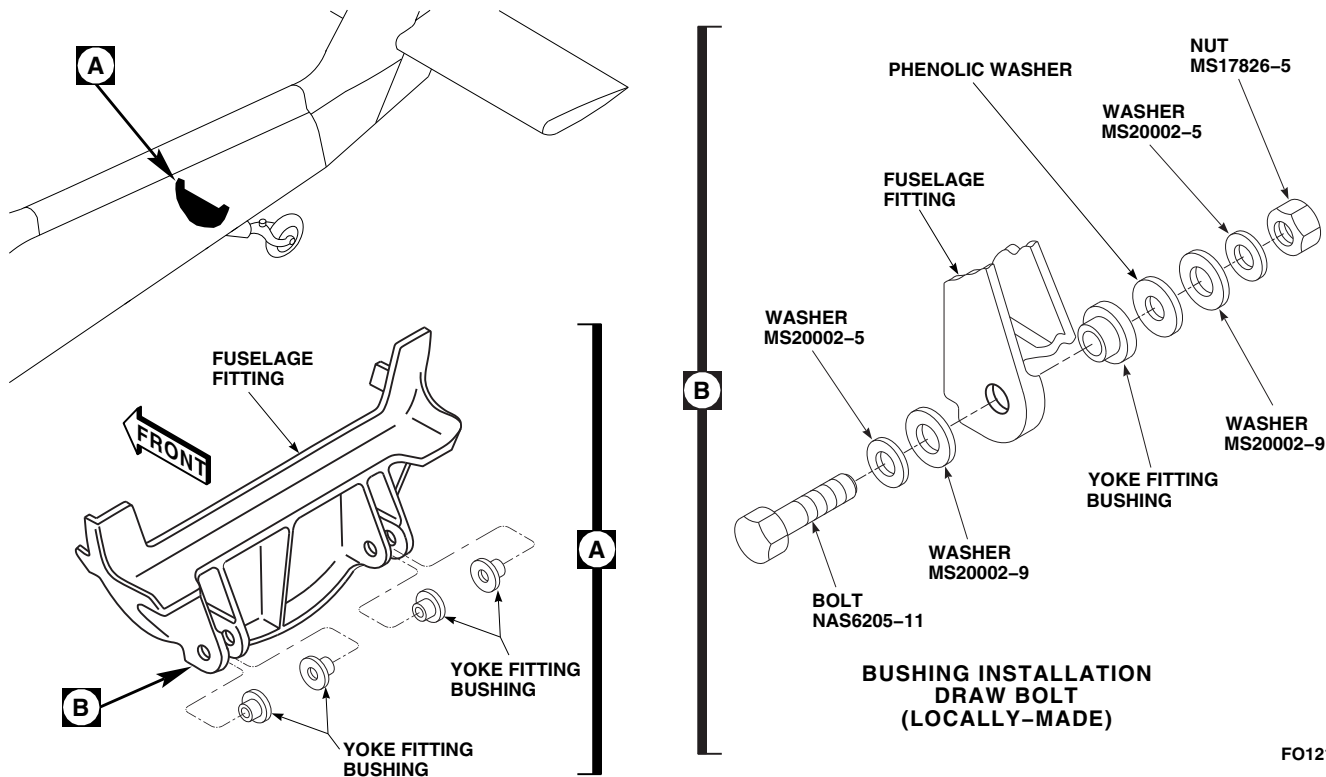
3-5-9.2.2 Install.

- a. Using pressure sensitive adhesive tape, Item 248, Appendix D, or equivalent, wrap threads of draw bolt, NAS6205-11 (Figure 3-5-9-2, Detail B).
- b. Using epoxy primer coating kit, Item 135, Appendix D, and paint brush, Item 86, Appendix D, coat bushing bore and outside surface of replacement yoke fitting bushing.
- c. With flange of yoke fitting bushing inside fuselage fitting, press yoke fitting bushing into hole using bolt, NAS6205-11, washers, MS20002-9 and MS20002-5, phenolic washer, and nut, MS17826-5, until bushing flange is bottomed on fitting.
- d. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe excess primer out of bushing bore.
- e. Remove bolt, washers, phenolic washer, and nut from fuselage fitting.
- f. Touch up paint as required (PARA 2-6-5).



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FIGURE 3-5-9-1. REPLACE YOKE ATTACHMENT BUSHINGS (BENCH)



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FIGURE 3-5-9-2. REPLACE FUSELAGE/YOKE FITTING BUSHINGS (BENCH)

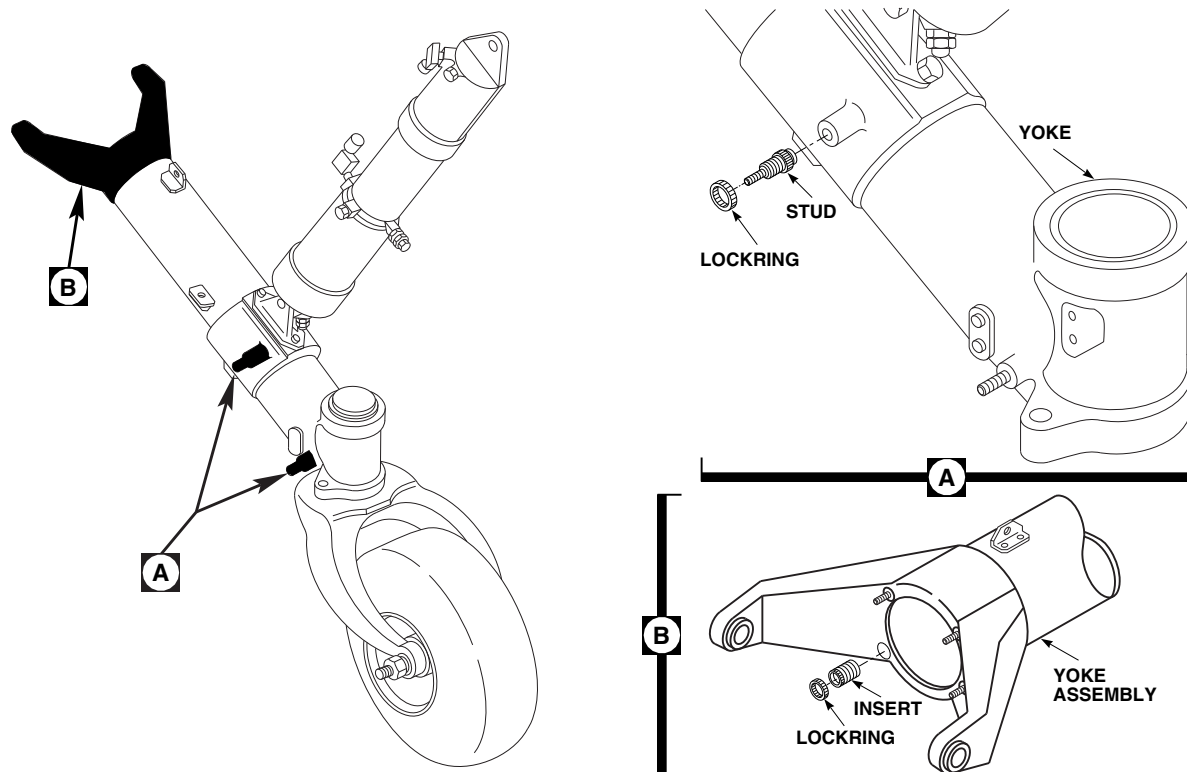
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FIGURE 3-5-9-3. REPLACE INSERTS, STUDS, AND LOCKRINGS (BENCH)

3-5-9.3 Replace Inserts, Studs, and Lockrings (BENCH).

3-5-9.3.1 Remove.

Remove lockring, stud, or insert from tail landing gear yoke using stud removal/insert kit (Figure 3-5-9-3, Detail A and Detail B).

3-5-9.3.2 Install.

Install lockring, stud, or insert in tail landing gear yoke using stud removal/insert kit (Figure 3-5-9-3, Detail A and Detail B).

3-5-10 TAIL LANDING GEAR FORK (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
3-5-10.1 Repair Fork Assembly Lockpin Bushing Hole (BENCH)	3-5-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Hand Reamer Set, Adjustable

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D

- Tail Landing Gear Lockpin Oversize Bushing, Locally-Made, Figure H-221, Appendix H

References

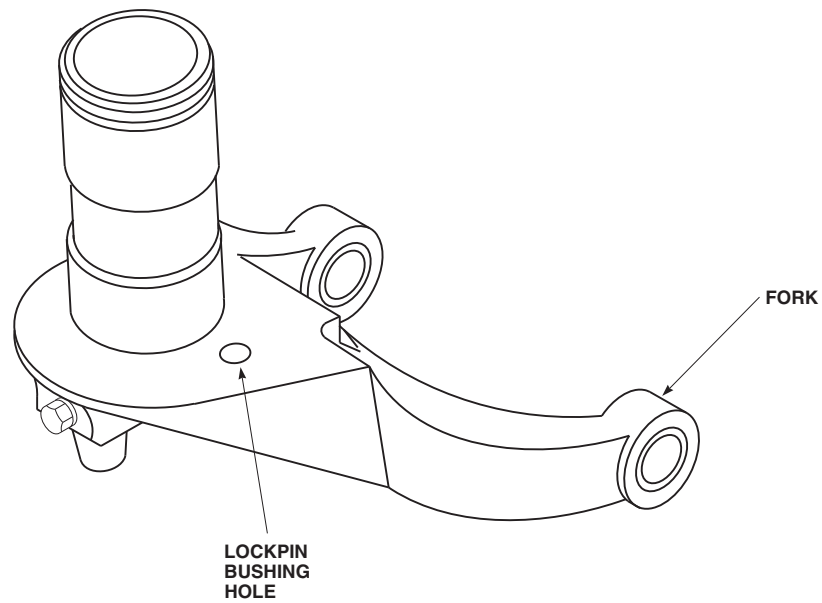
- Appendix D
- Appendix H

3-5-10.1 Repair Fork Assembly Lockpin Bushing Hole (BENCH).

3-5-10.1.1 Repair.

- a. Locally make tail landing gear lockpin oversize bushing (Figure H-221, Appendix H).
- b. Using adjustable hand reamer set, clean up hole in tail wheel fork by reaming hole to 0.870 - 0.875 inch-diameter (Figure 3-5-10-1).

- c. Using epoxy primer coating kit, Item 135, Appendix D, coat lockpin bushing hole.
- d. Using arbor press, install locally-made bushing while primer is still wet.

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3-5-11 TAIL LANDING GEAR WHEEL (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
3-5-11.1	Repair Tail Landing Gear Wheel (BENCH)	3-5-11-1	3-5-11.2	Replace Wheel Bearing Cup (BENCH)	3-5-11-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Mallet
- Nonmetallic Scraper, Locally-Made
- Oven
- Phenolic Drift Pin
- Protection, Eye

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Paint, Polyurethane, Item 222A, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- ASTM E1417
- ASTM E1444
- PARA 2-6-5
- PARA 3-4-28

3-5-11.1 Repair Tail Landing Gear Wheel (BENCH).

3-5-11.1.1 Disassemble.

NOTE

Inspect and repair wheel whenever wheel is disassembled.

Remove tire from tail landing gear wheel (PARA 3-4-28).

3-5-11.1.2 Inspect/Repair.

- Using paint remover, Item 230, Appendix D, and nonmetallic scraper, remove paint from wheel halves.

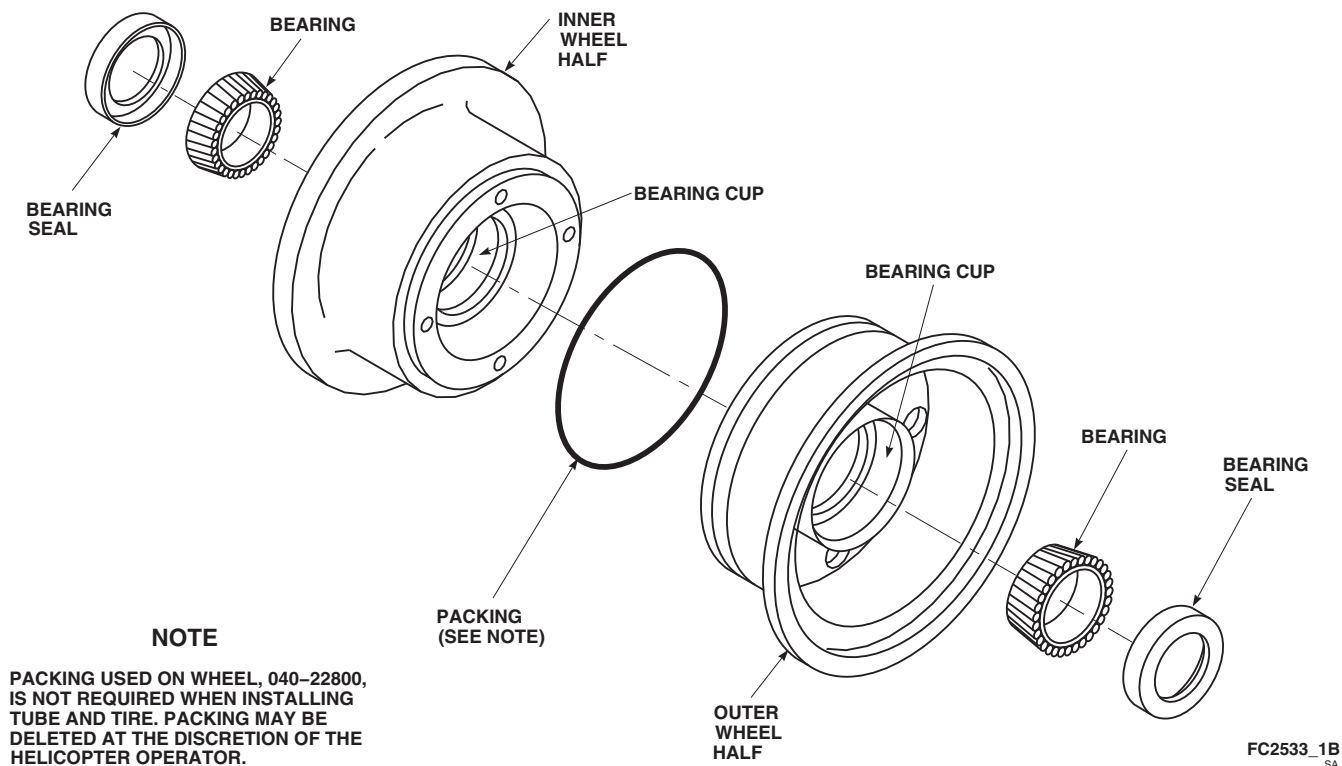
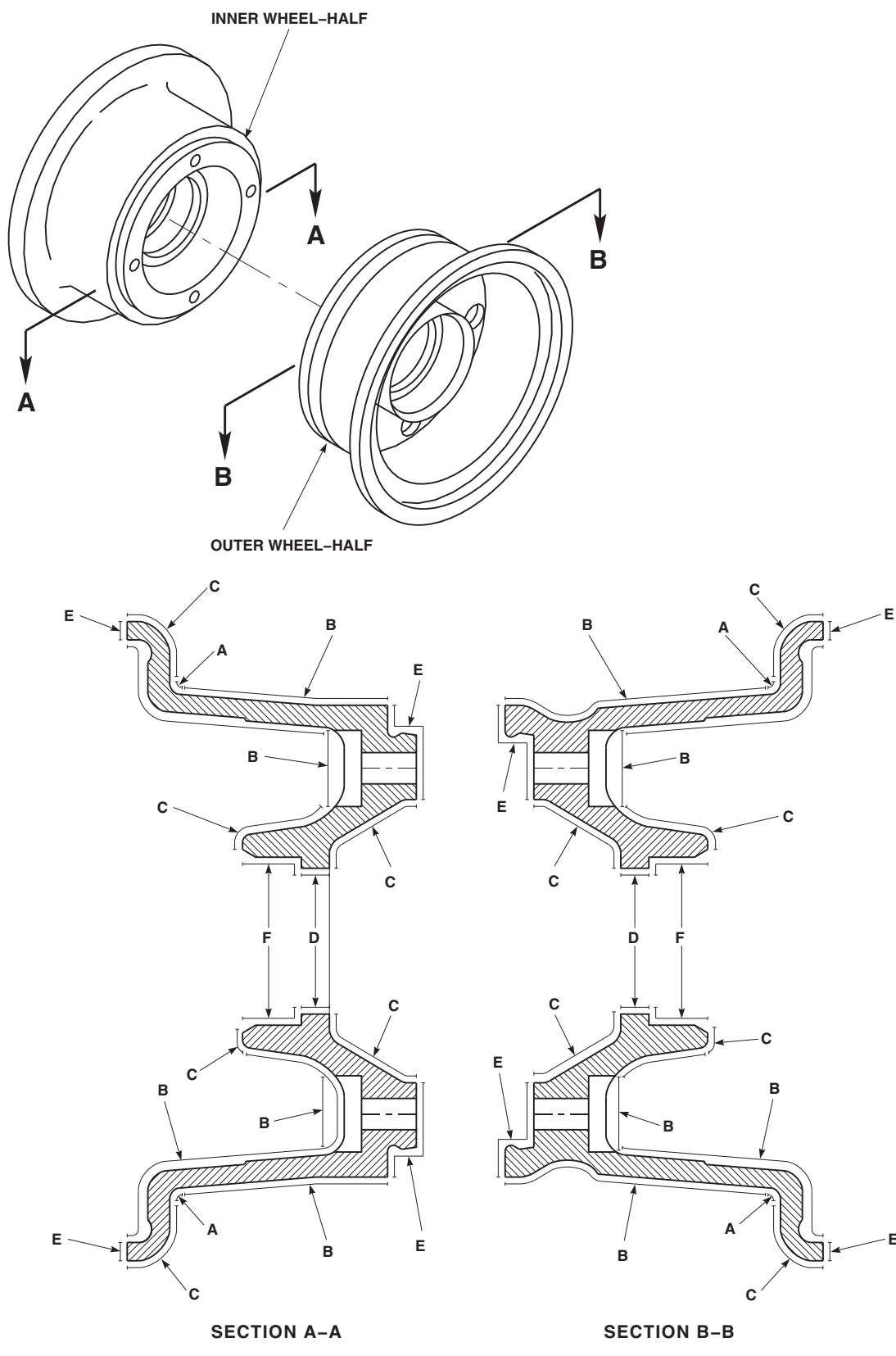


FIGURE 3-5-11-1. REPAIR TAIL LANDING GEAR WHEEL (BENCH) (SHEET 1 OF 2)

- b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect wheel halves for cracks and structural damage (ASTM E1417). **NONE ALLOWED.** Pay particular attention to bead seat, bolt boss, male/female mating surfaces, and valve hole areas. If crack is found, replace wheel.
- c. Inspect wheel halves for scratches, nicks, or pits (Figure 3-5-11-1, Sheet 2), and repair using abrasive cloth, Item 8, Appendix D, and abrasive cloth, Item 10, Appendix D, as follows:
 - (1) Repair AREA A as follows:
 - (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP.**
 - (b) If damage or repairs are more than limit, replace wheel (PARA 3-4-28).
 - (2) Repair AREA B as follows:
 - (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP AND 1.00 SQUARE-INCH IN AREA.**
 - (b) If damage or repairs are more than limits, replace wheel (PARA 3-4-28).
 - (3) Repair AREA C as follows:
 - (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.031-INCH DEEP AND 1.00 INCH-LONG.**
 - (b) If damage or repairs are more than limits, replace wheel (PARA 3-4-28).
 - (4) Repair AREA D as follows:
 - (a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.062-INCH DEEP AND 1.00 SQUARE-INCH IN AREA.**
 - (b) If damage or repairs are more than limits, replace wheel (PARA 3-4-28).



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FIGURE 3-5-11-1. REPAIR TAIL LANDING GEAR WHEEL (BENCH) (SHEET 2 OF 2)

(5) Repair AREA E as follows:

(a) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.015-INCH DEEP AND 1 INCH-LONG.**

(b) If damage or repairs are more than limits, replace wheel (PARA 3-4-28).

(6) Repair AREA F as follows:

(a) Remove bearing cups (PARA 3-5-11.2.1).

(b) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS.**

(c) **REPAIR AFTER BLENDING AND POLISHING MUST NOT AFFECT BEARING CUP RETENTION.**

(d) If damage or repairs effect bearing cup retention, replace wheel (PARA 3-4-28).

(e) Install bearing cups (PARA 3-5-11.2.2).

d. Using fluorescent penetrant inspection kit , fluorescent penetrant inspect wheel halves for cracks and structural damage (ASTM E1417). **NONE ALLOWED.** Pay particular attention to bead seat, bolt boss, male/female mating surfaces, and valve hole areas. If crack is found, replace wheel.

e. Using corrosion resistant coating, Item 118, Appendix D, touch up blended area.

f. Using epoxy primer coating kit, Item 135, Appendix D, prime wheel.

NOTE

Male/female mating surfaces and bolt bosses need primer only.

g. Using polyurethane paint, Item 222A, Appendix D, paint wheel (PARA 2-6-5).

h. Check bearing cups for scratches, pitting, corrosion, and signs of overheating (Figure 3-5-11-1, Sheet 1). **NONE ALLOWED.** Replace damaged cup (PARA 3-5-11.2).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Do not allow bearings to spin while drying.

i. Using dry-cleaning solvent, Item 124, Appendix D, clean bearings. Dry bearings with compressed air, if available, or let air-dry.

j. Inspect bearings for scratches, water stains, heat discoloration, worn rollers, or cracked or distorted bearing cage. **NONE ALLOWED.** If any of these conditions are found, replace bearing.

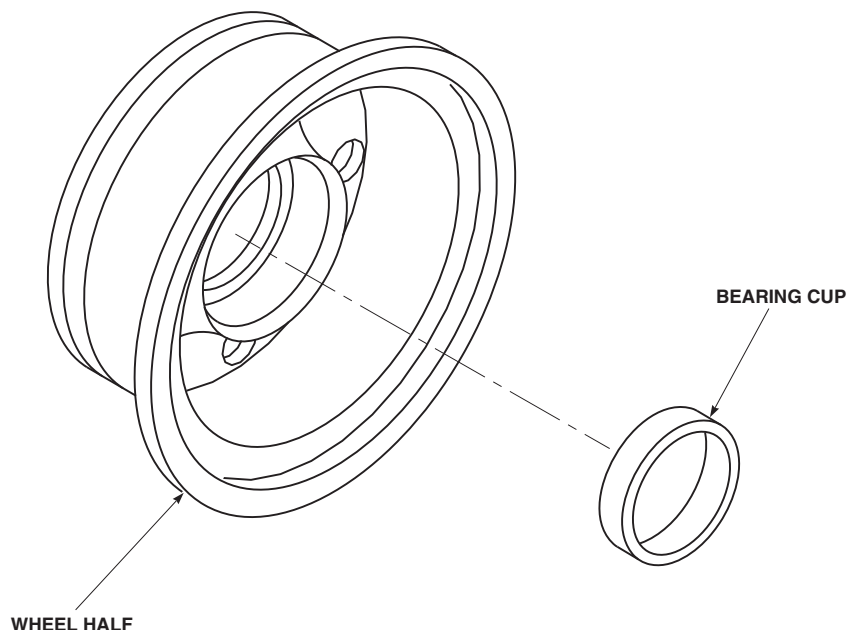
k. If bearing is determined serviceable, repack it with grease, Item 149, Appendix D.

NOTE

New bearings packaged in preservative do not need degreasing before being packed with grease.

l. Inspect bearing seals for damaged reinforcing rings, visible damage, or deformation. Replace any seal found unserviceable.

m. Inspect wheel bolts for thread damage and cracks under boltheads. Using magnetic



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FIGURE 3-5-11-2. REPLACE WHEEL BEARING CUP (BENCH)

particle inspection kit (ASTM E1444), inspect all wheel bolts. Replace any cracked or damaged bolts.

- n. Inspect nuts for obvious damage. Thread bolts into nuts by hand. Replace any nut that allows bolt to thread past top of nut.

3-5-11.1.3 Assemble.

Install tire on tail landing gear wheel (PARA 3-4-28).

3-5-11.2 Replace Wheel Bearing Cup (BENCH).

3-5-11.2.1 Remove.

CAUTION

Damage to wheel half will result if overheated. Overheating will affect service life of wheel. Do not overheat wheel half or heat longer than required.

- a. Heat wheel half in an oven not over 300°F (149°C), or place in boiling water for 30 minutes.

CAUTION

- Be careful when removing bearing cup to avoid raising burrs in hub bore.
- Support wheel half on hub and not on flange.
- b. Remove wheel half from oven and immediately remove cup from bore (Figure 3-5-11-2). After heating wheel, bearing cup should be loose enough to fall out of bore when wheel is turned upside down. If cup does not fall out, tap cup from bore evenly with phenolic drift pin and mallet.
- c. Carefully remove any primer or other foreign matter from hub bore. Make sure bearing bore and shoulder are free of burrs.

3-5-11.2.2 Install.

- a. Place replacement bearing cup in a container of dry ice, Item 125, Appendix D.

CAUTION

Damage to wheel half will result if overheated. Overheating will affect service life of wheel. Do not overheat wheel half or heat longer than required.

- b. Heat wheel half in an oven not over 300°F (149°C), or place in boiling water for 30 minutes.
- c. Remove wheel half from heat source.
- d. Remove bearing cup from dry-ice. Make sure cup is thoroughly dry.

CAUTION

Damage to wheel half will result if supported on flange. Support wheel half on hub and not on flange.

- e. Using primer, Item 257, Appendix D, coat wheel contacting surface of bearing cup. Install cup into hub bore while primer is still wet. Use phenolic drift pin and a mallet to tap cup into bore. Make sure cup is against shoulder and avoid cocking cup during installation.
- f. Allow wheel to cool to room temperature. Apply light coat of grease, Item 149, Appendix D, to bearing cup to prevent rust and corrosion.